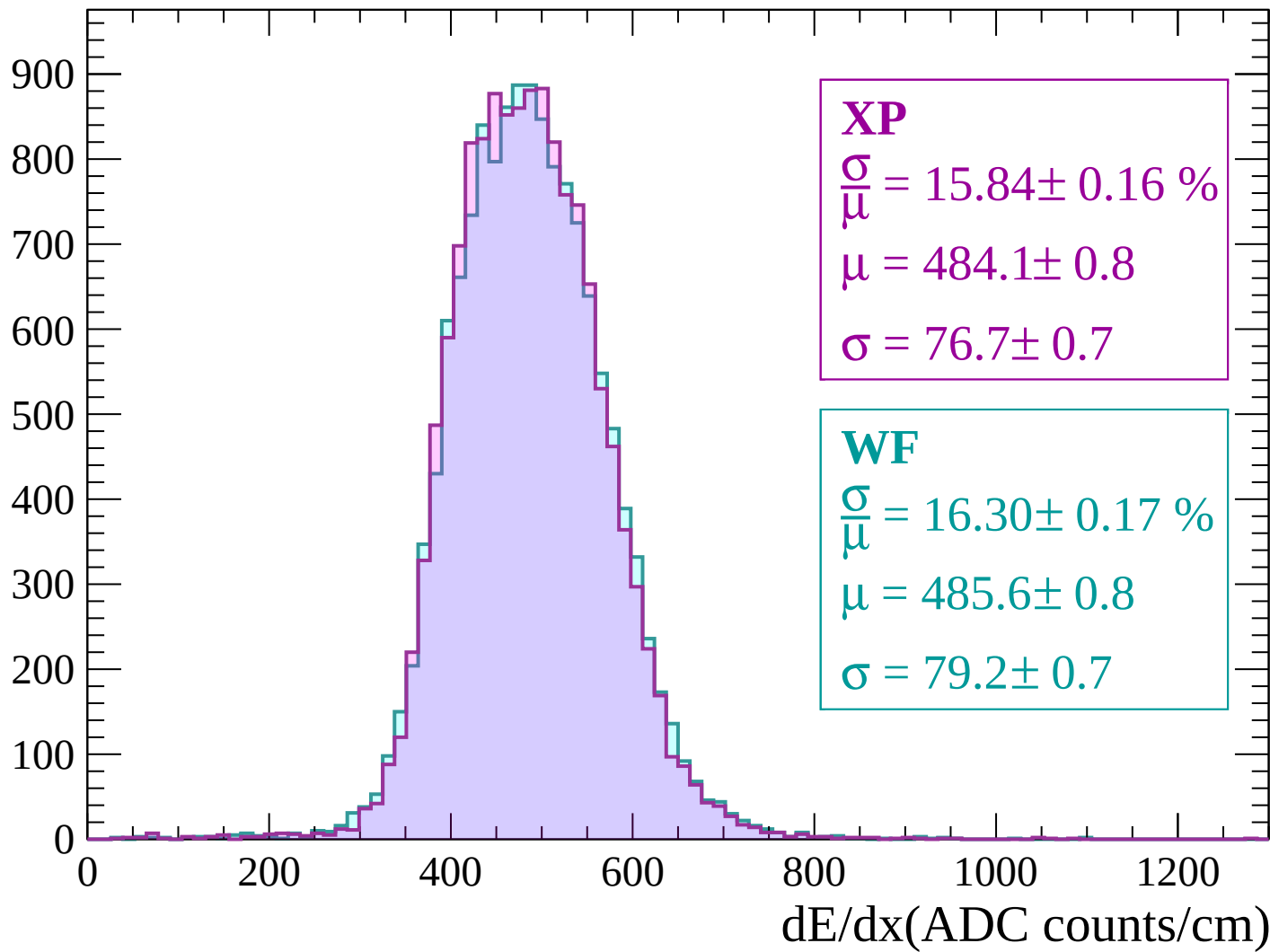
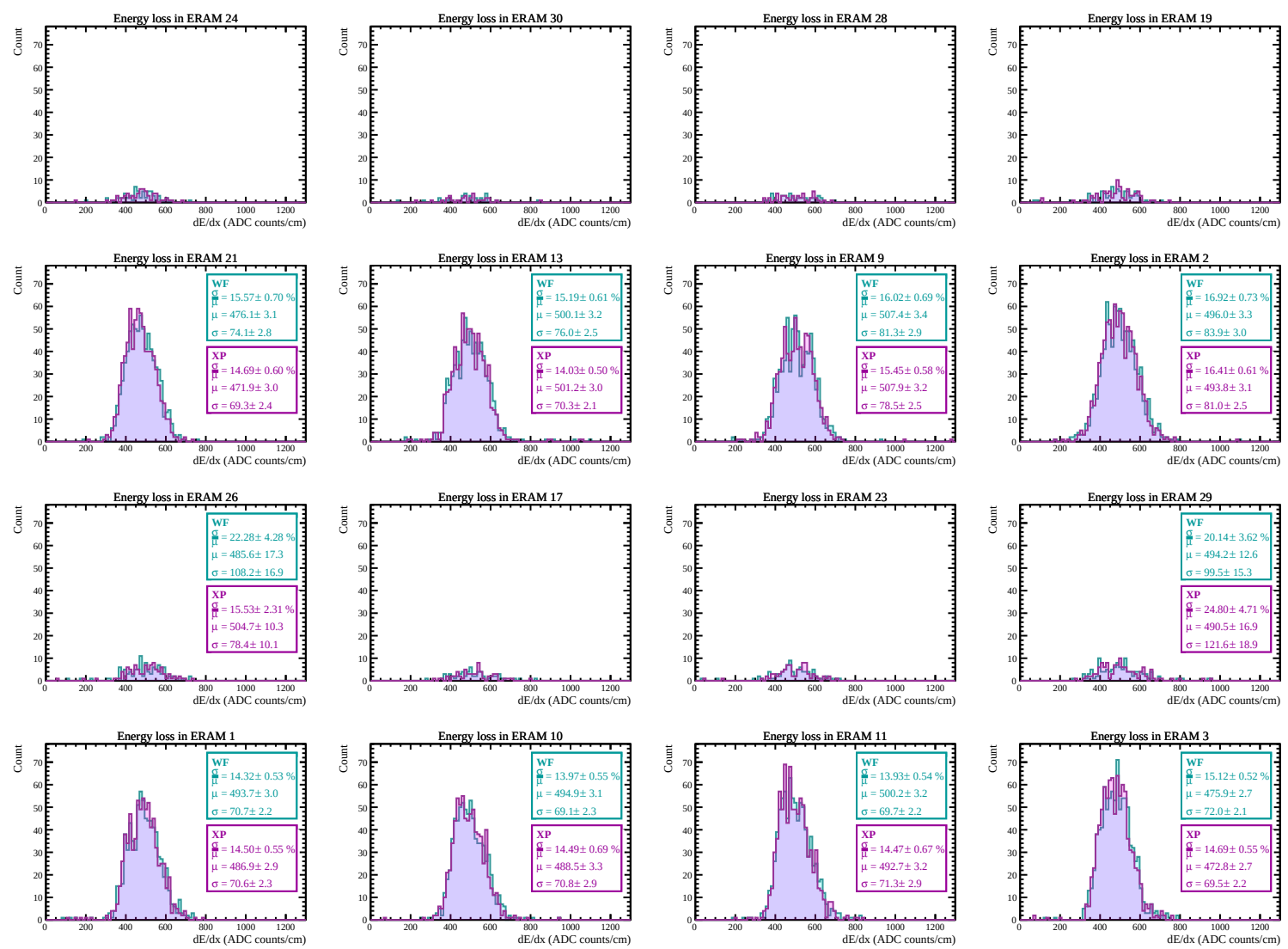
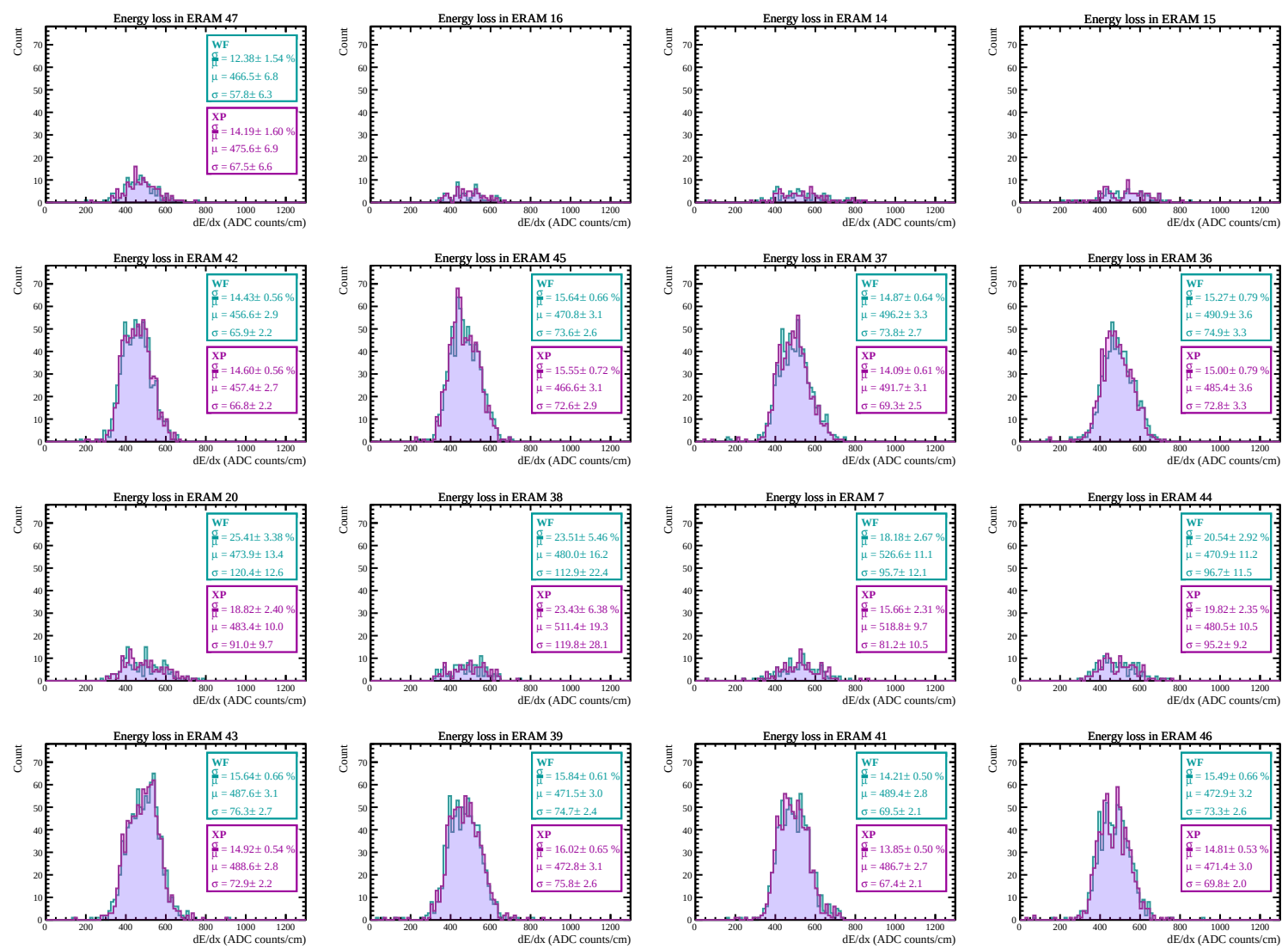
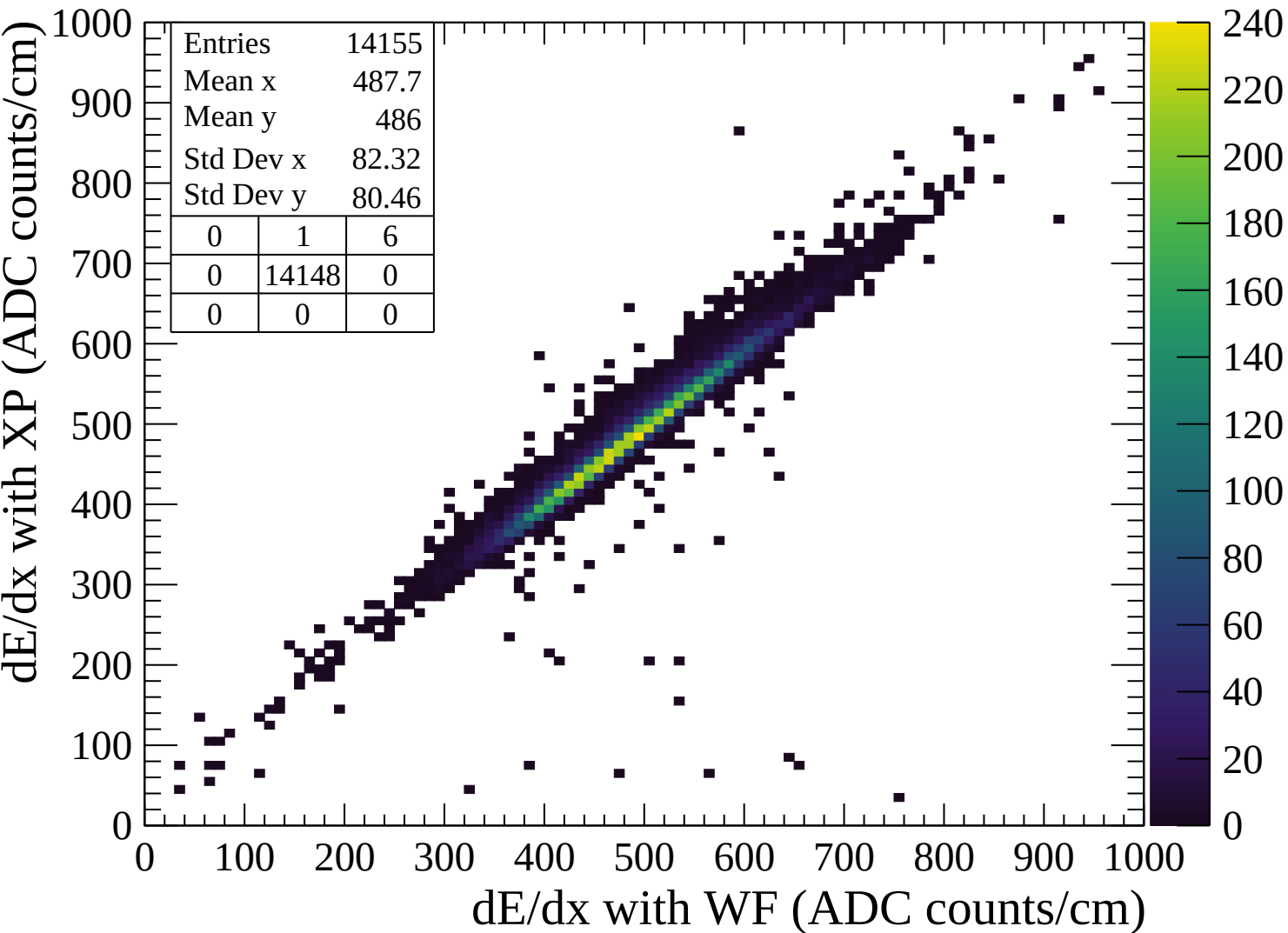


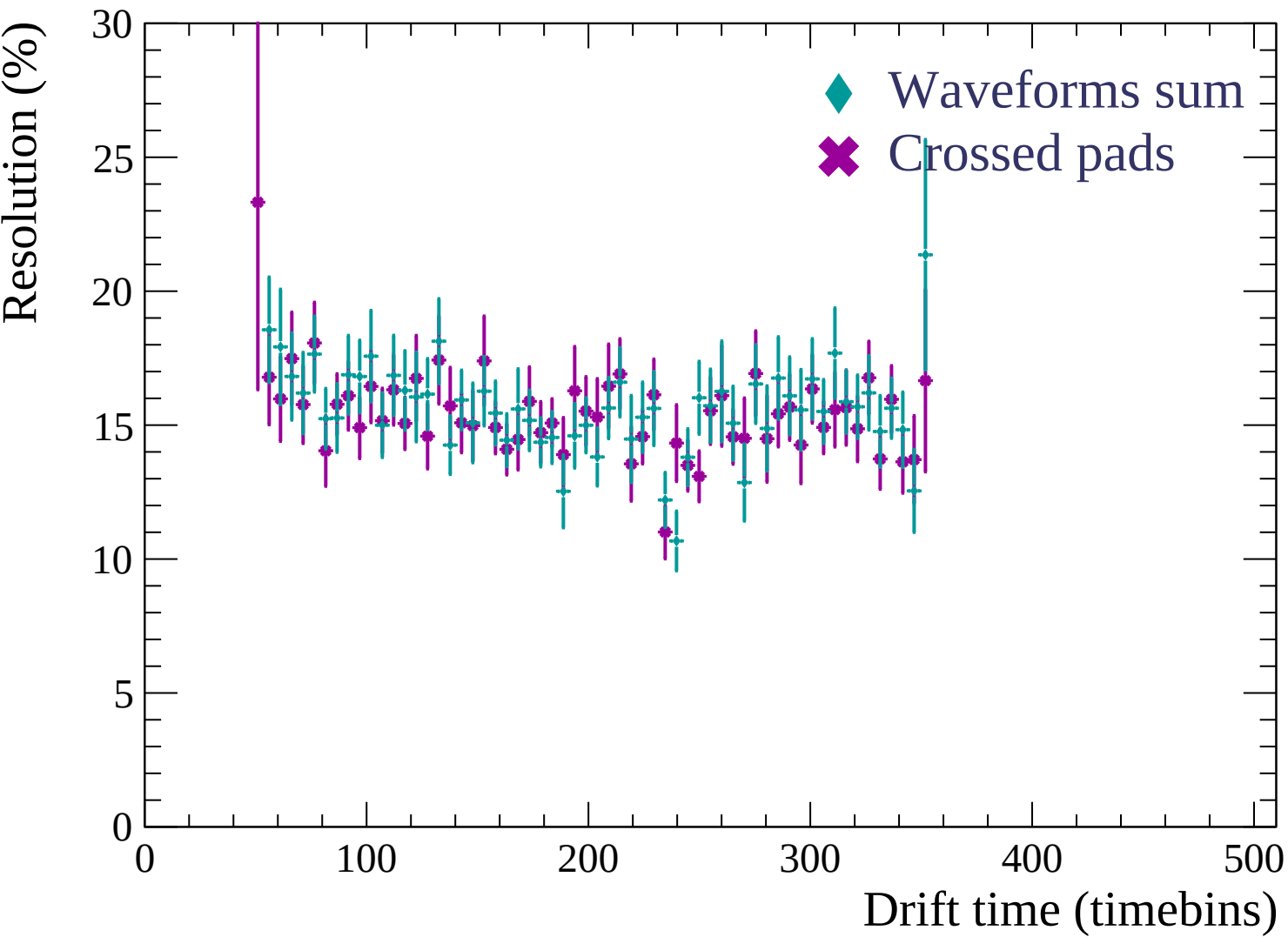
Count

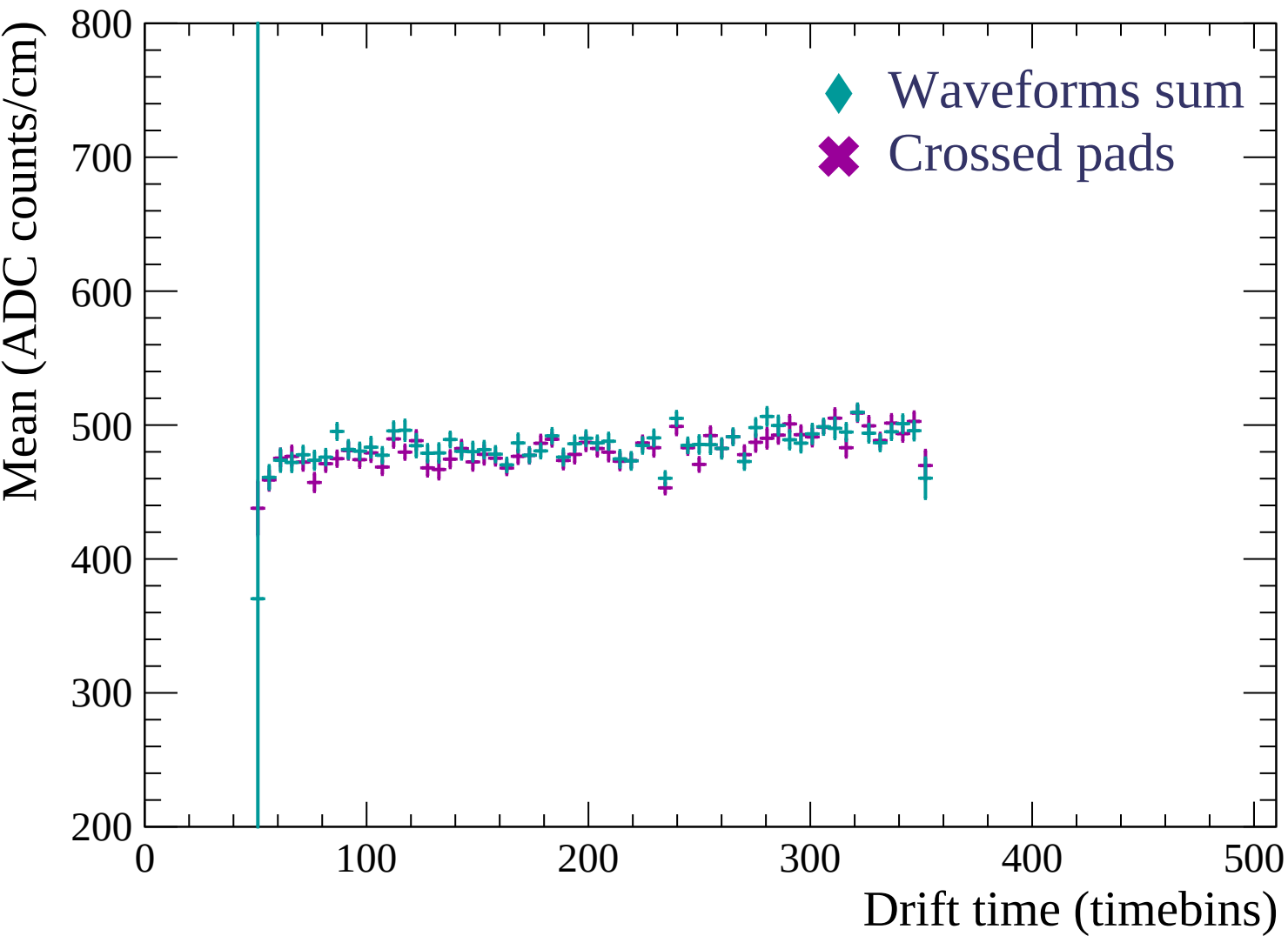


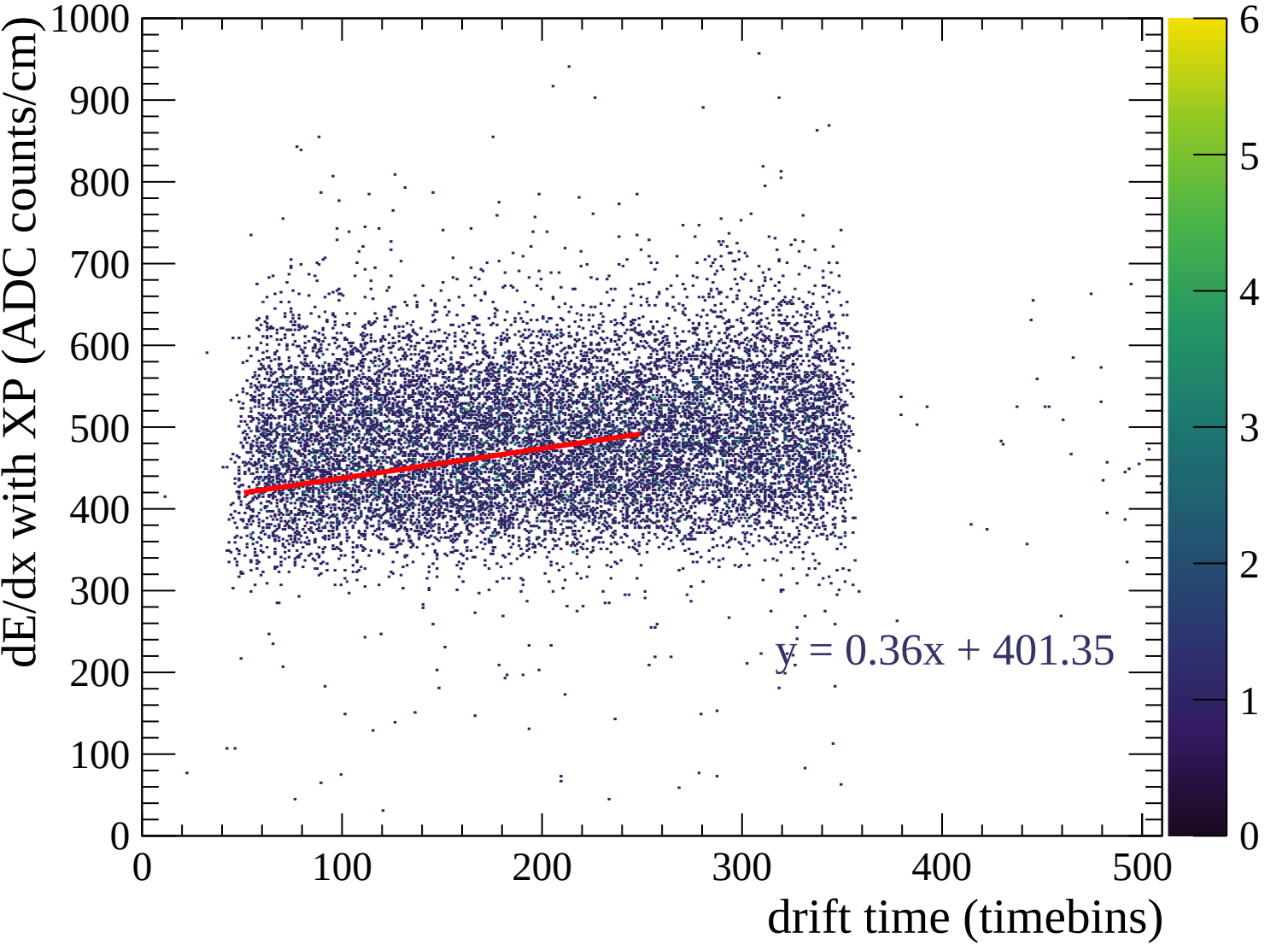


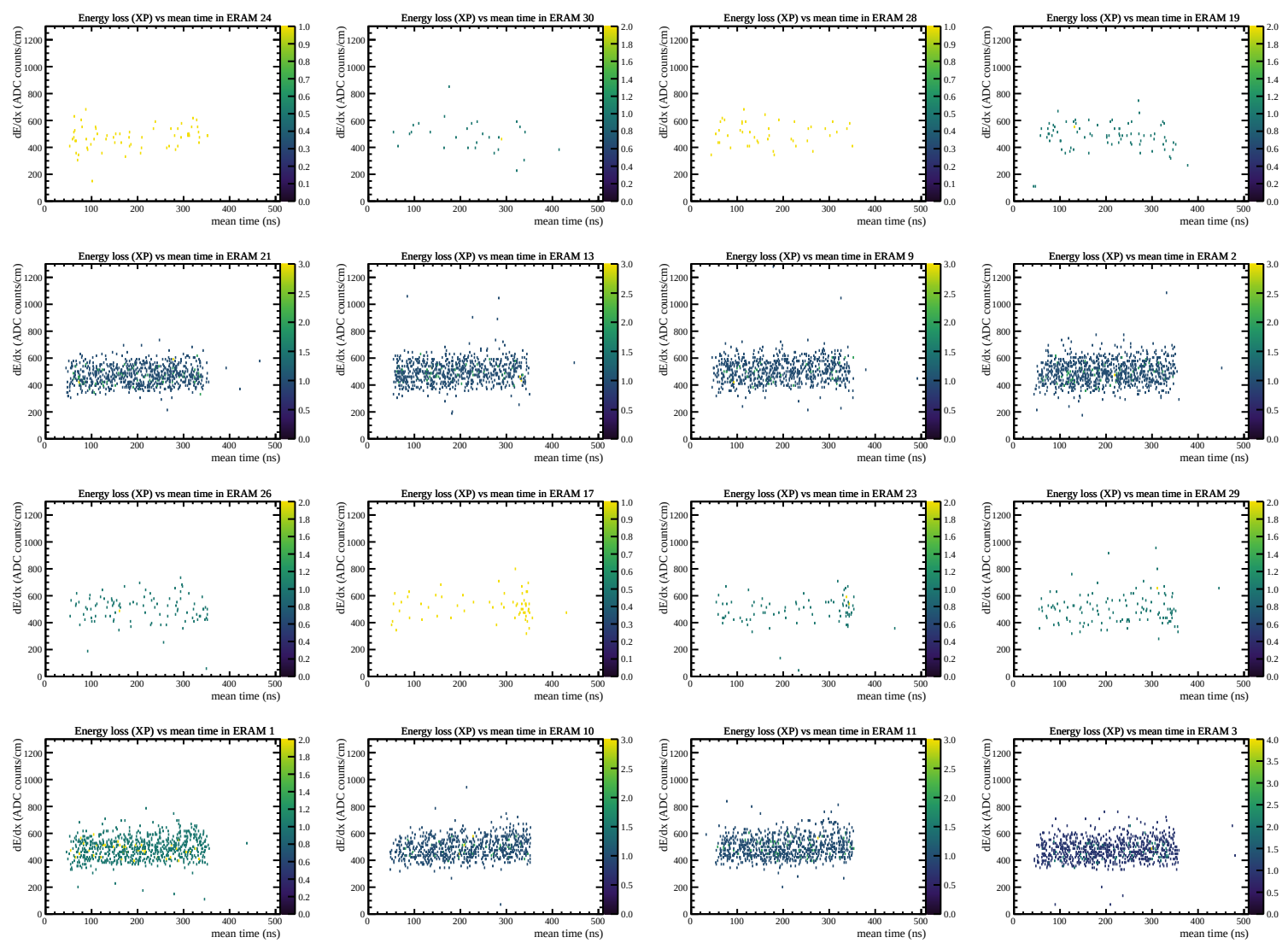


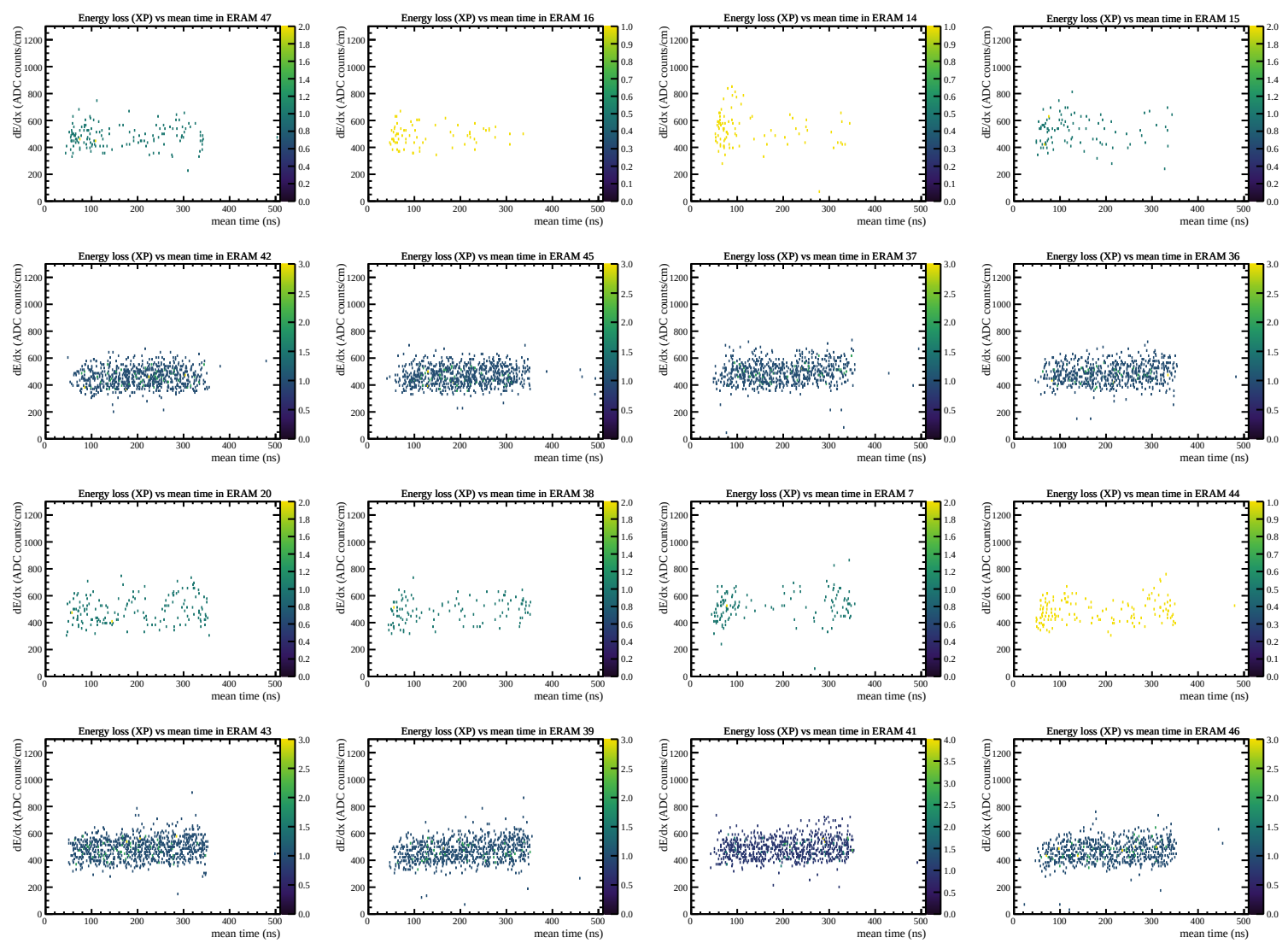


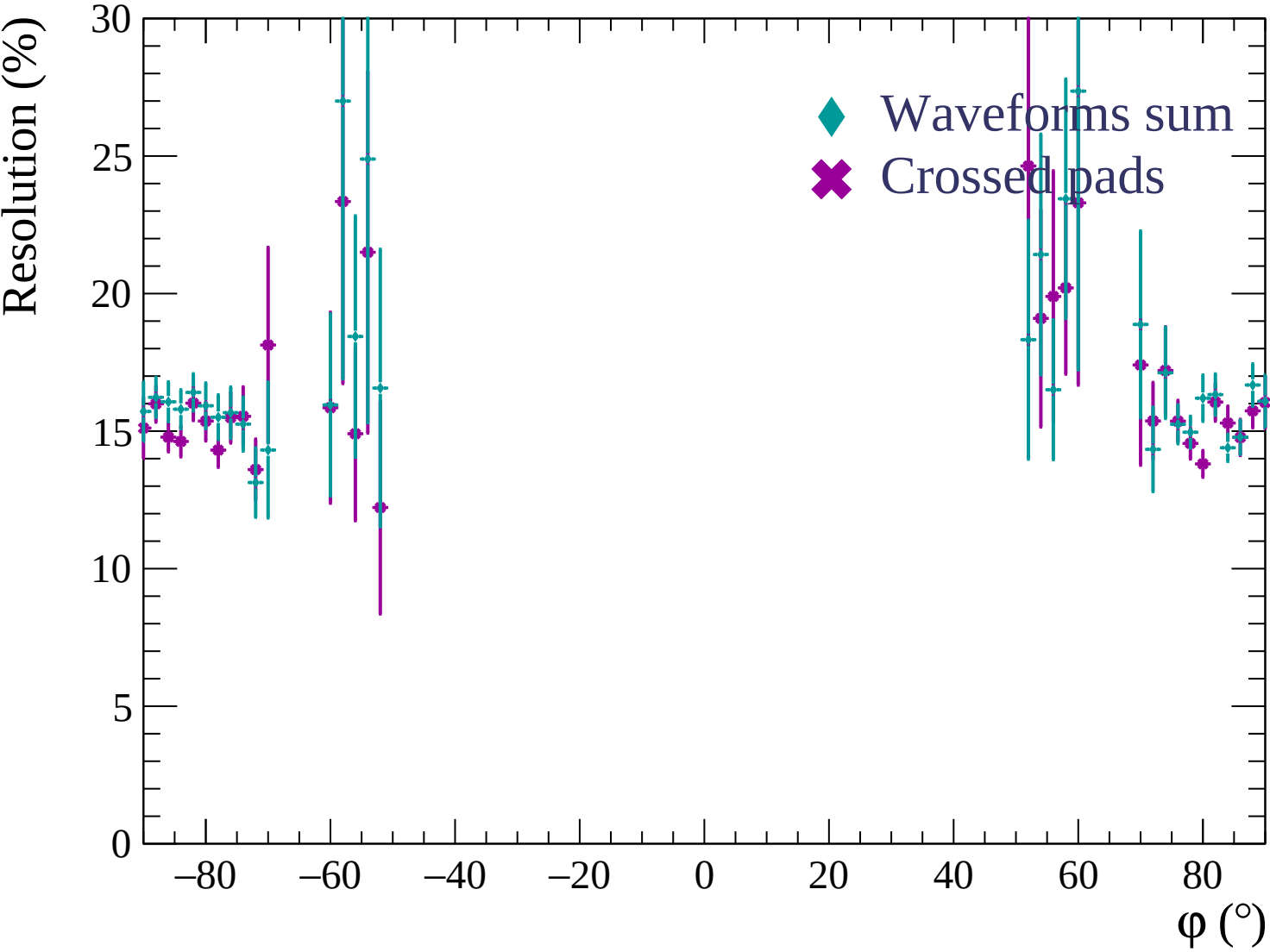


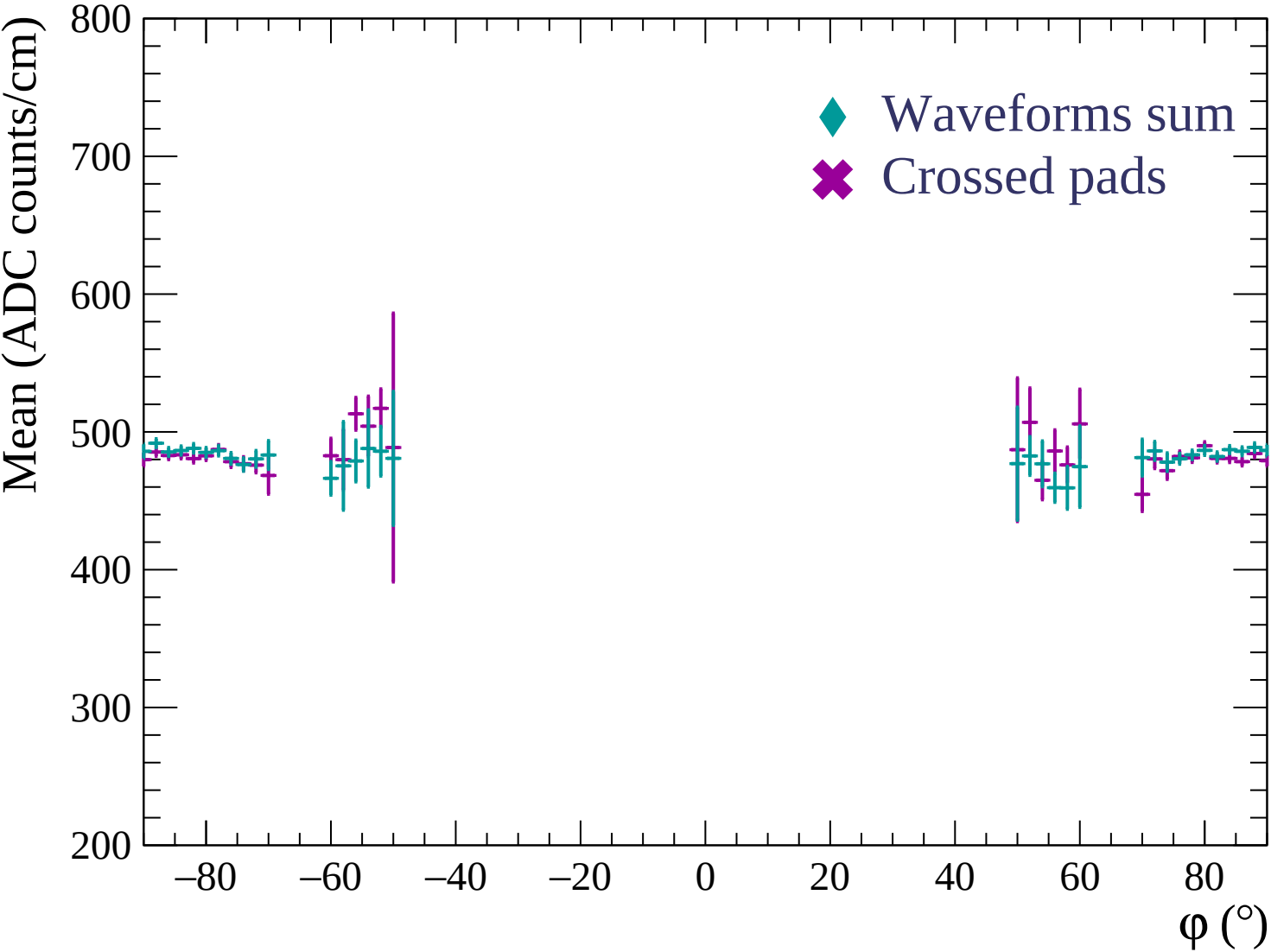


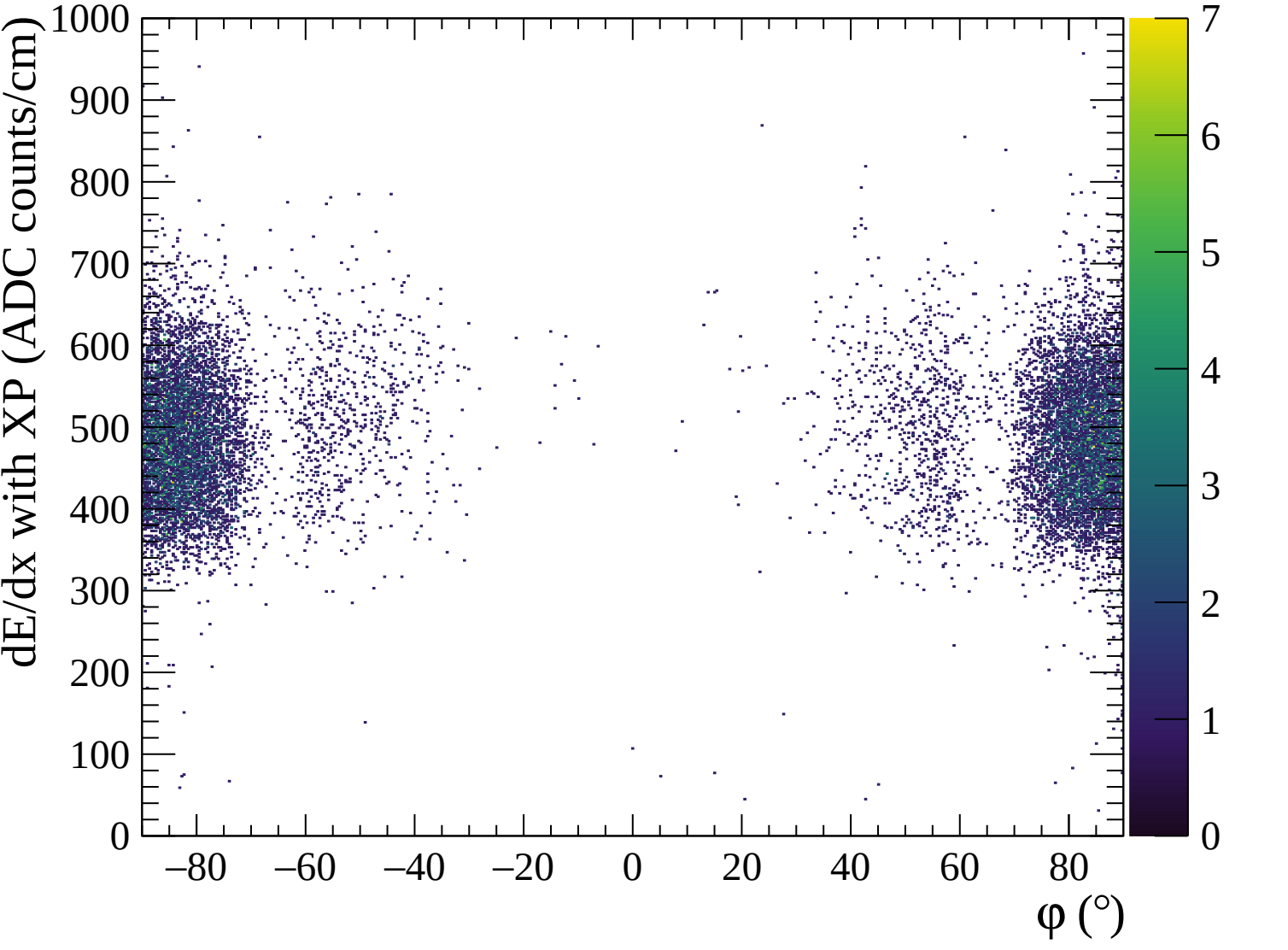


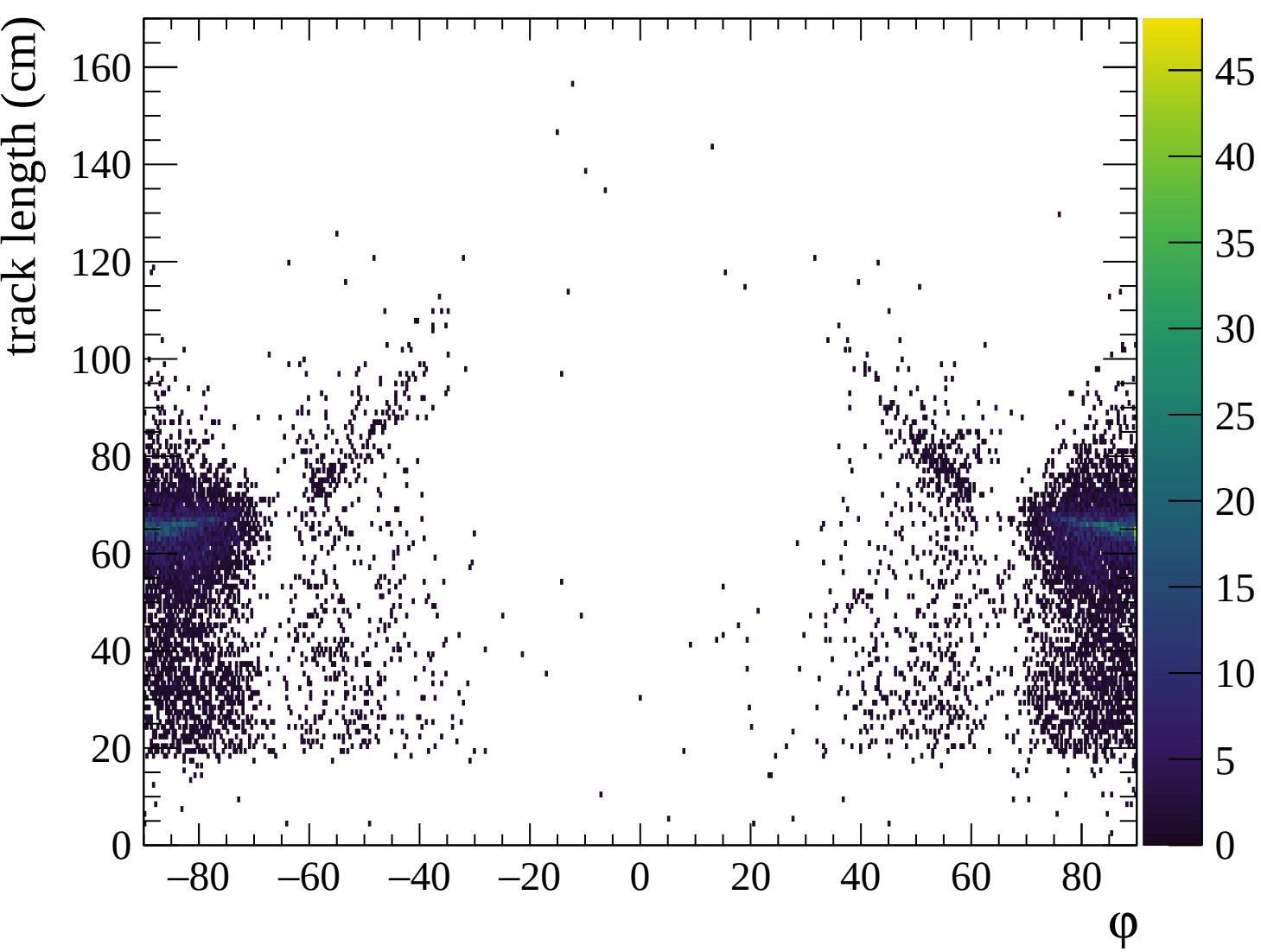


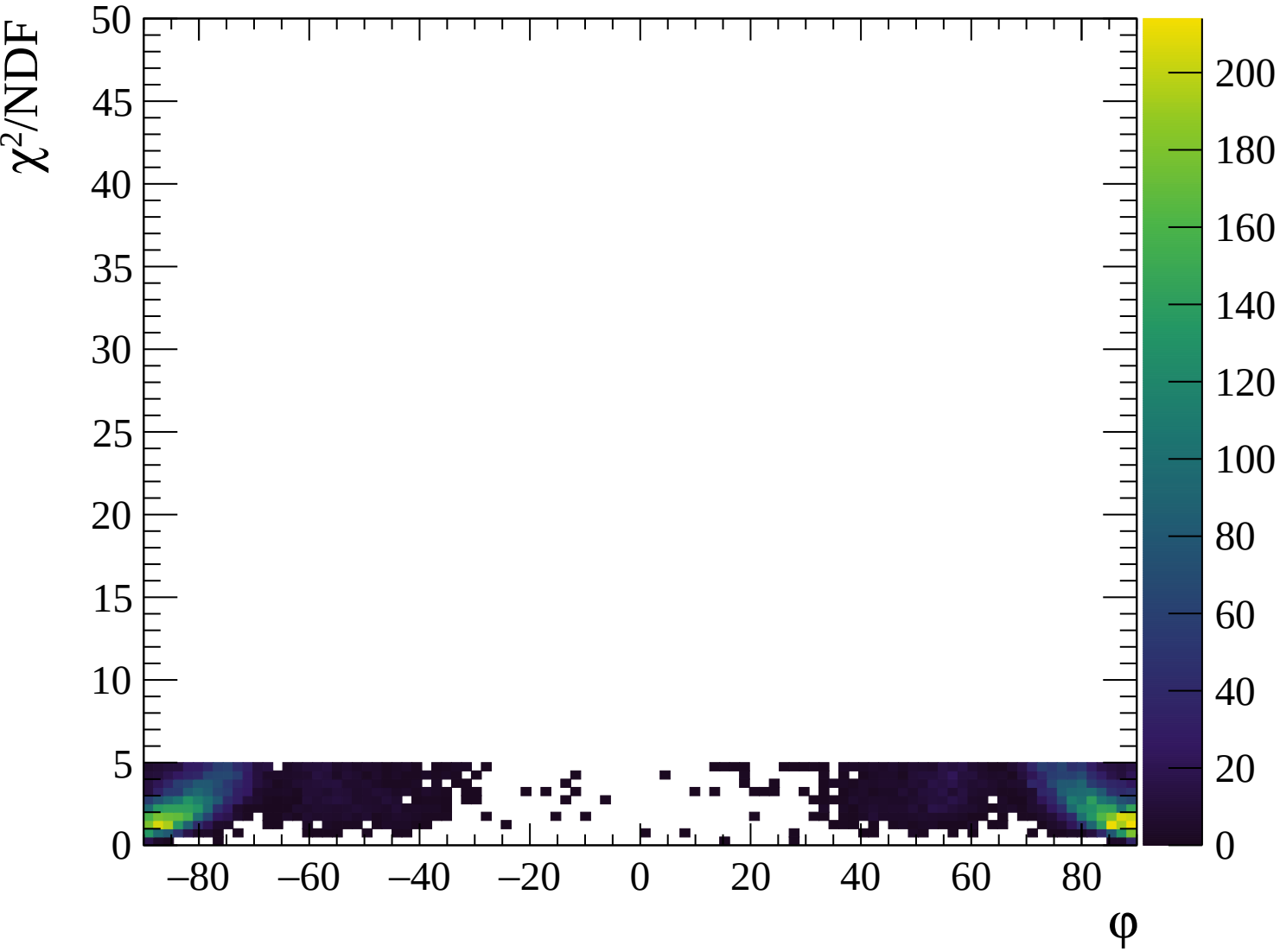


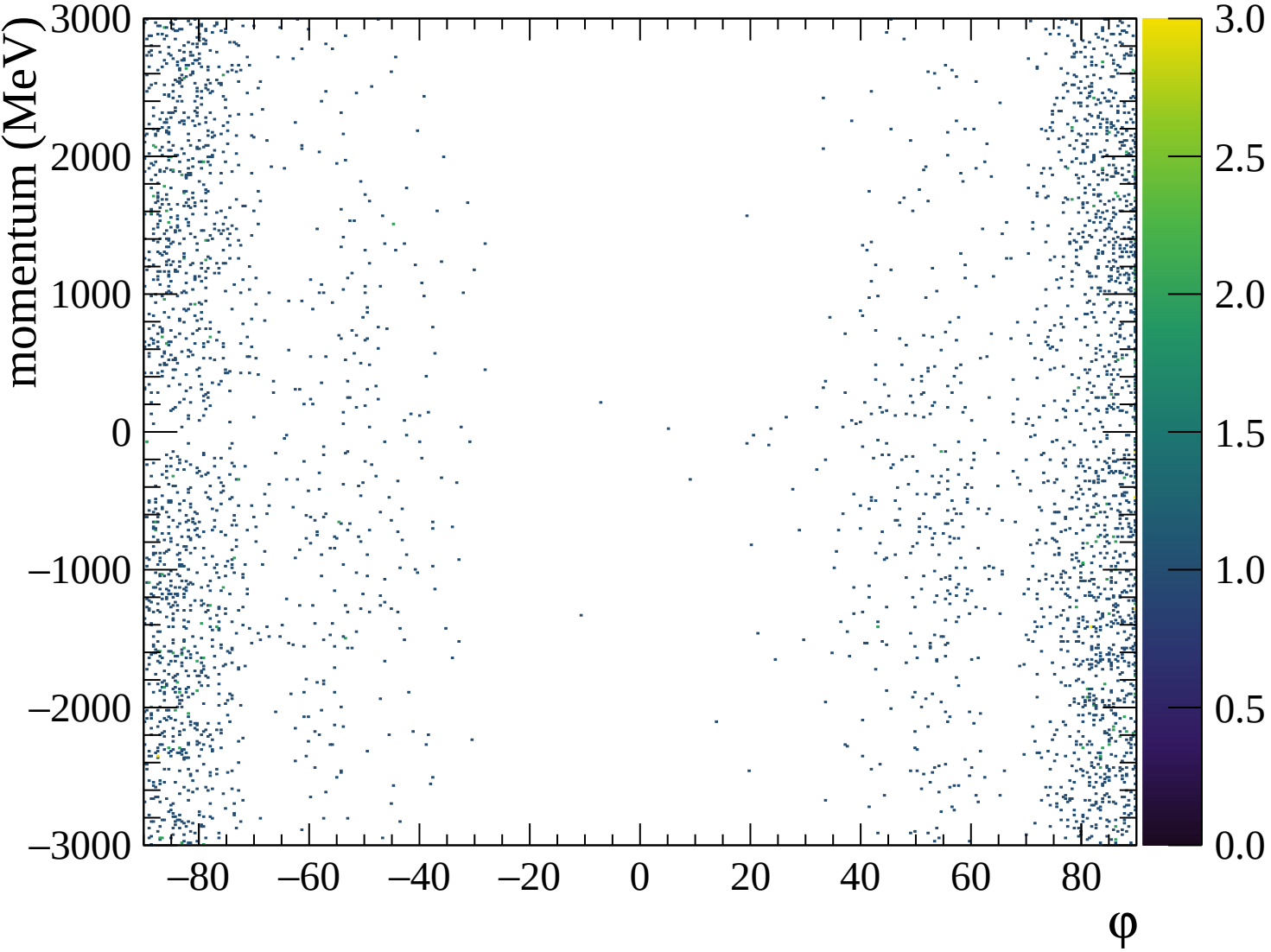


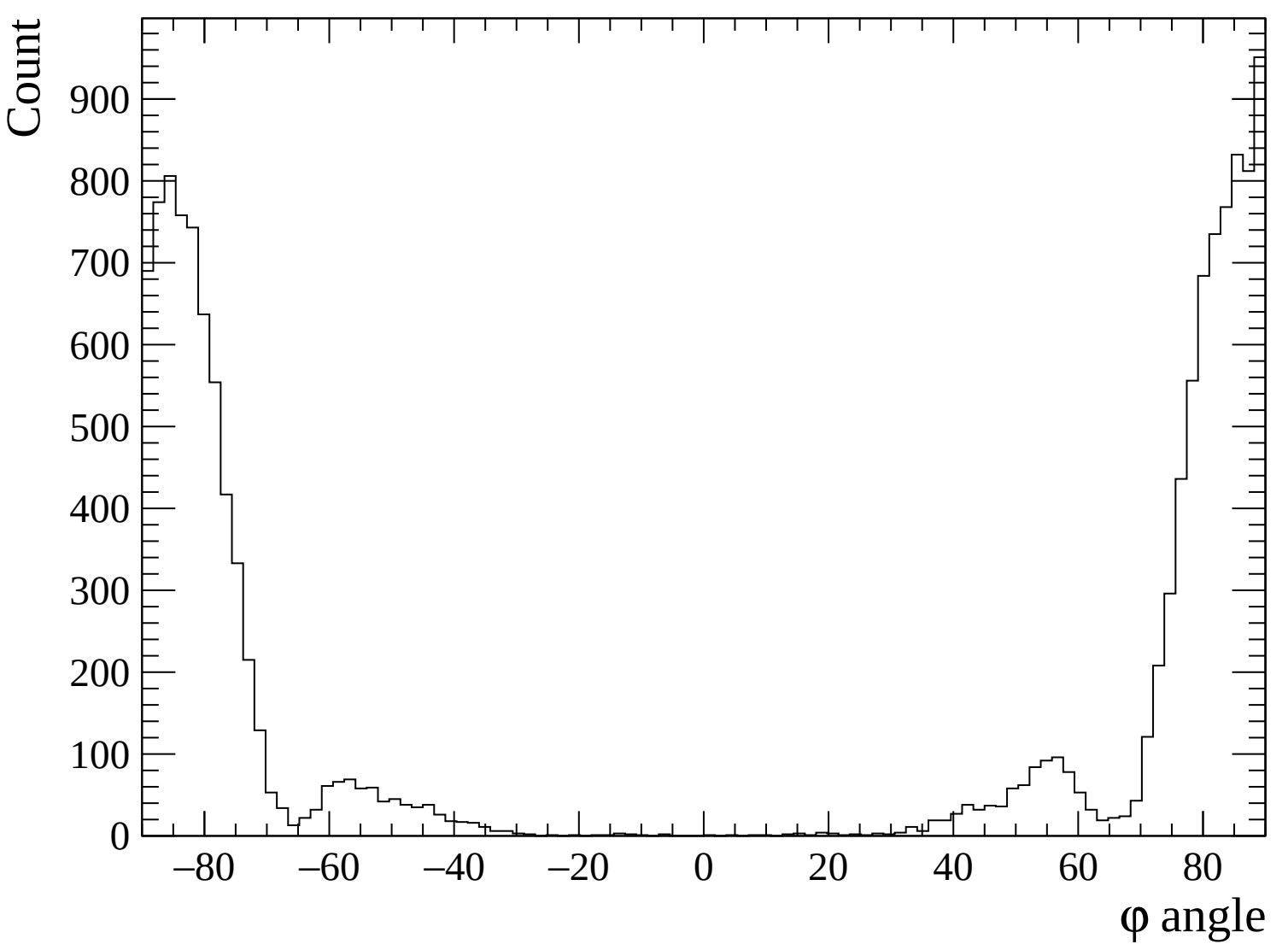


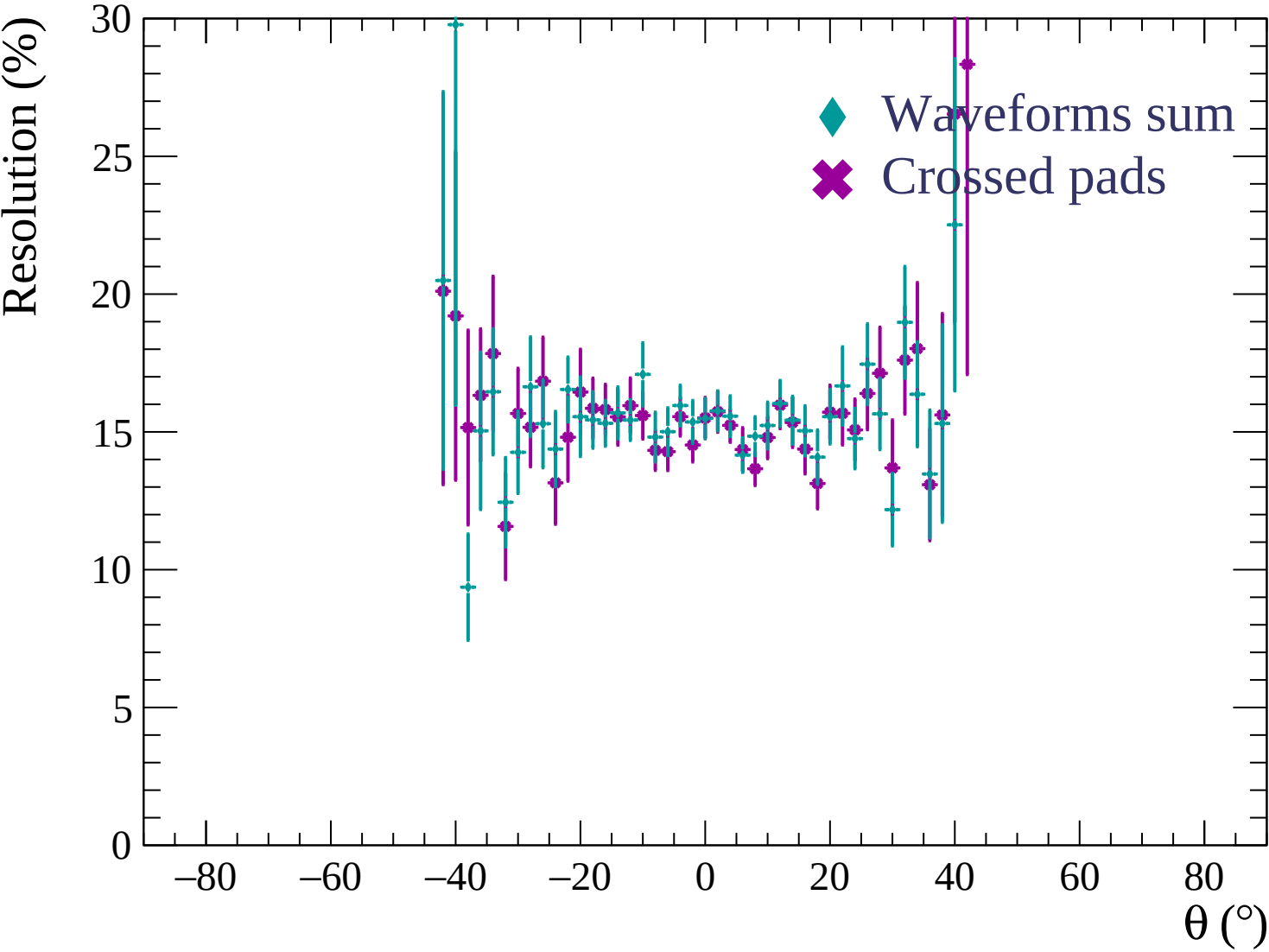


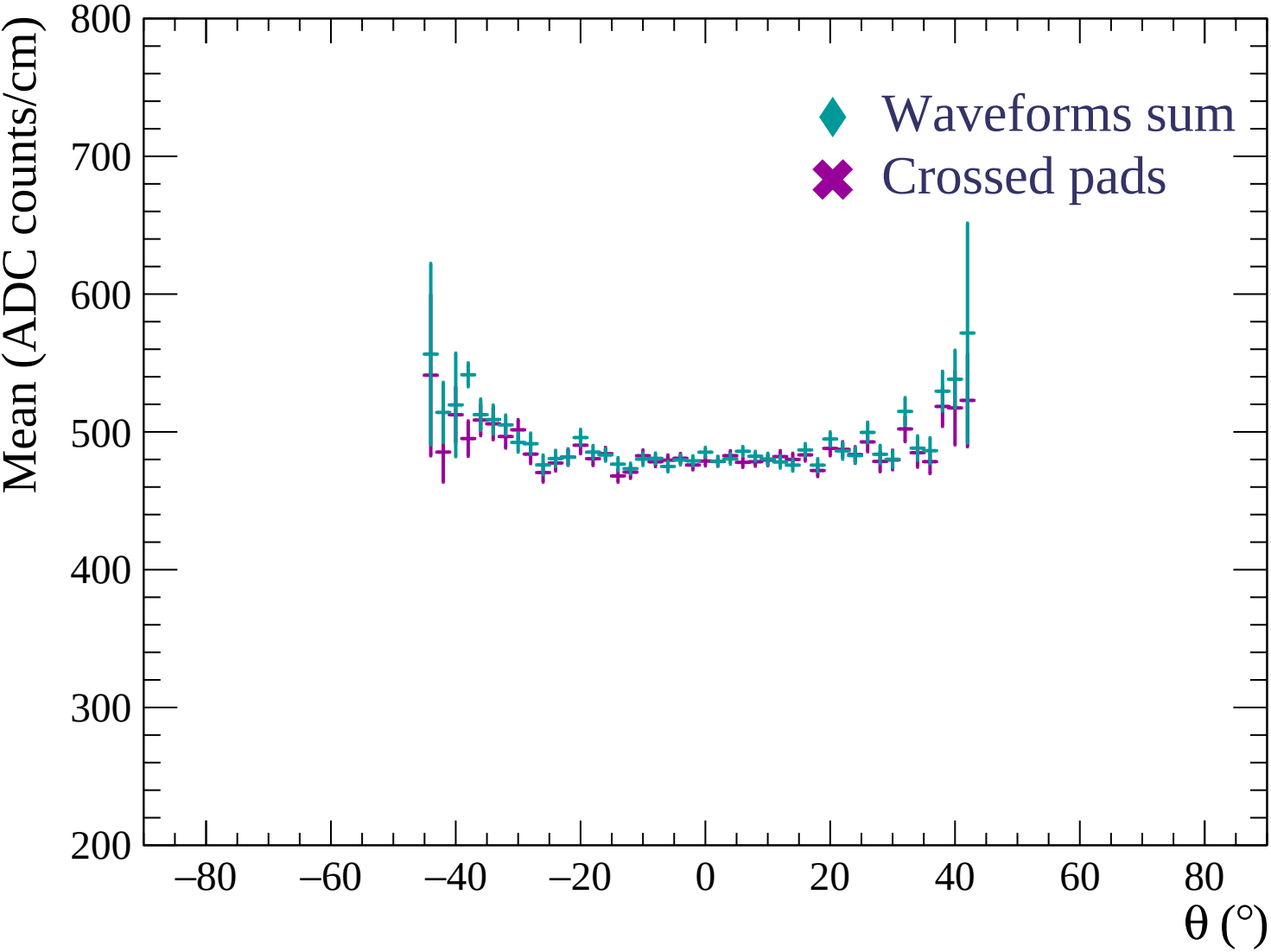


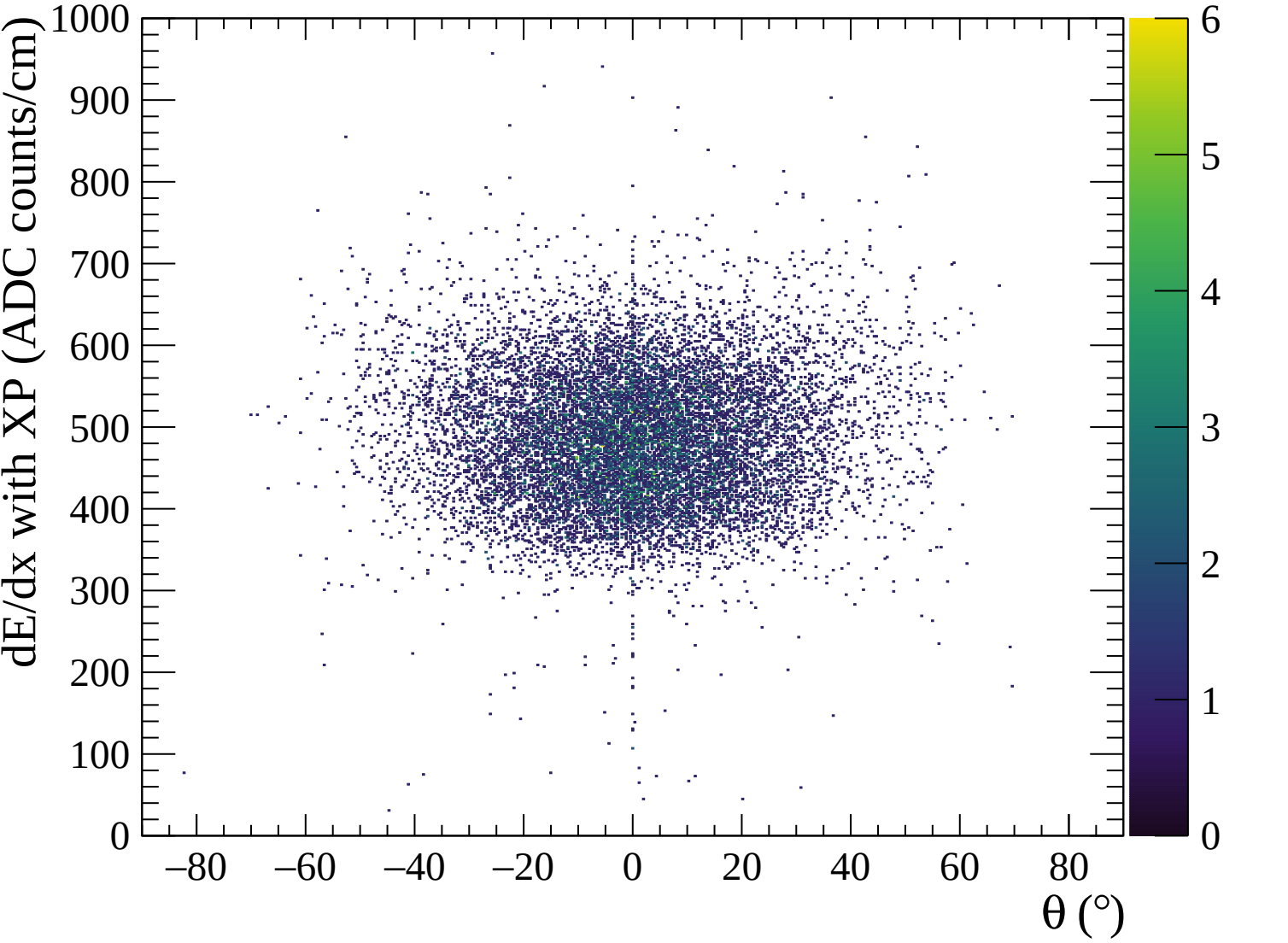


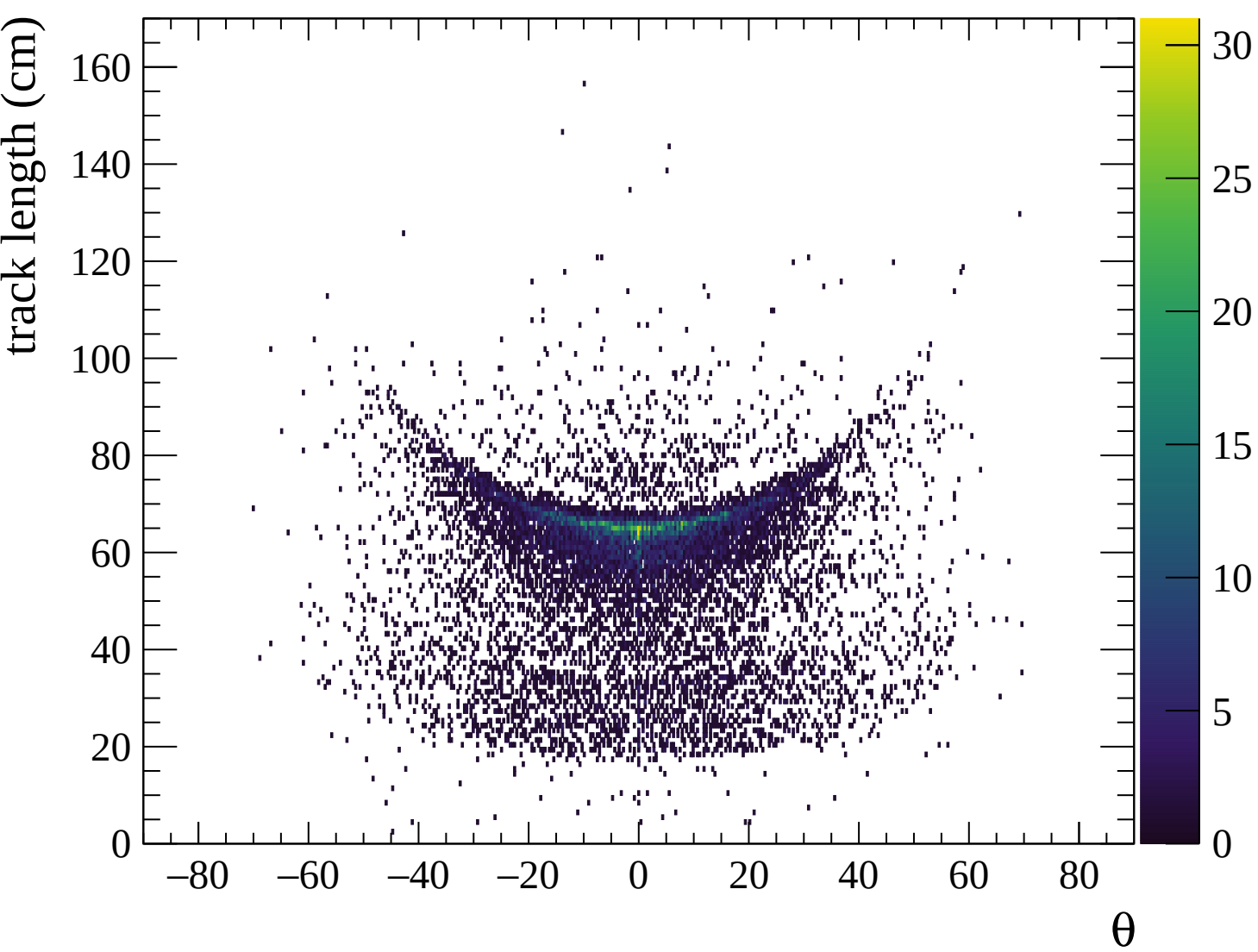


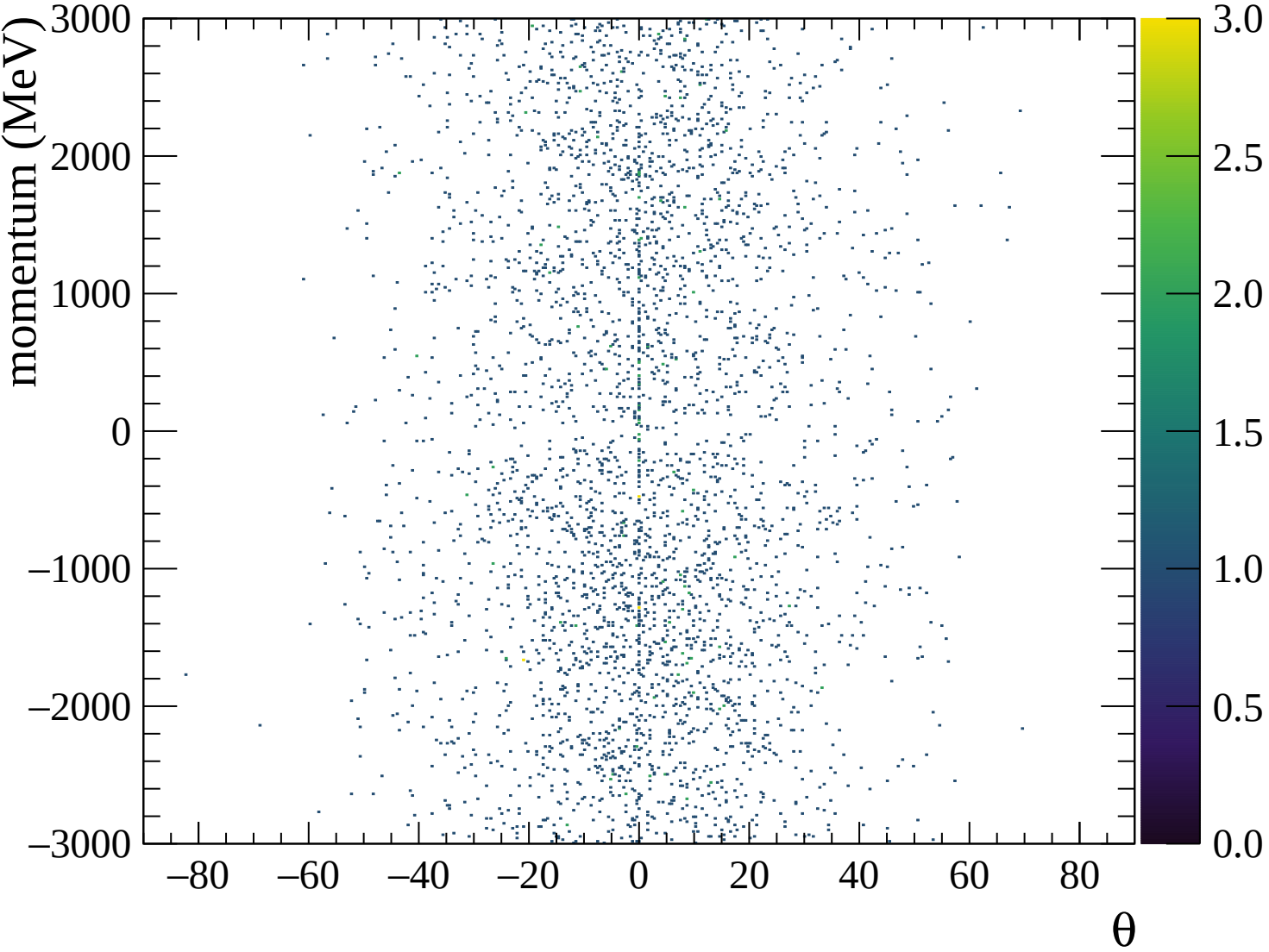


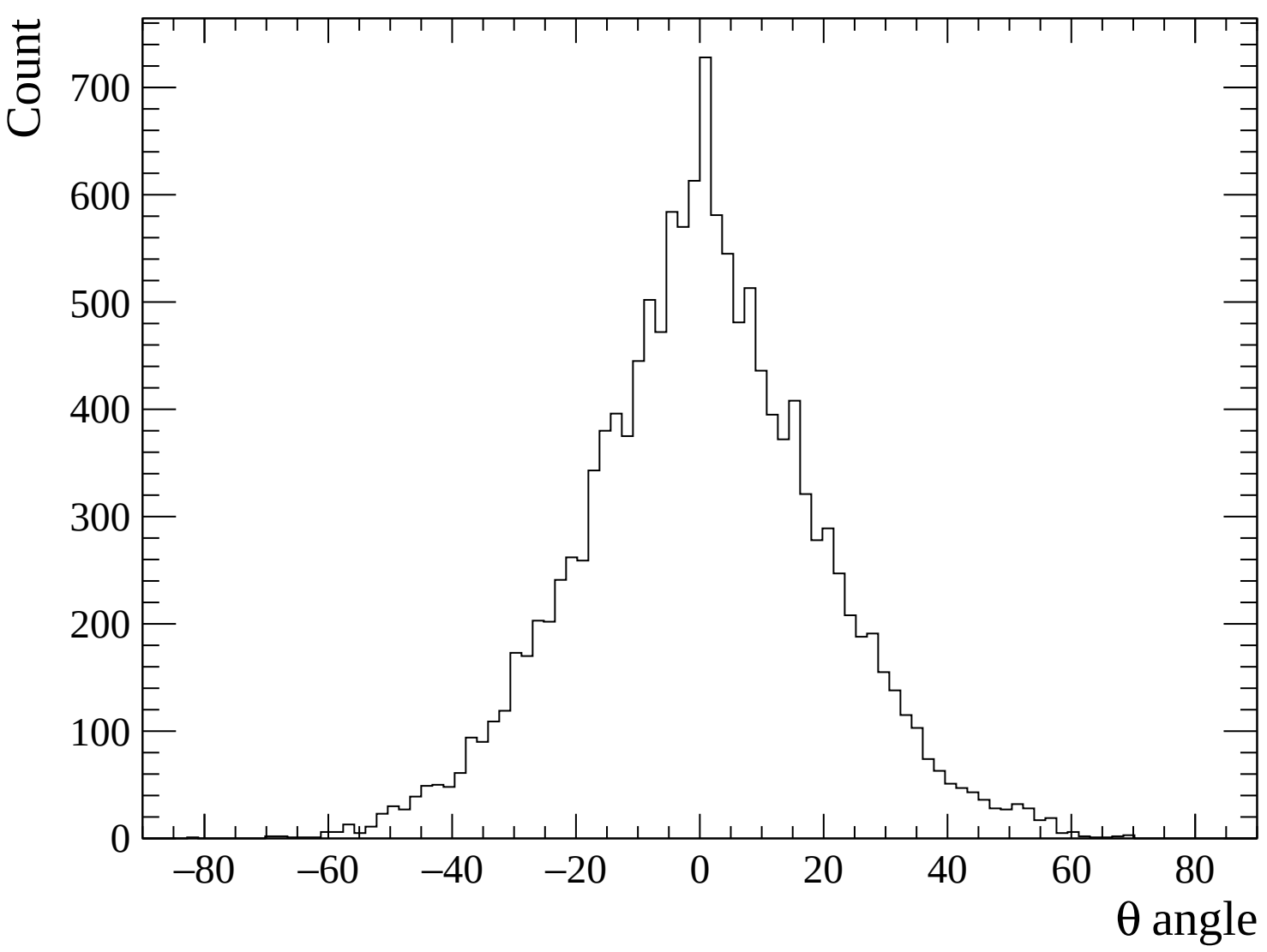


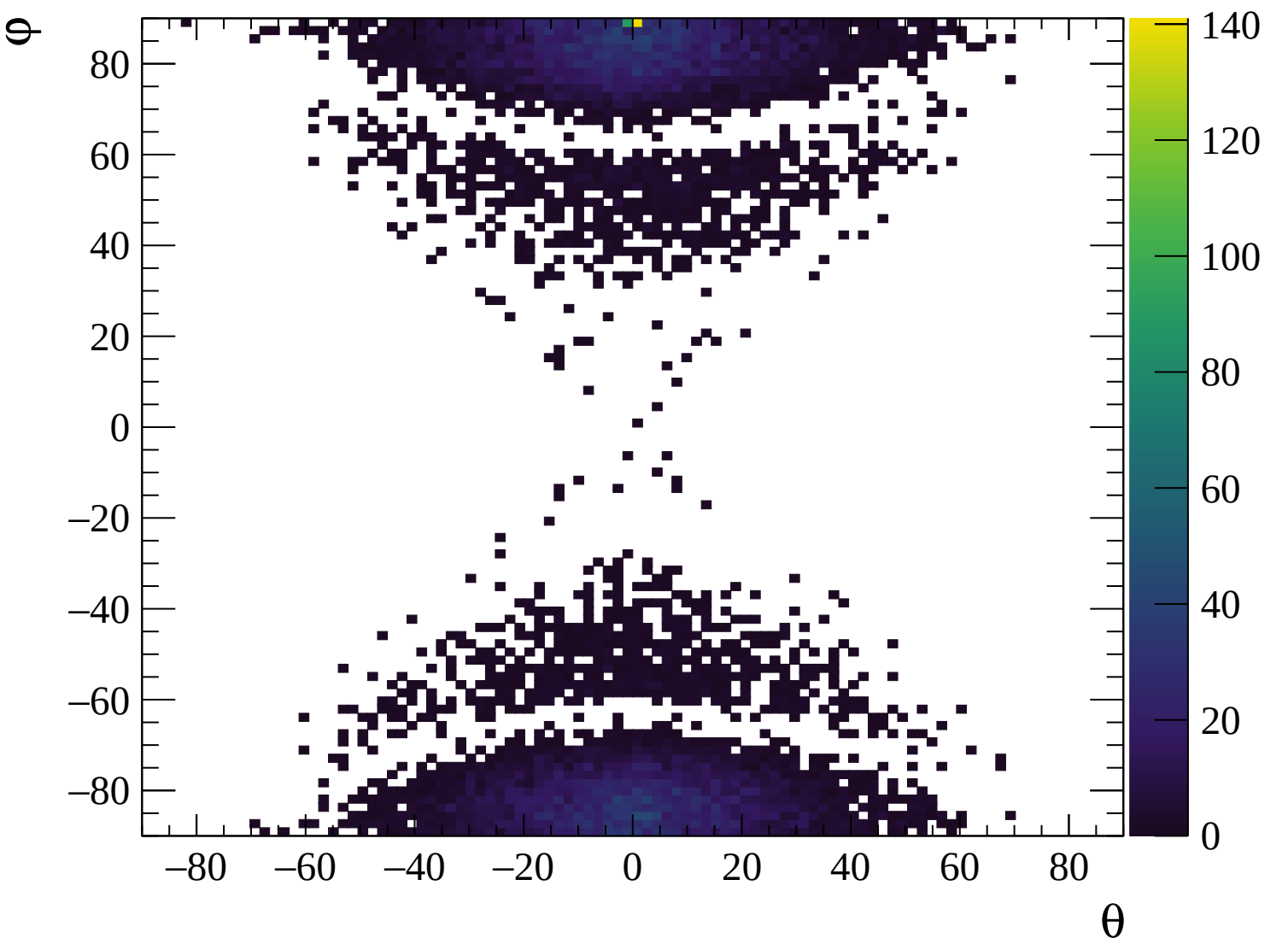


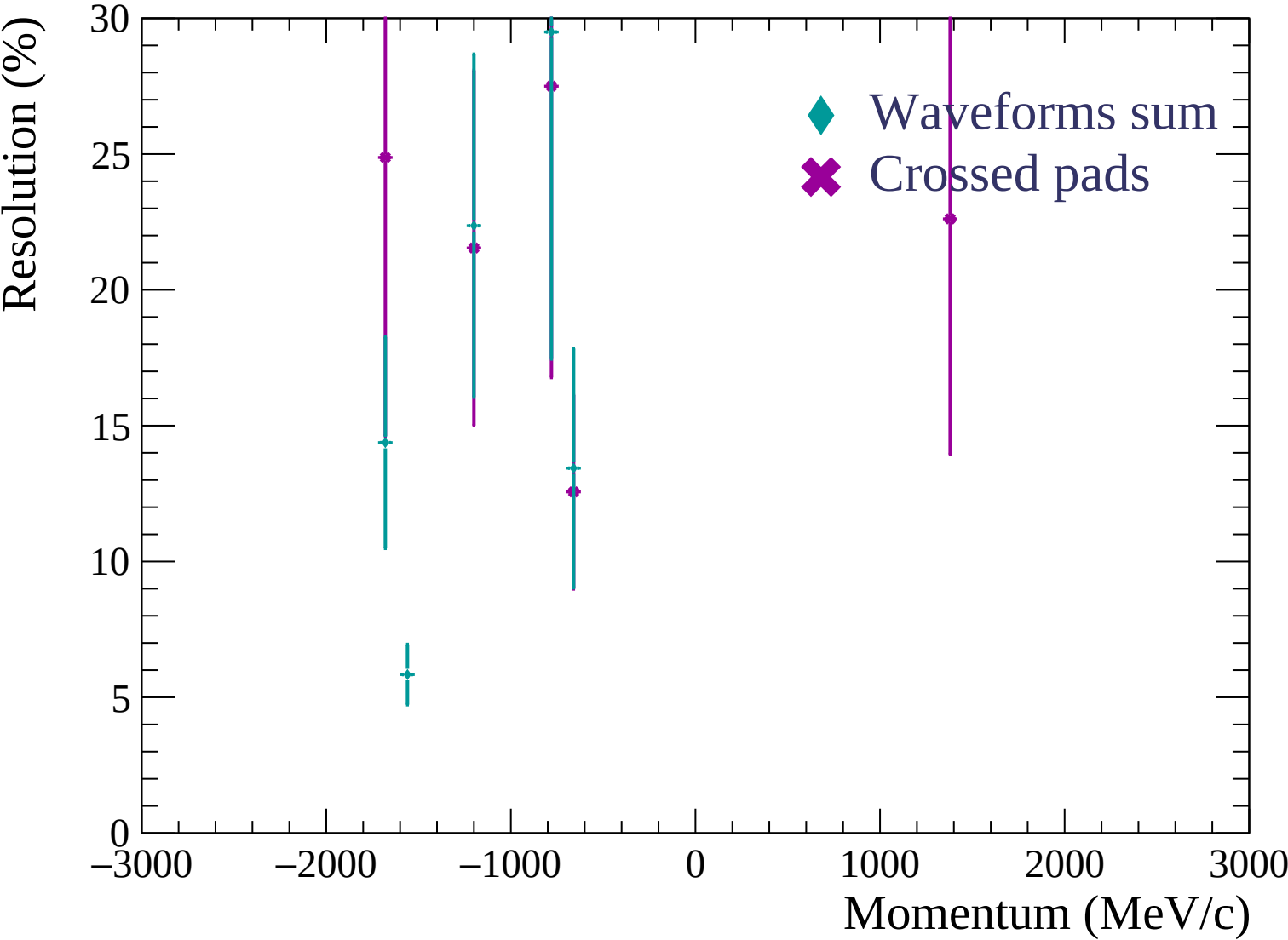


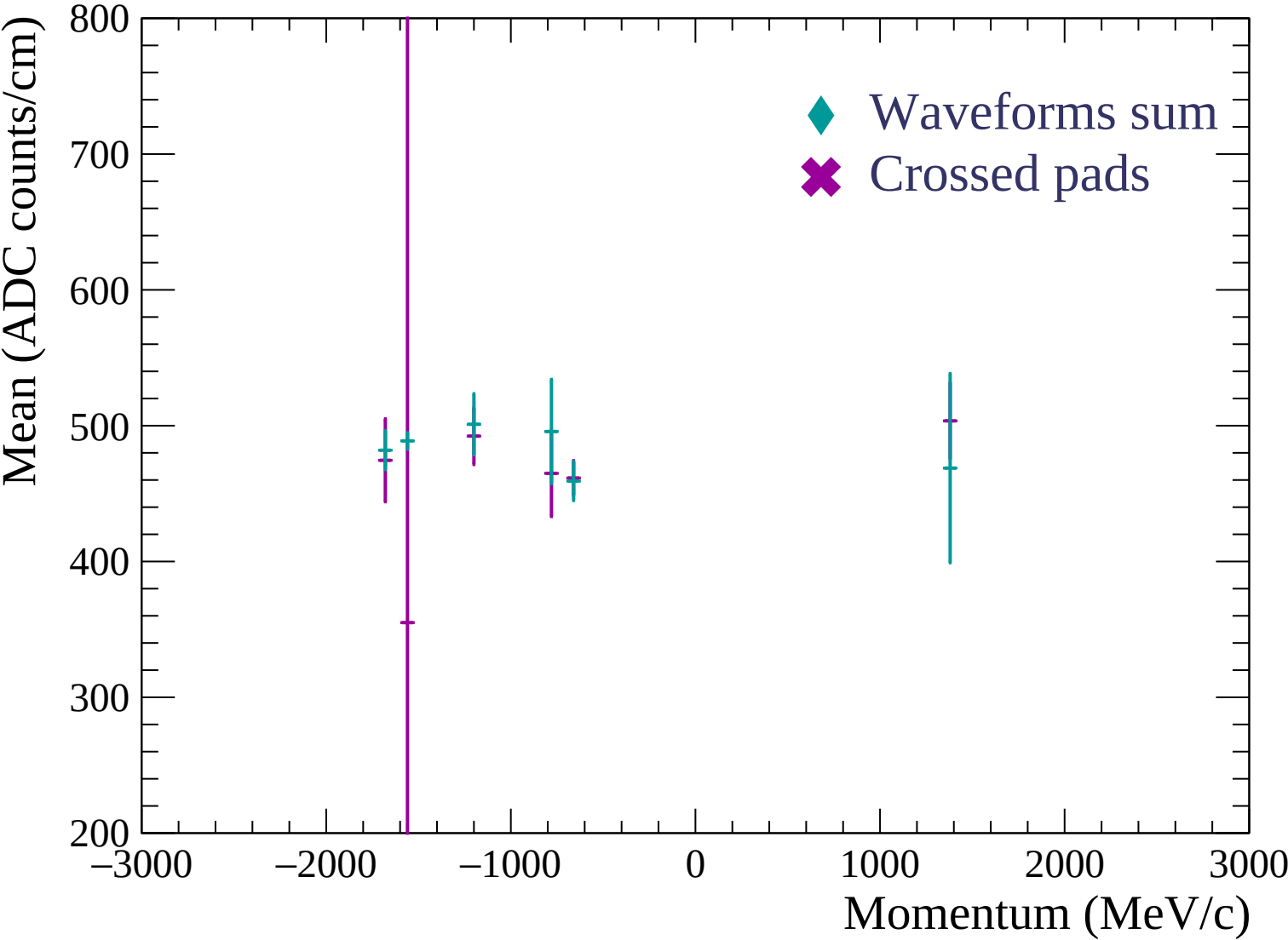


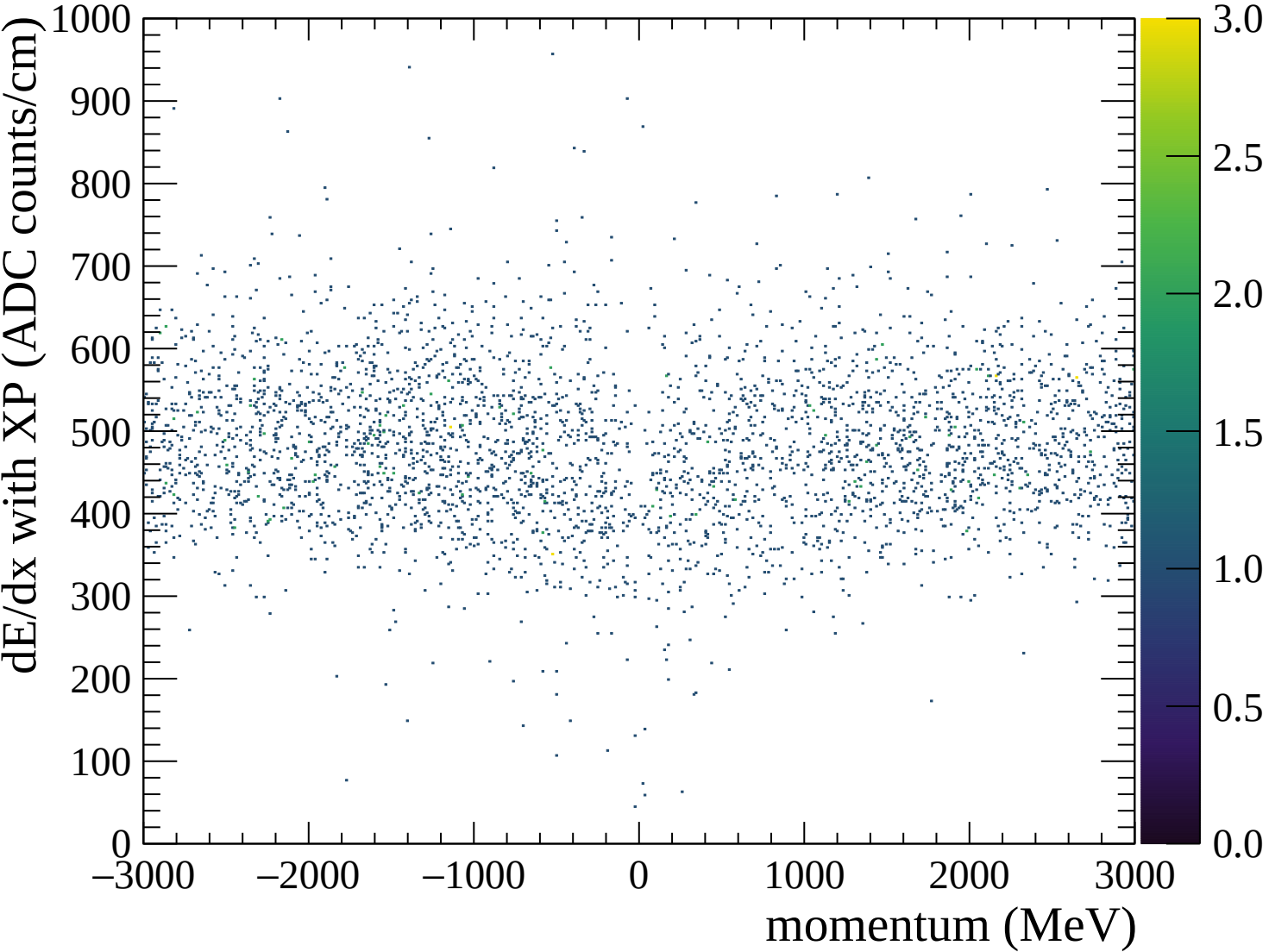


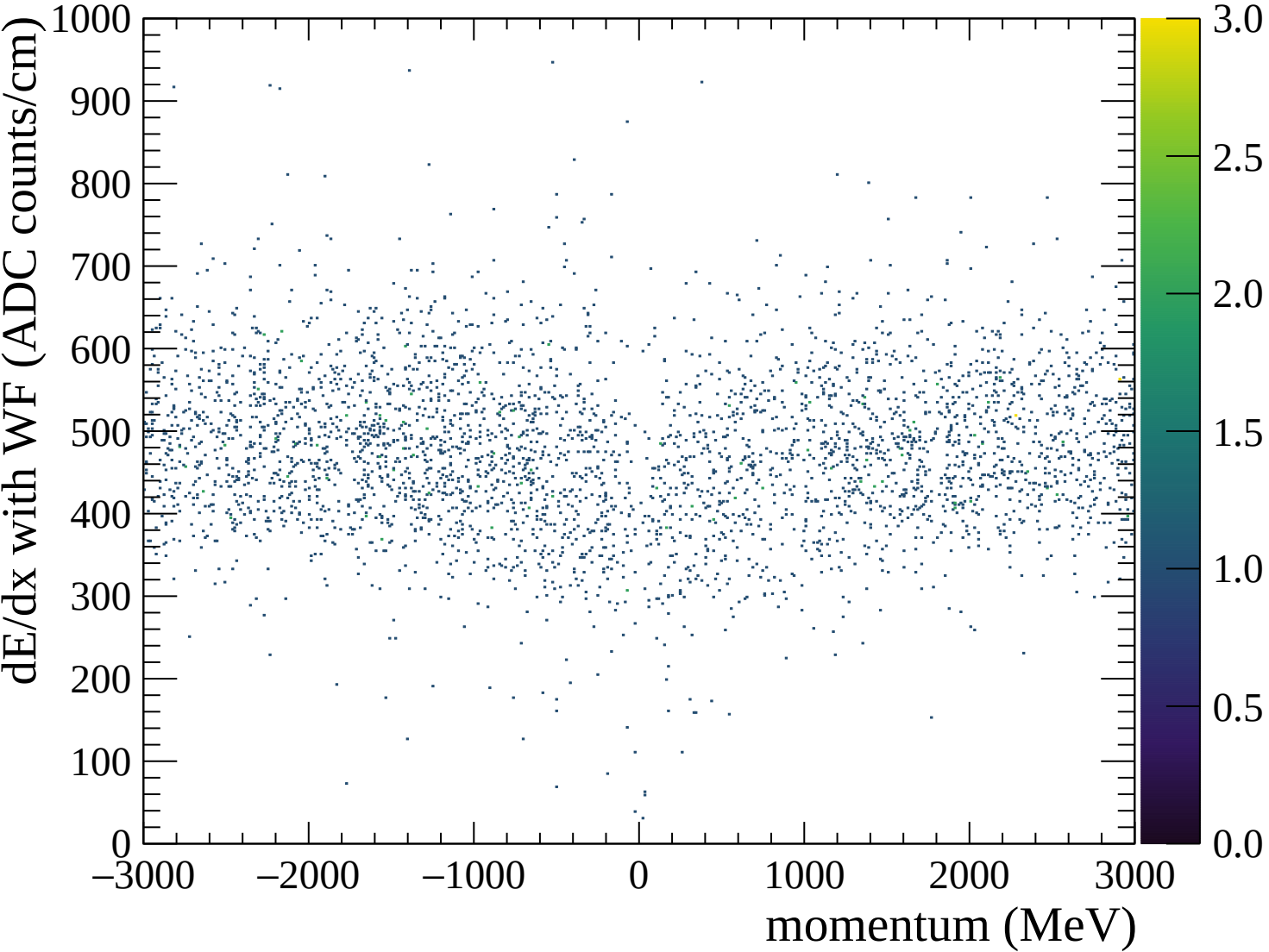


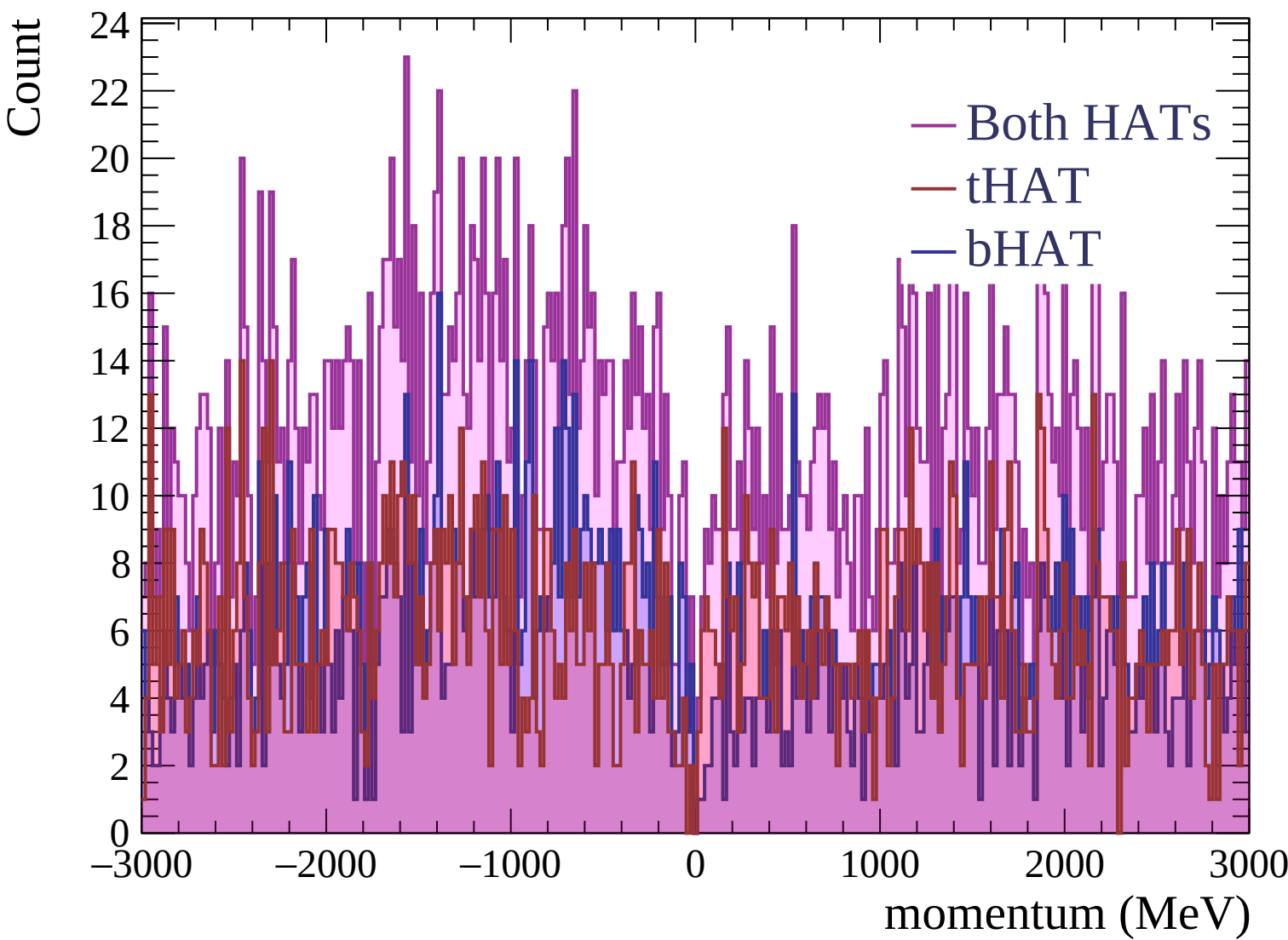


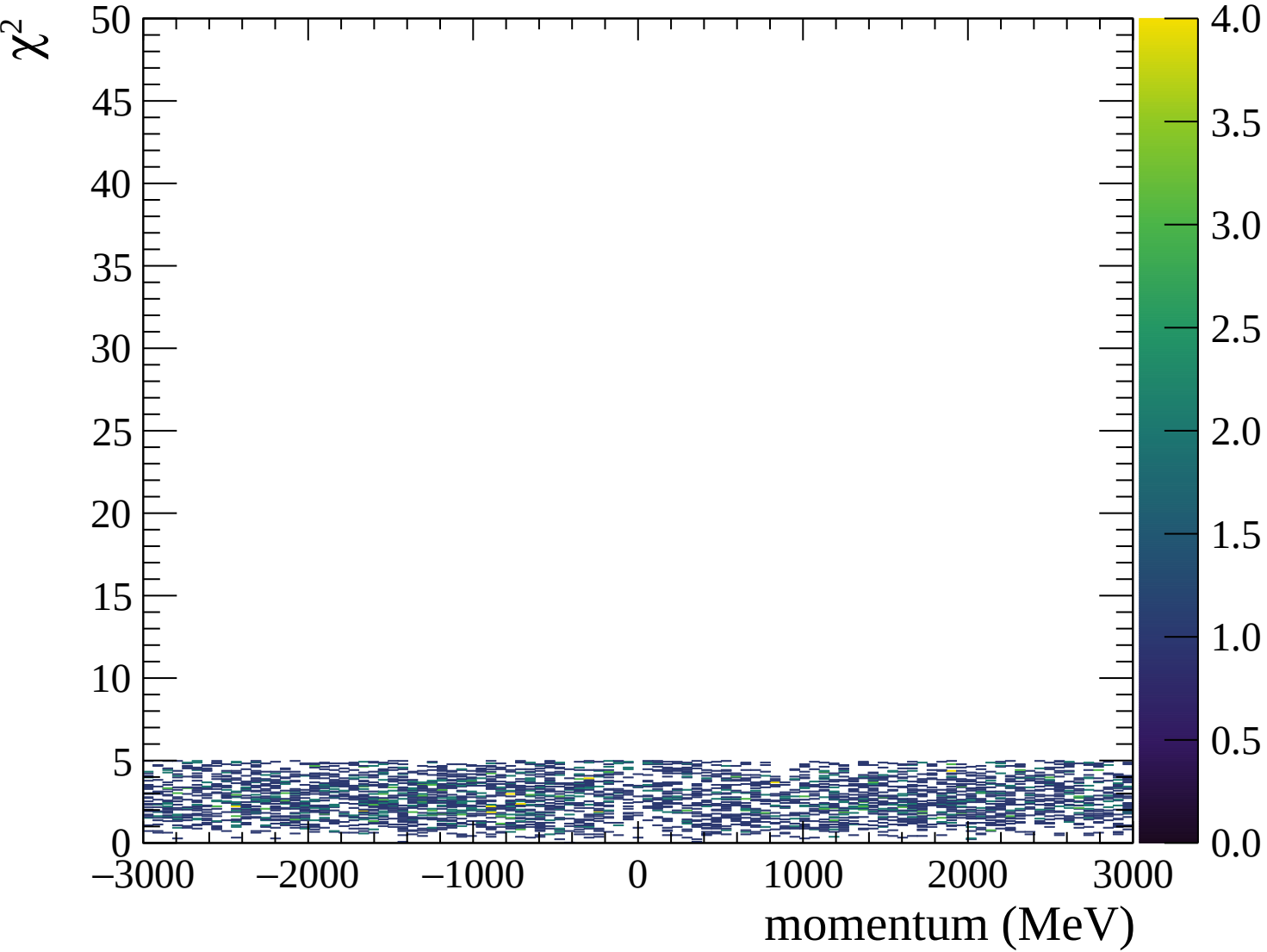


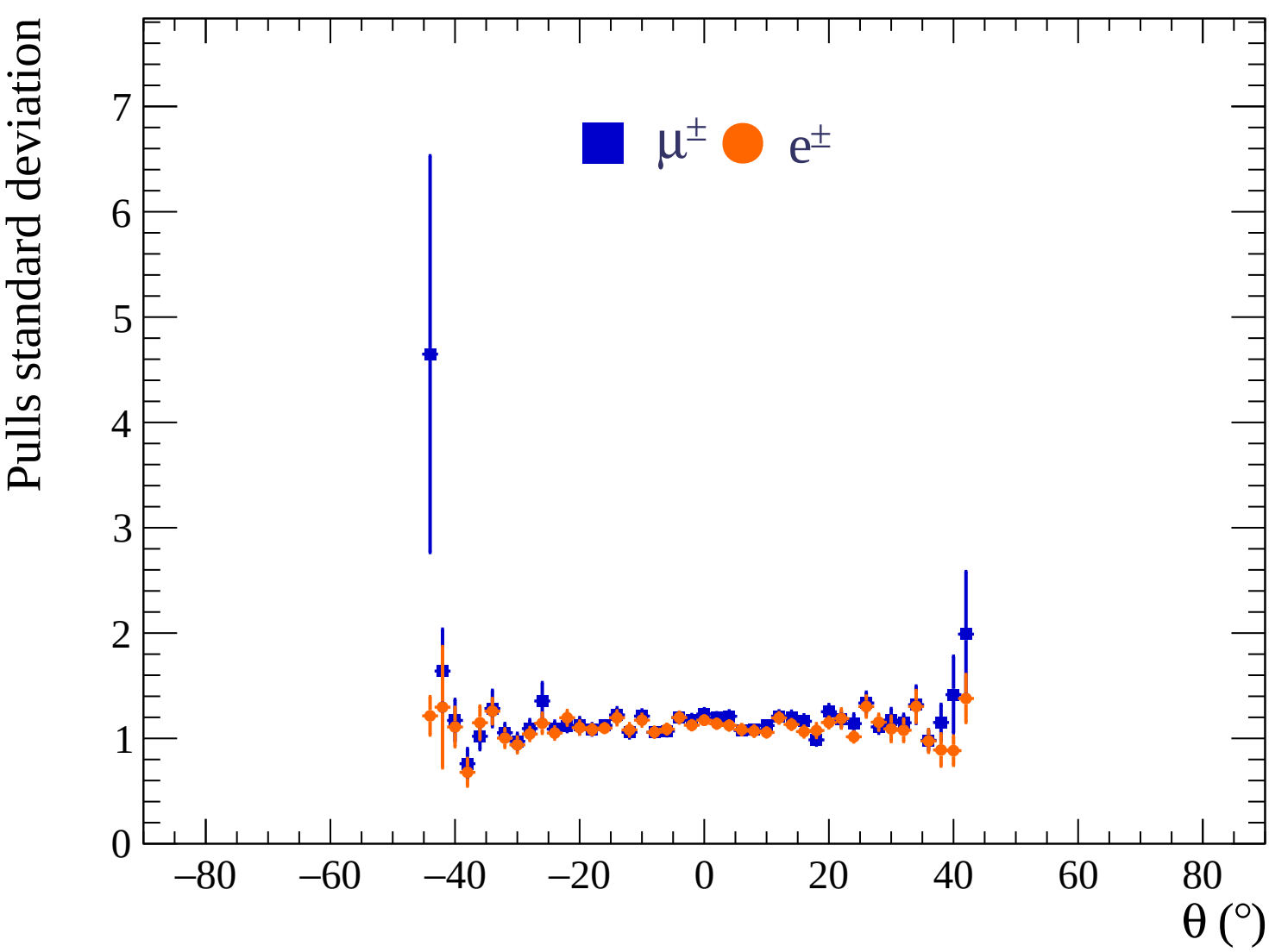


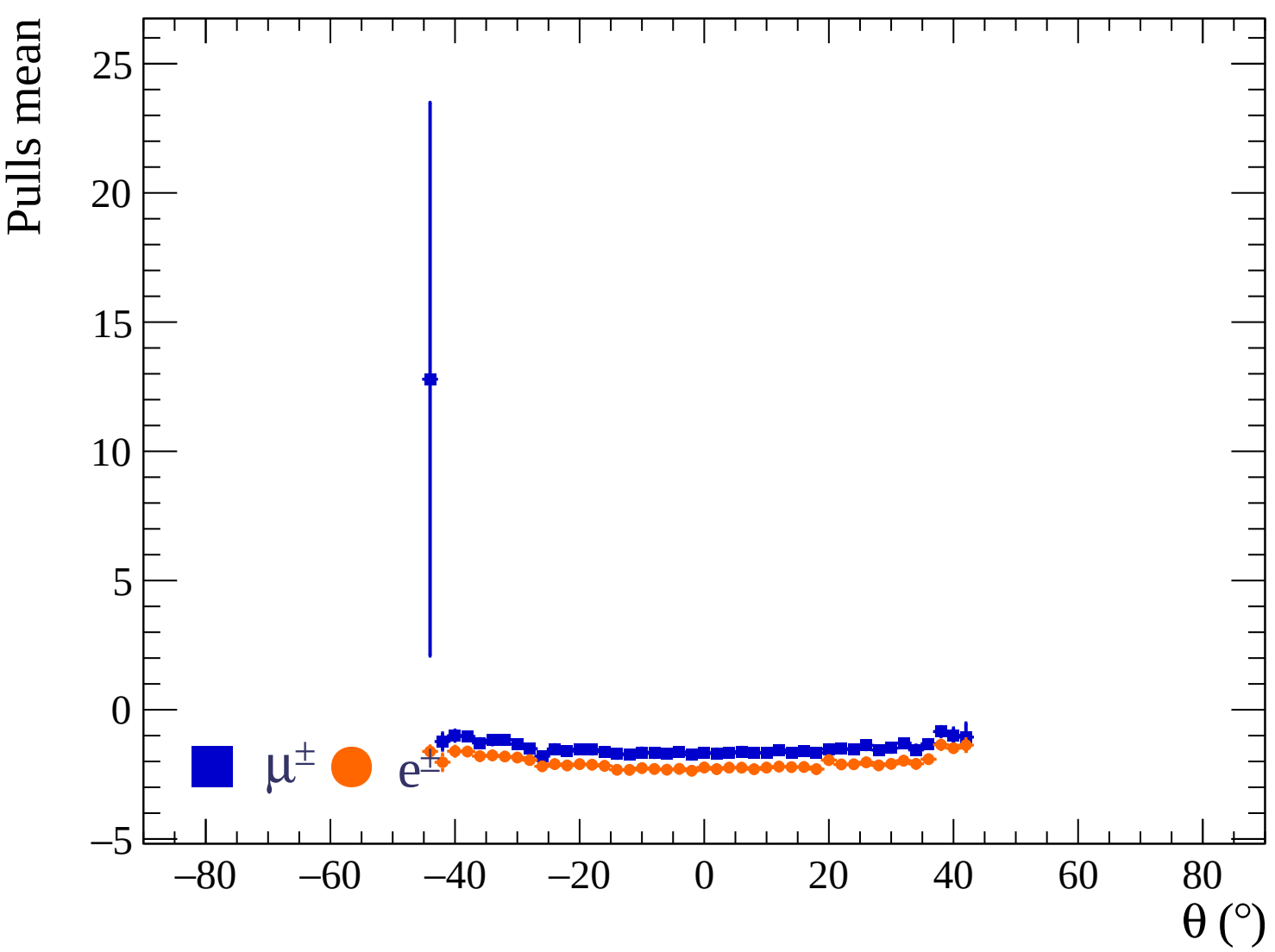


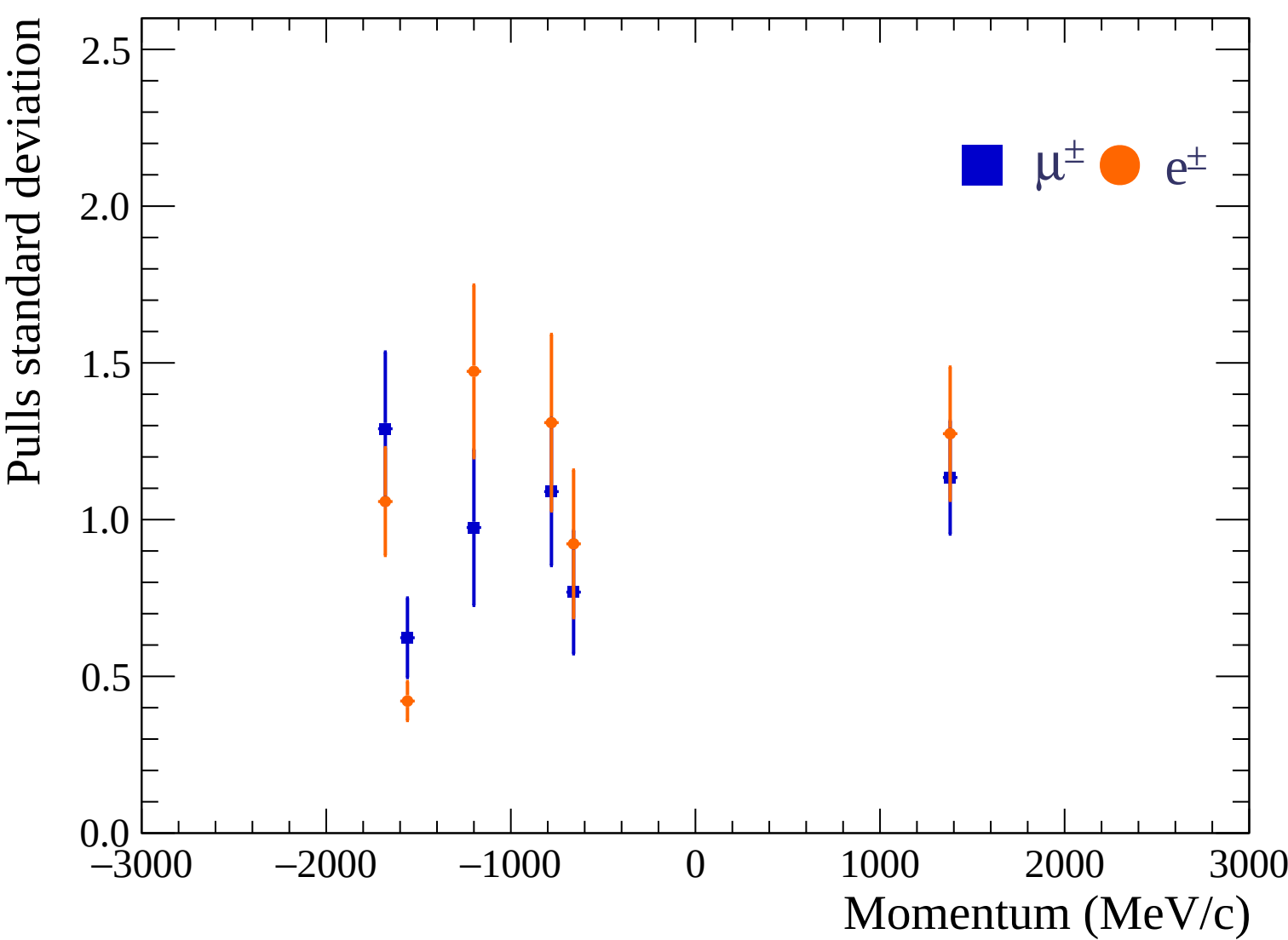


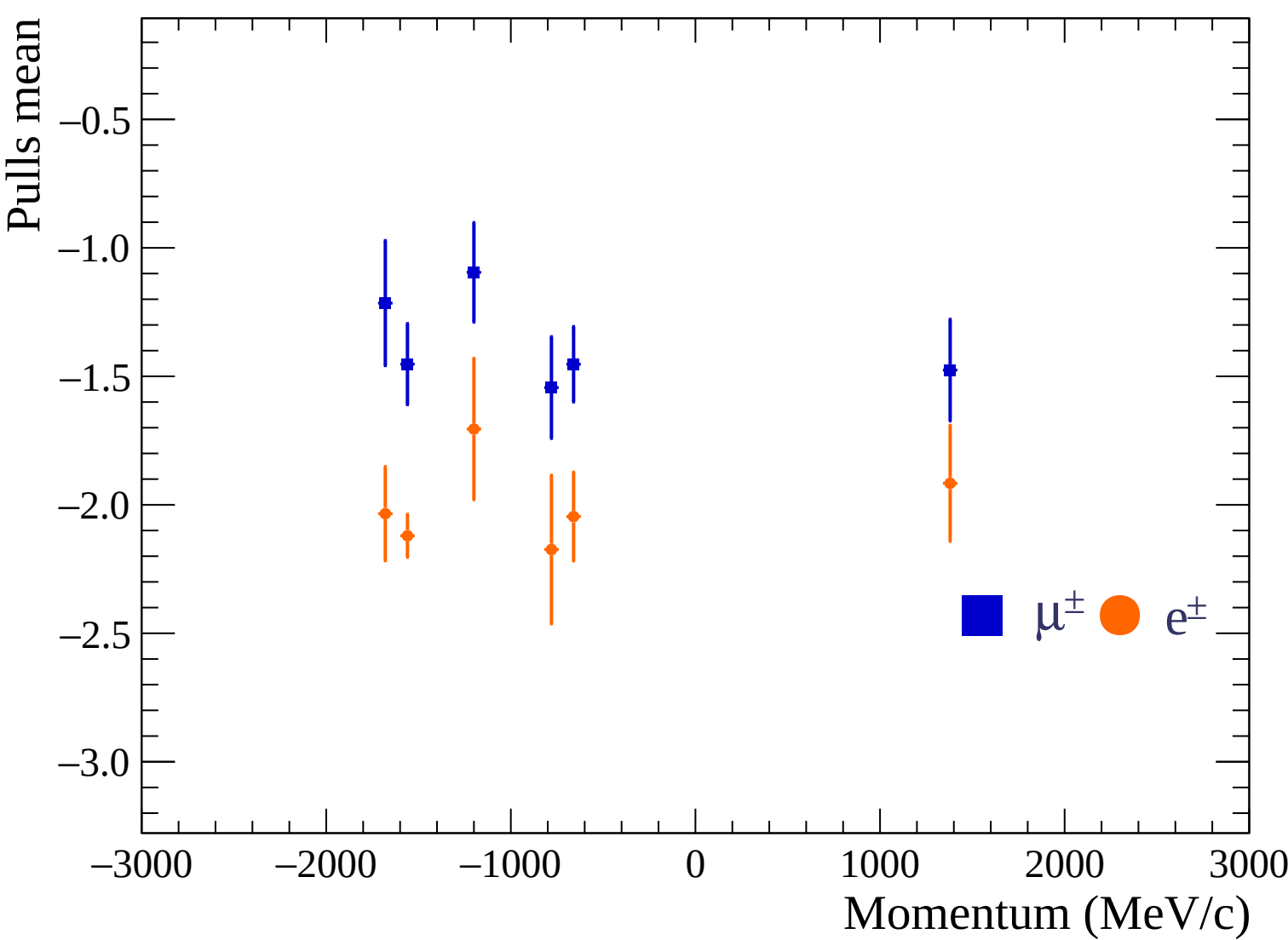


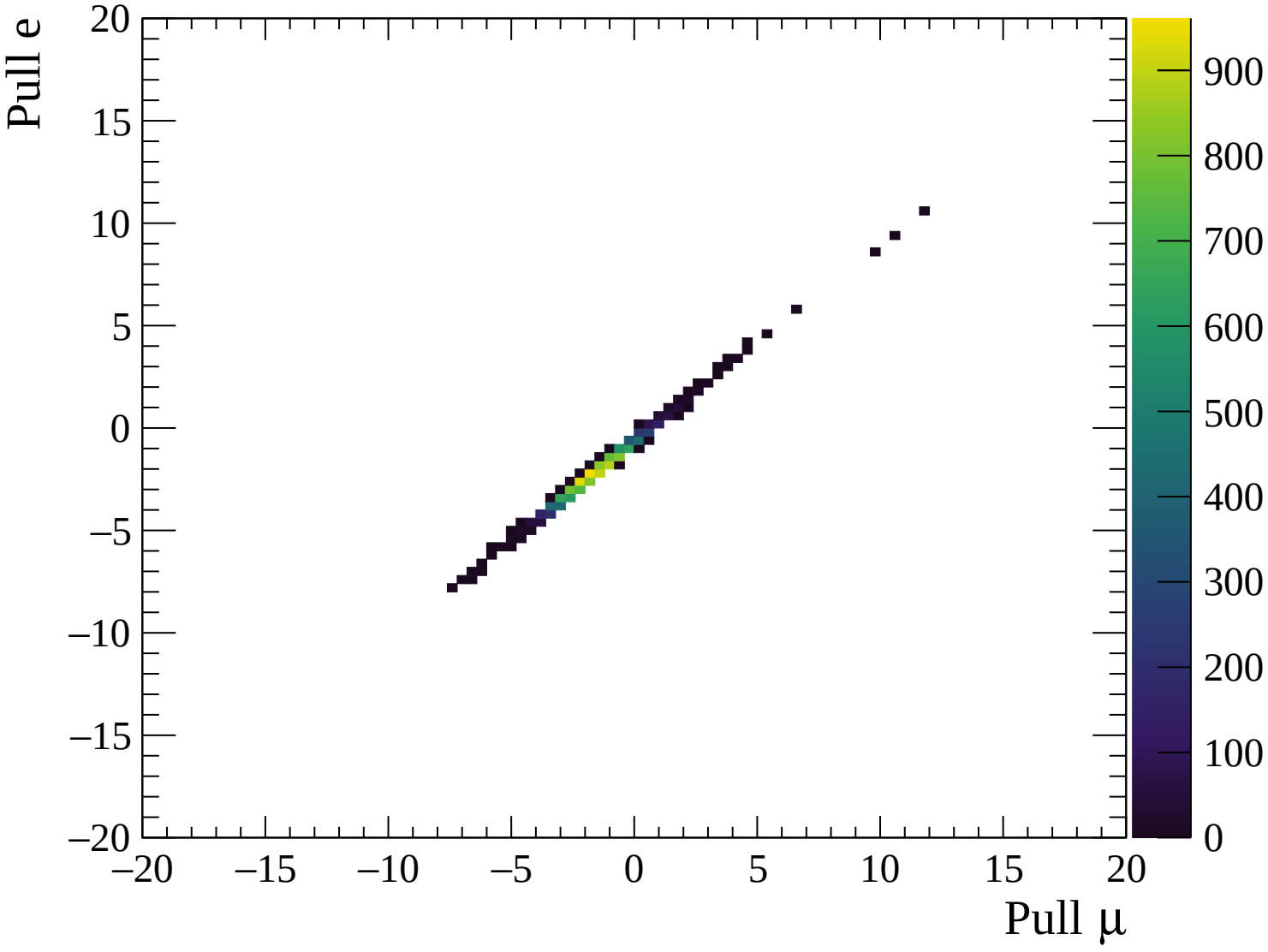


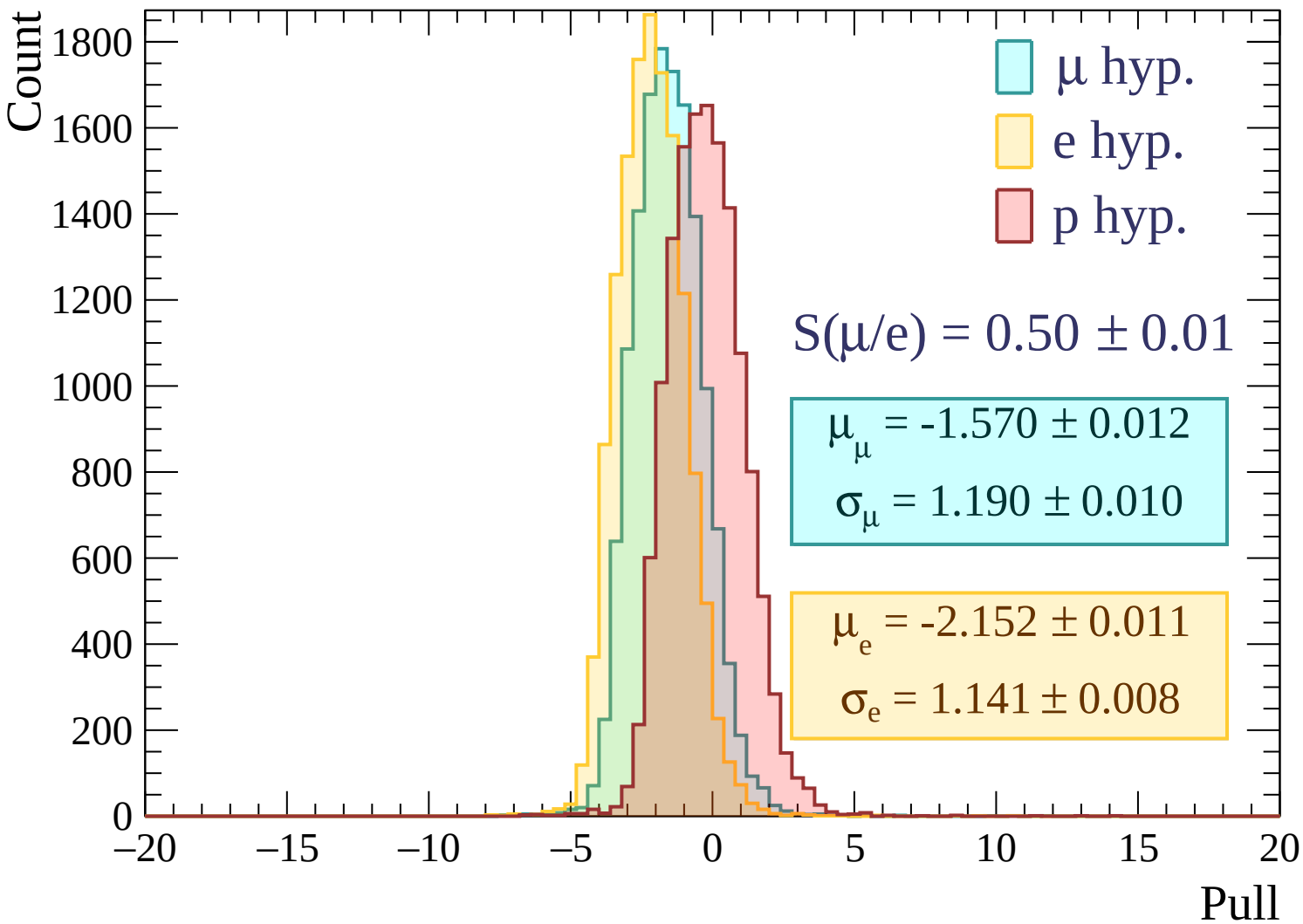


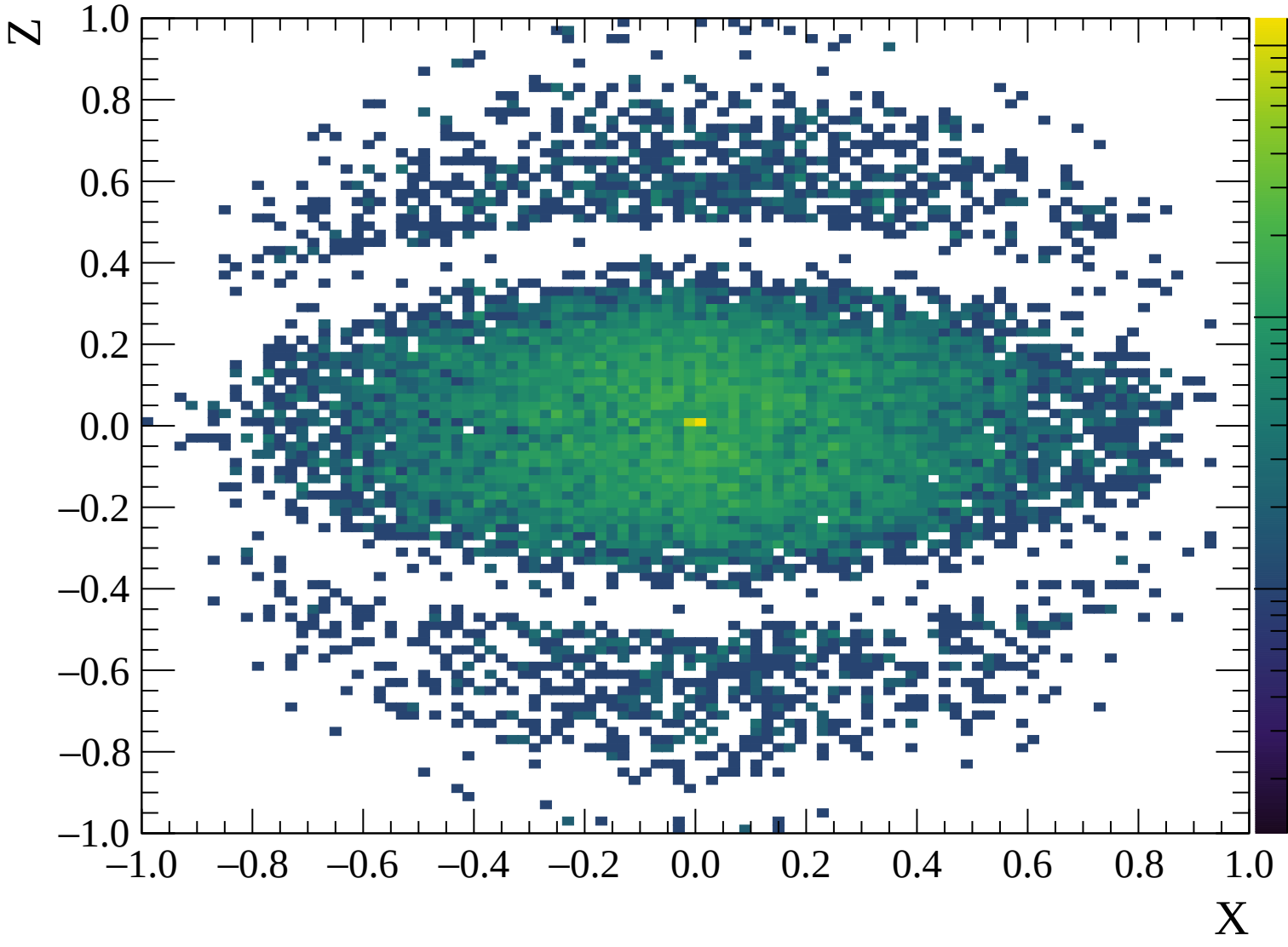


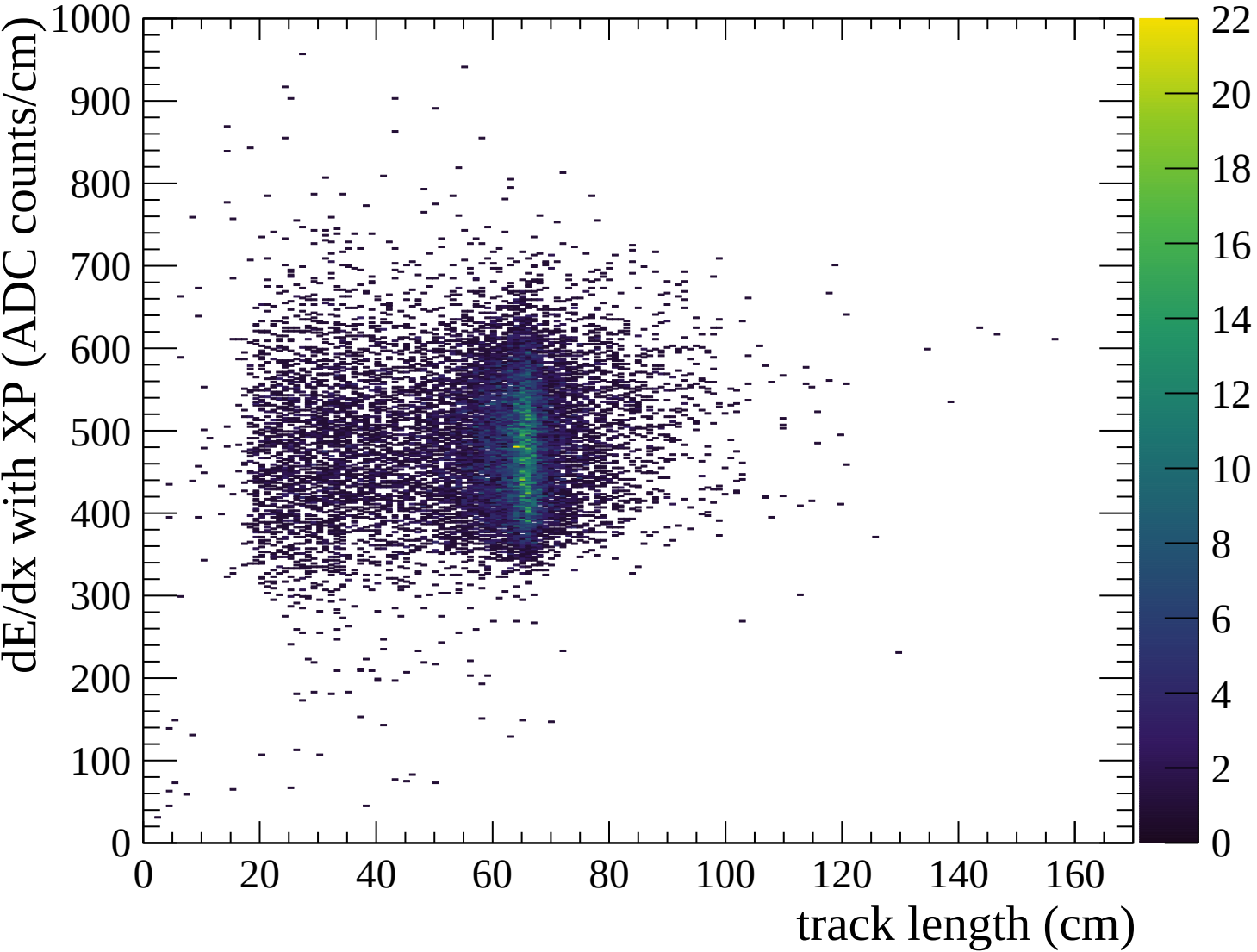


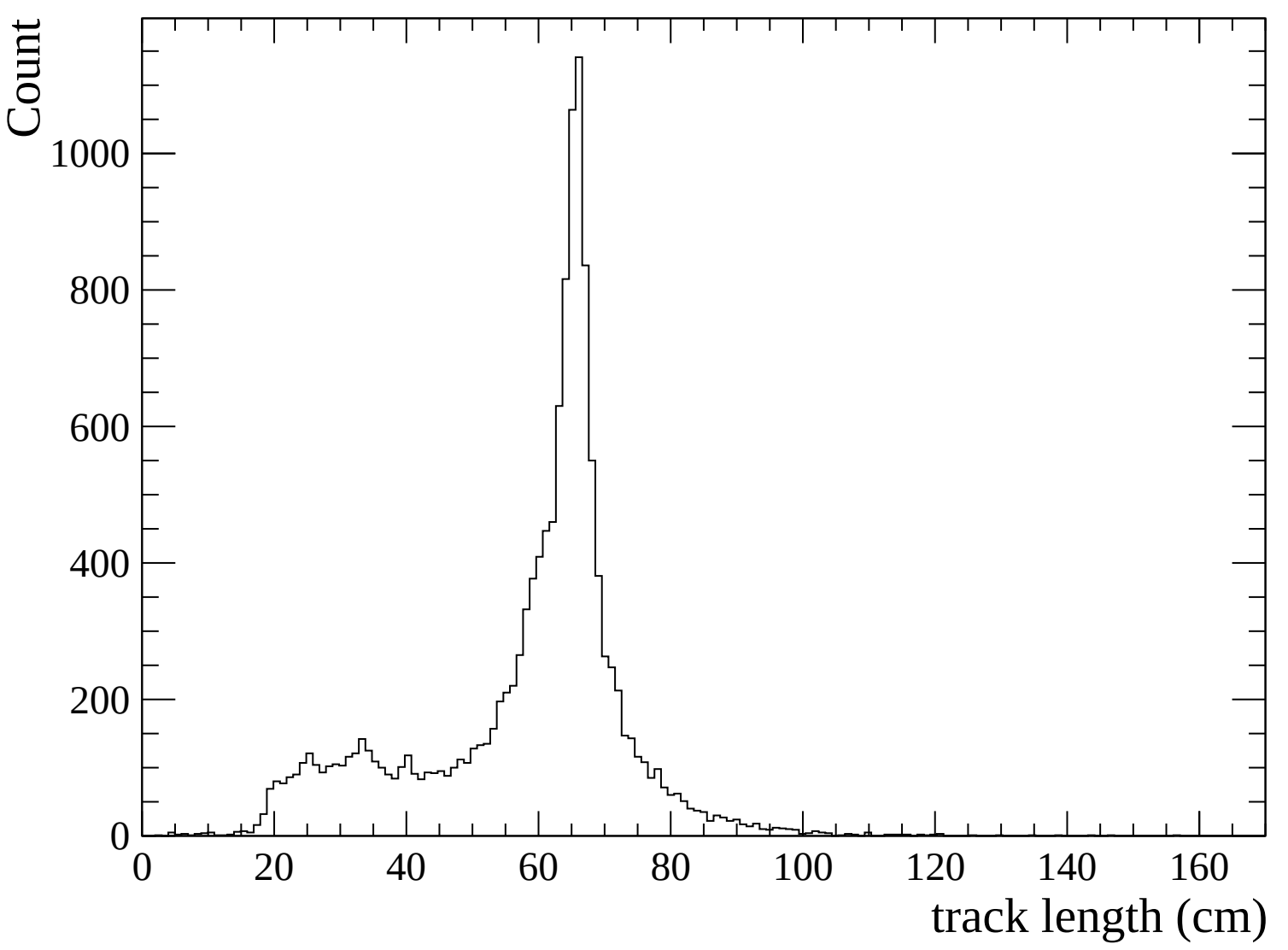


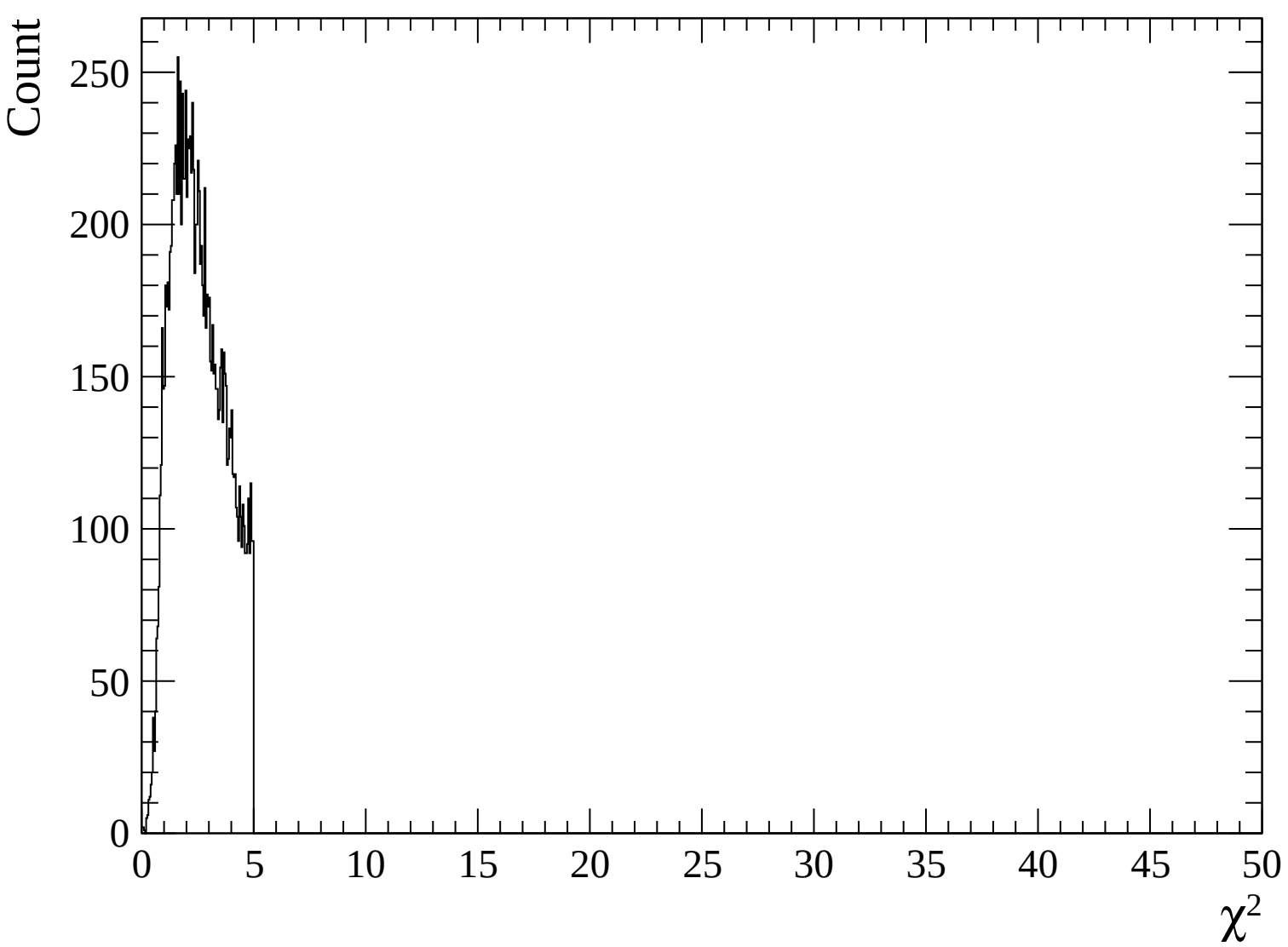


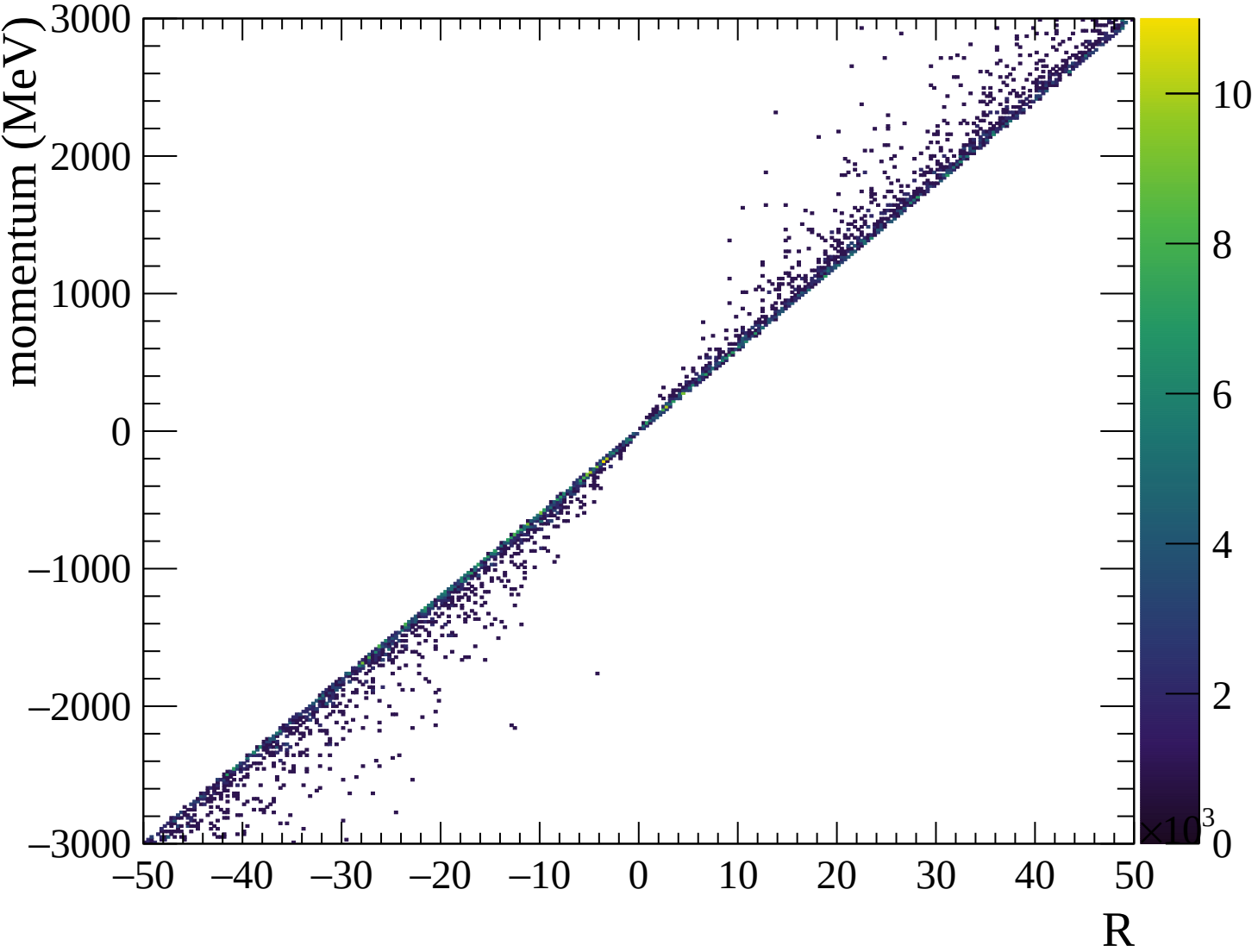


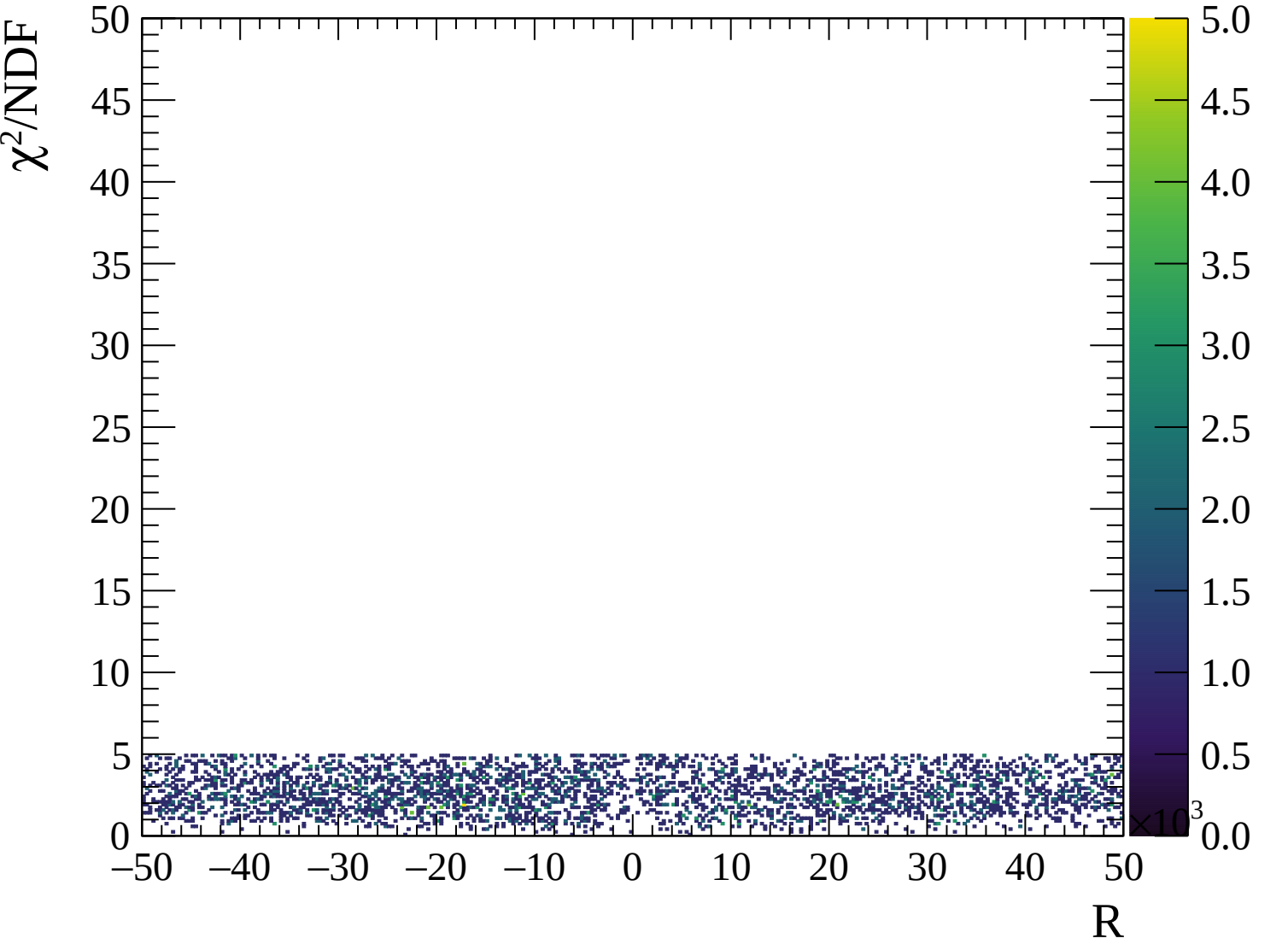






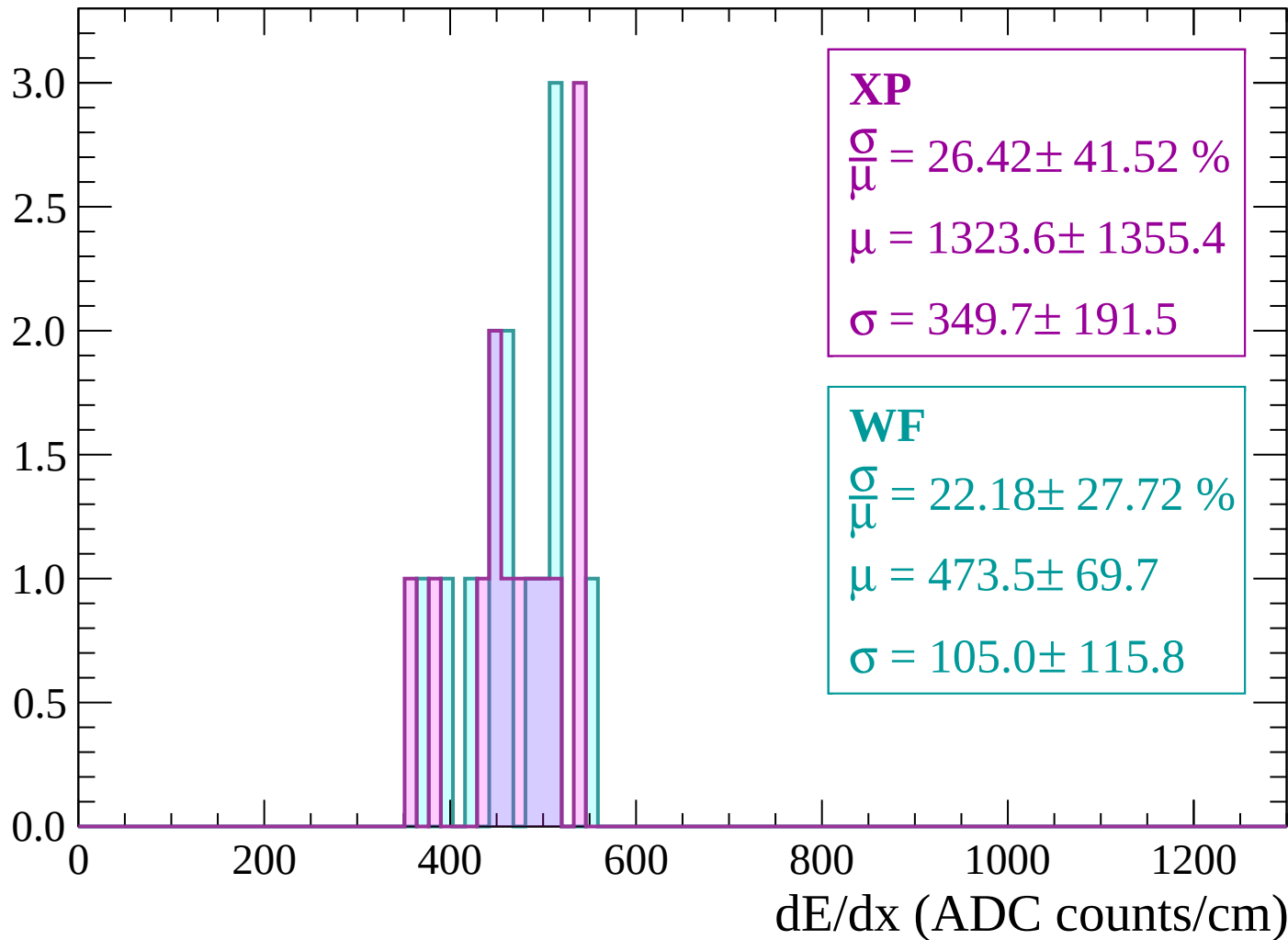






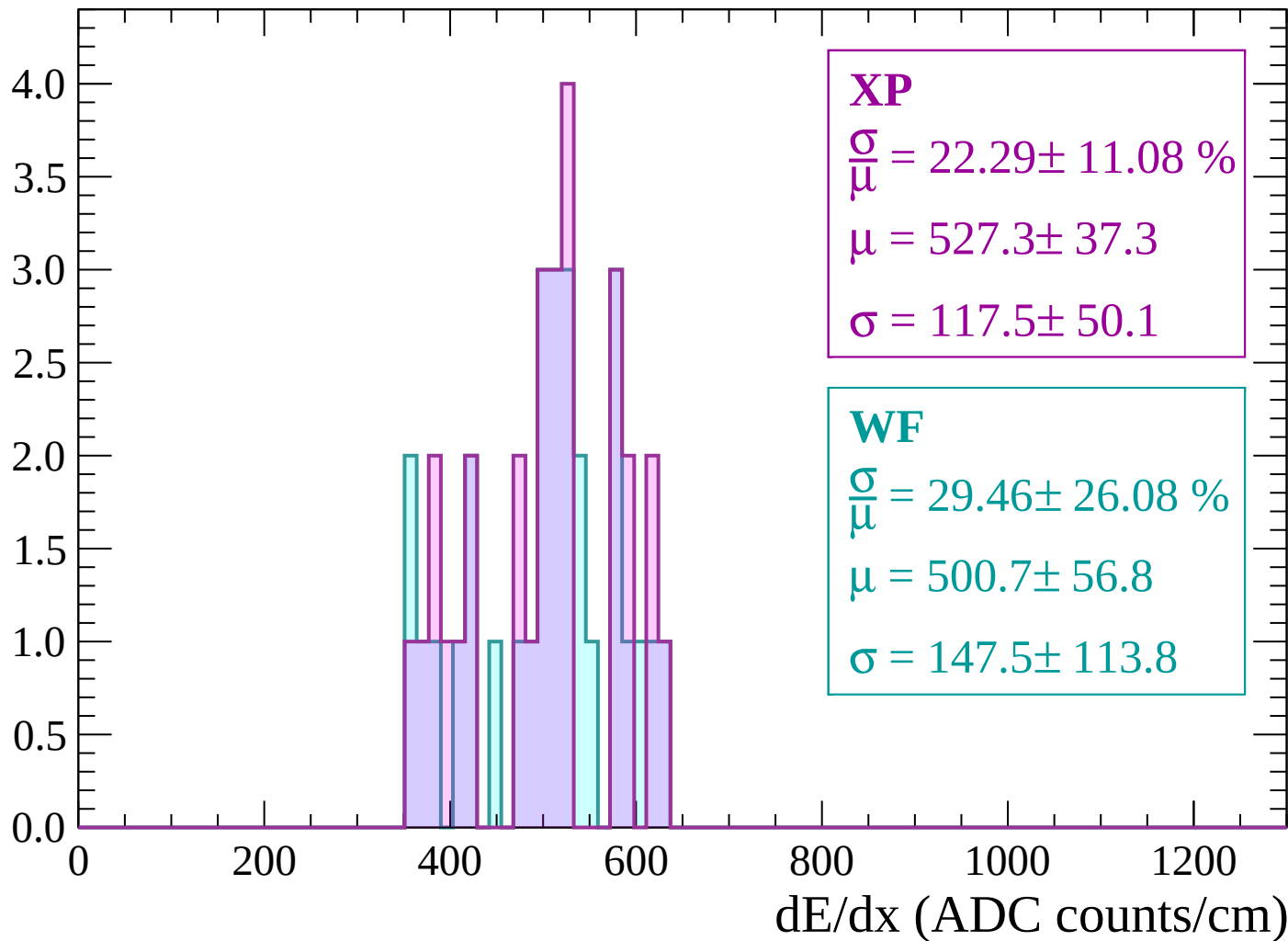
Energy loss | -3000 < p < -2940

Count



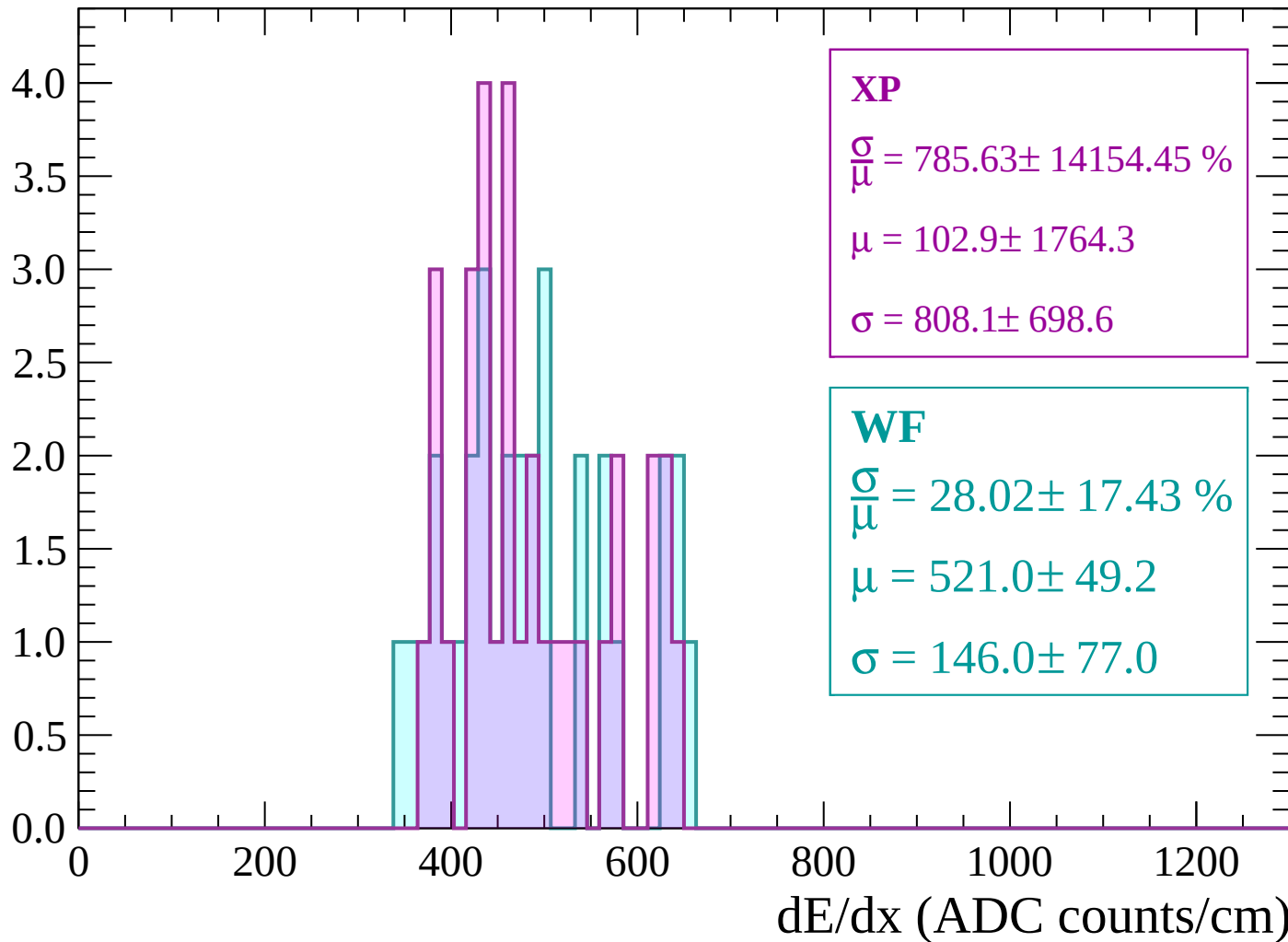
Energy loss | $-2940 < p < -2880$

Count



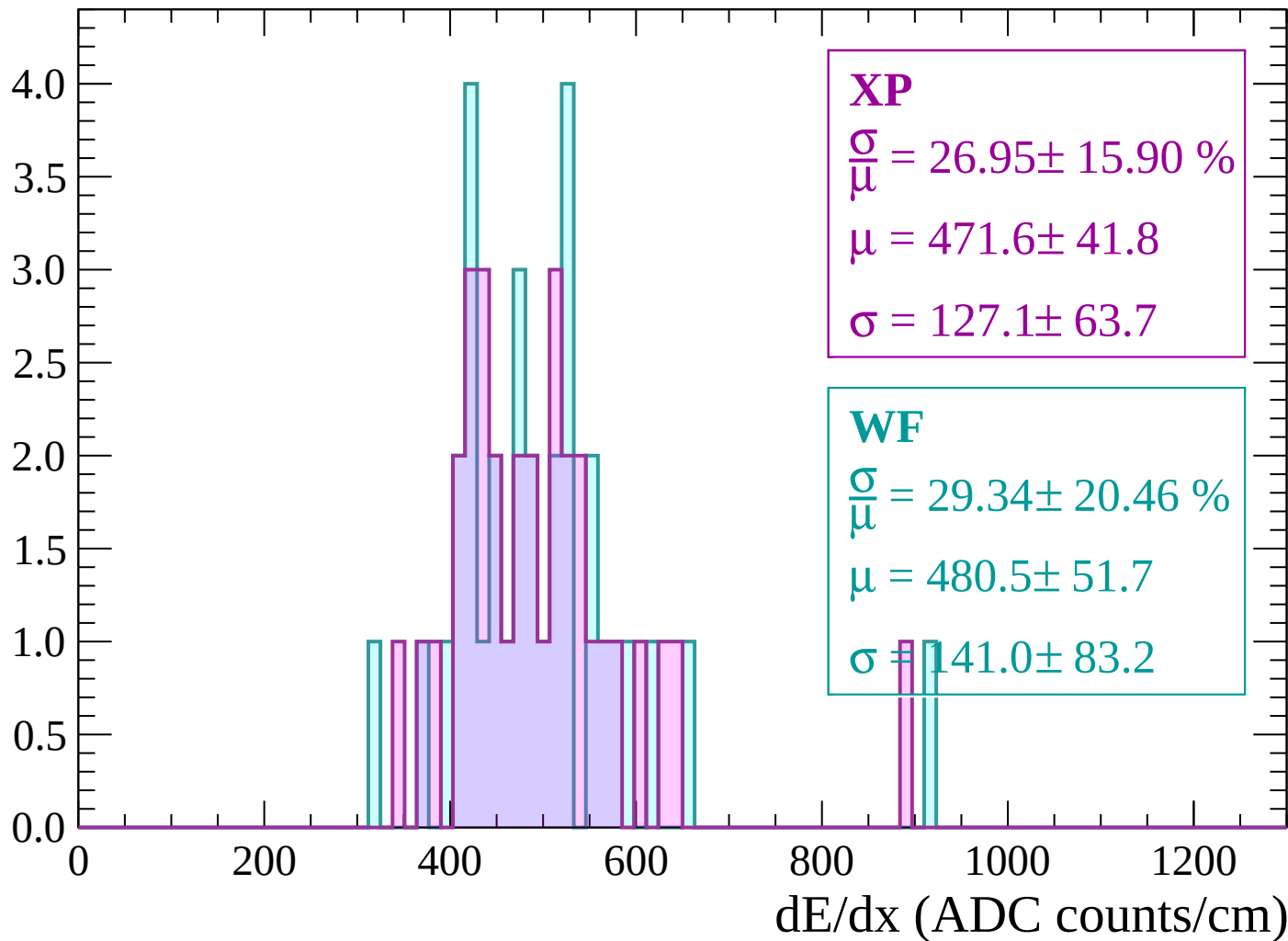
Energy loss | -2880 < p < -2820

Count



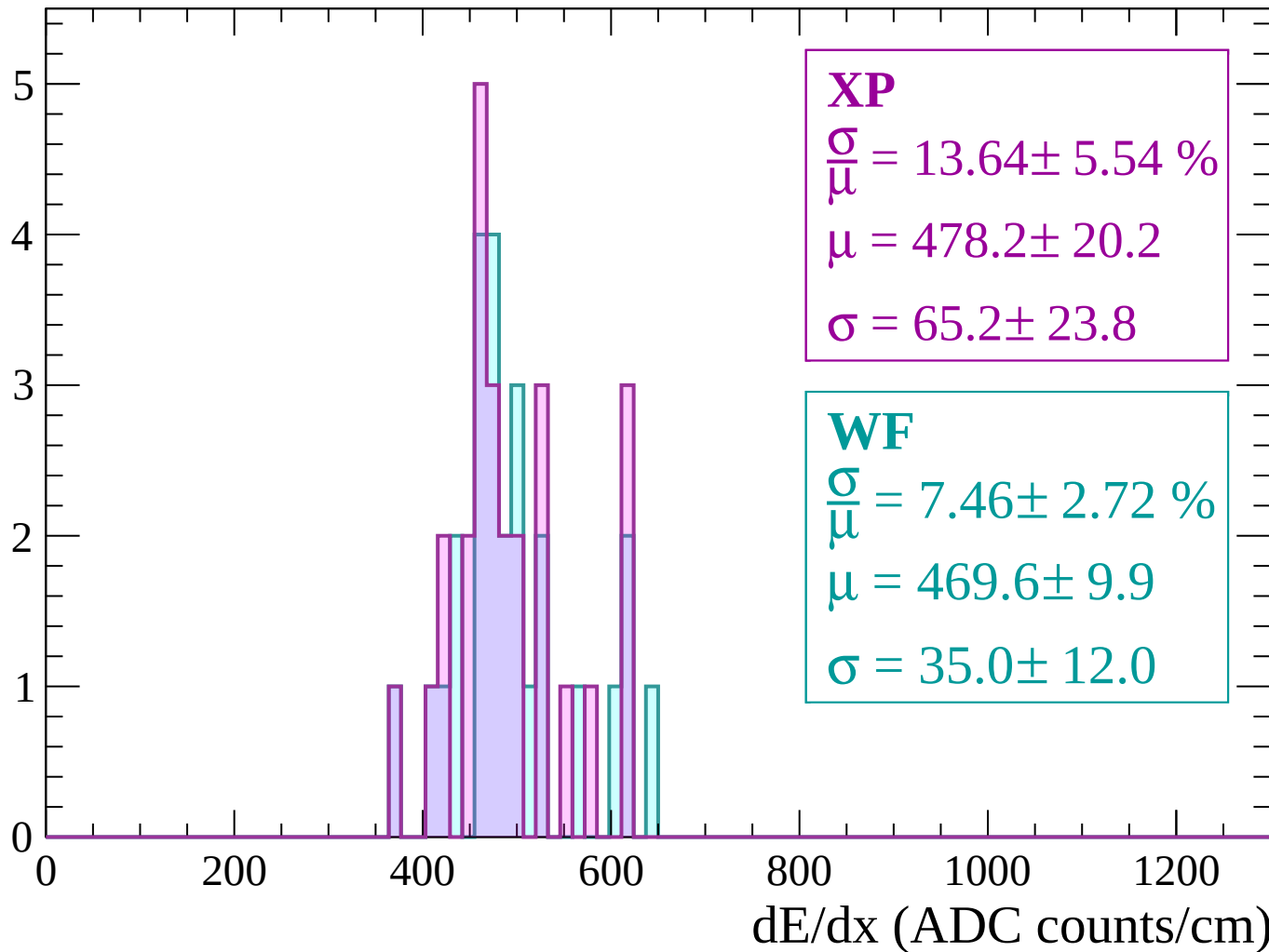
Energy loss | $-2820 < p < -2760$

Count



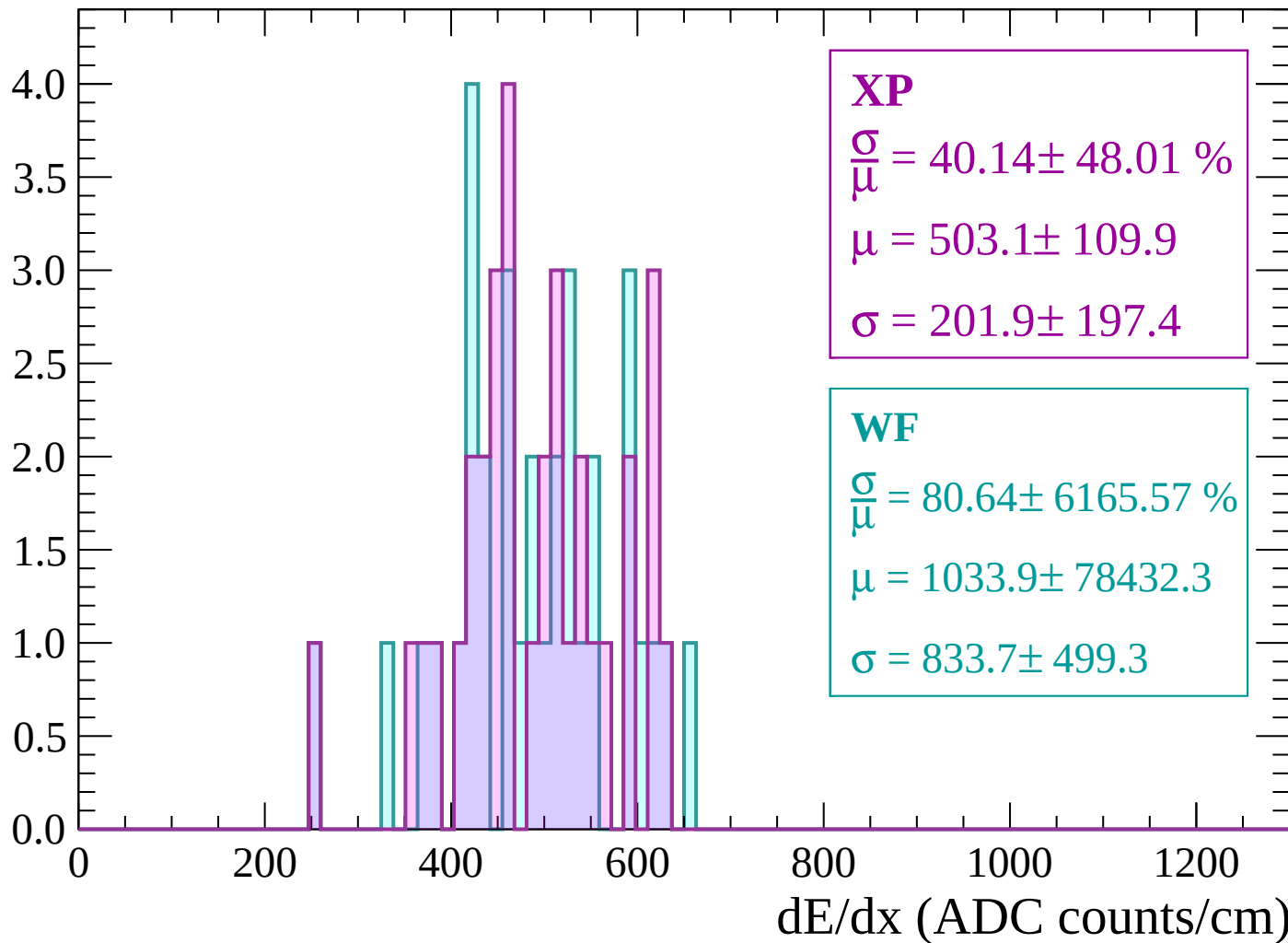
Energy loss | $-2760 < p < -2700$

Count



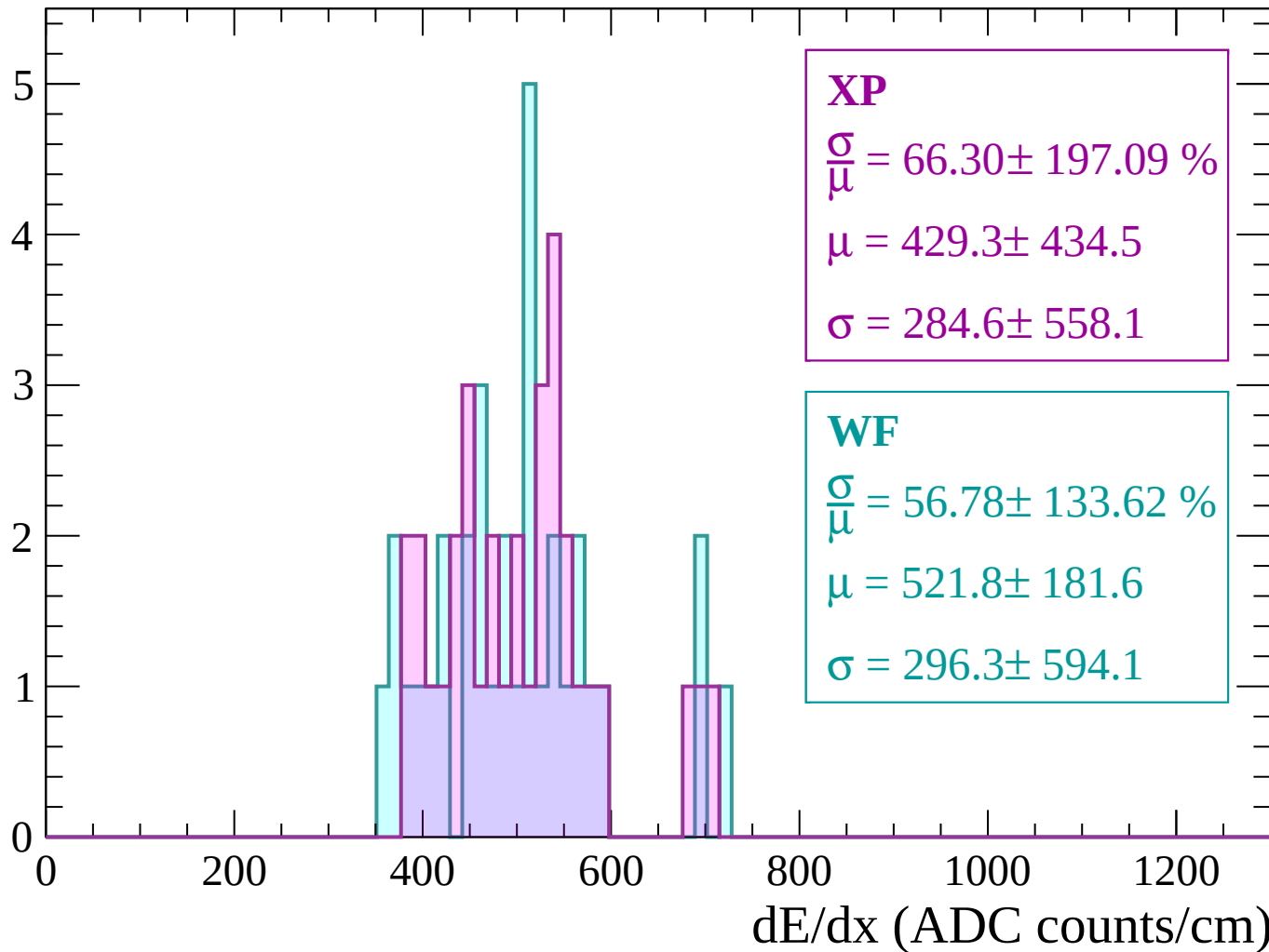
Energy loss | -2700 < p < -2640

Count



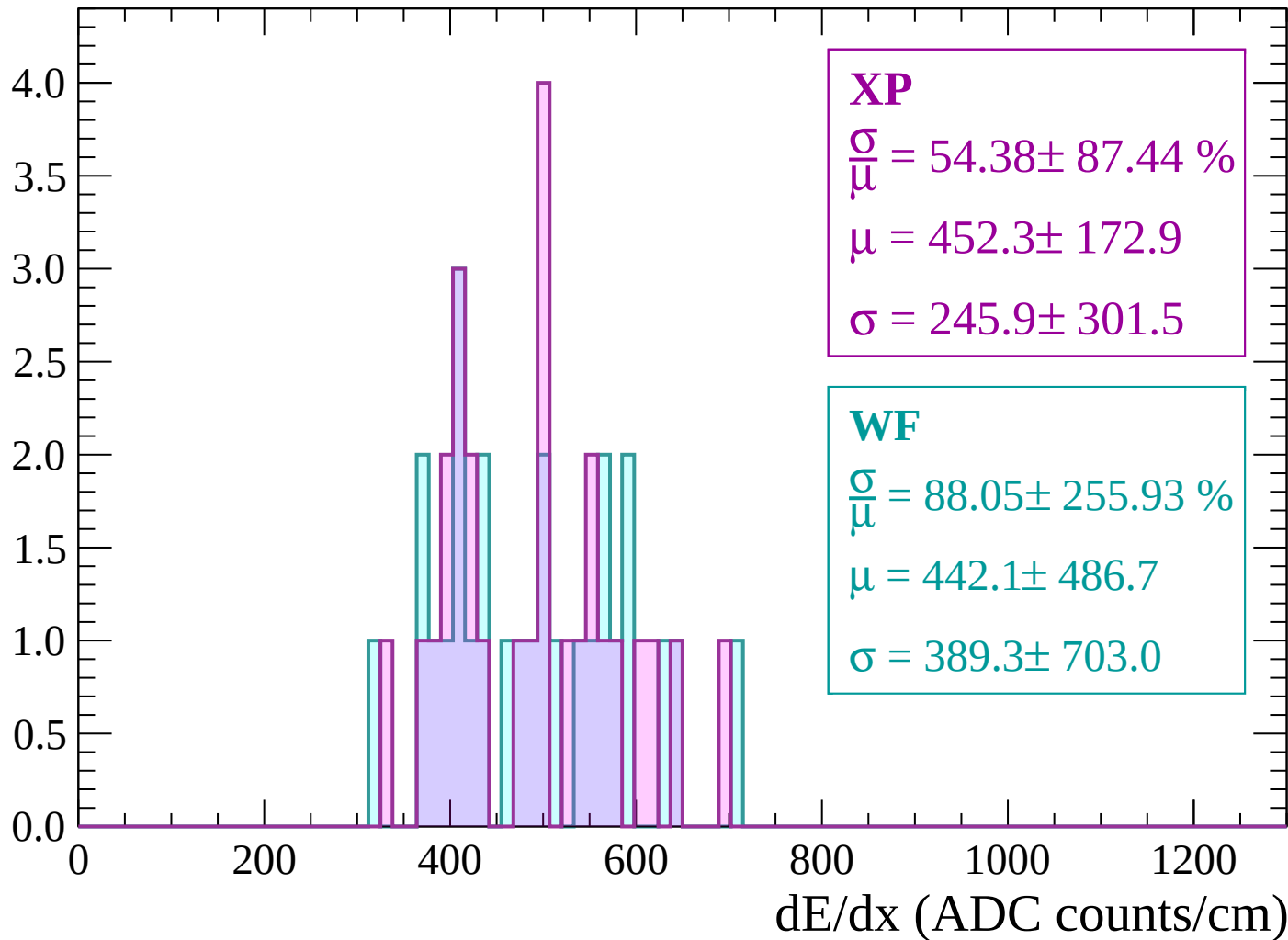
Energy loss | $-2640 < p < -2580$

Count



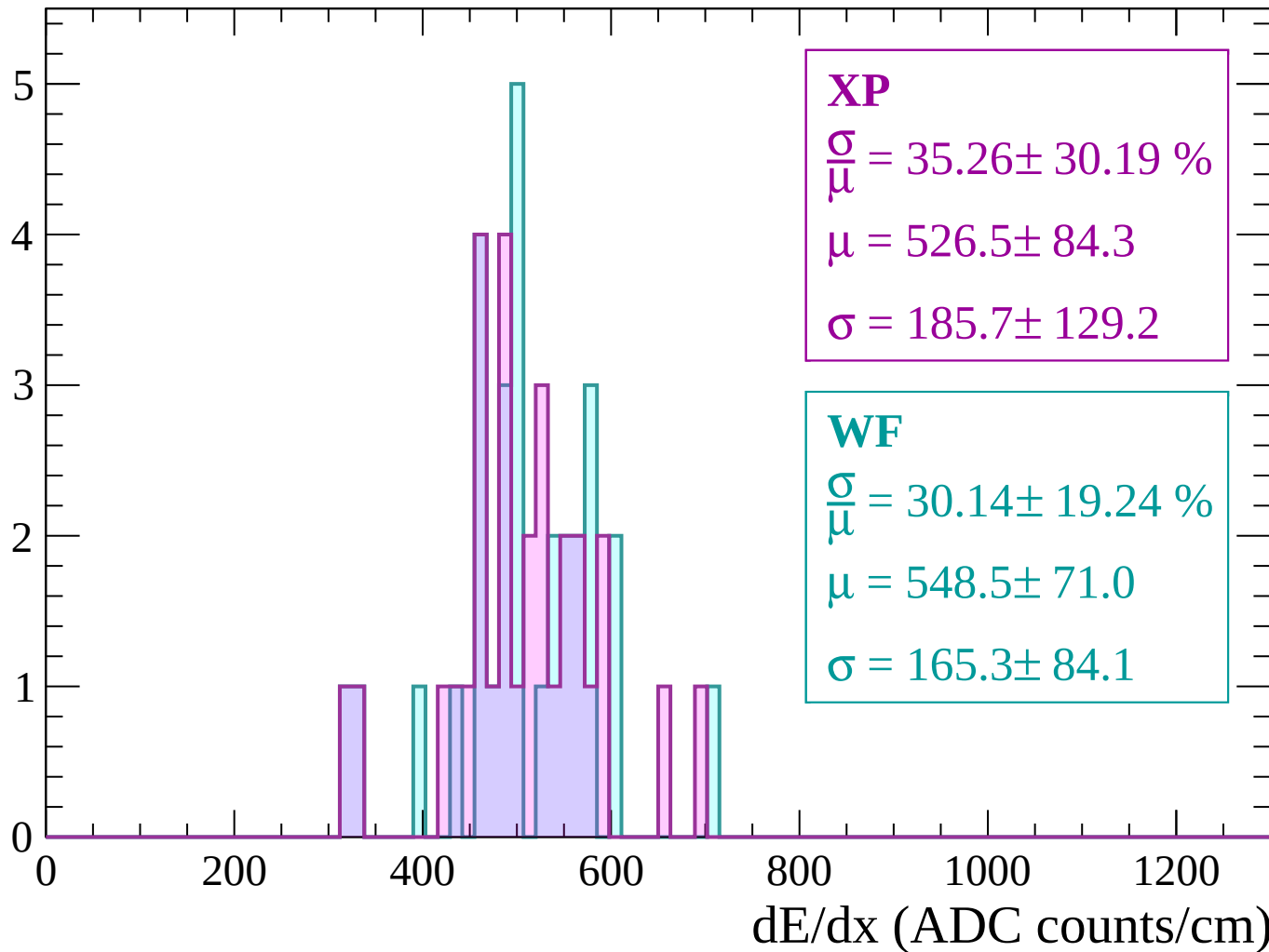
Energy loss | -2580 < p < -2520

Count



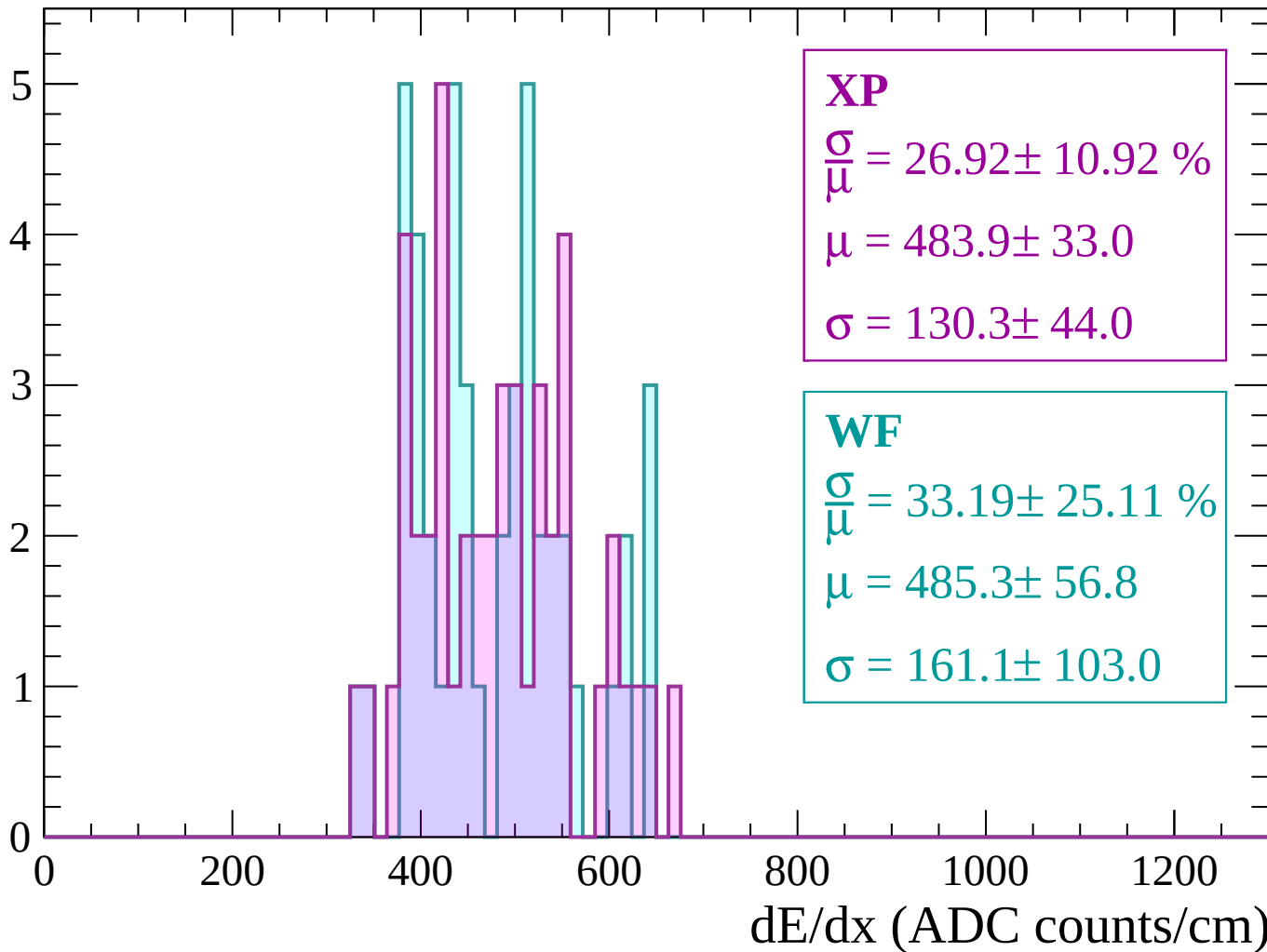
Energy loss | -2520 < p < -2460

Count



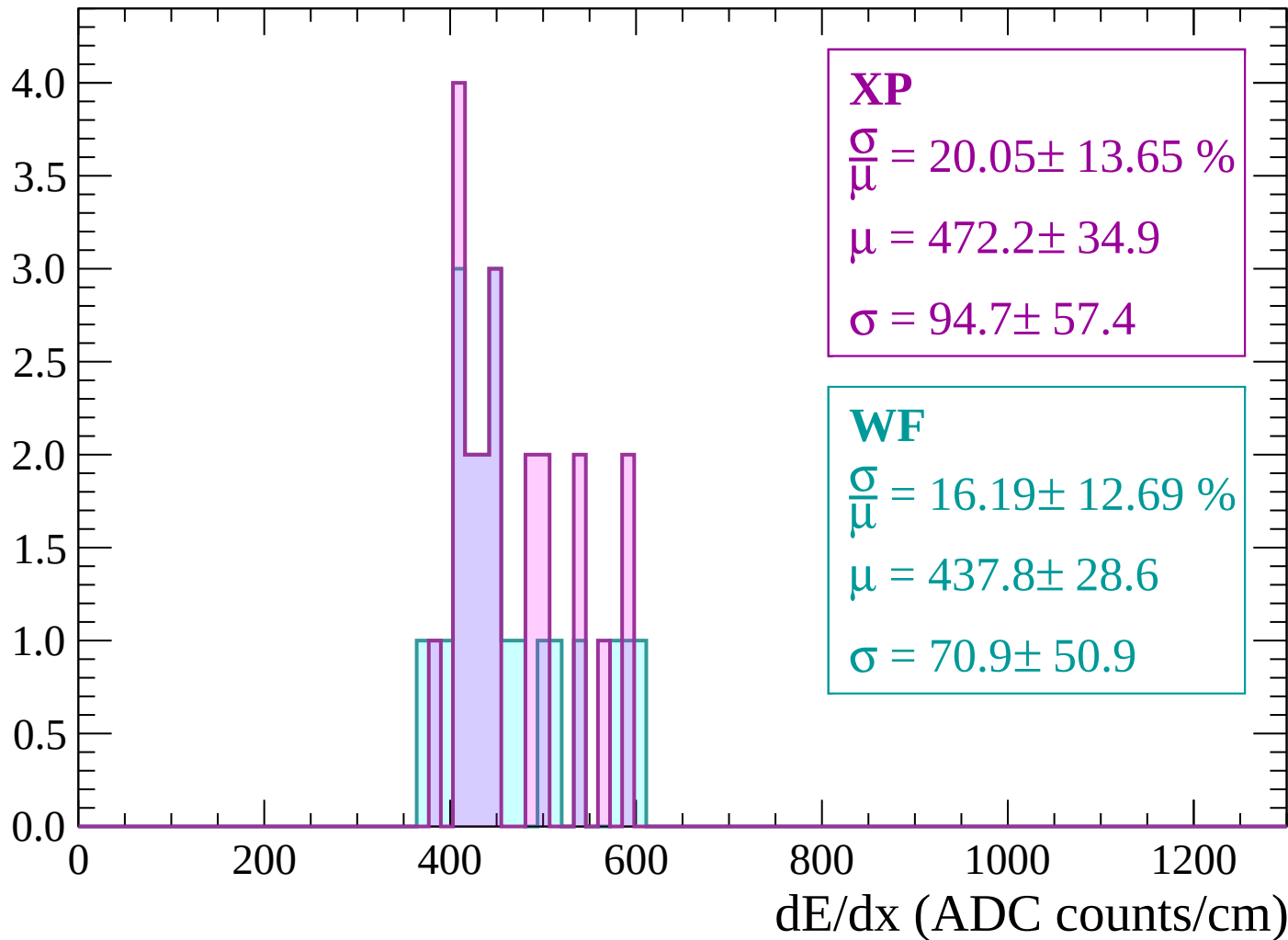
Energy loss | $-2460 < p < -2400$

Count



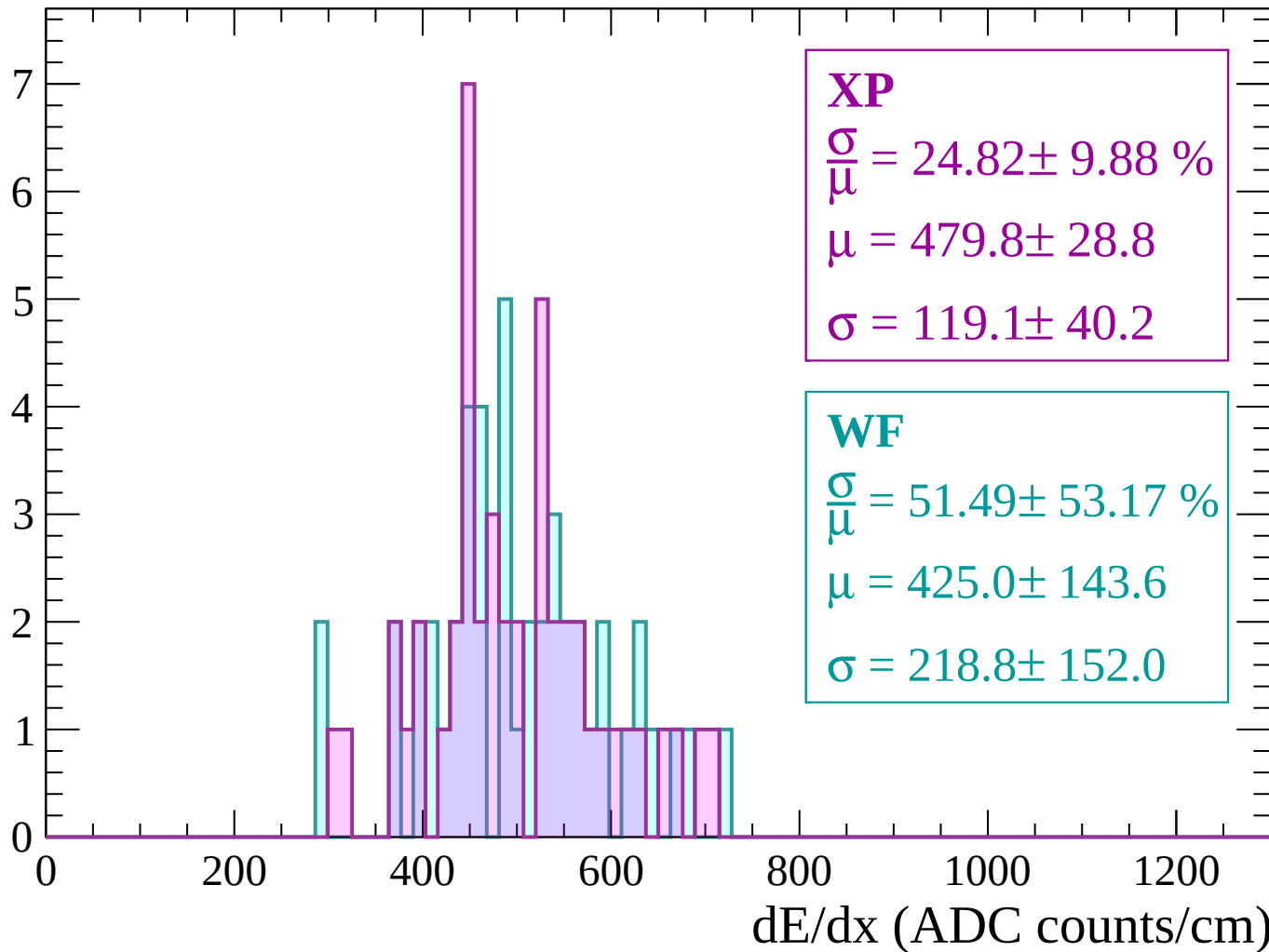
Energy loss | -2400 < p < -2340

Count



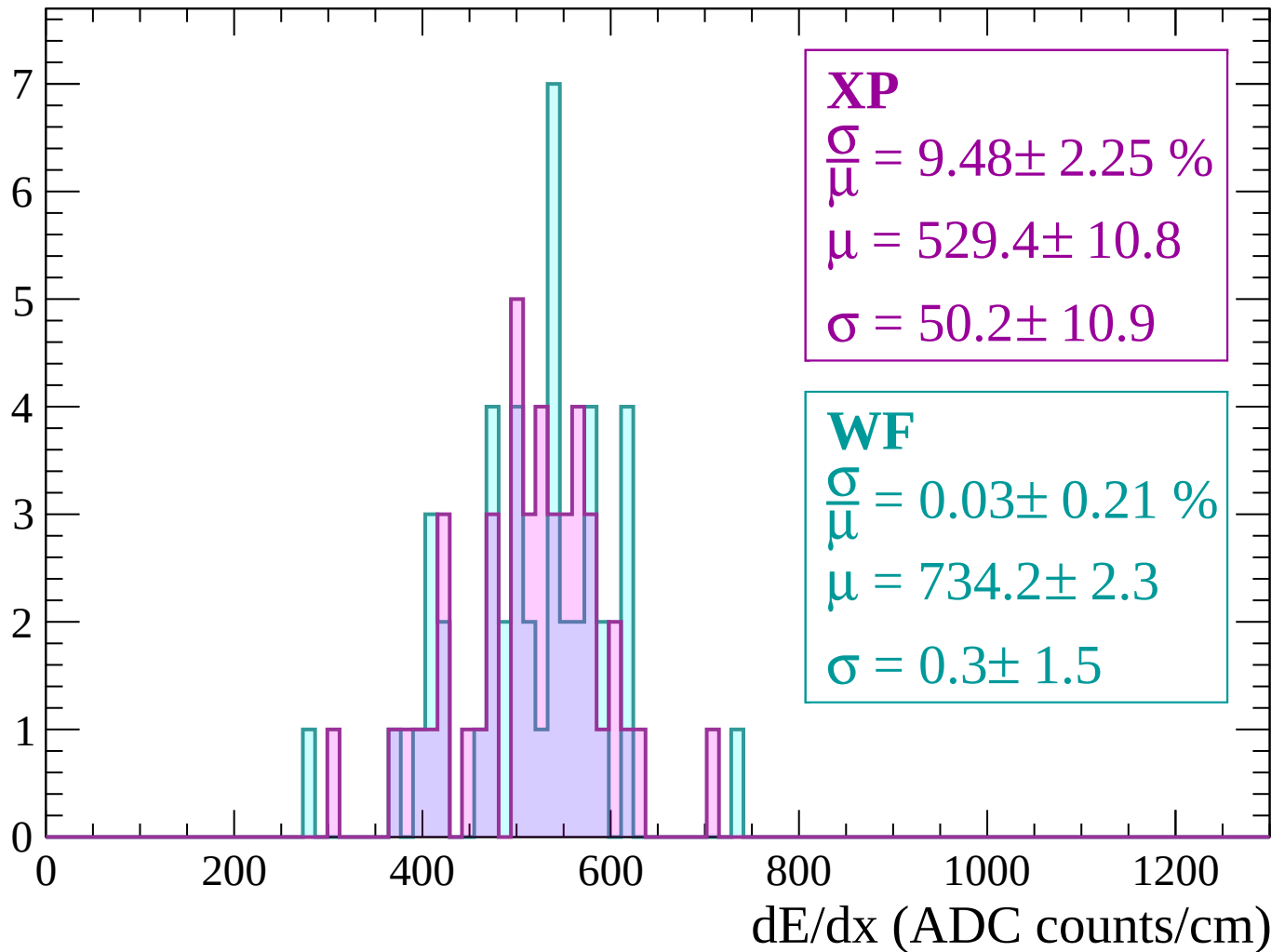
Energy loss | $-2340 < p < -2280$

Count



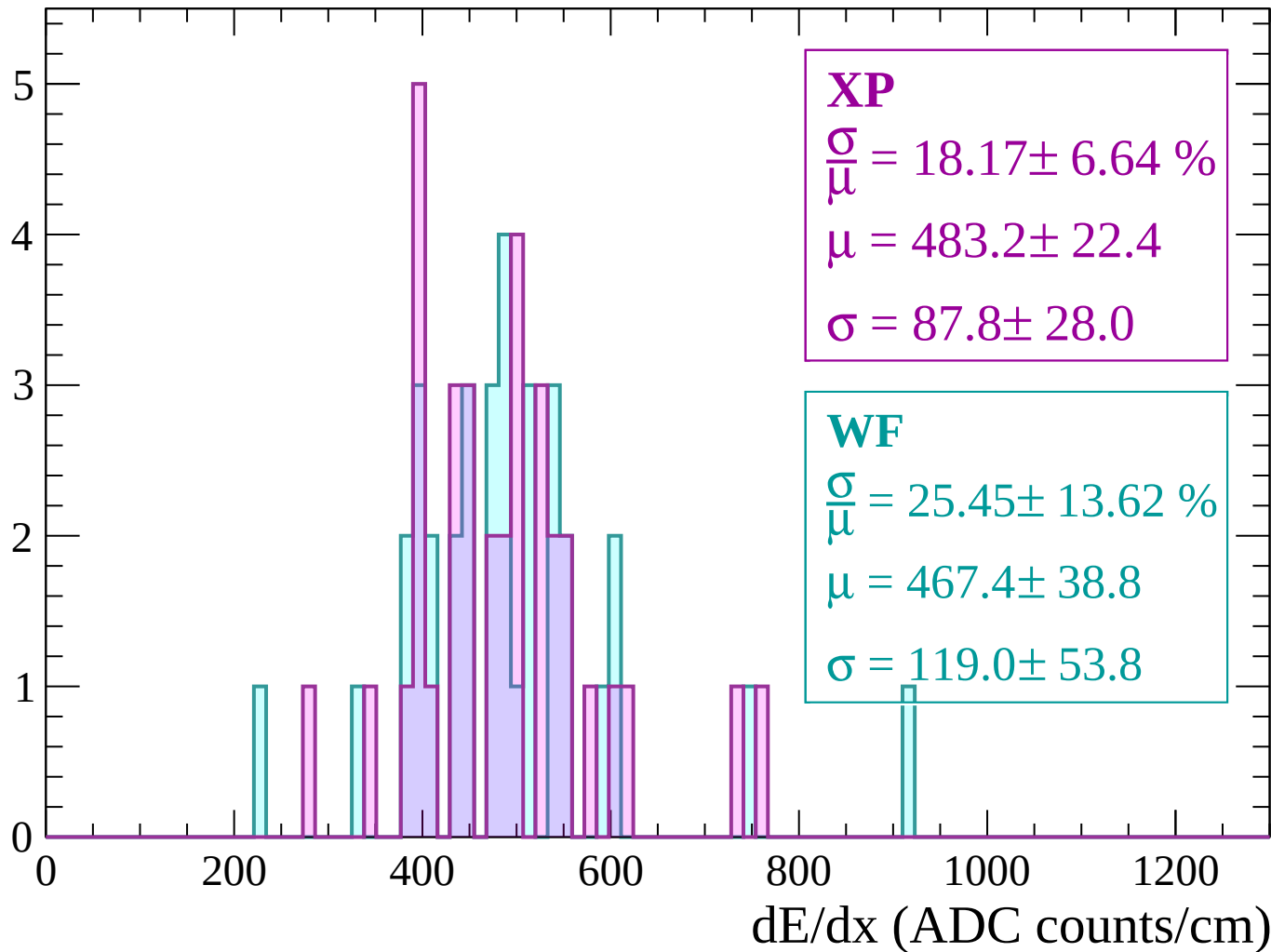
Energy loss | $-2280 < p < -2220$

Count



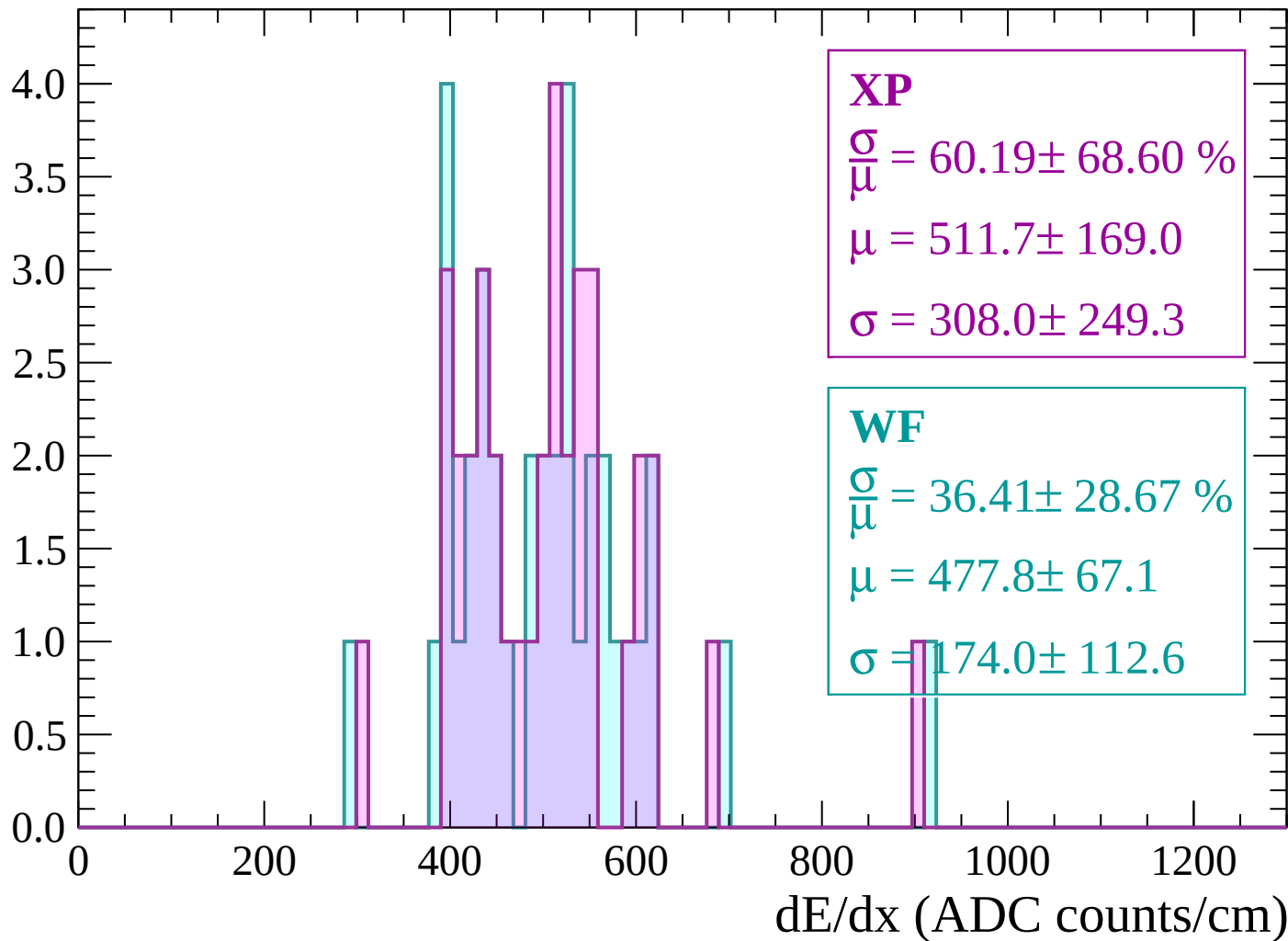
Energy loss | -2220 < p < -2160

Count



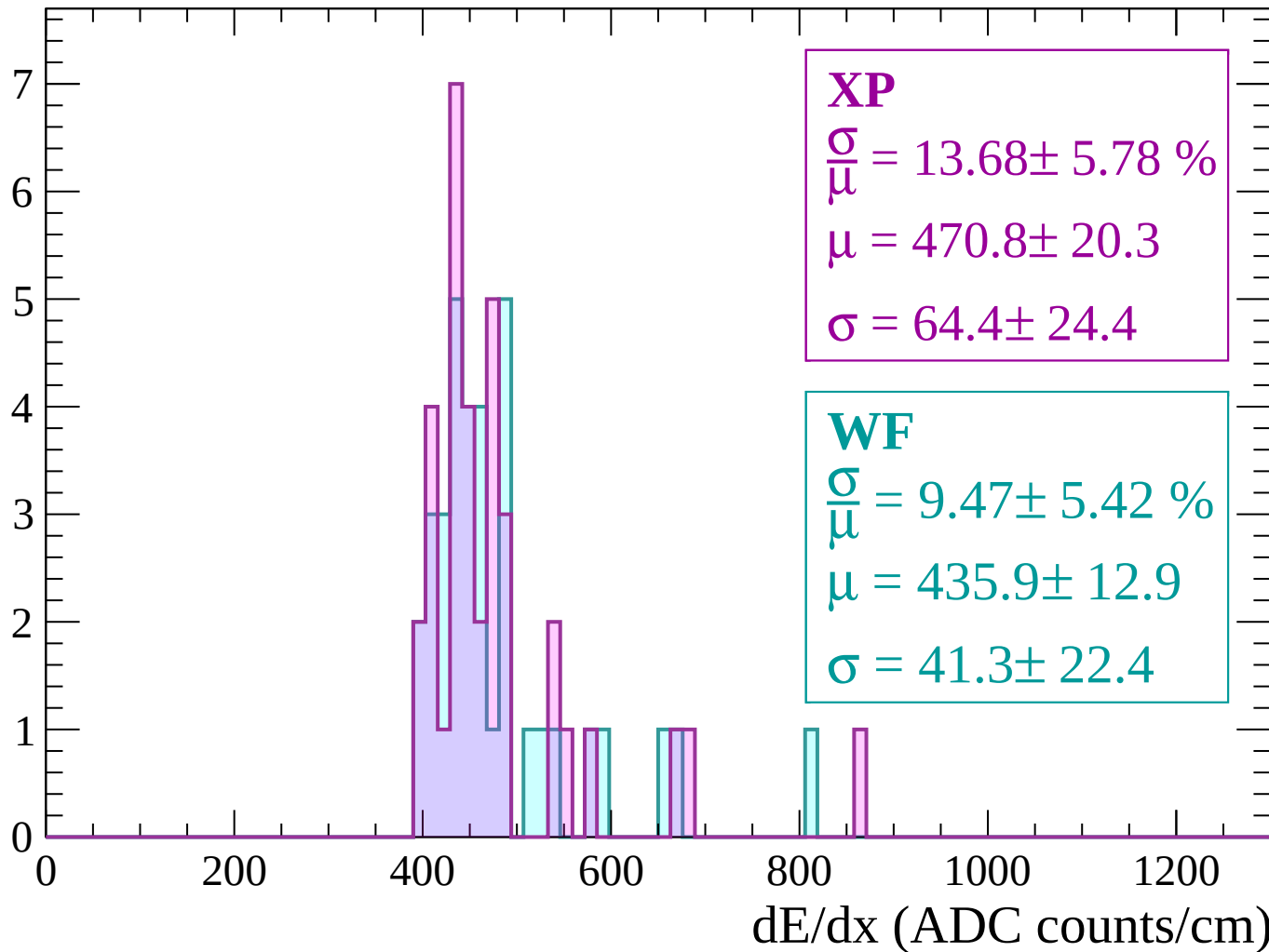
Energy loss | -2160 < p < -2100

Count



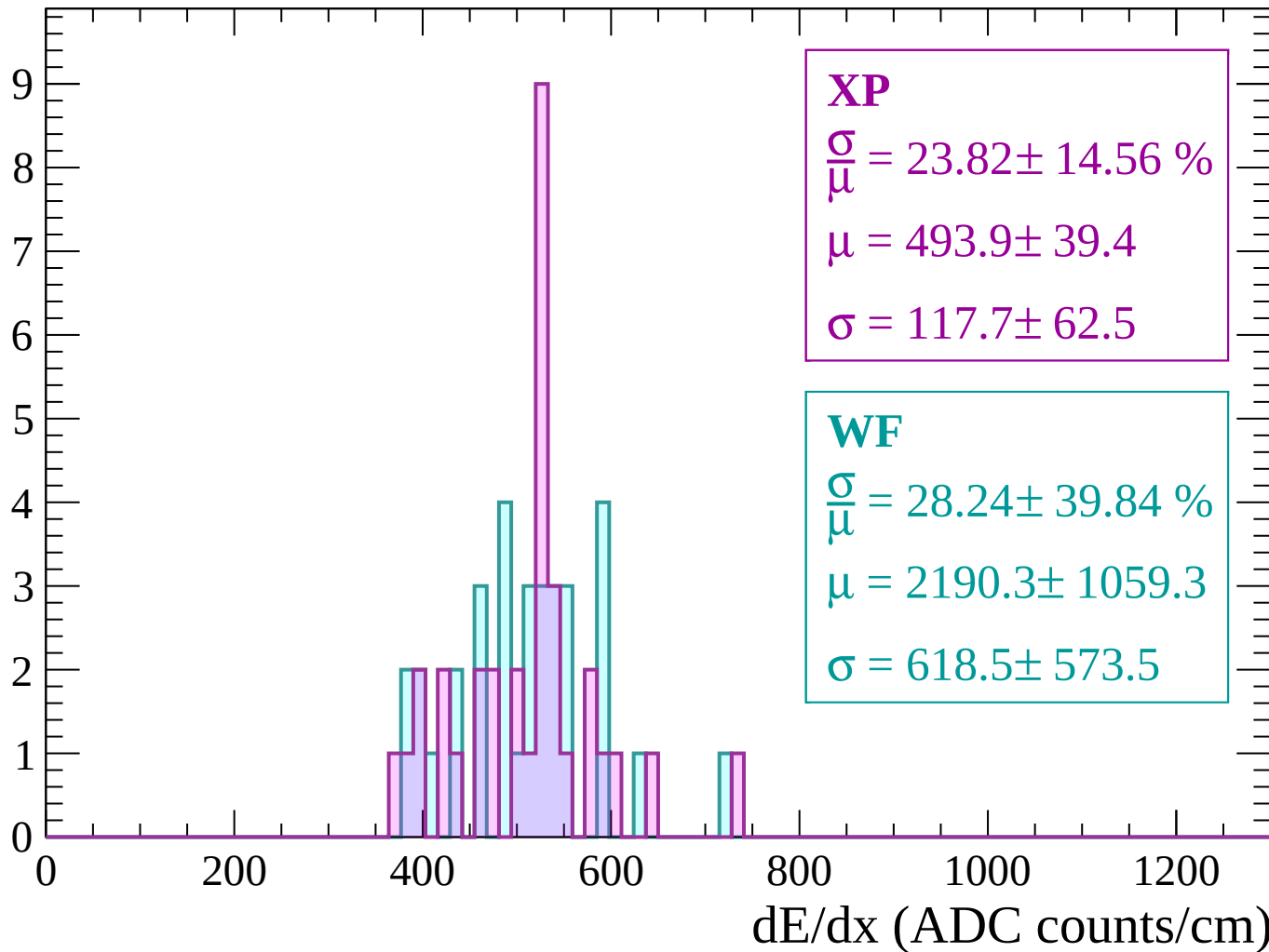
Energy loss | -2100 < p < -2040

Count



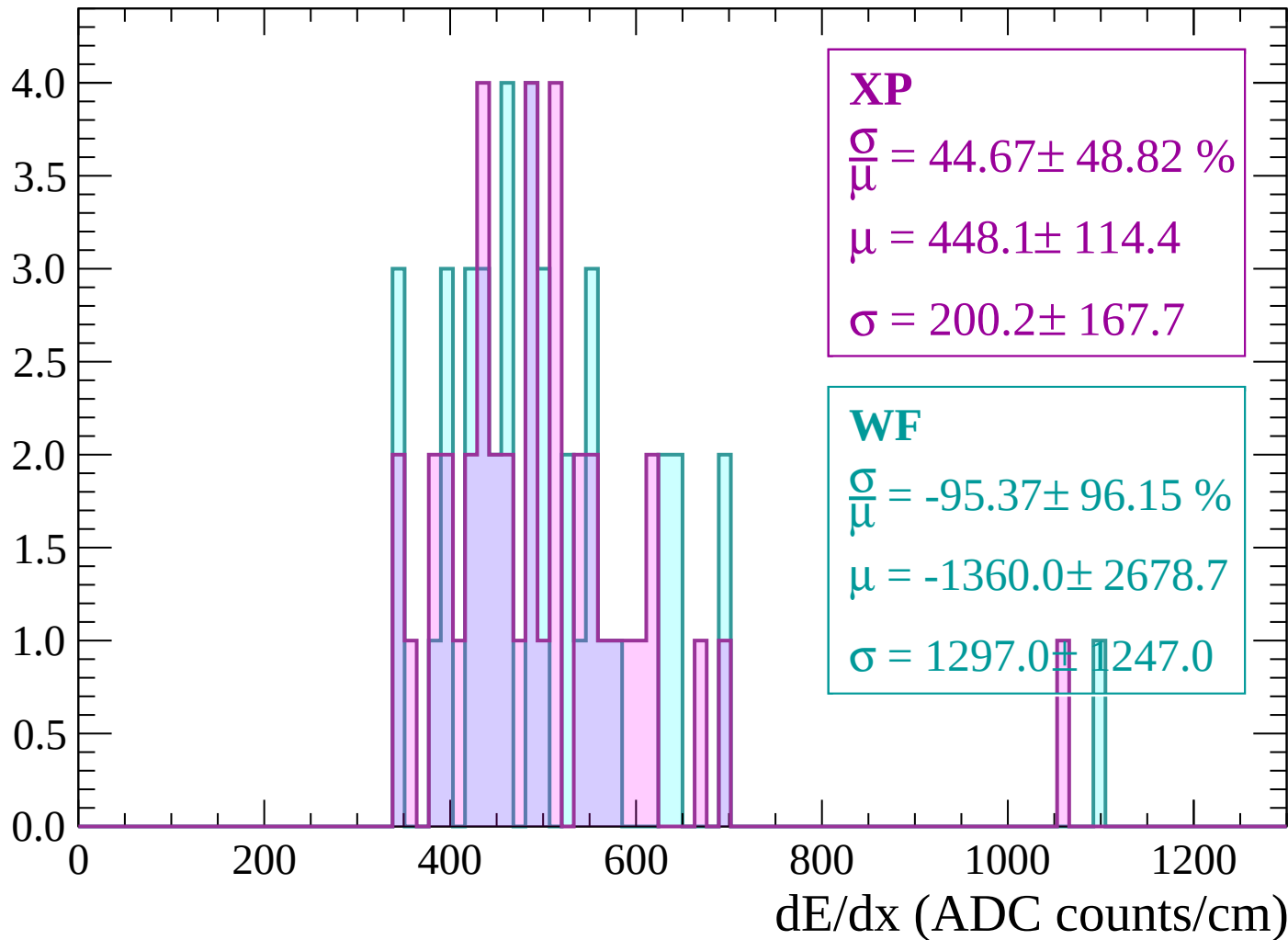
Energy loss | -2040 < p < -1980

Count



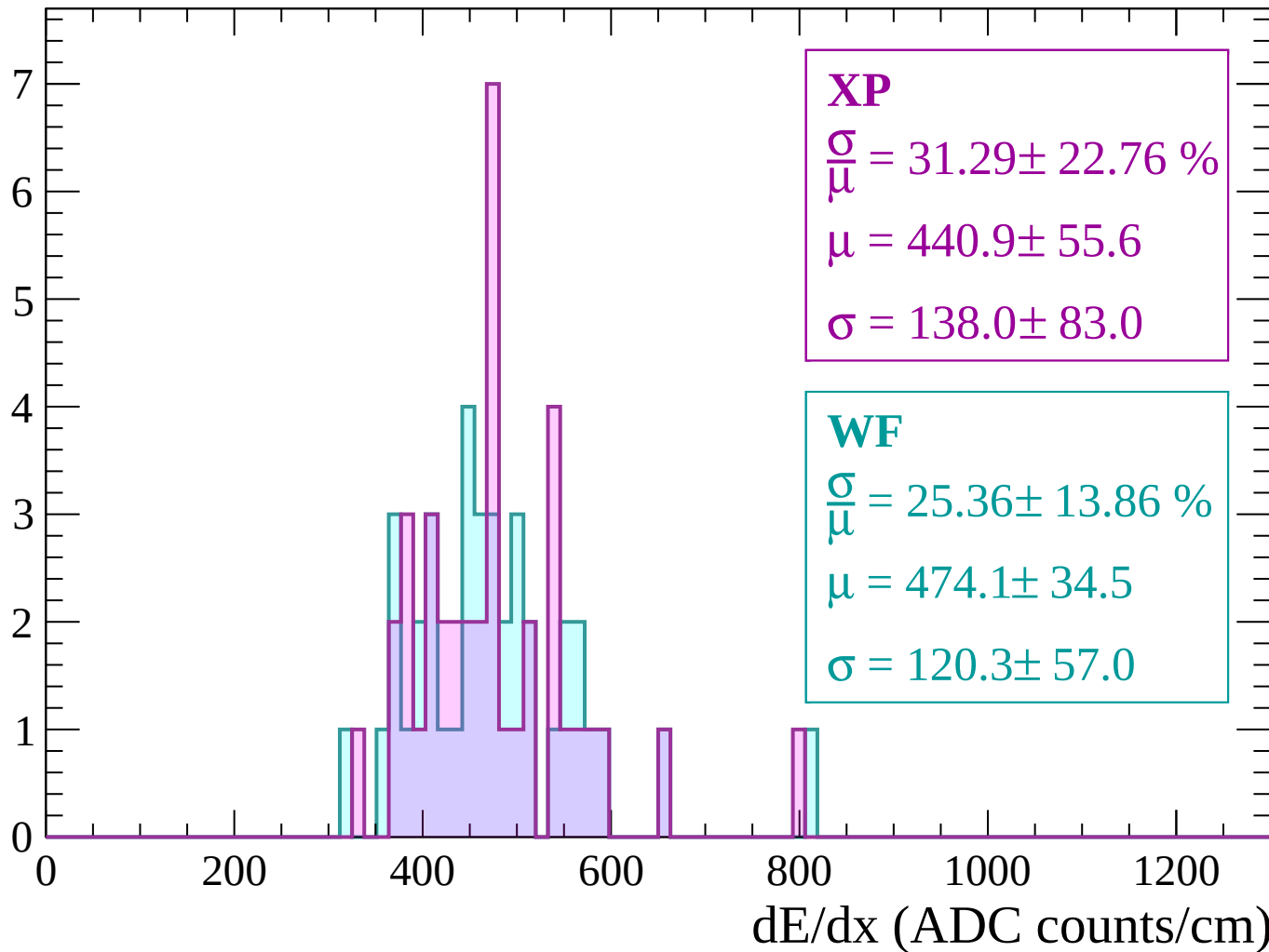
Energy loss | -1980 < p < -1920

Count



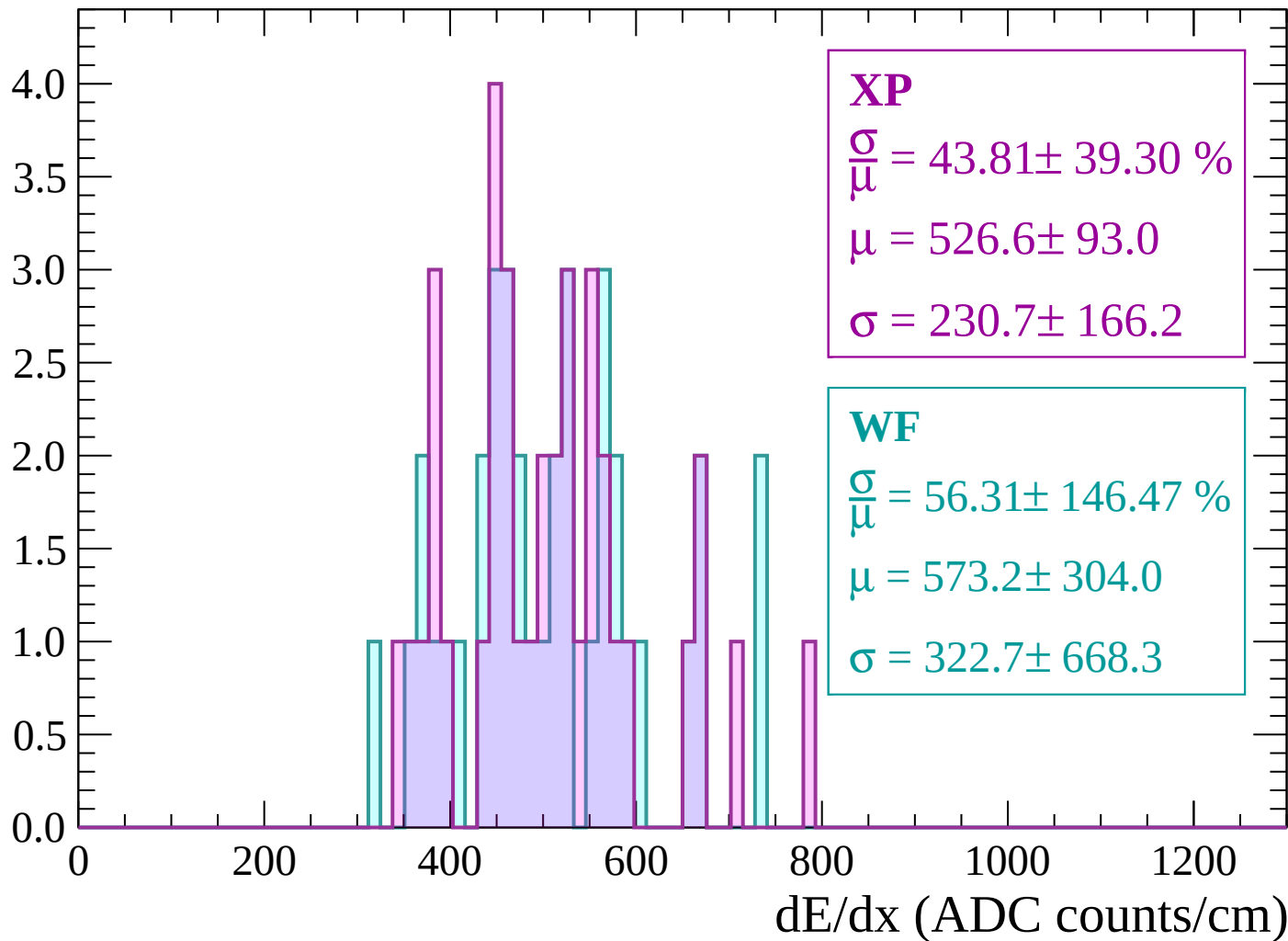
Energy loss | -1920 < p < -1860

Count



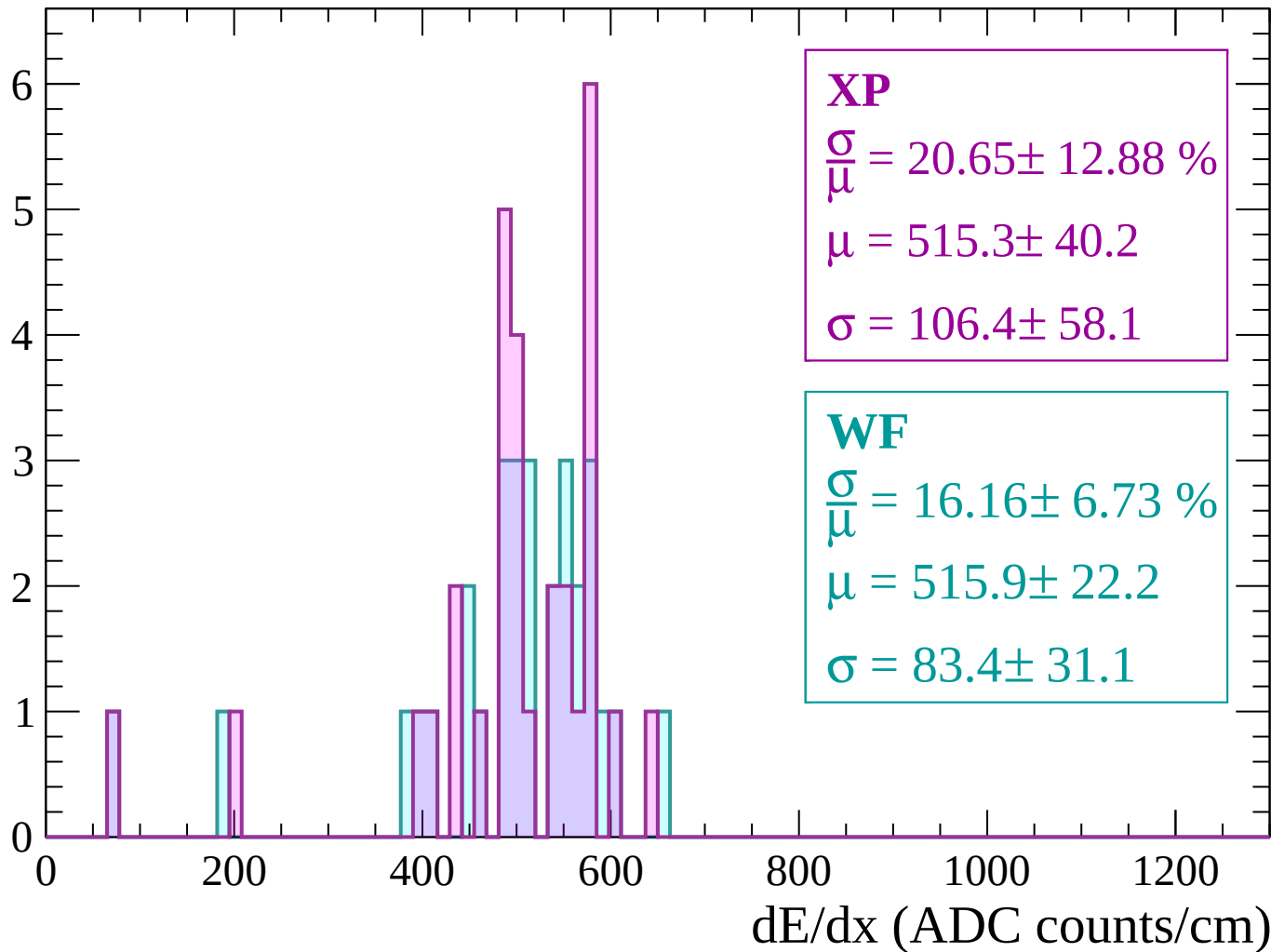
Energy loss | -1860 < p < -1800

Count



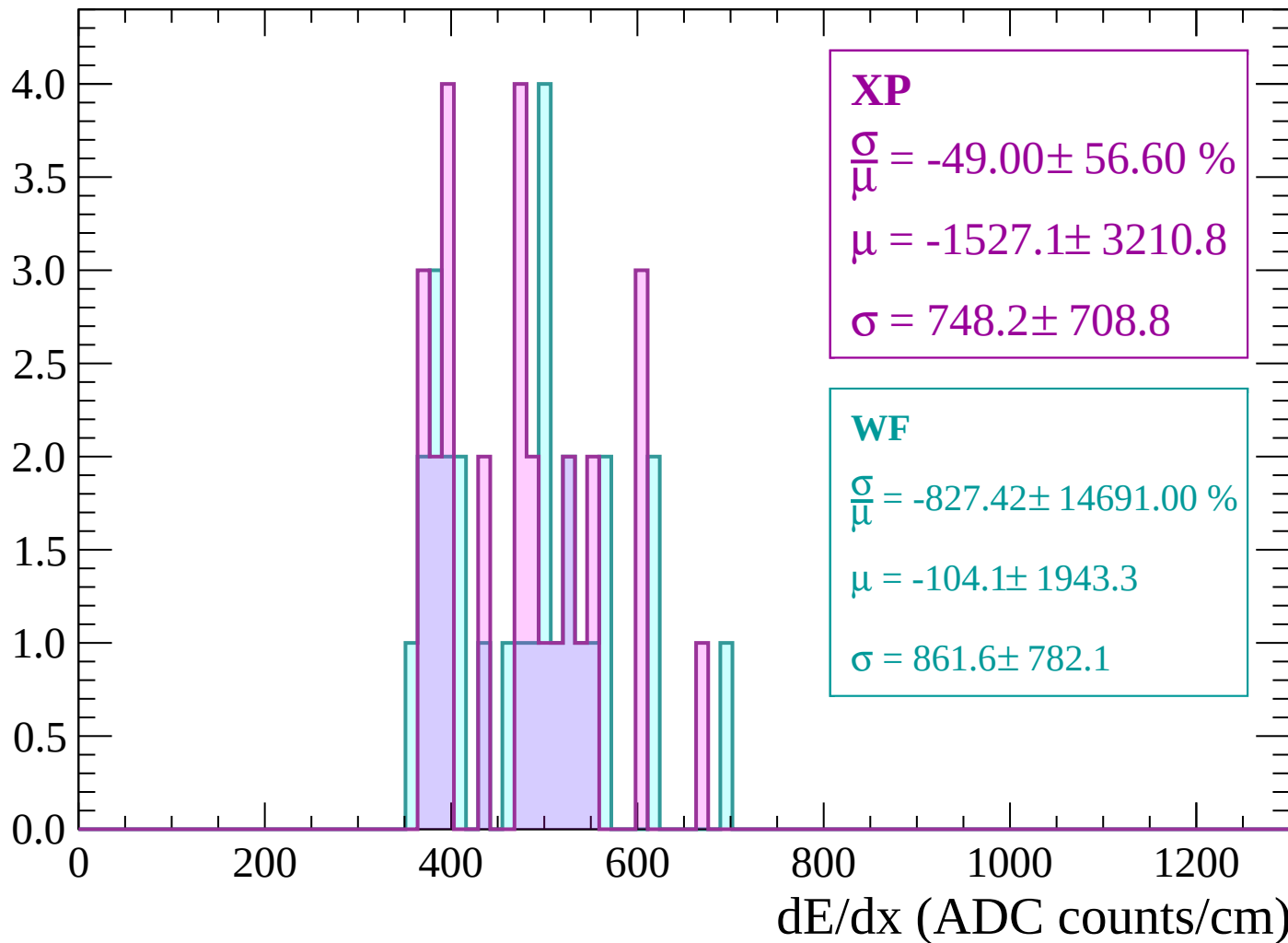
Energy loss | -1800 < p < -1740

Count



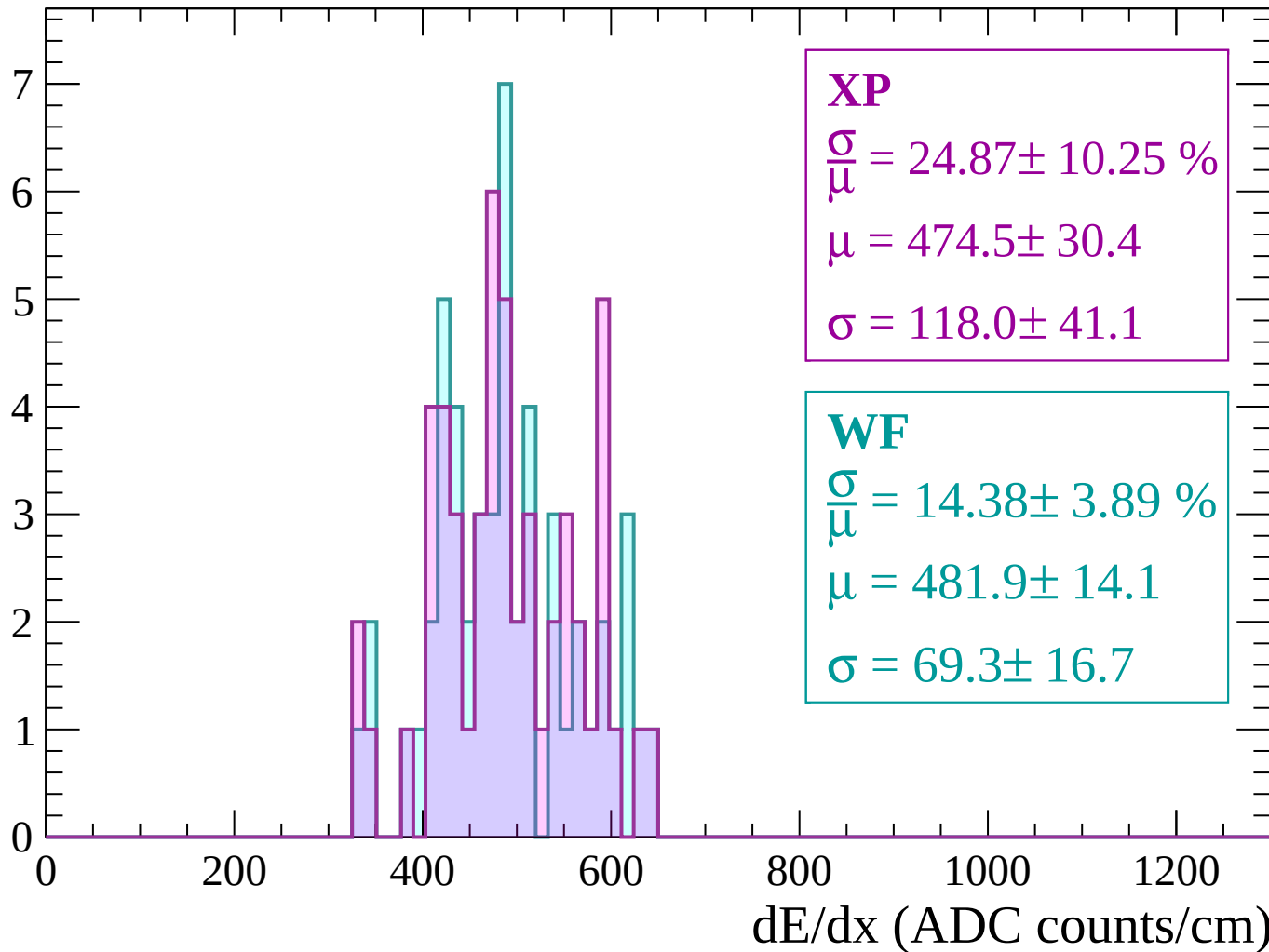
Energy loss | -1740 < p < -1680

Count



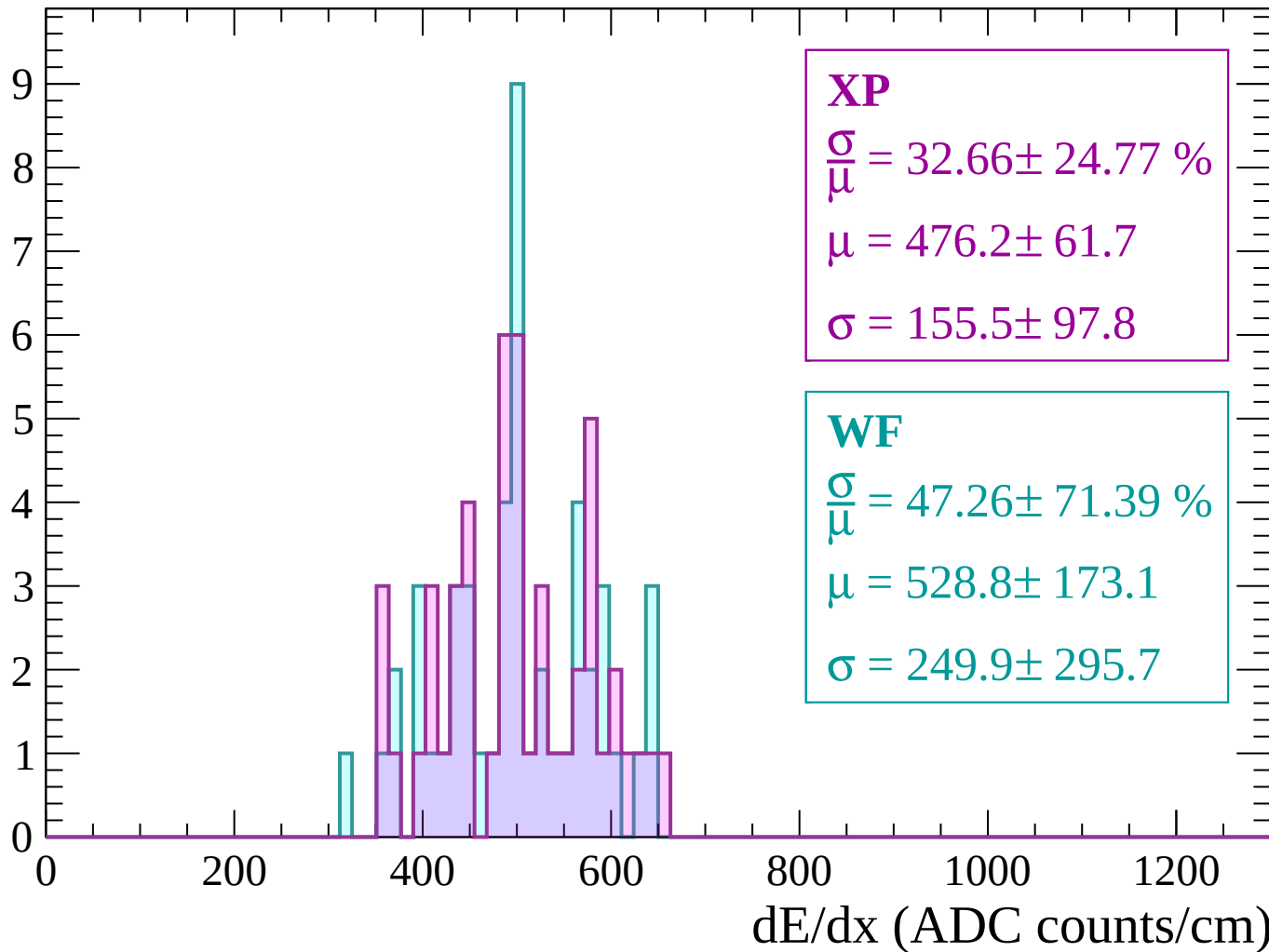
Energy loss | $-1680 < p < -1620$

Count



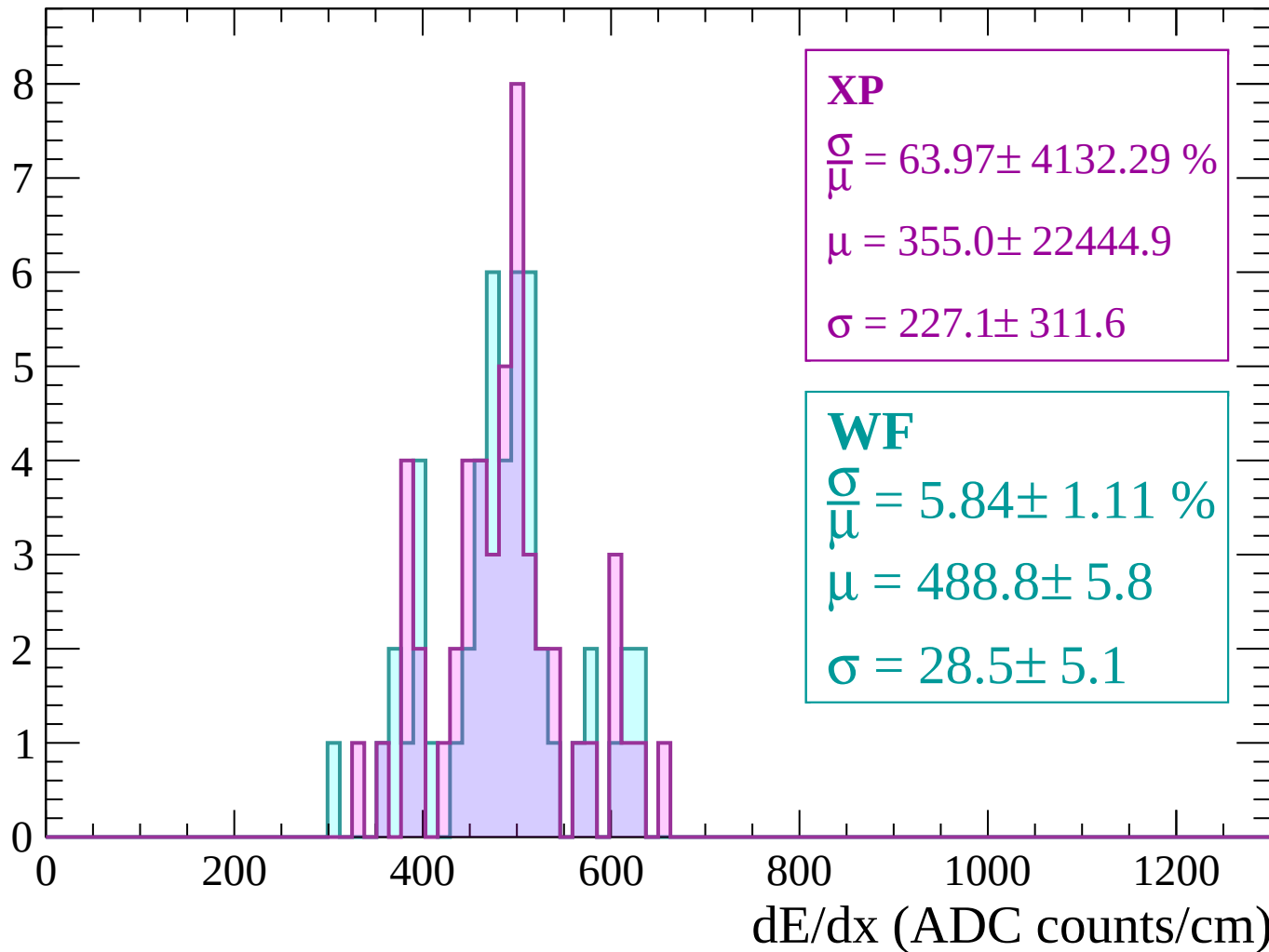
Energy loss | -1620 < p < -1560

Count



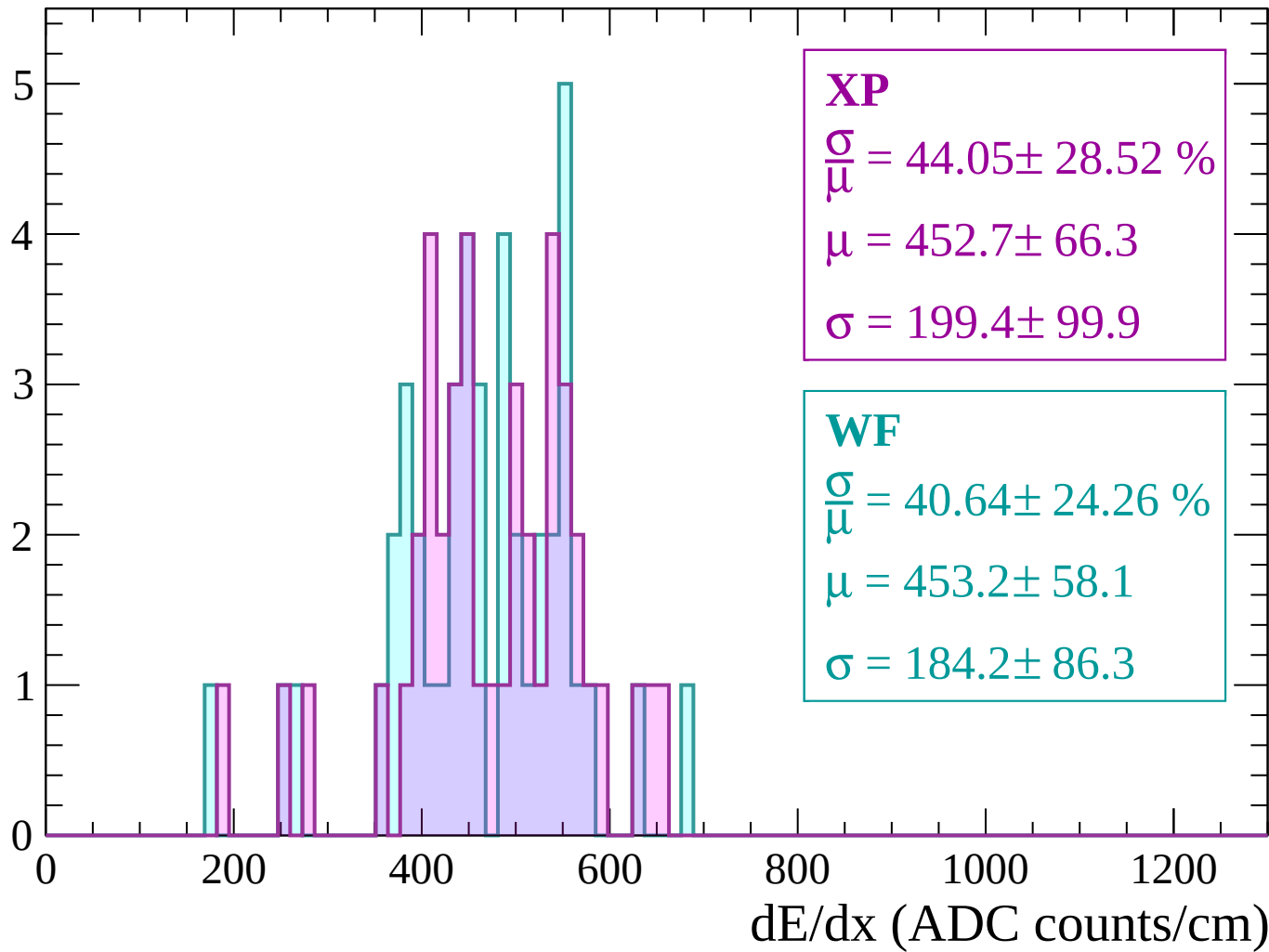
Energy loss | -1560 < p < -1500

Count



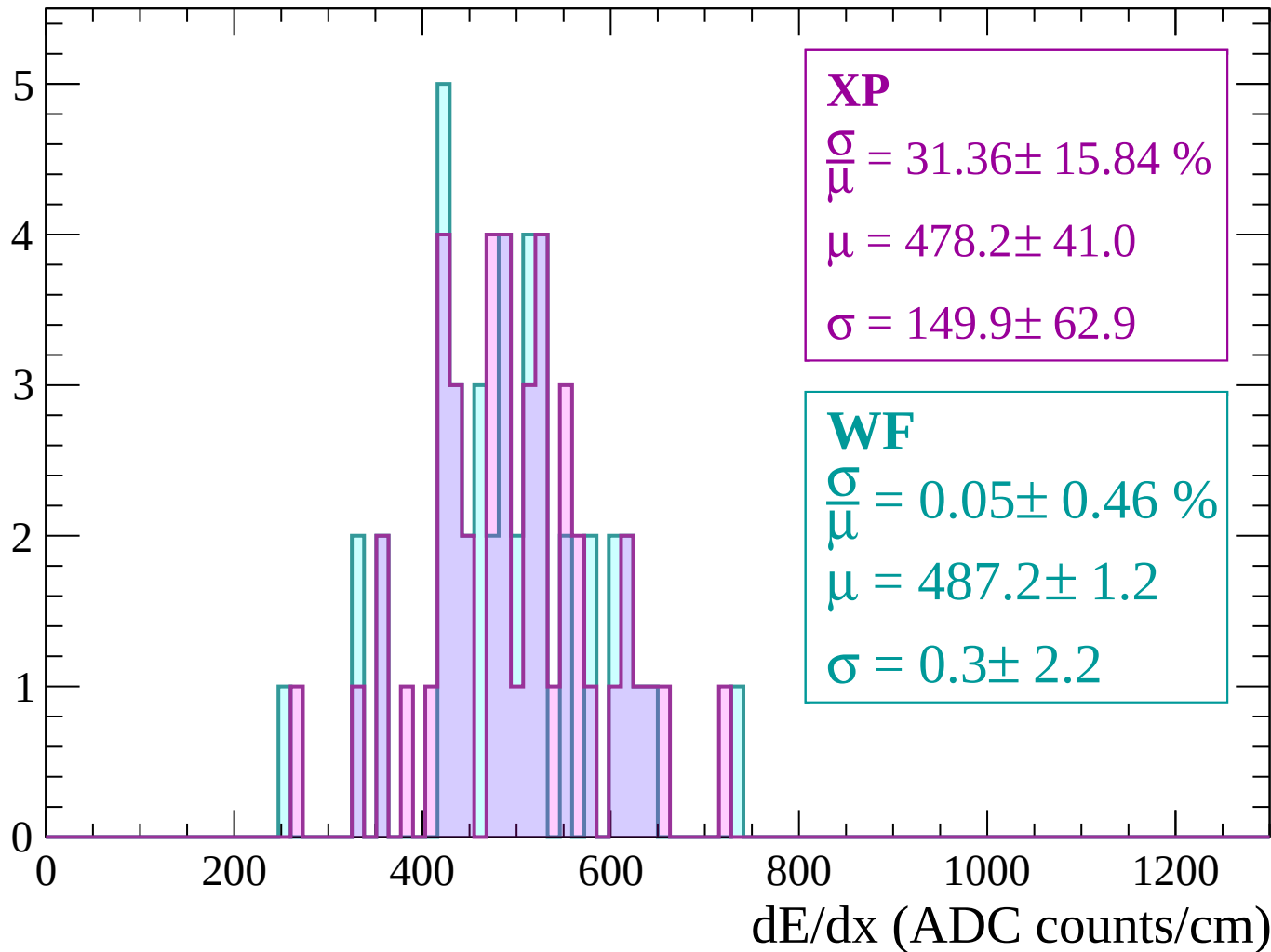
Energy loss | -1500 < p < -1440

Count



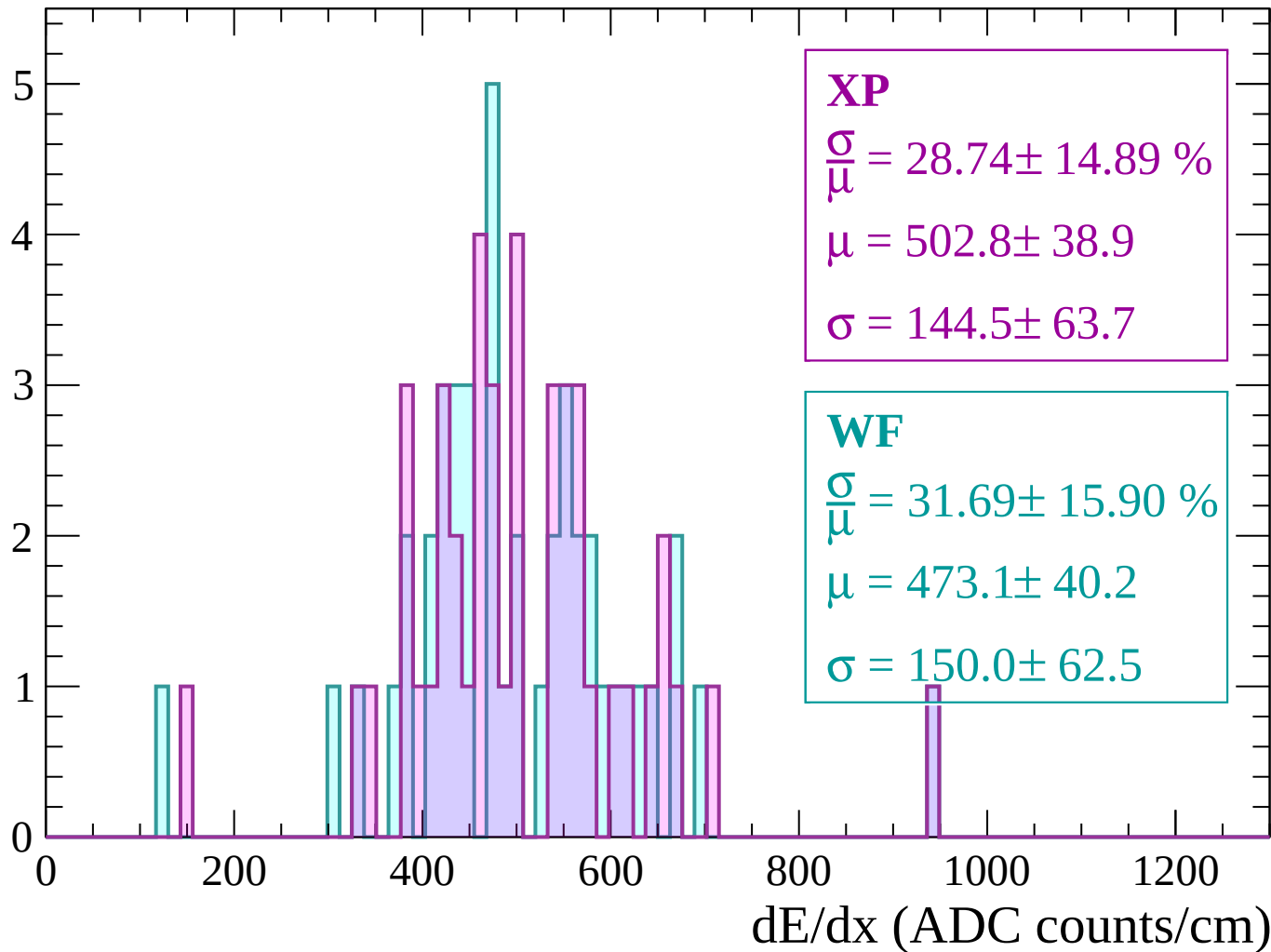
Energy loss | $-1440 < p < -1380$

Count



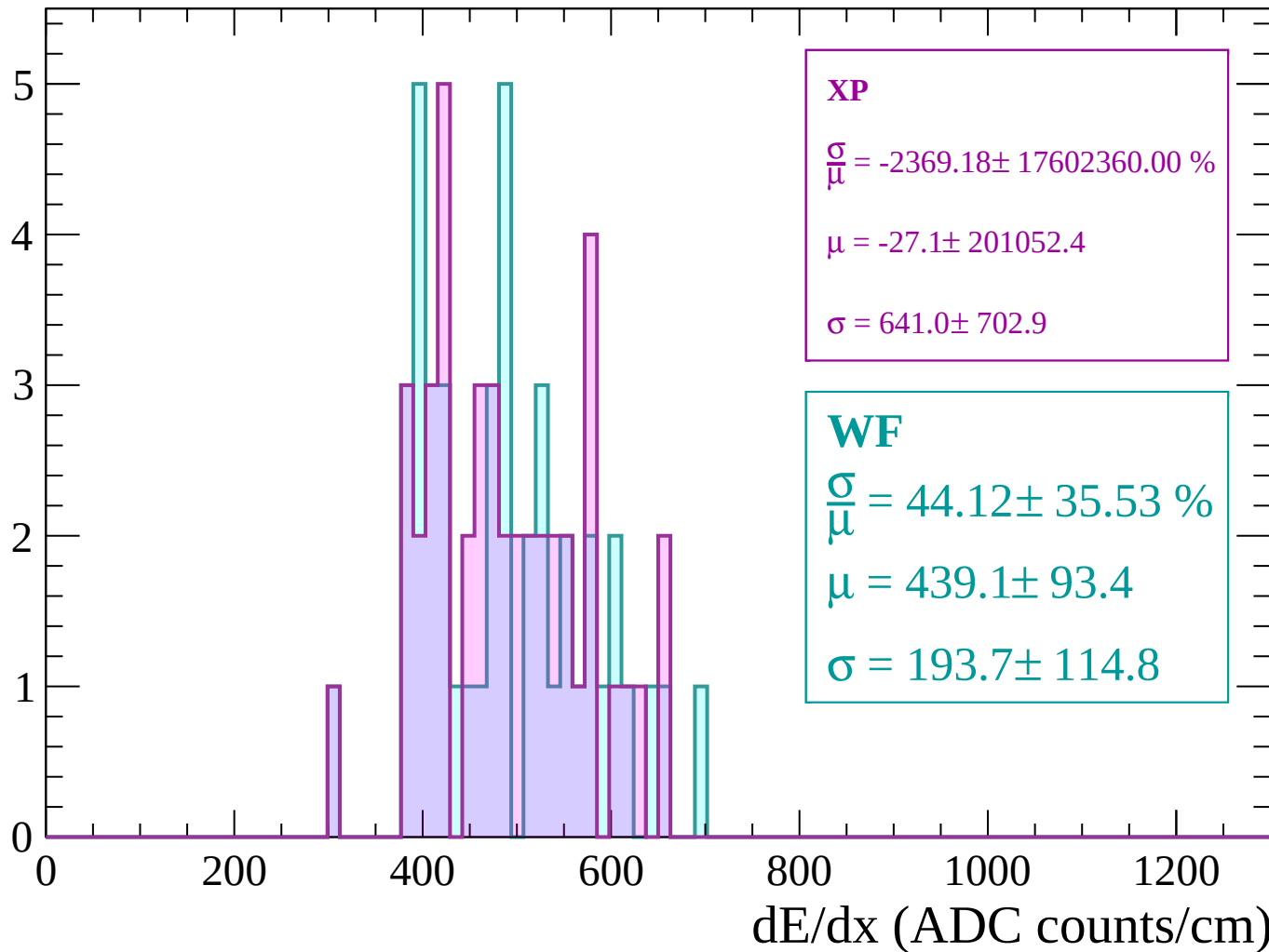
Energy loss | $-1380 < p < -1320$

Count



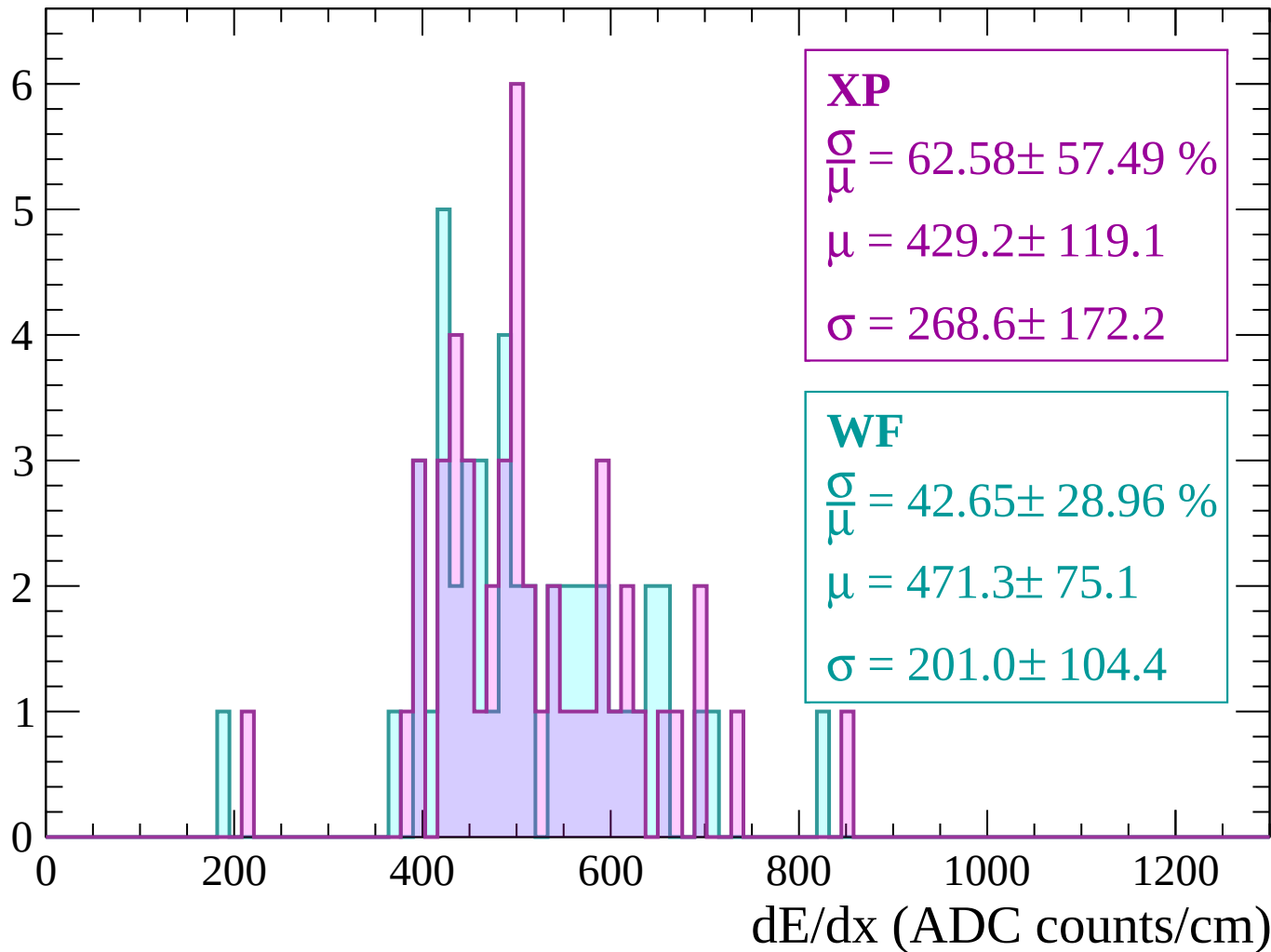
Energy loss | -1320 < p < -1260

Count



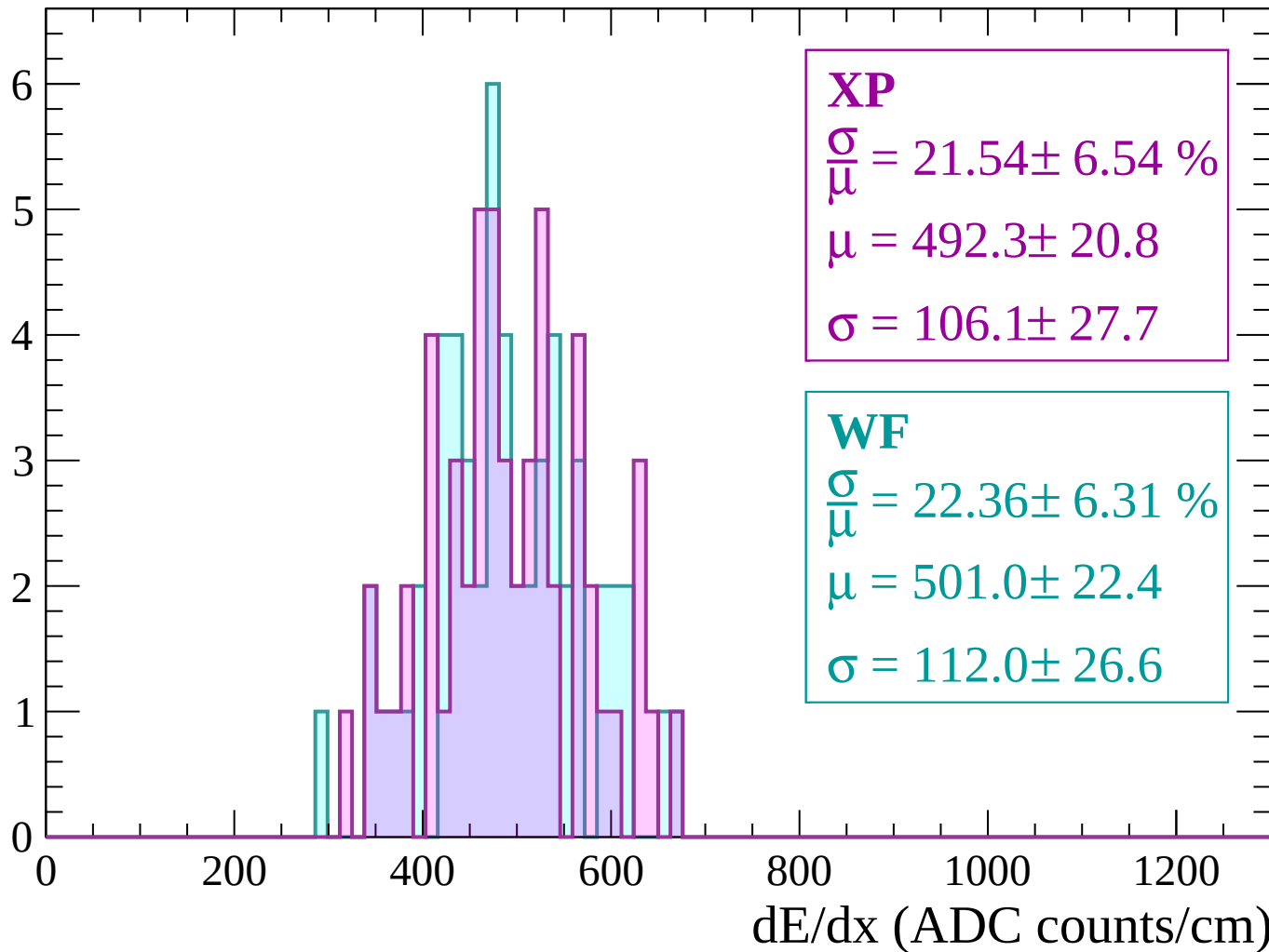
Energy loss | -1260 < p < -1200

Count



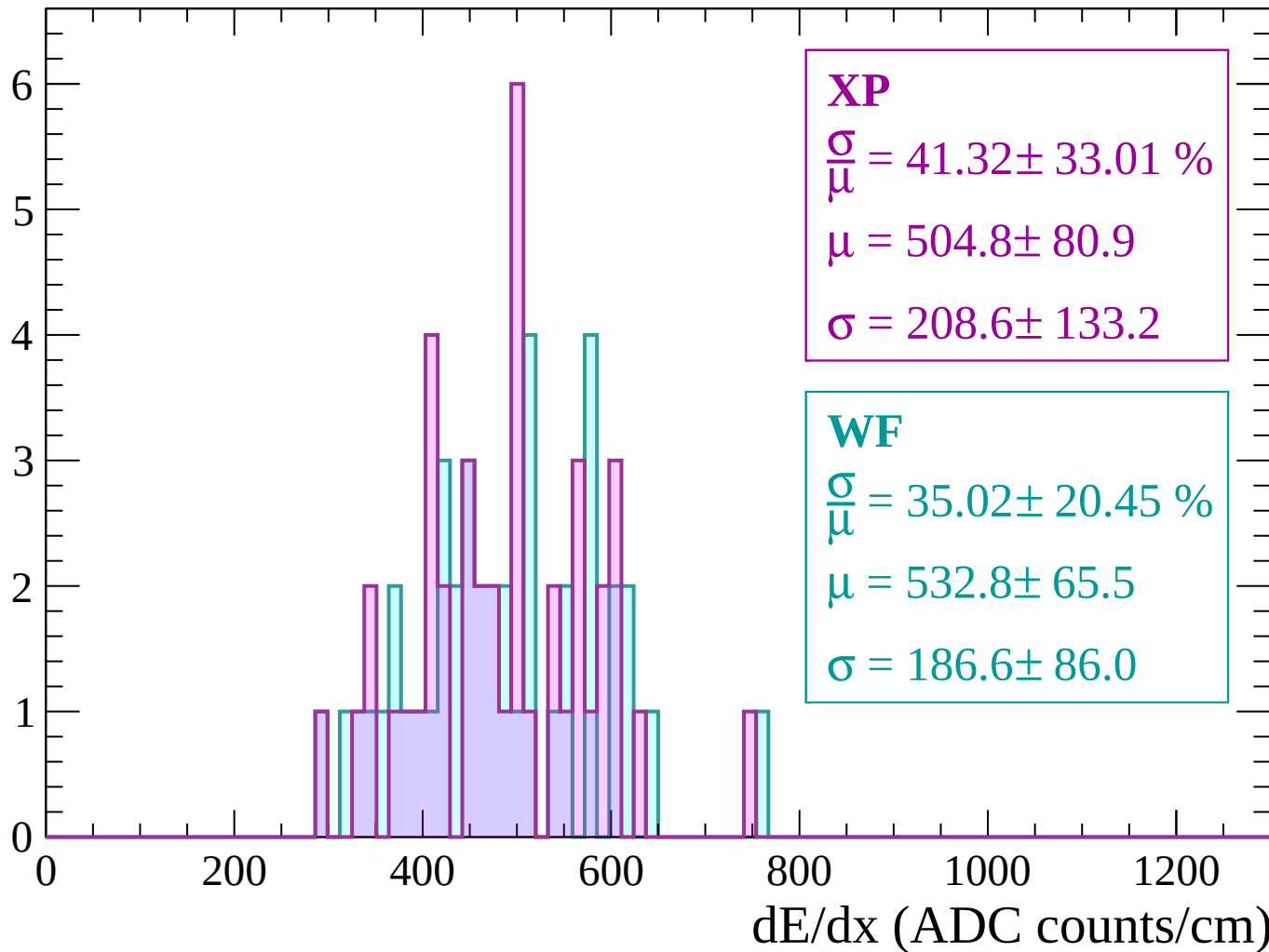
Energy loss | -1200 < p < -1140

Count



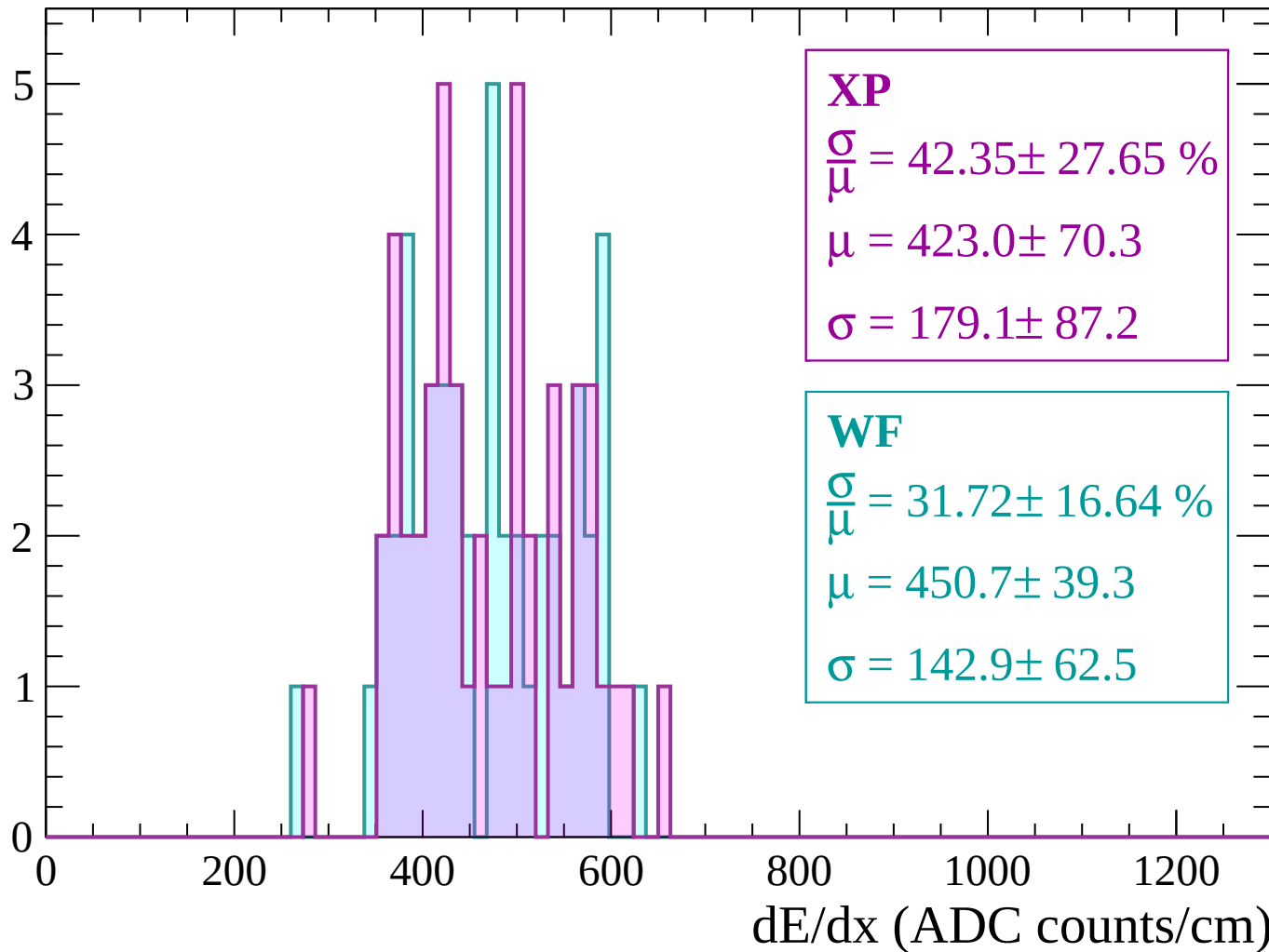
Energy loss | $-1140 < p < -1080$

Count



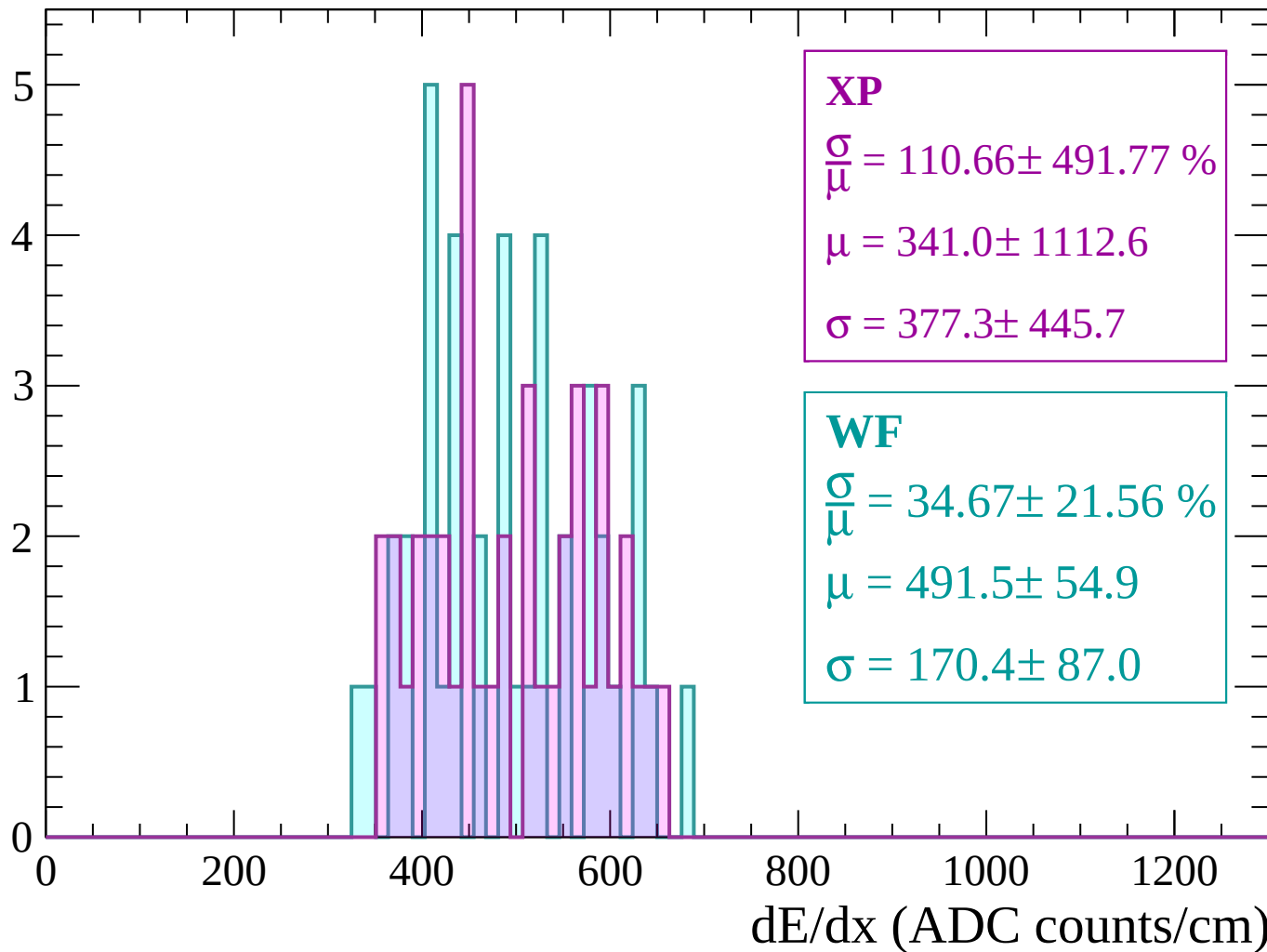
Energy loss | -1080 < p < -1020

Count



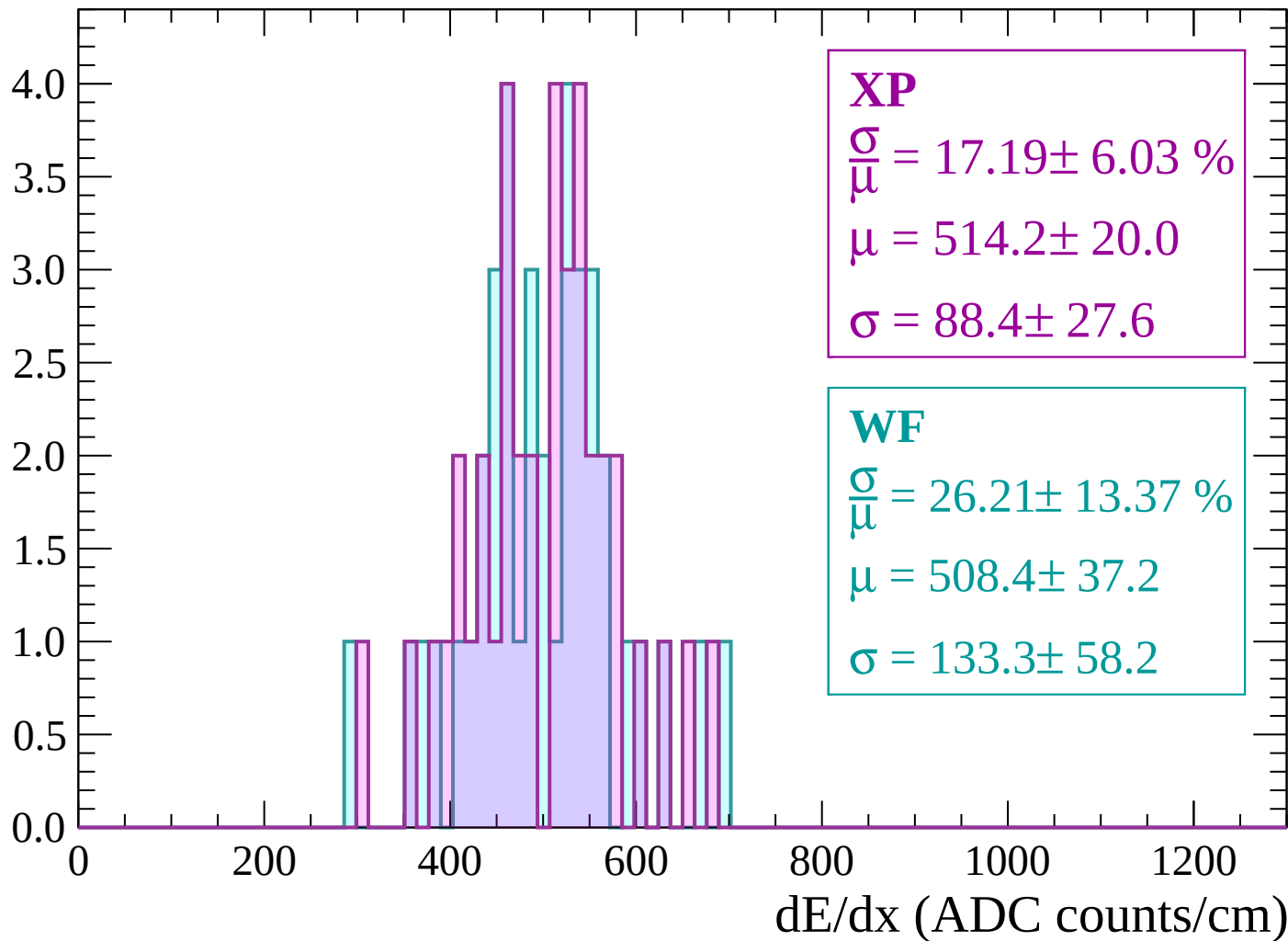
Energy loss | $-1020 < p < -960$

Count



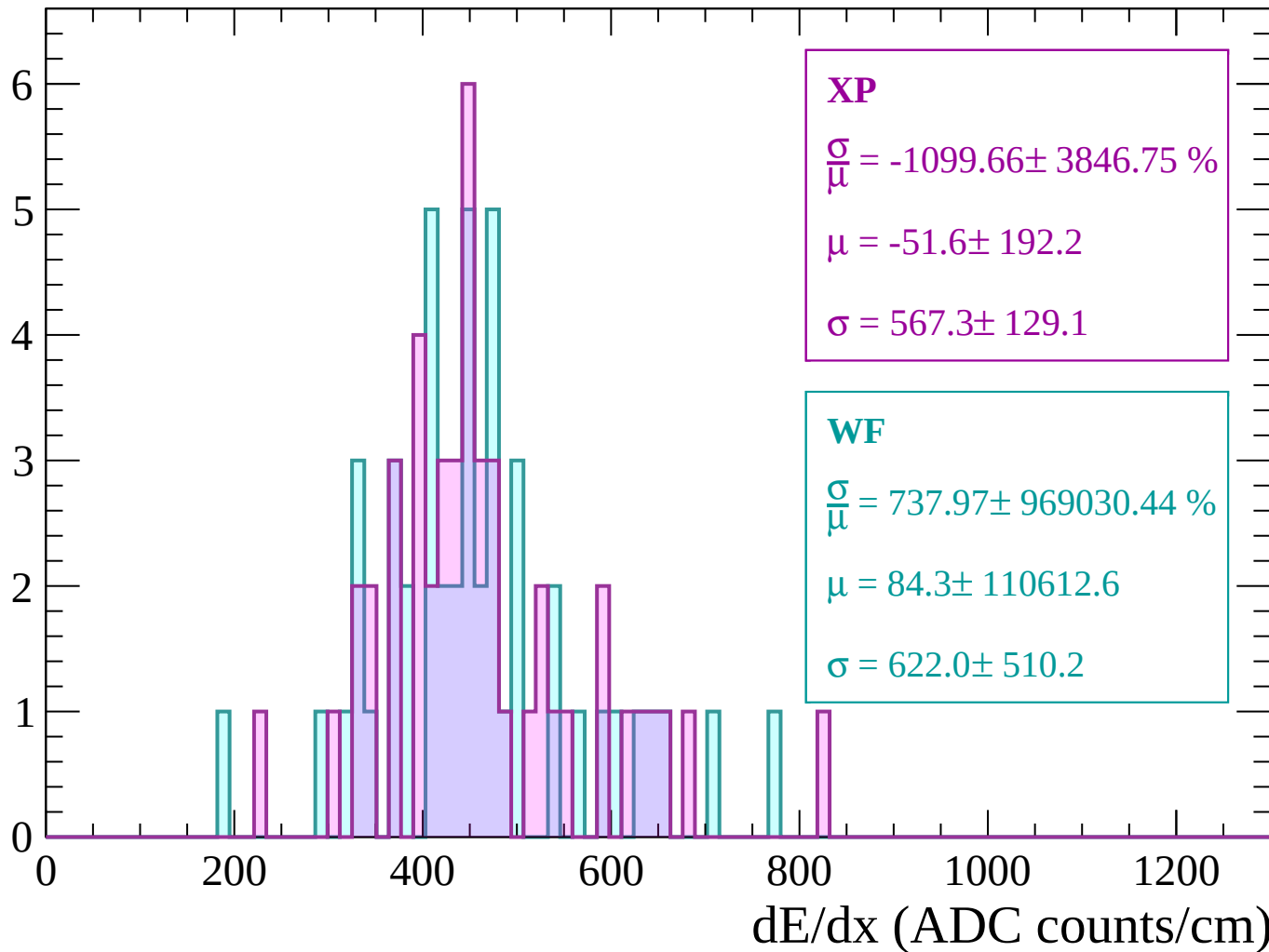
Energy loss | $-960 < p < -900$

Count



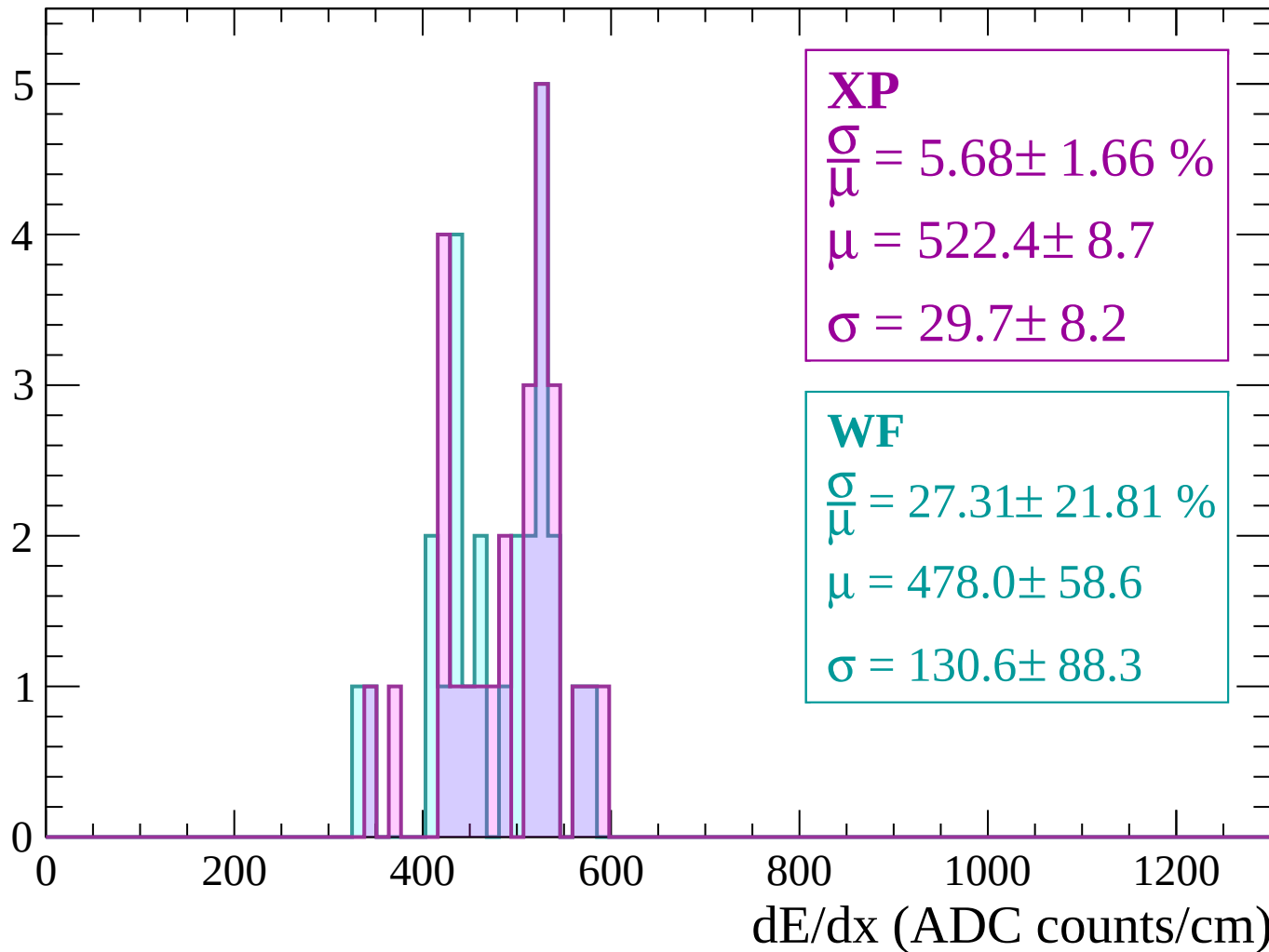
Energy loss | $-900 < p < -840$

Count



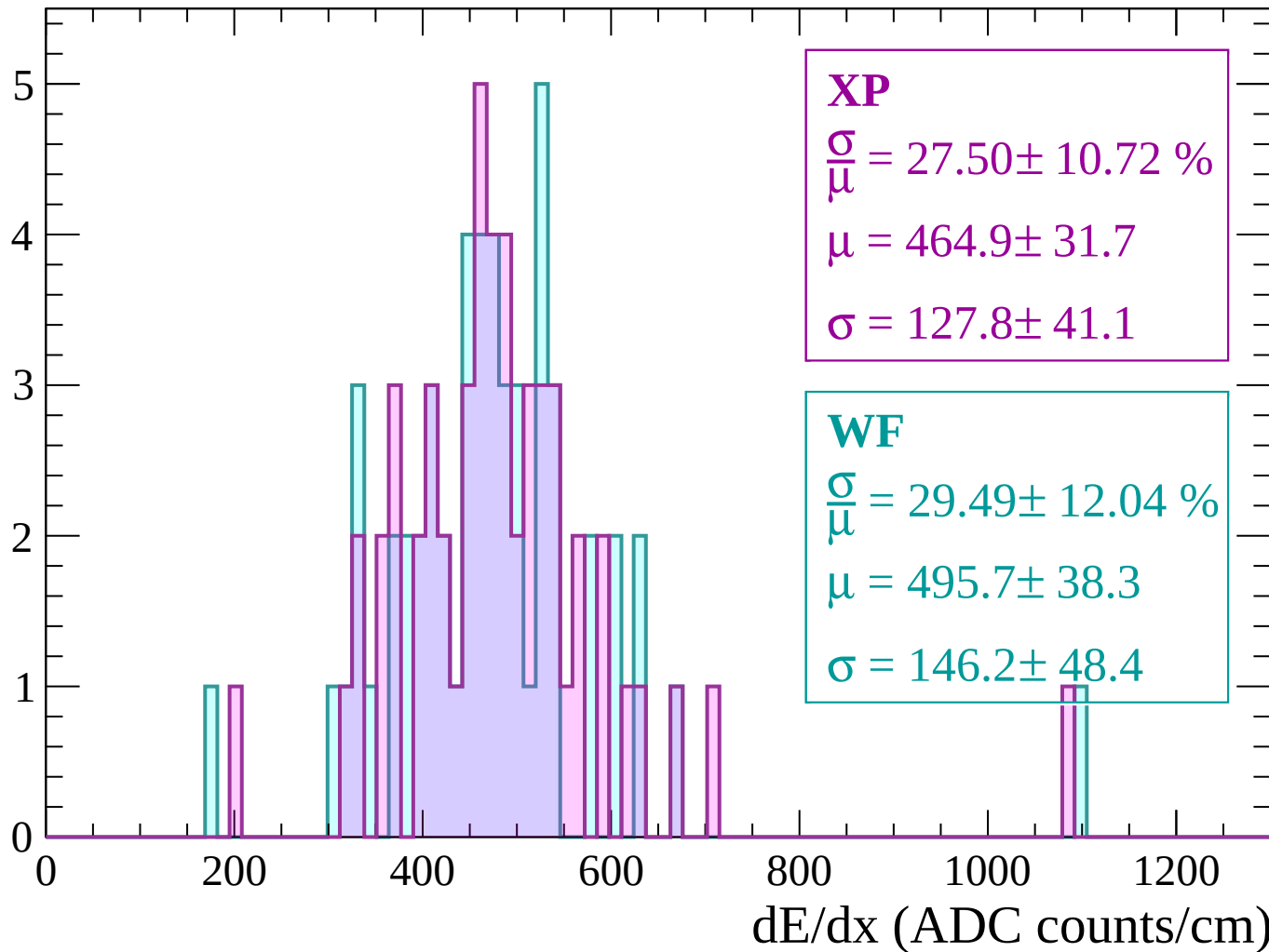
Energy loss | $-840 < p < -780$

Count



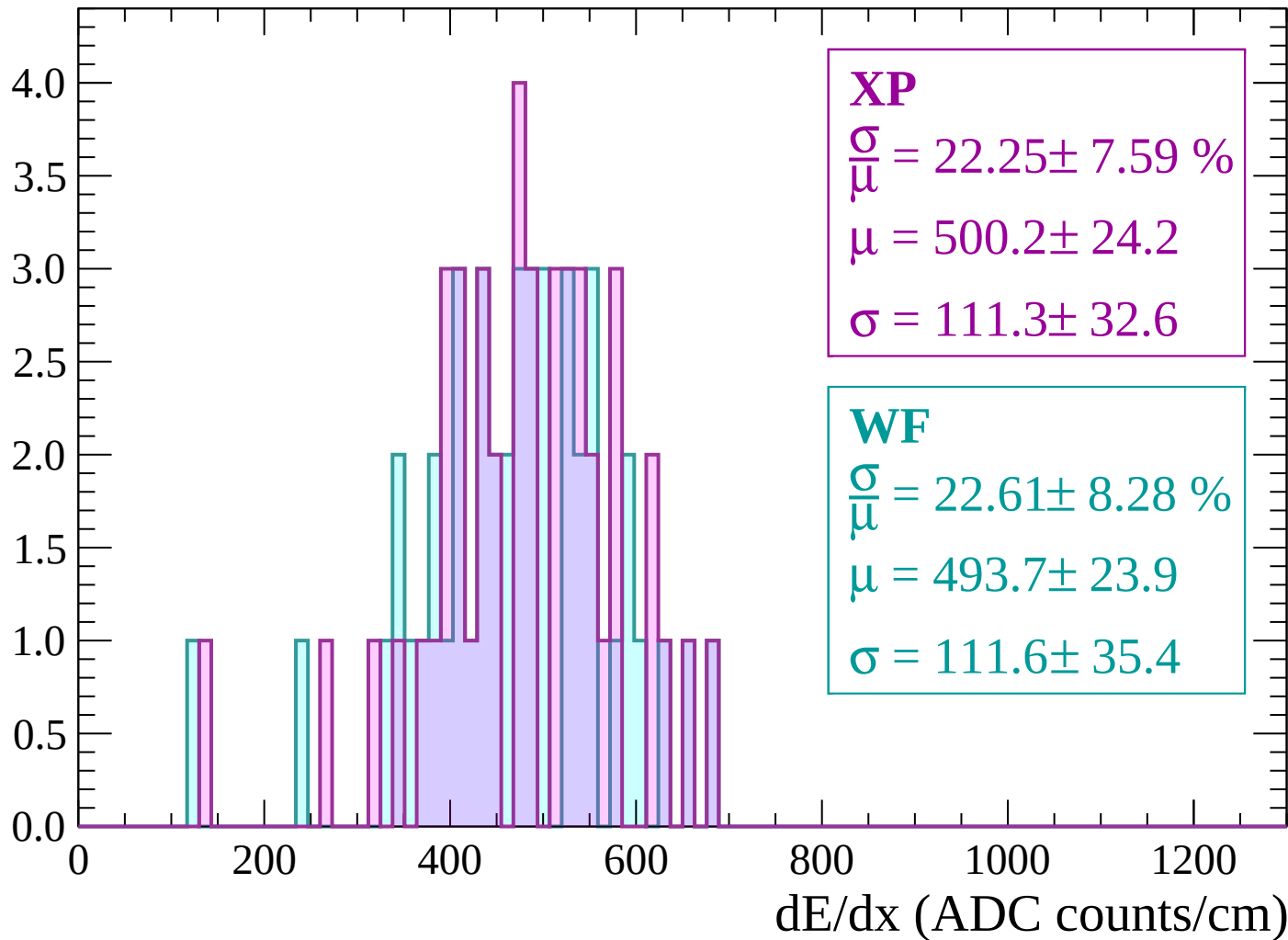
Energy loss | $-780 < p < -720$

Count



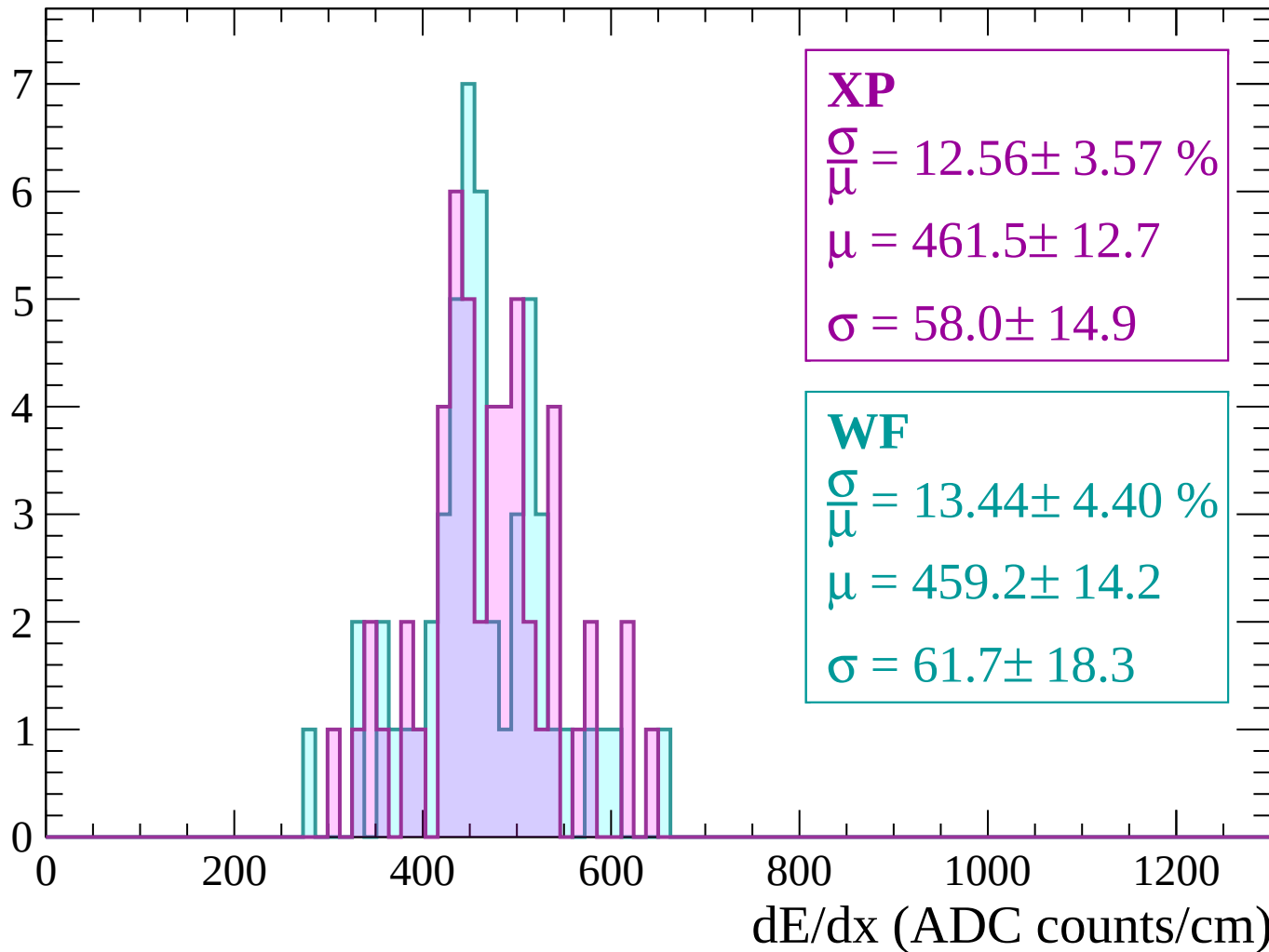
Energy loss | -720 < p < -660

Count



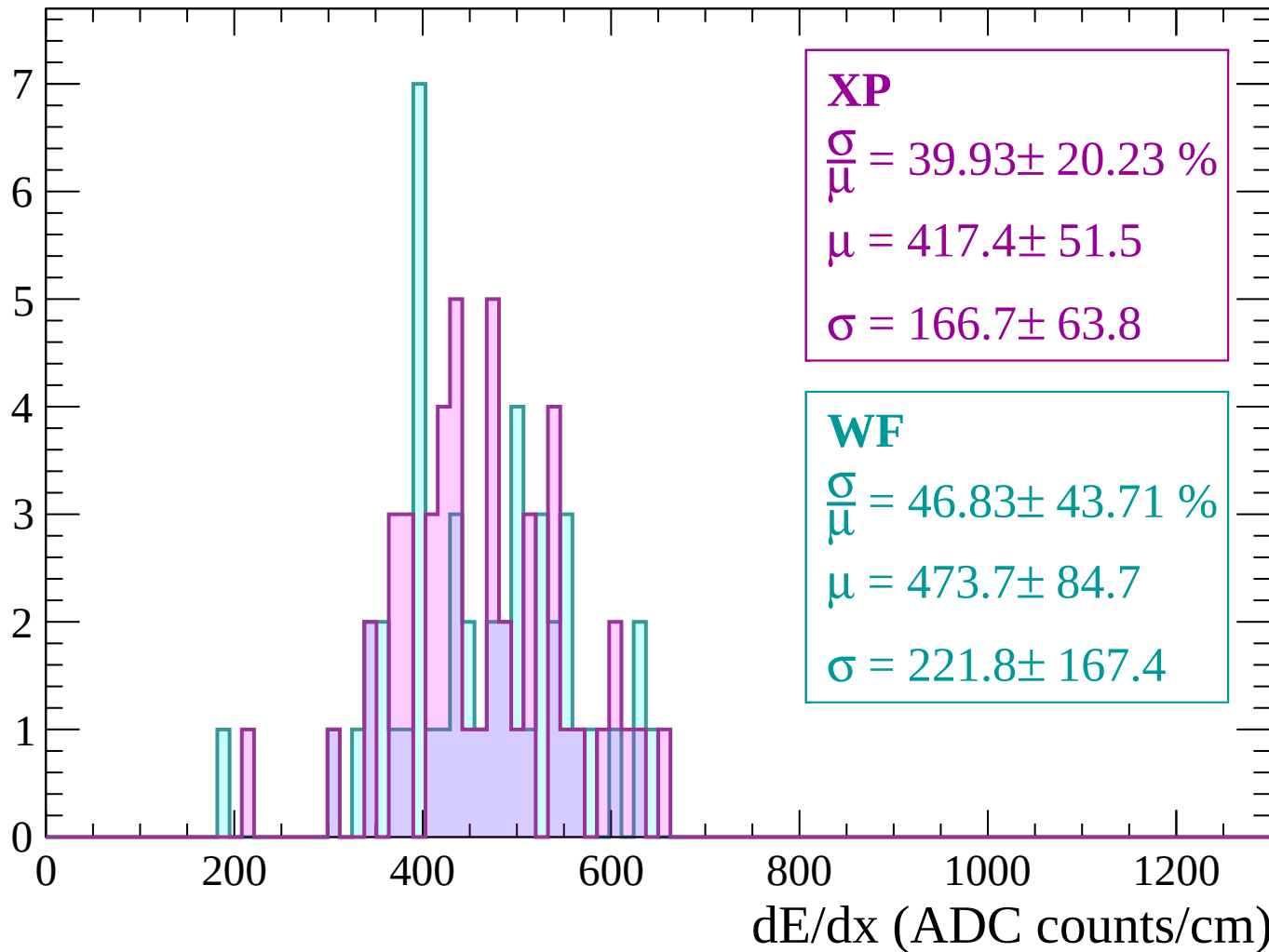
Energy loss | $-660 < p < -600$

Count



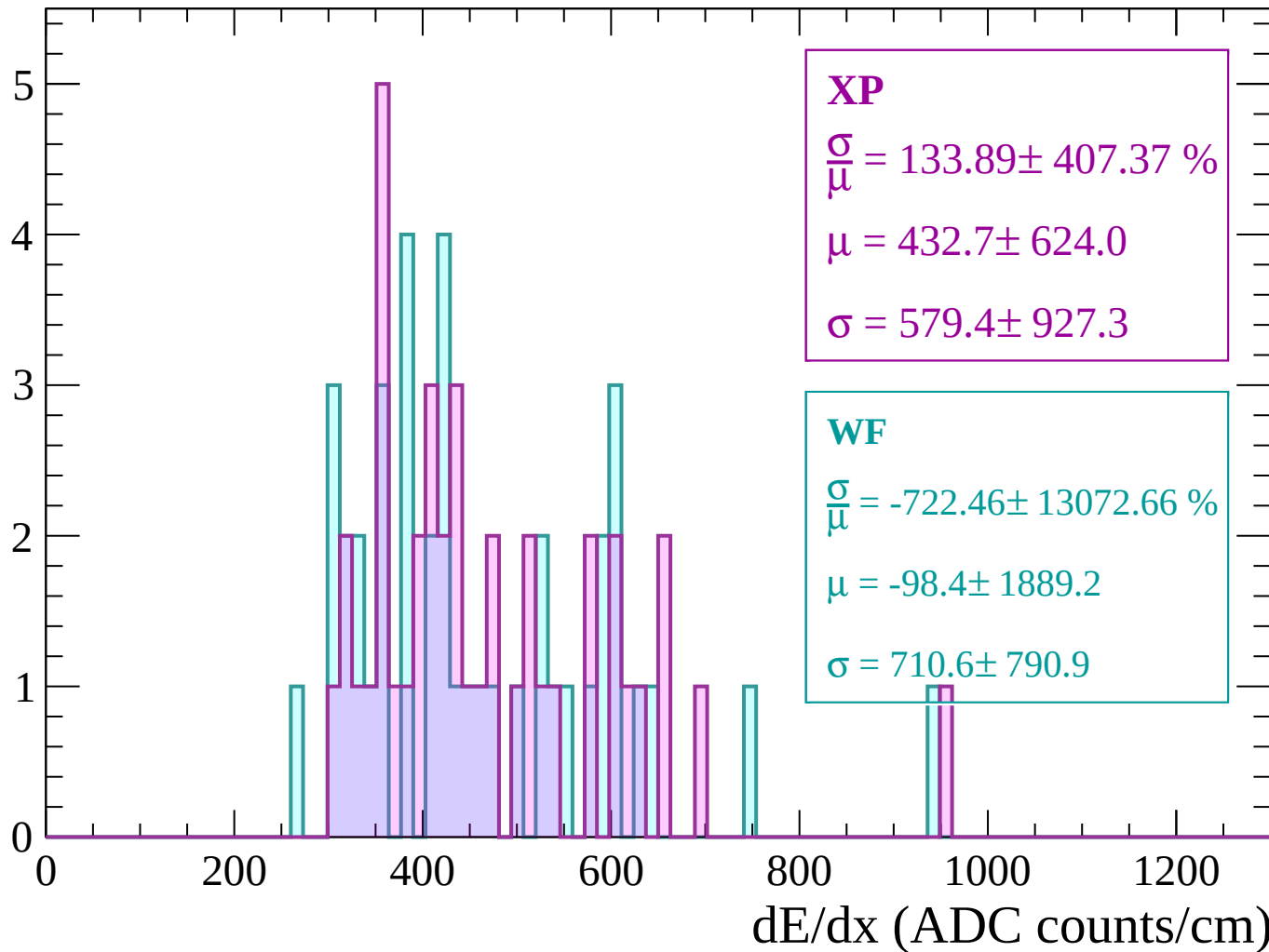
Energy loss | $-600 < p < -540$

Count



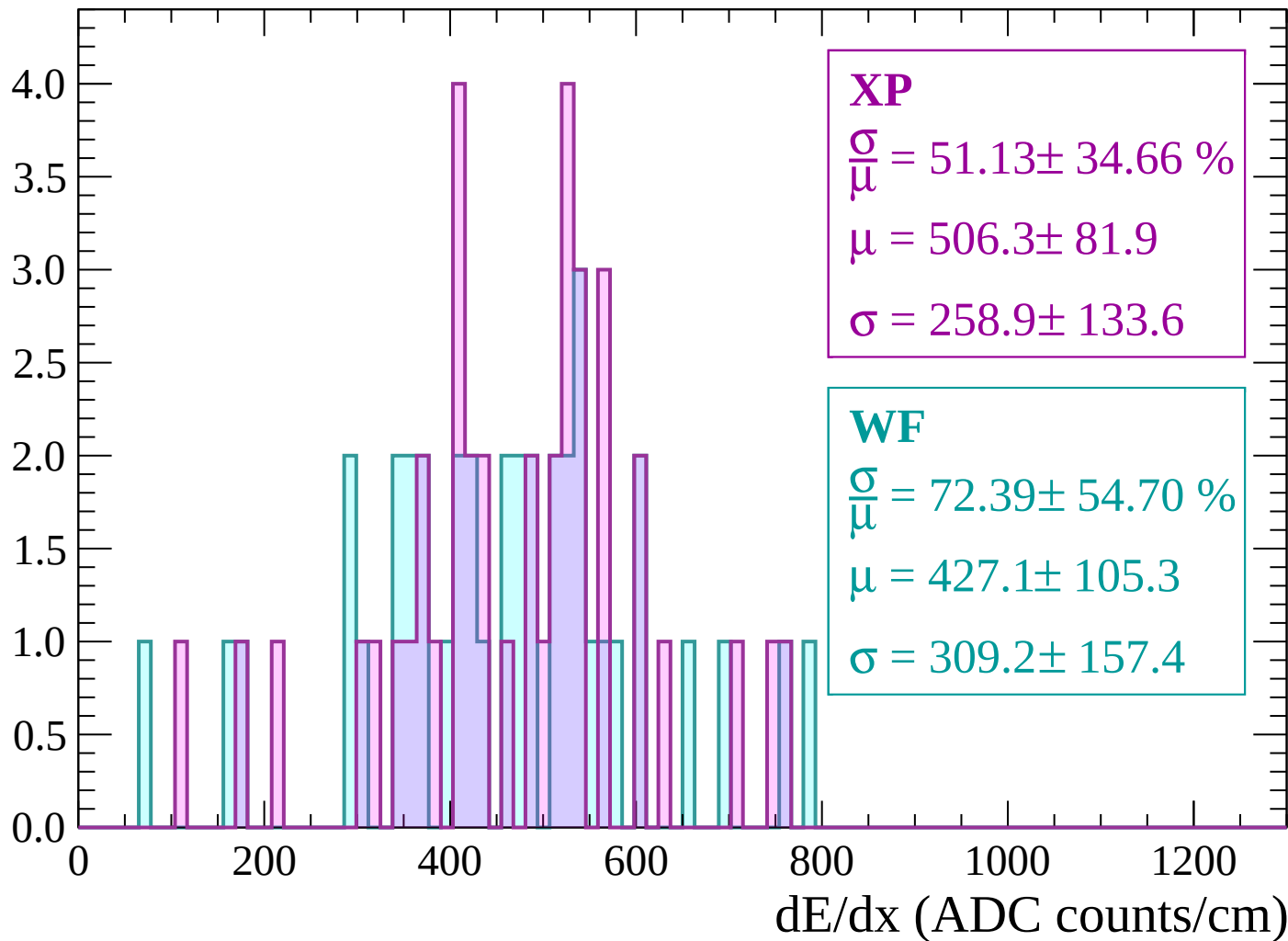
Energy loss | $-540 < p < -480$

Count



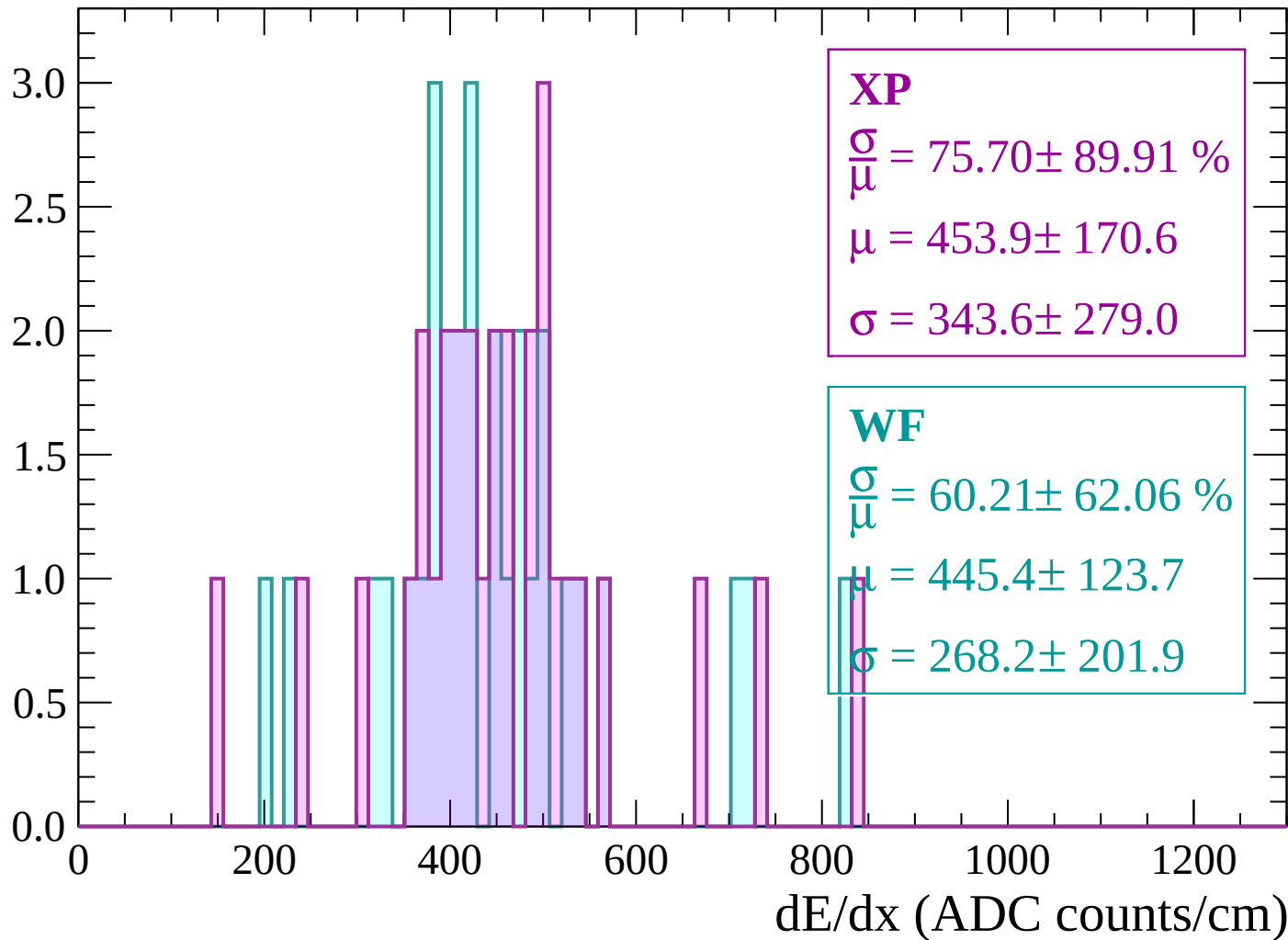
Energy loss | $-480 < p < -420$

Count



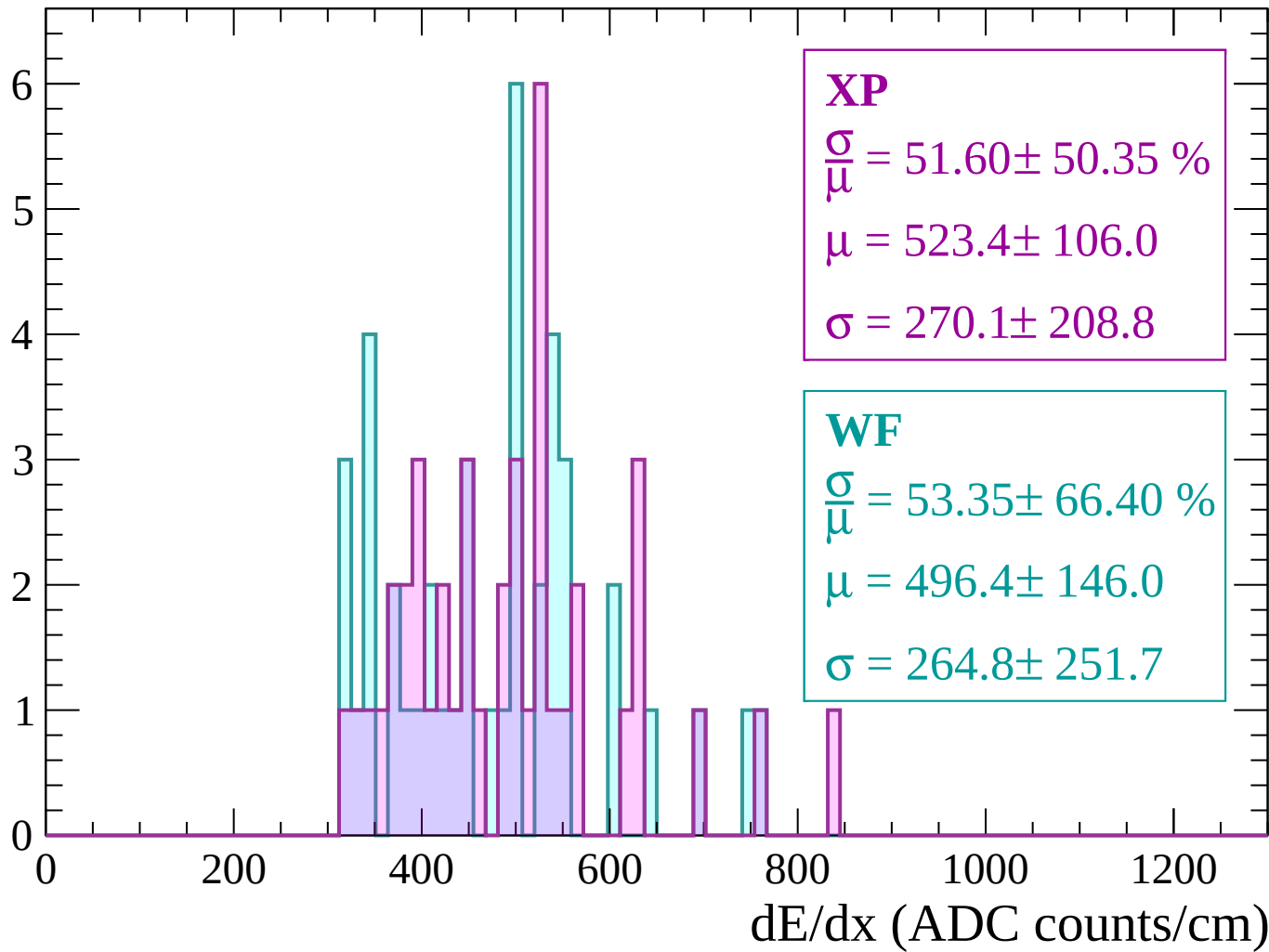
Energy loss | $-420 < p < -360$

Count



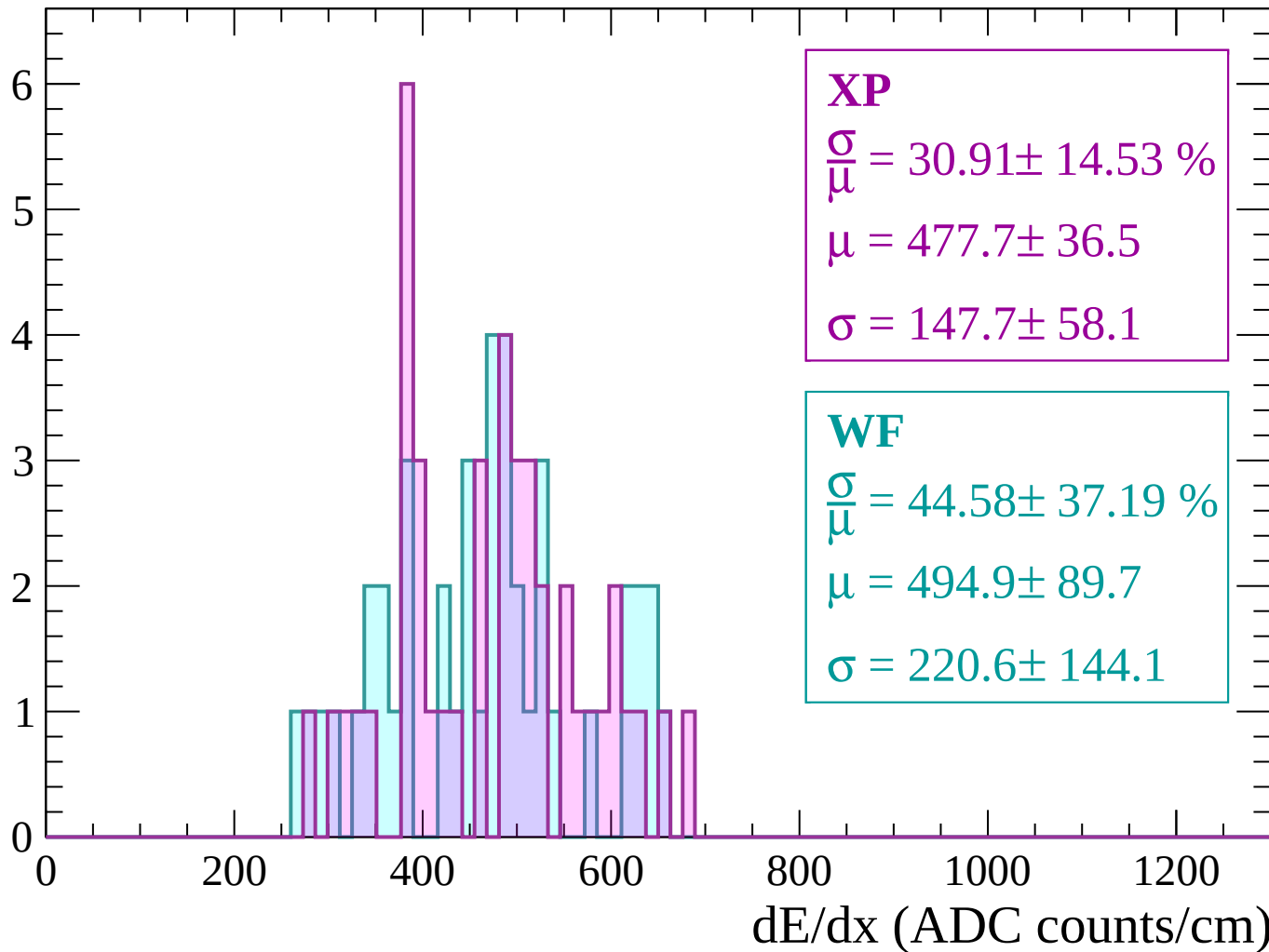
Energy loss | -360 < p < -300

Count



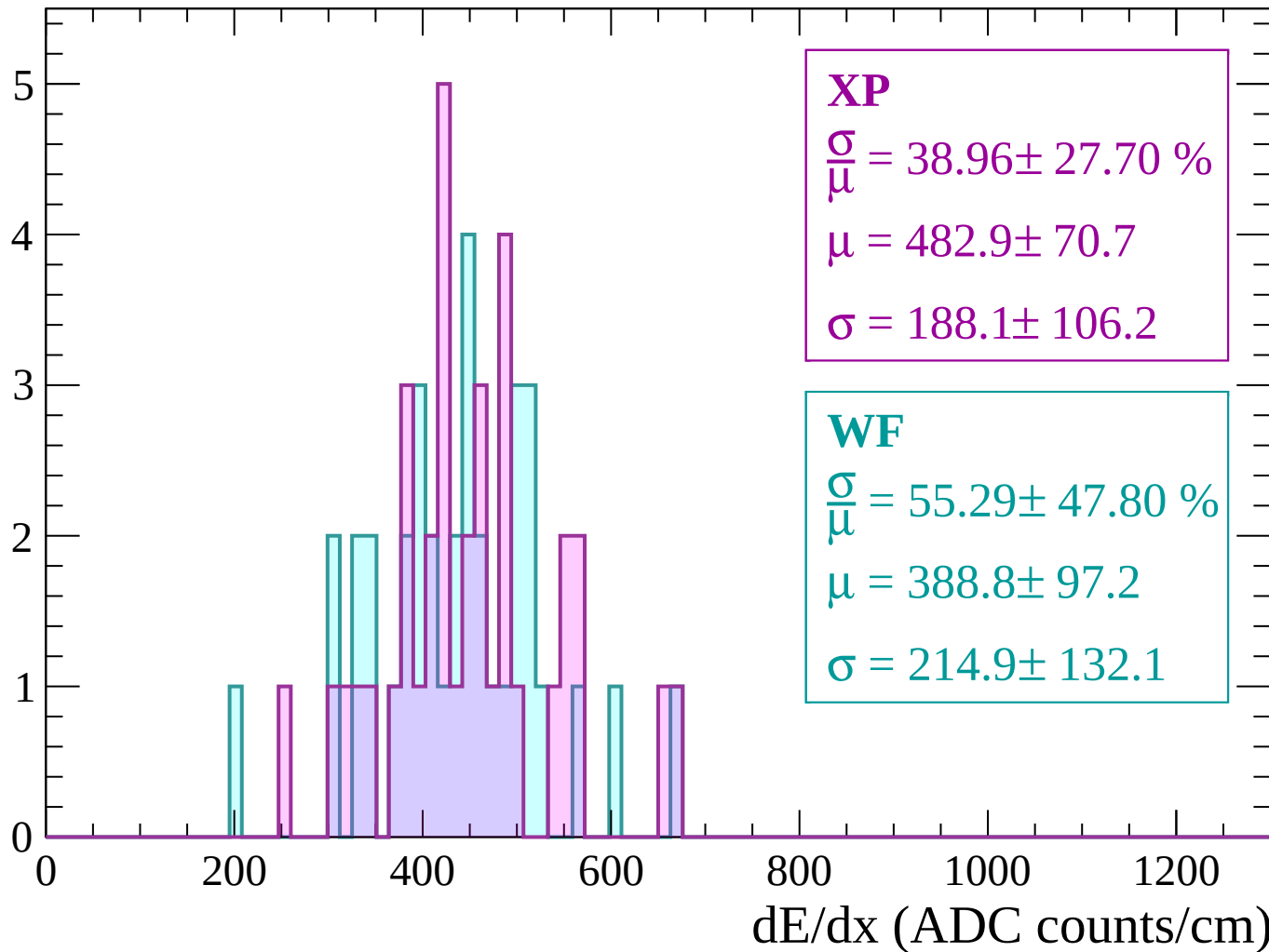
Energy loss | $-300 < p < -240$

Count



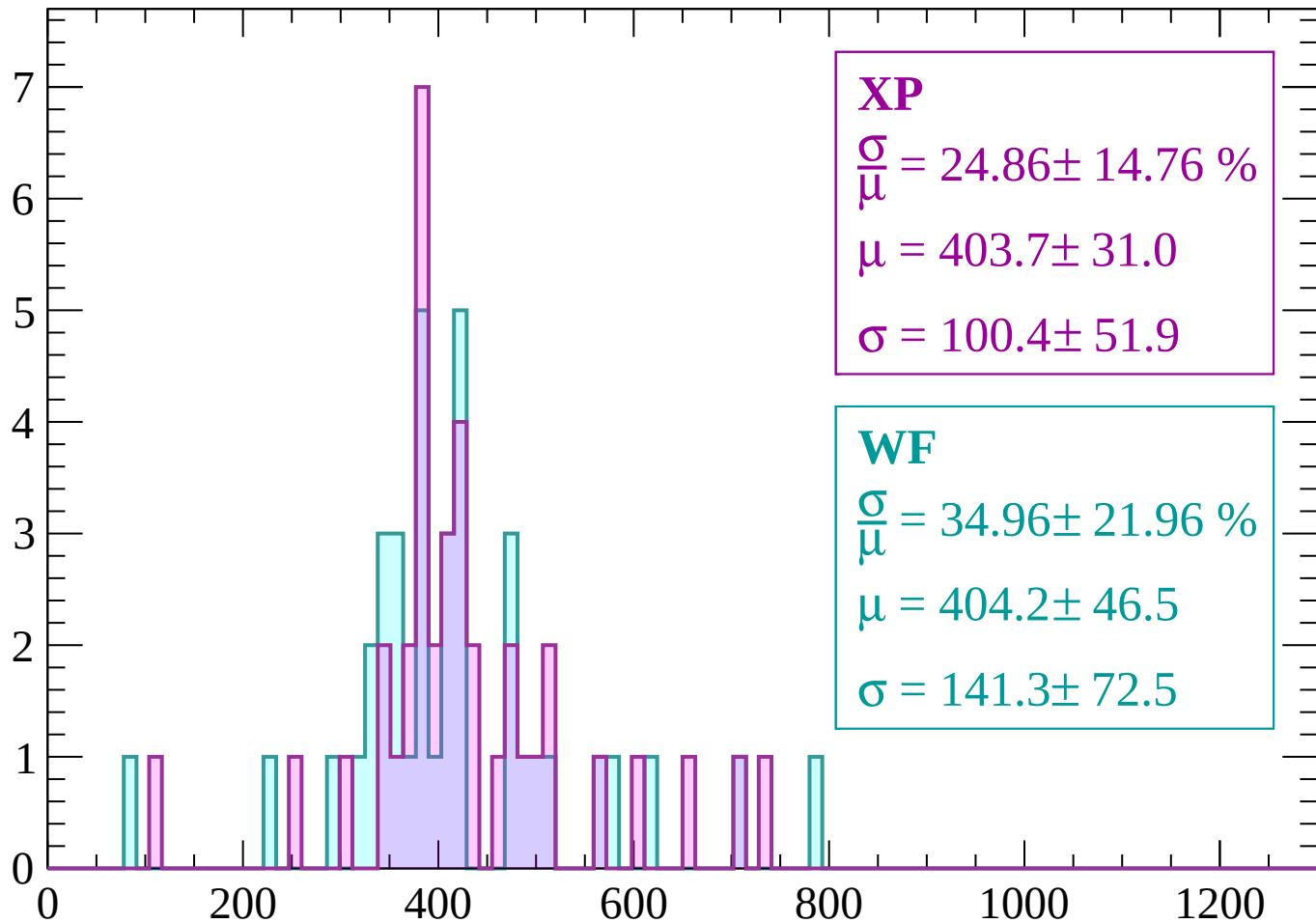
Energy loss | $-240 < p < -180$

Count



Energy loss | $-180 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 24.86 \pm 14.76 \%$$

$$\mu = 403.7 \pm 31.0$$

$$\sigma = 100.4 \pm 51.9$$

WF

$$\frac{\sigma}{\mu} = 34.96 \pm 21.96 \%$$

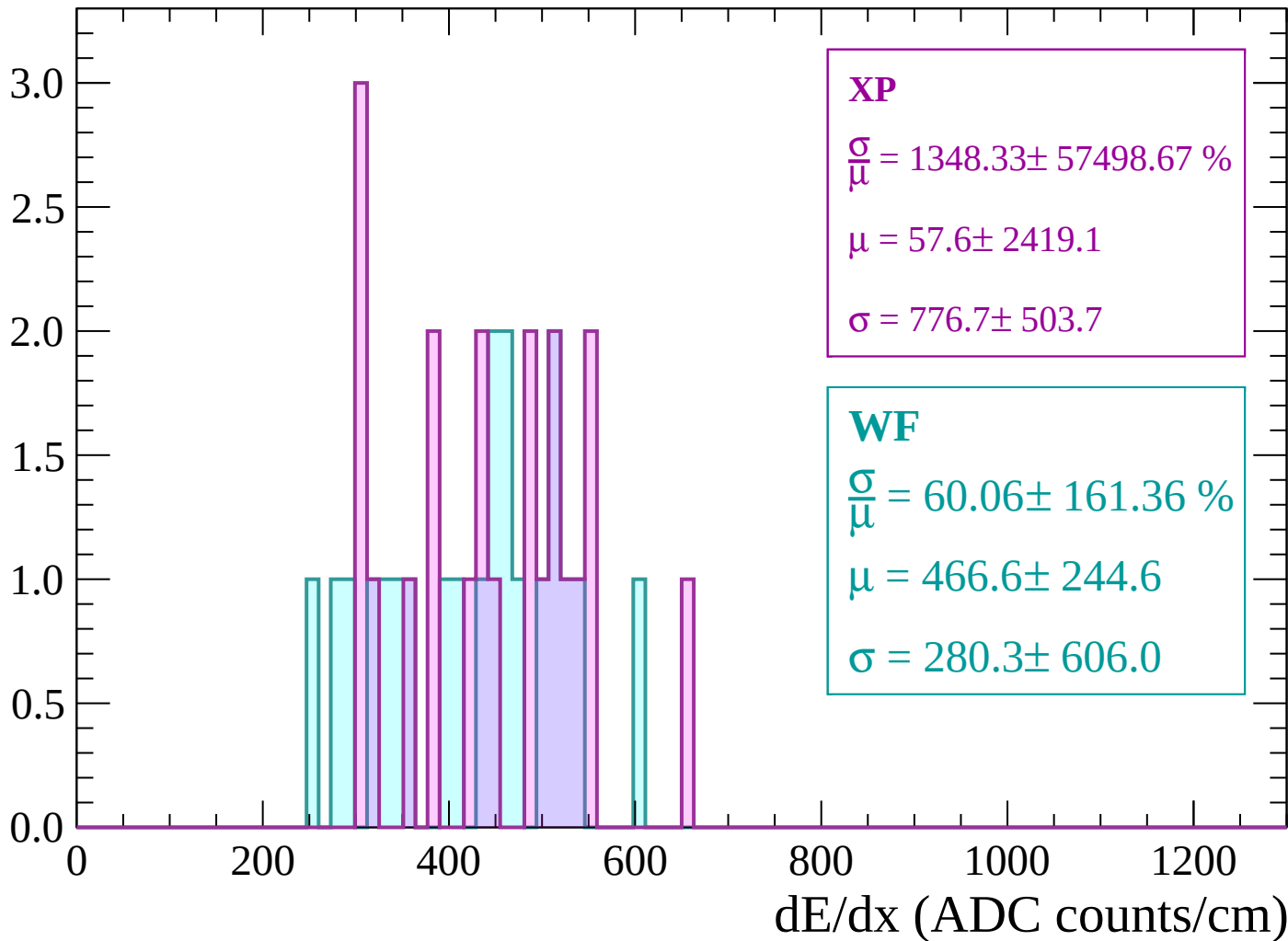
$$\mu = 404.2 \pm 46.5$$

$$\sigma = 141.3 \pm 72.5$$

dE/dx (ADC counts/cm)

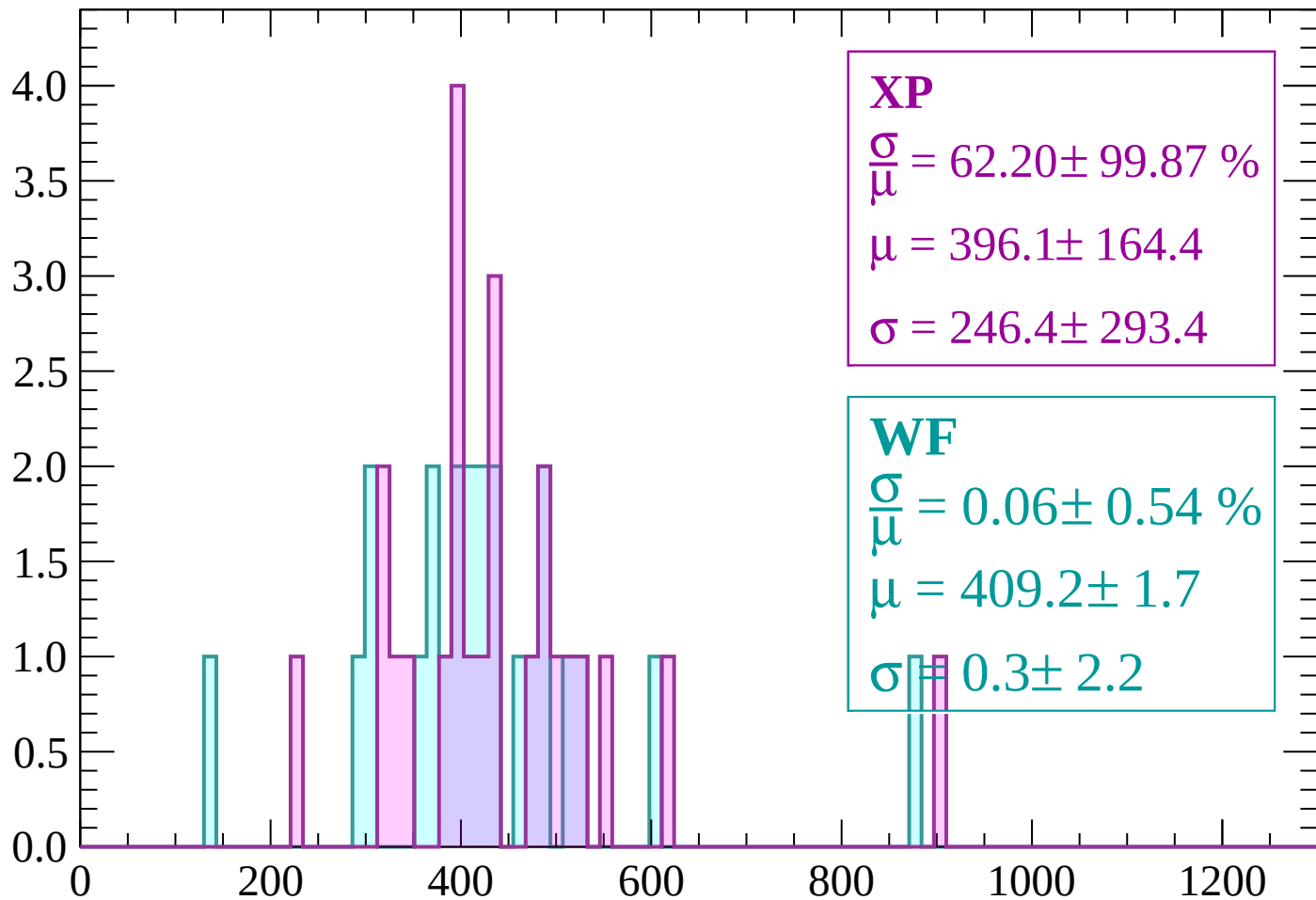
Energy loss | $-120 < p < -60$

Count



Energy loss | $-60 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 62.20 \pm 99.87 \%$$

$$\mu = 396.1 \pm 164.4$$

$$\sigma = 246.4 \pm 293.4$$

WF

$$\frac{\sigma}{\mu} = 0.06 \pm 0.54 \%$$

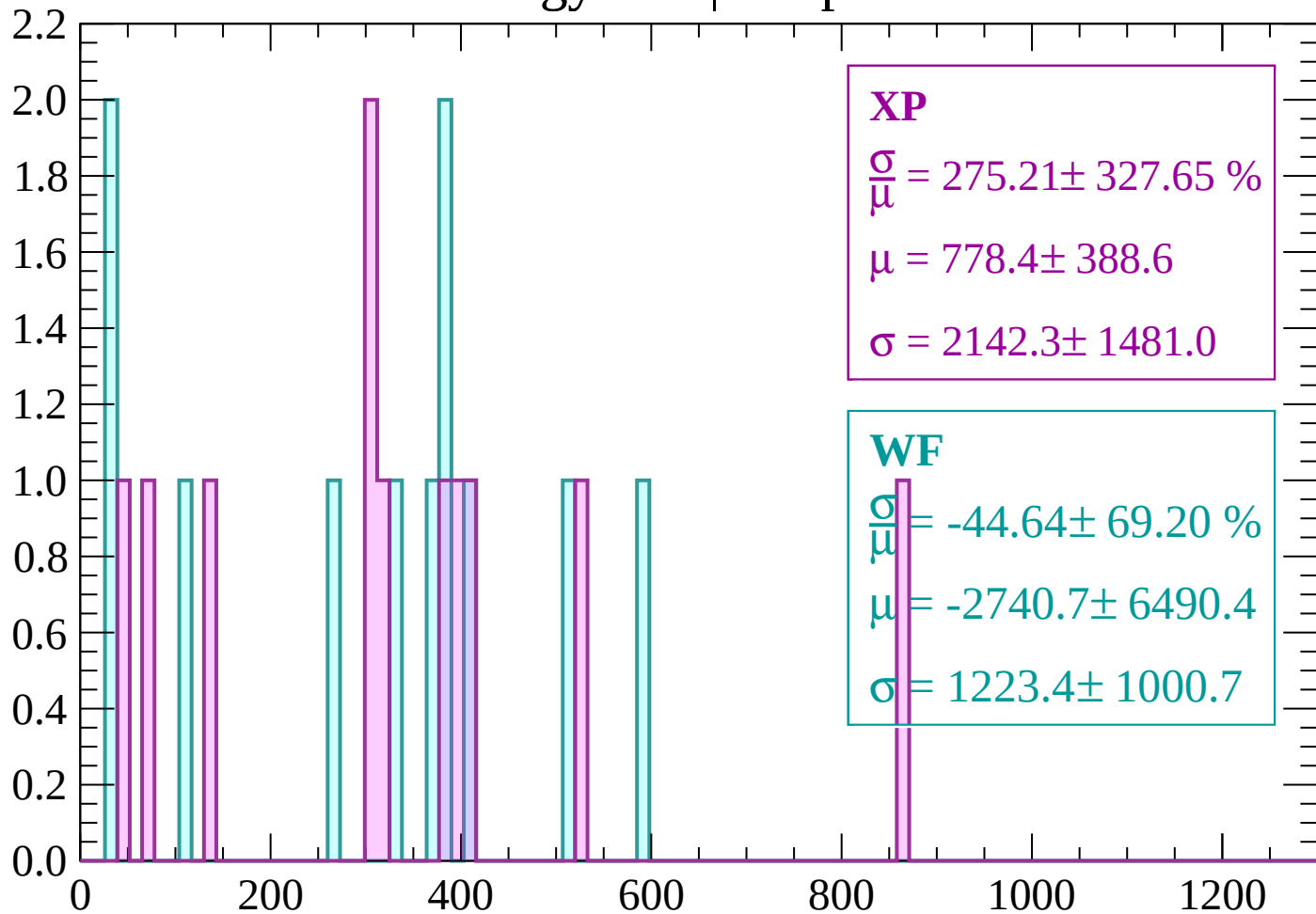
$$\mu = 409.2 \pm 1.7$$

$$\sigma = 0.3 \pm 2.2$$

dE/dx (ADC counts/cm)

Energy loss | $0 < p < 60$

Count



XP

$$\frac{\sigma}{\mu} = 275.21 \pm 327.65 \%$$

$$\mu = 778.4 \pm 388.6$$

$$\sigma = 2142.3 \pm 1481.0$$

WF

$$\frac{\sigma}{\mu} = -44.64 \pm 69.20 \%$$

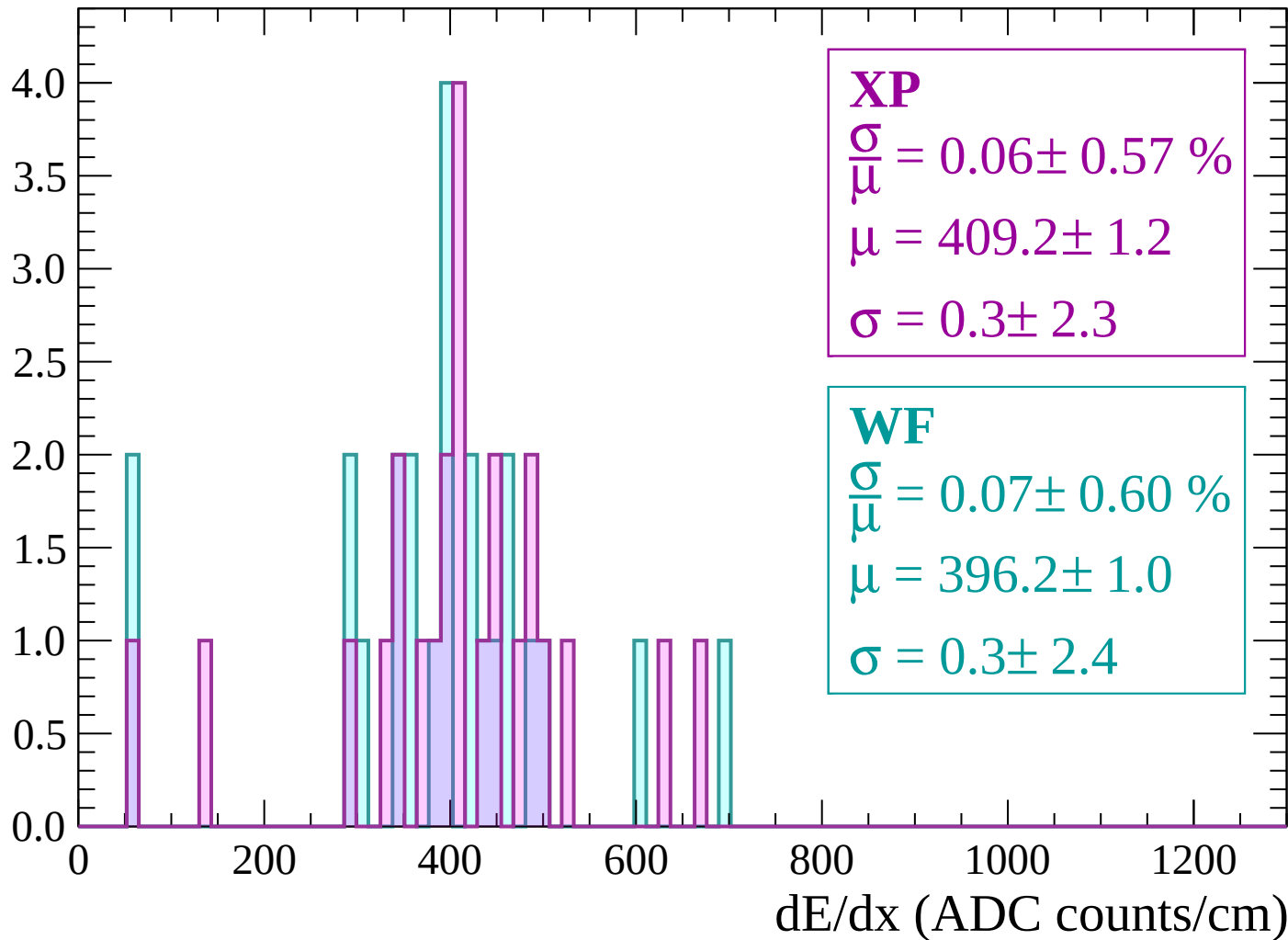
$$\mu = -2740.7 \pm 6490.4$$

$$\sigma = 1223.4 \pm 1000.7$$

dE/dx (ADC counts/cm)

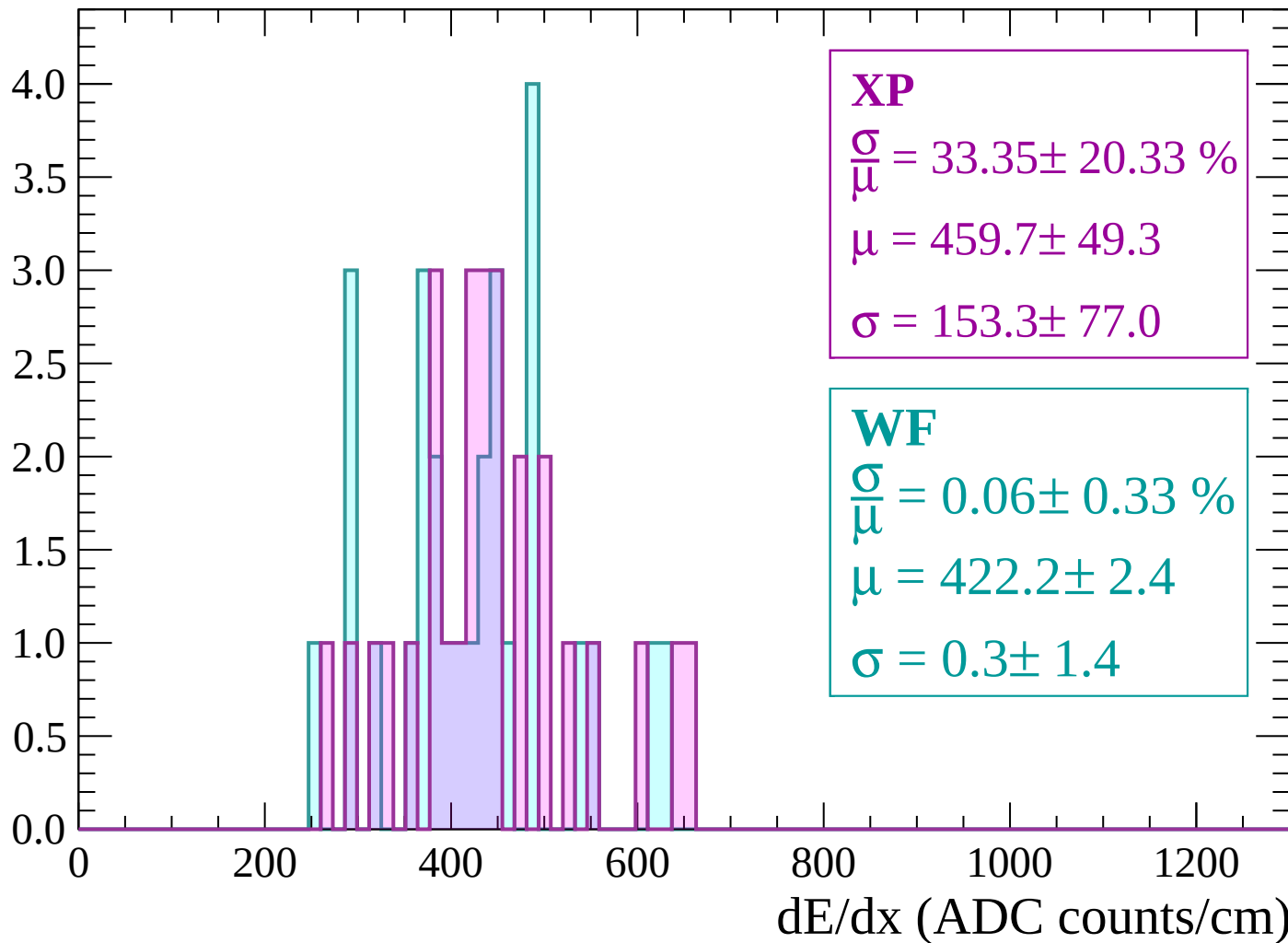
Energy loss | $60 < p < 120$

Count



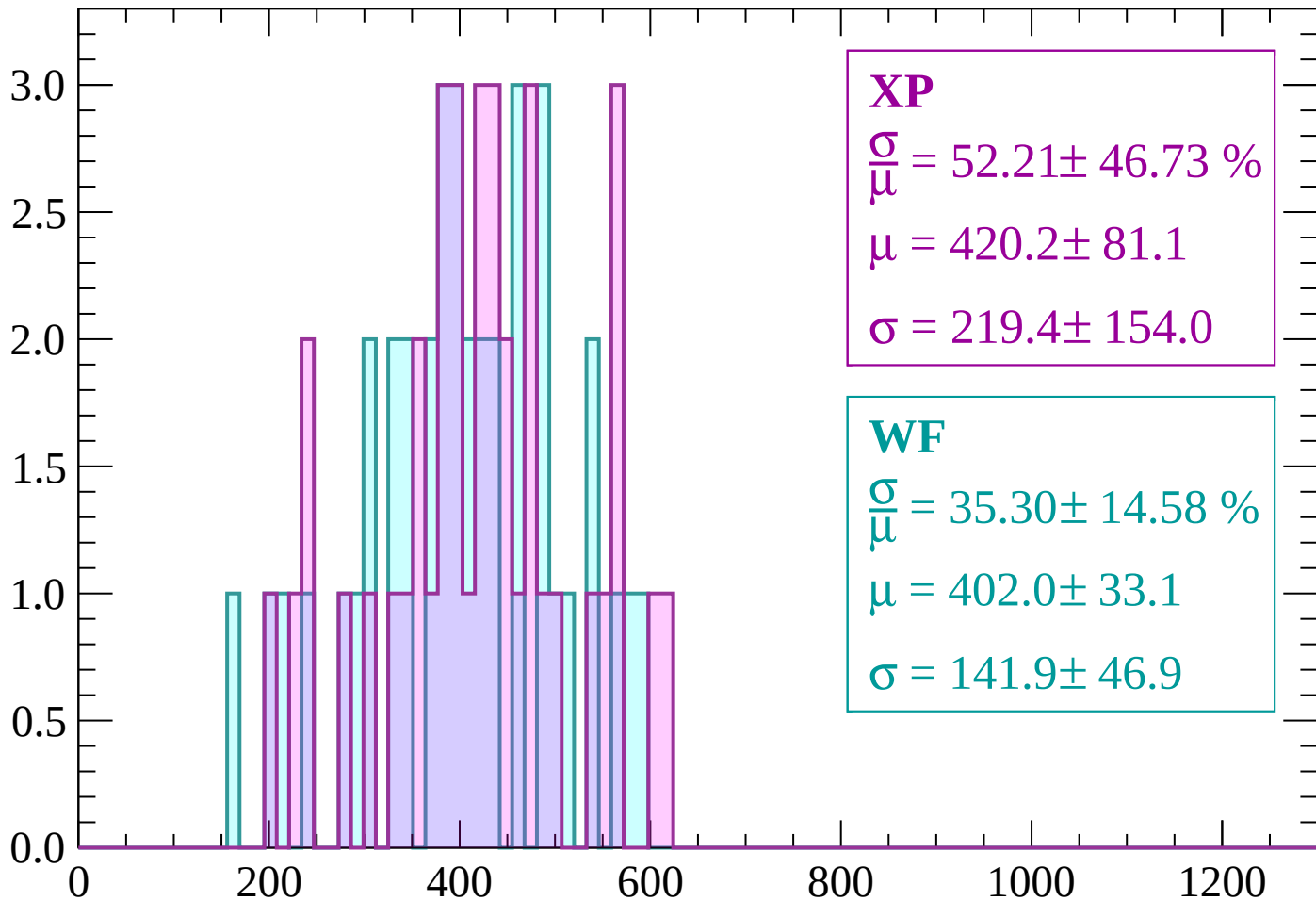
Energy loss | $120 < p < 180$

Count



Energy loss | $180 < p < 240$

Count



XP

$$\frac{\sigma}{\mu} = 52.21 \pm 46.73 \%$$

$$\mu = 420.2 \pm 81.1$$

$$\sigma = 219.4 \pm 154.0$$

WF

$$\frac{\sigma}{\mu} = 35.30 \pm 14.58 \%$$

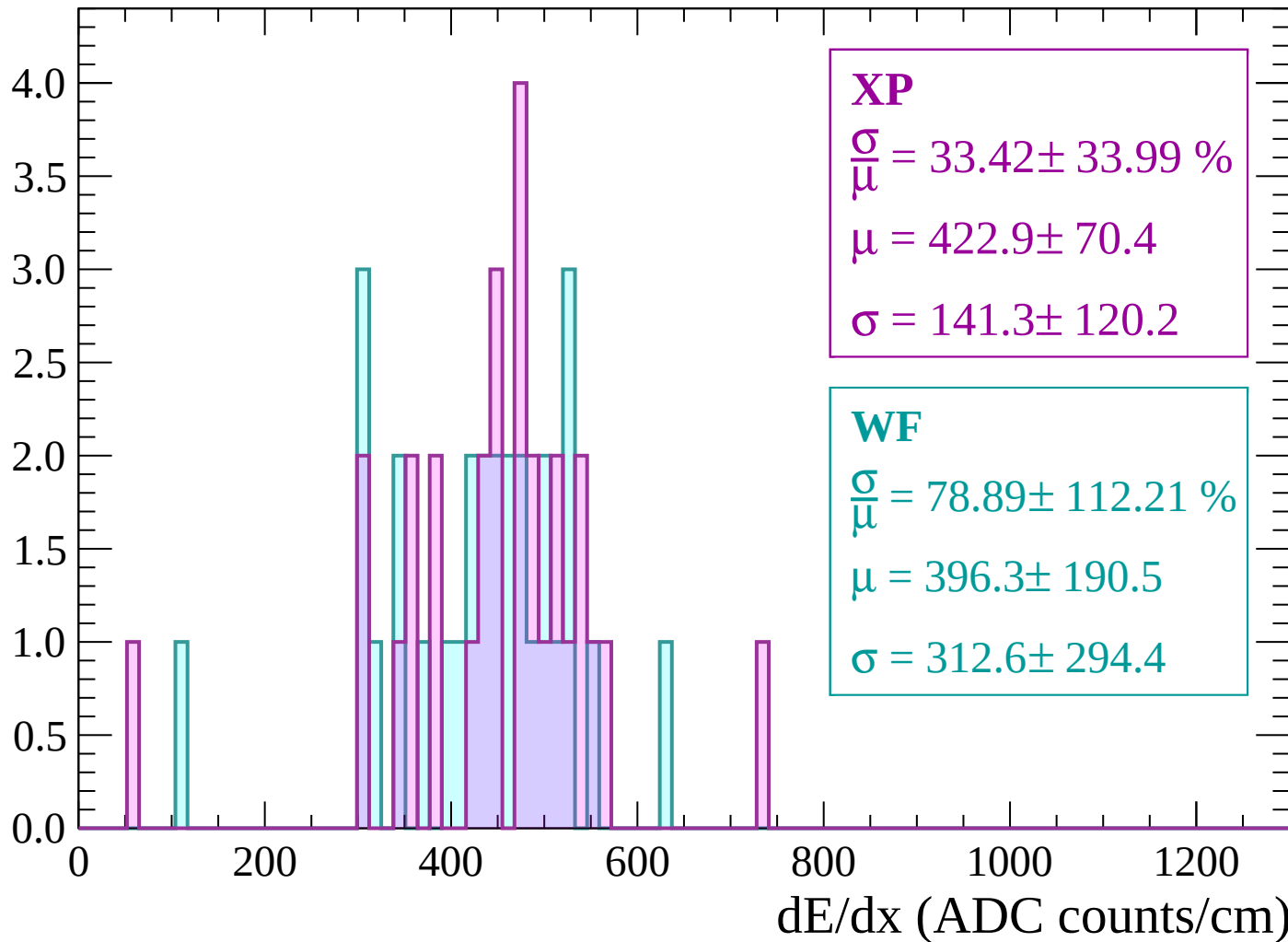
$$\mu = 402.0 \pm 33.1$$

$$\sigma = 141.9 \pm 46.9$$

dE/dx (ADC counts/cm)

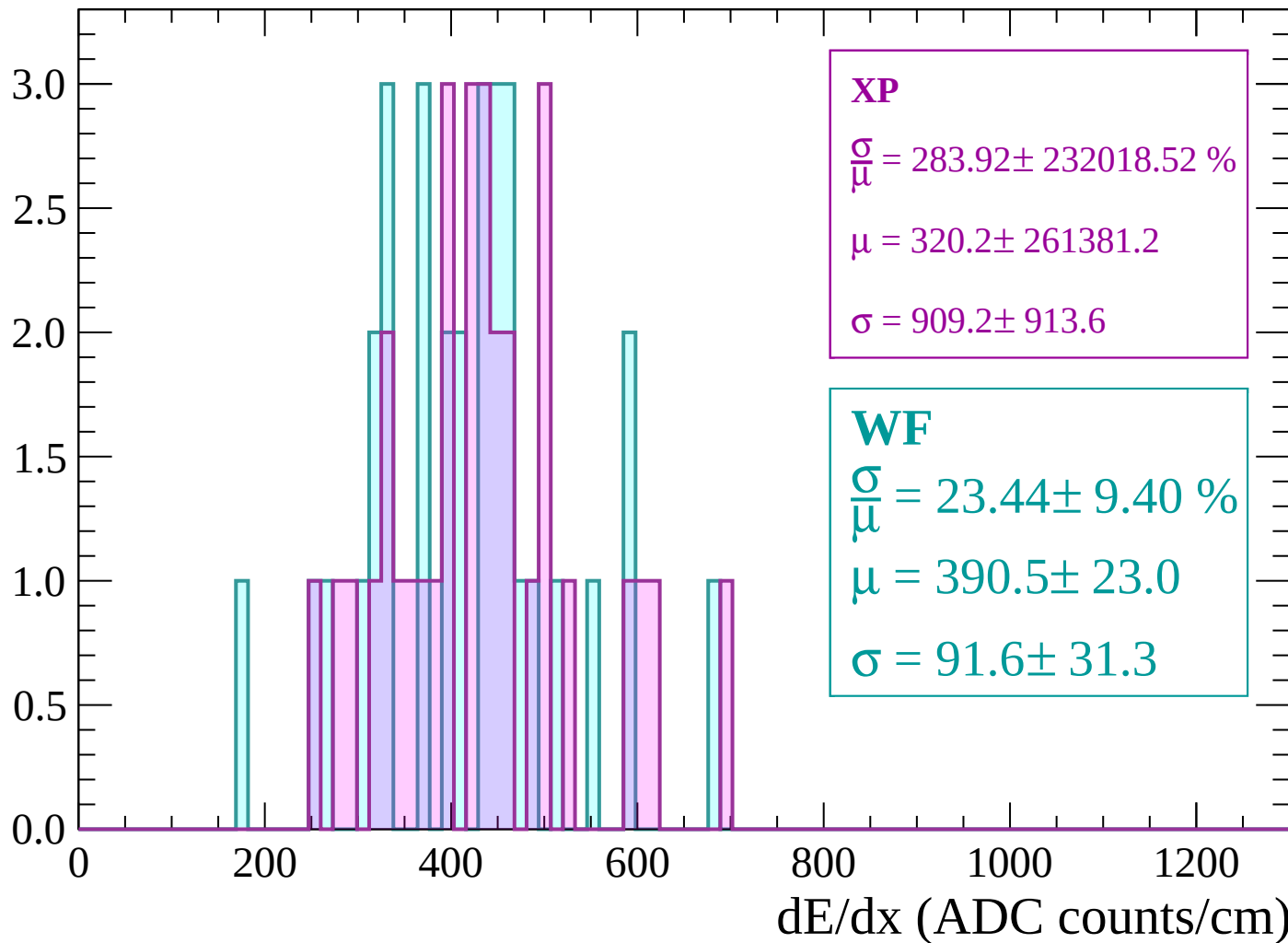
Energy loss | $240 < p < 300$

Count



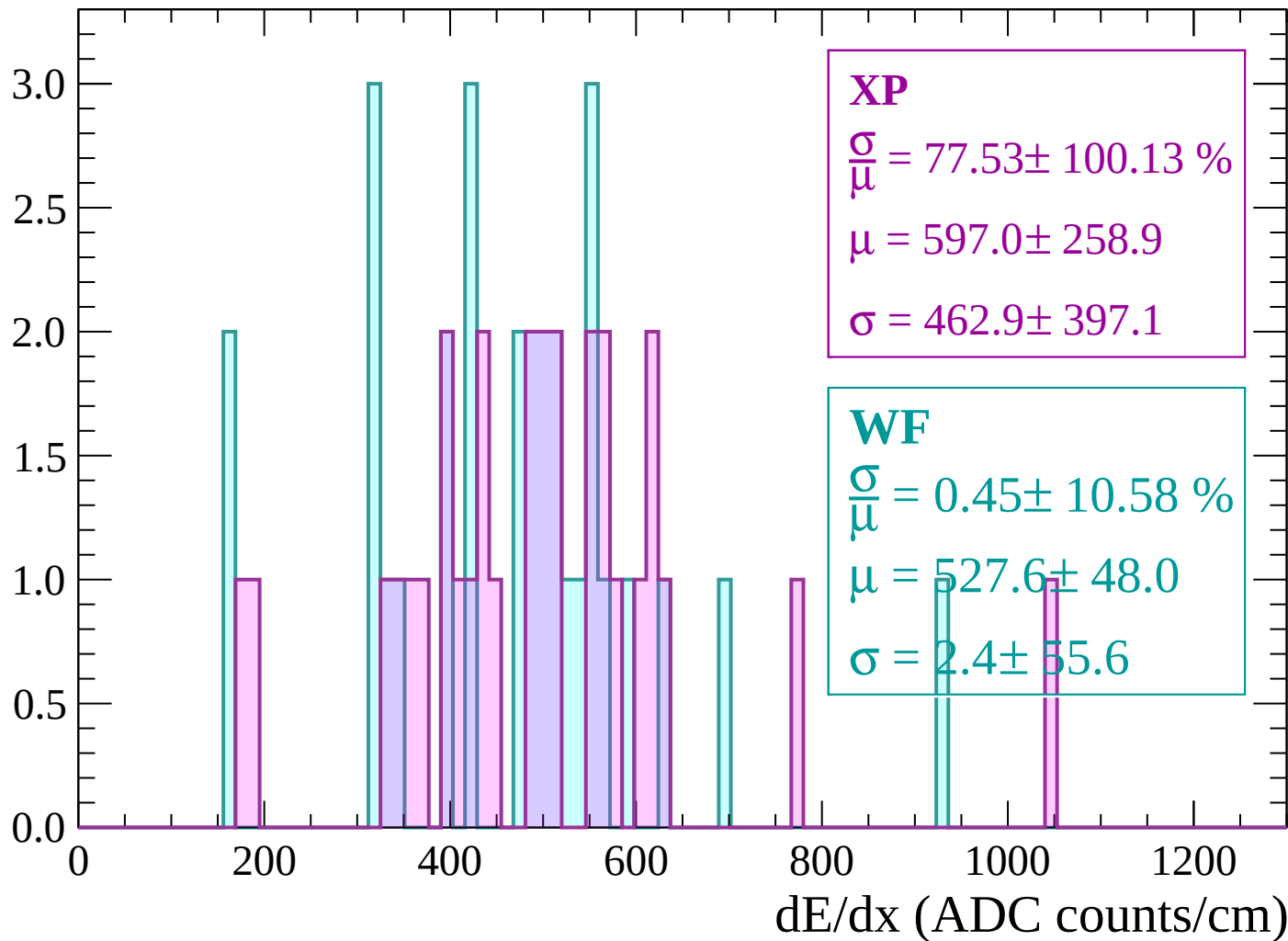
Energy loss | $300 < p < 360$

Count



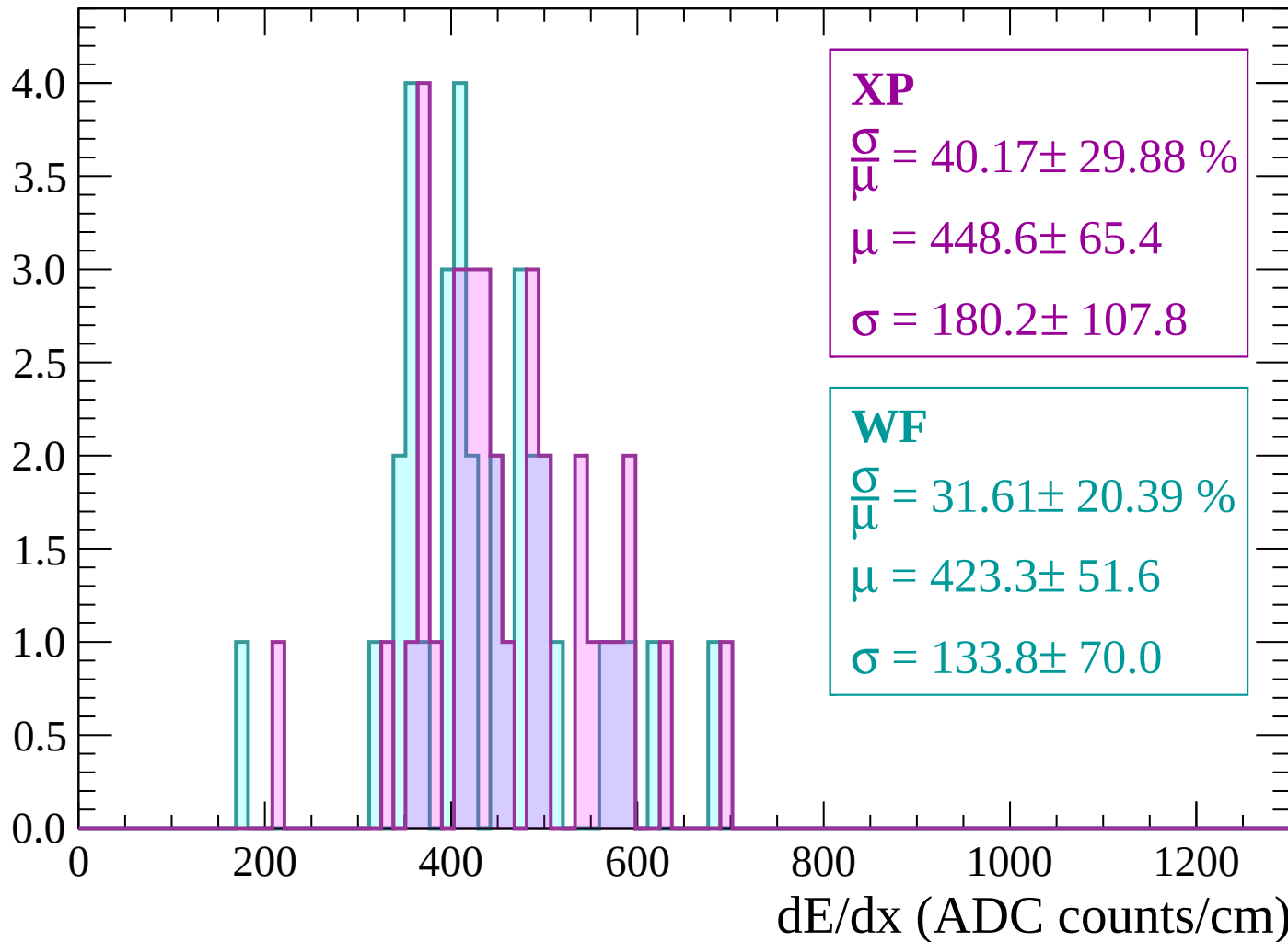
Energy loss | $360 < p < 420$

Count



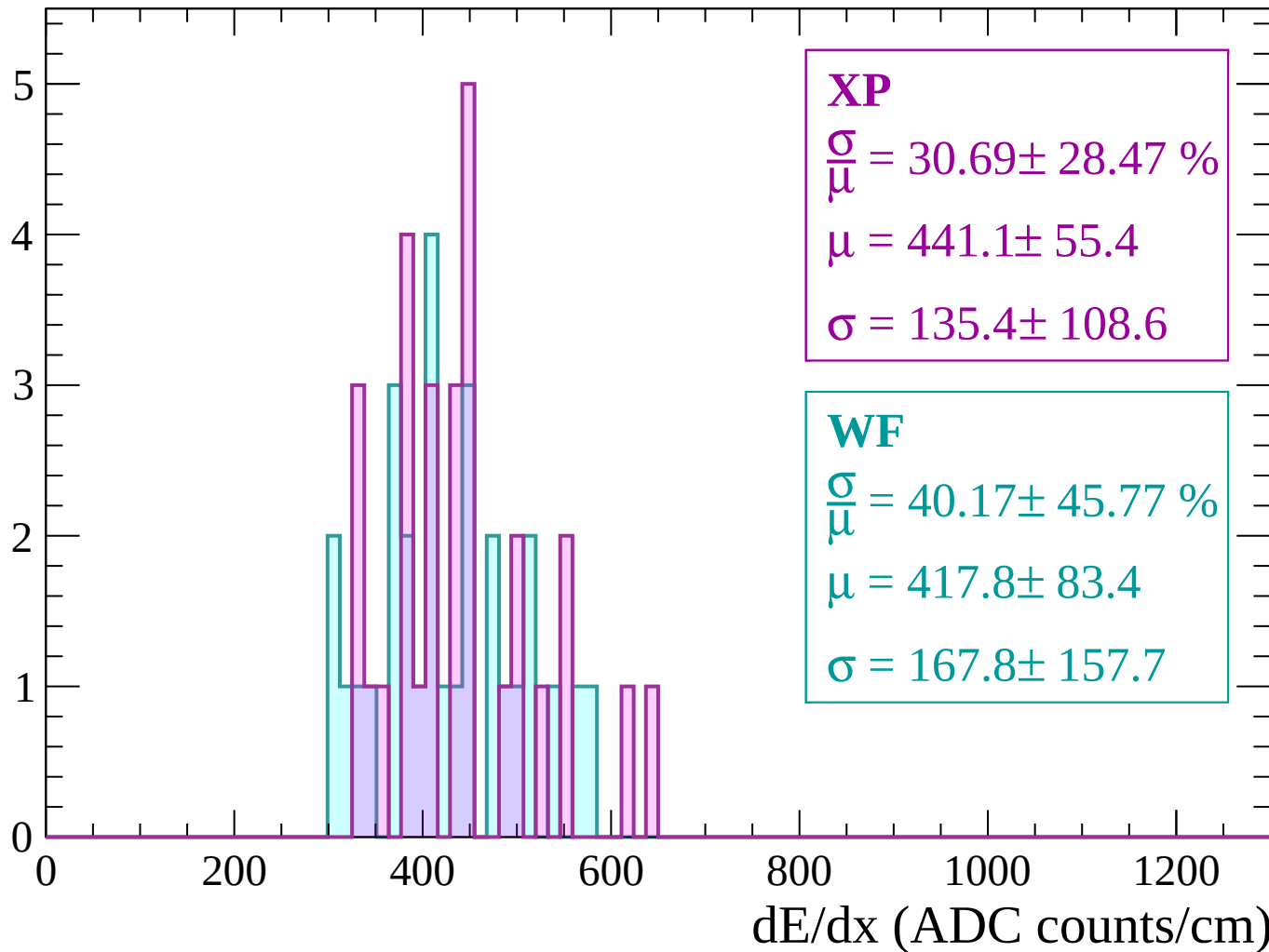
Energy loss | $420 < p < 480$

Count



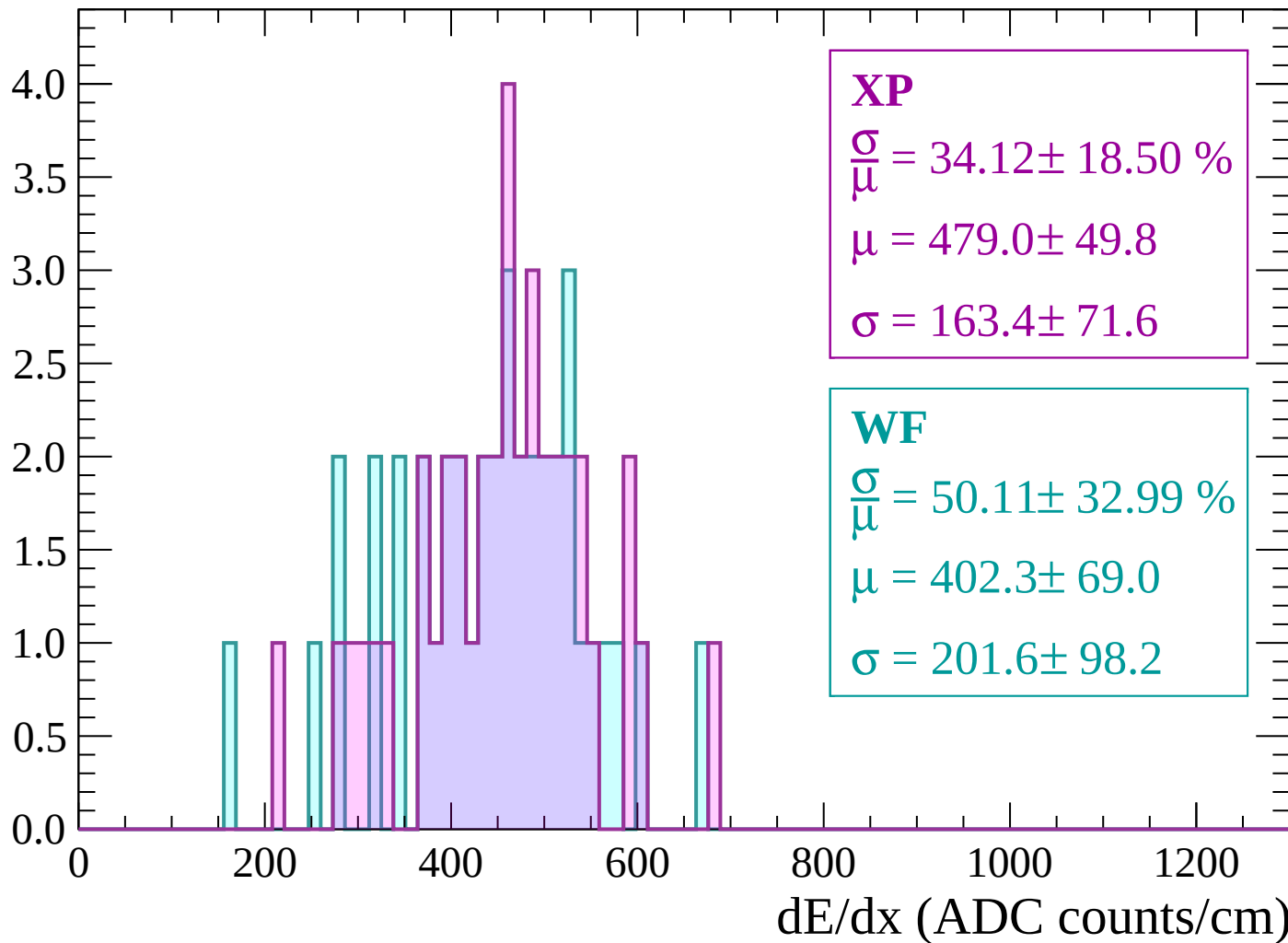
Energy loss | $480 < p < 540$

Count



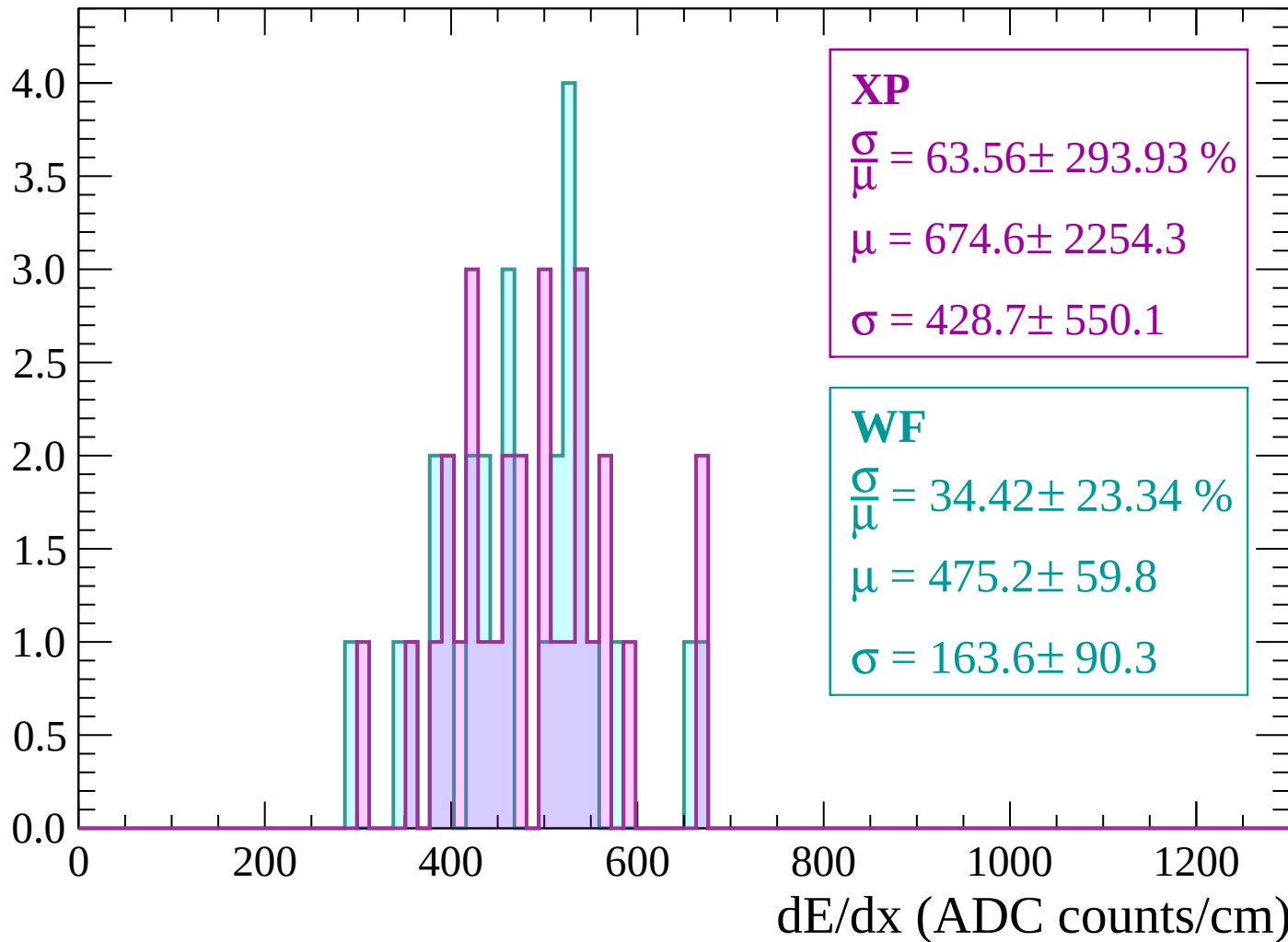
Energy loss | $540 < p < 600$

Count



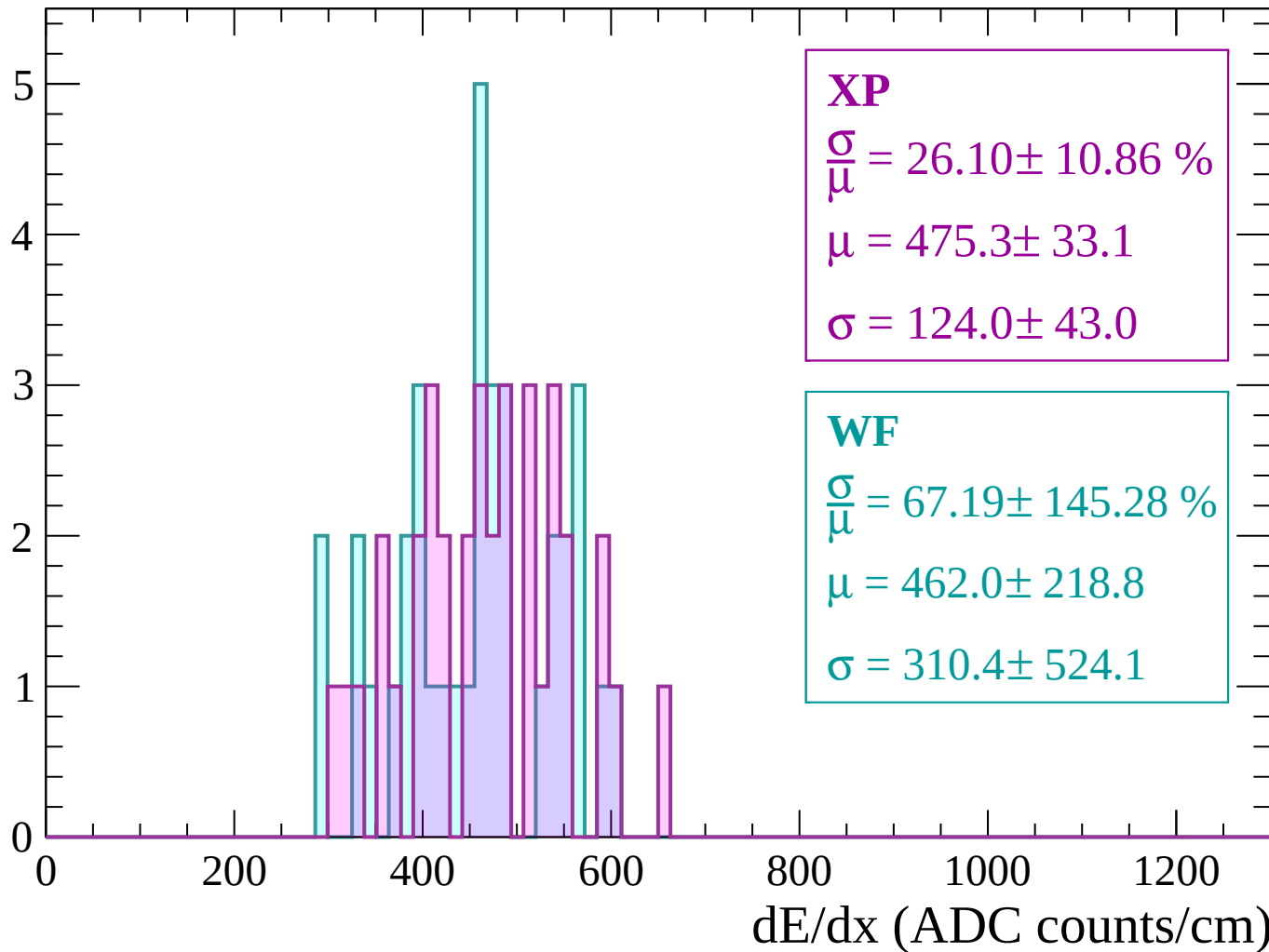
Energy loss | $600 < p < 660$

Count



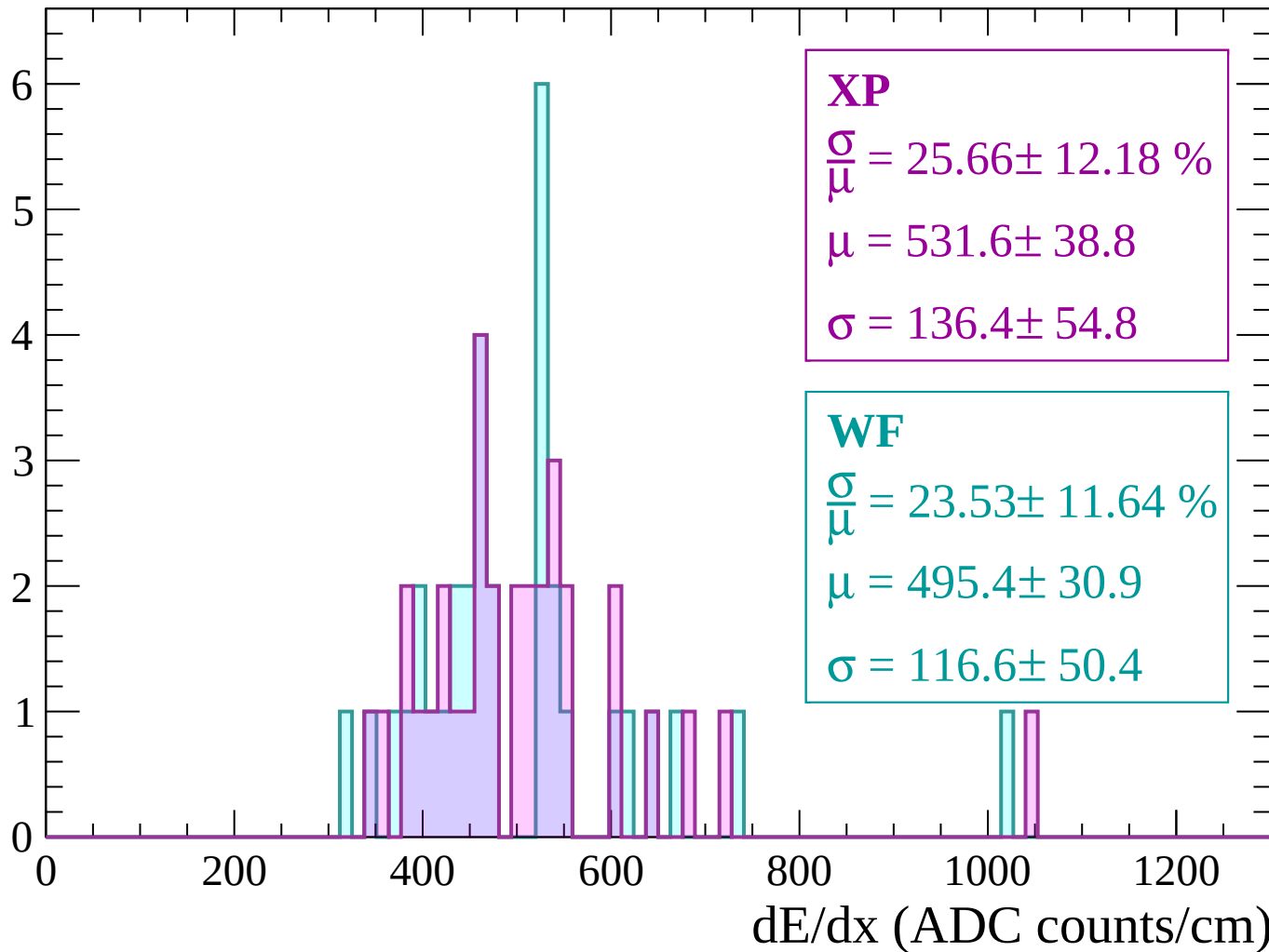
Energy loss | $660 < p < 720$

Count



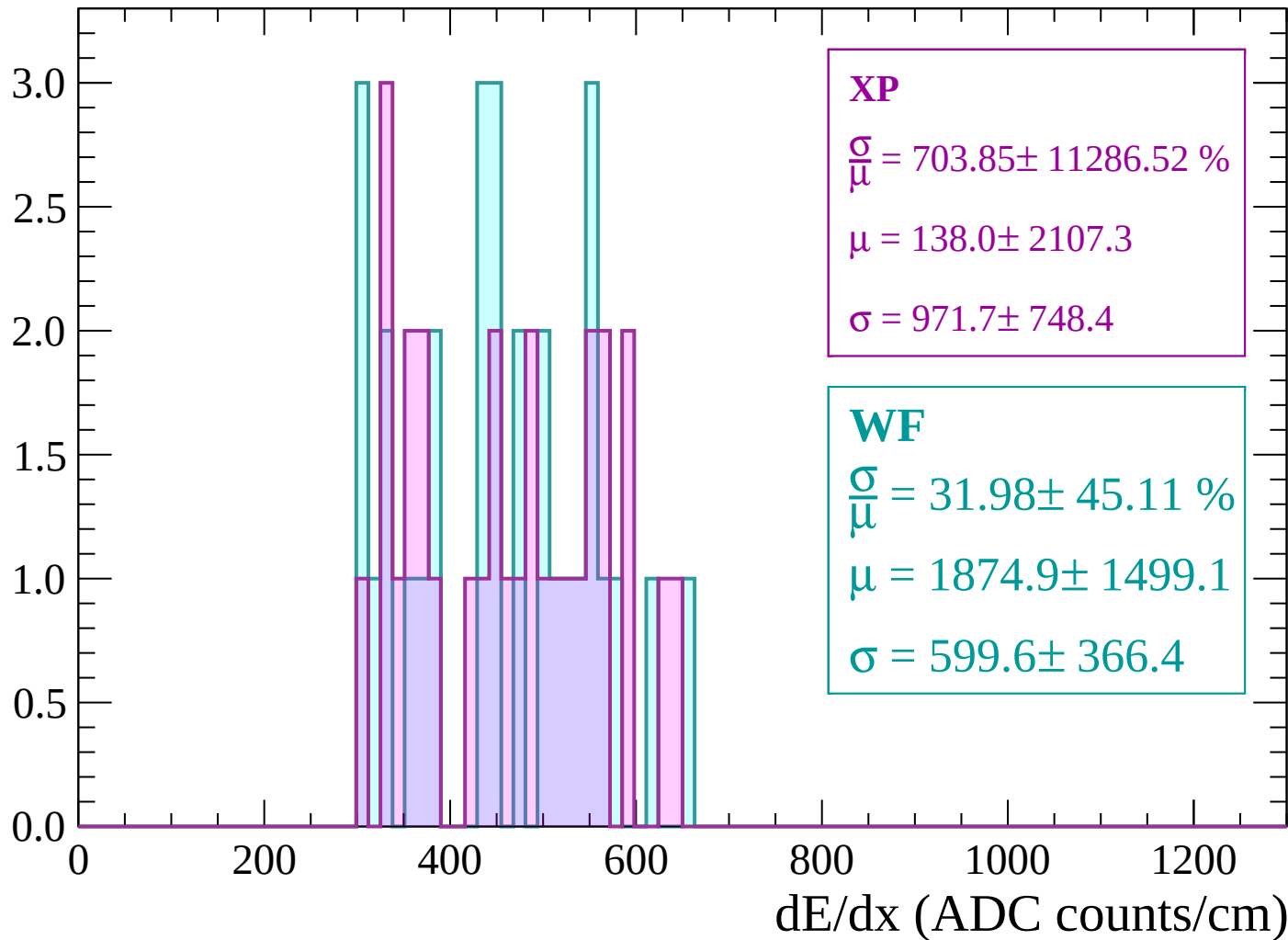
Energy loss | $720 < p < 780$

Count



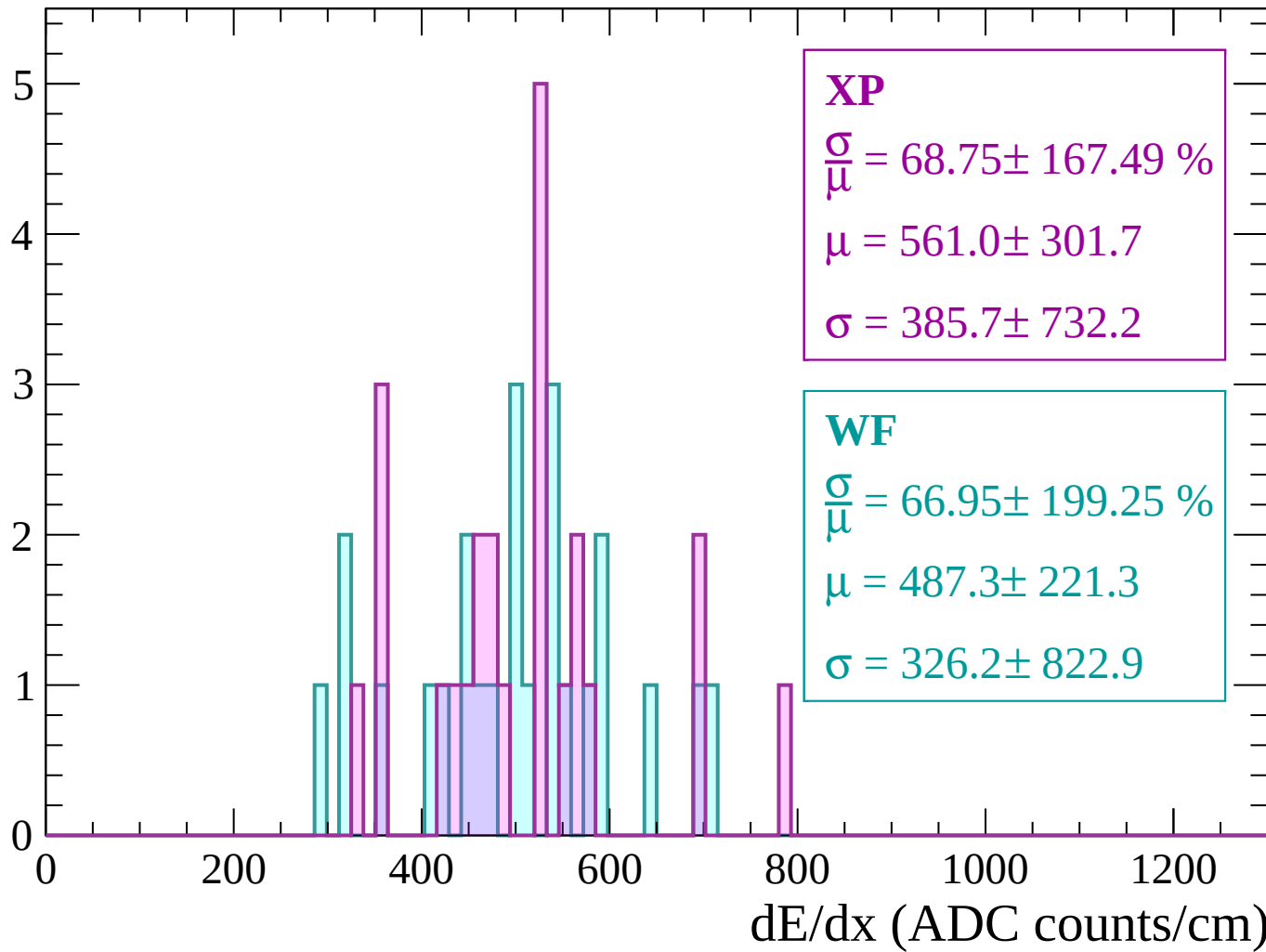
Energy loss | $780 < p < 840$

Count



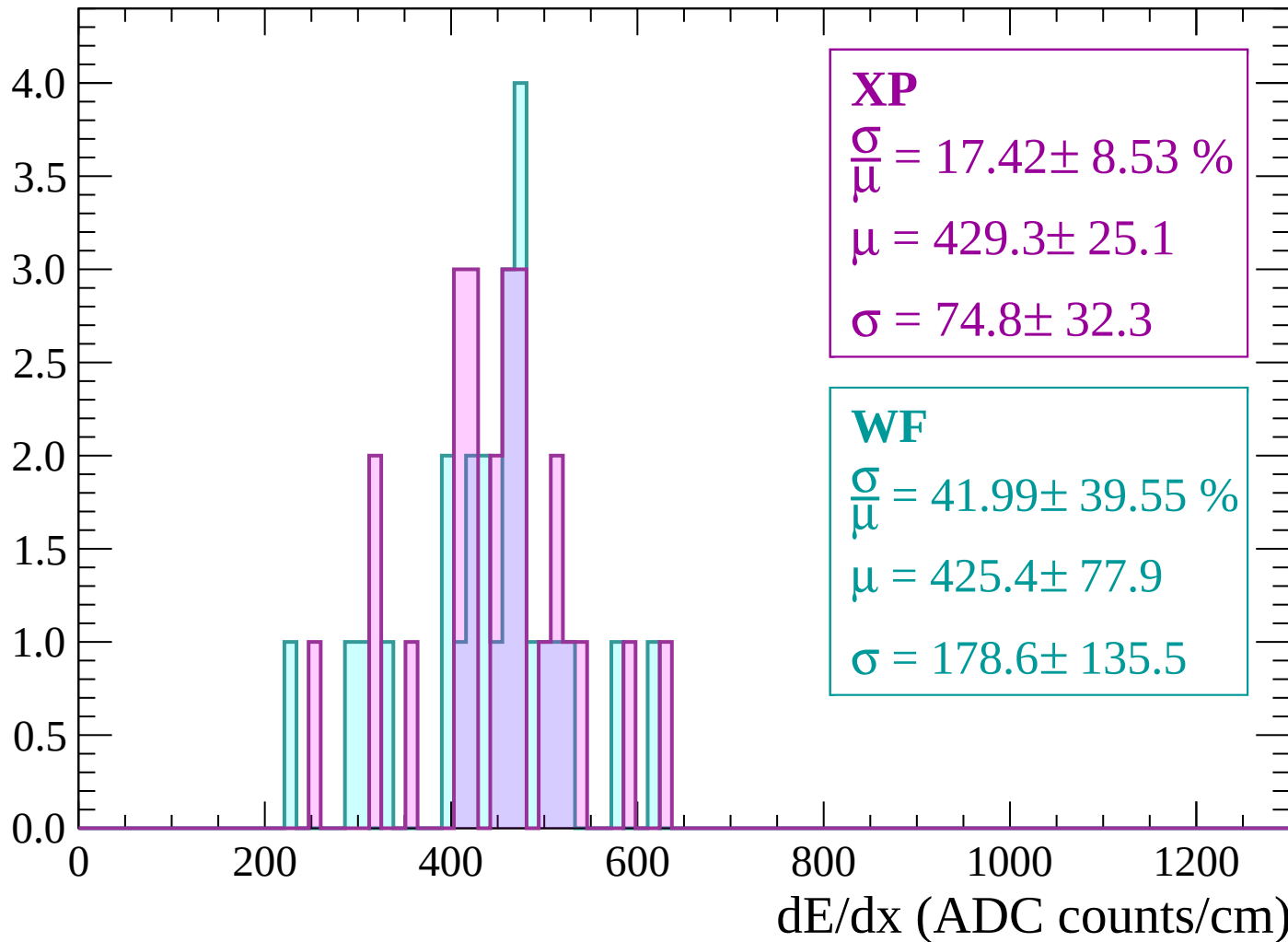
Energy loss | $840 < p < 900$

Count



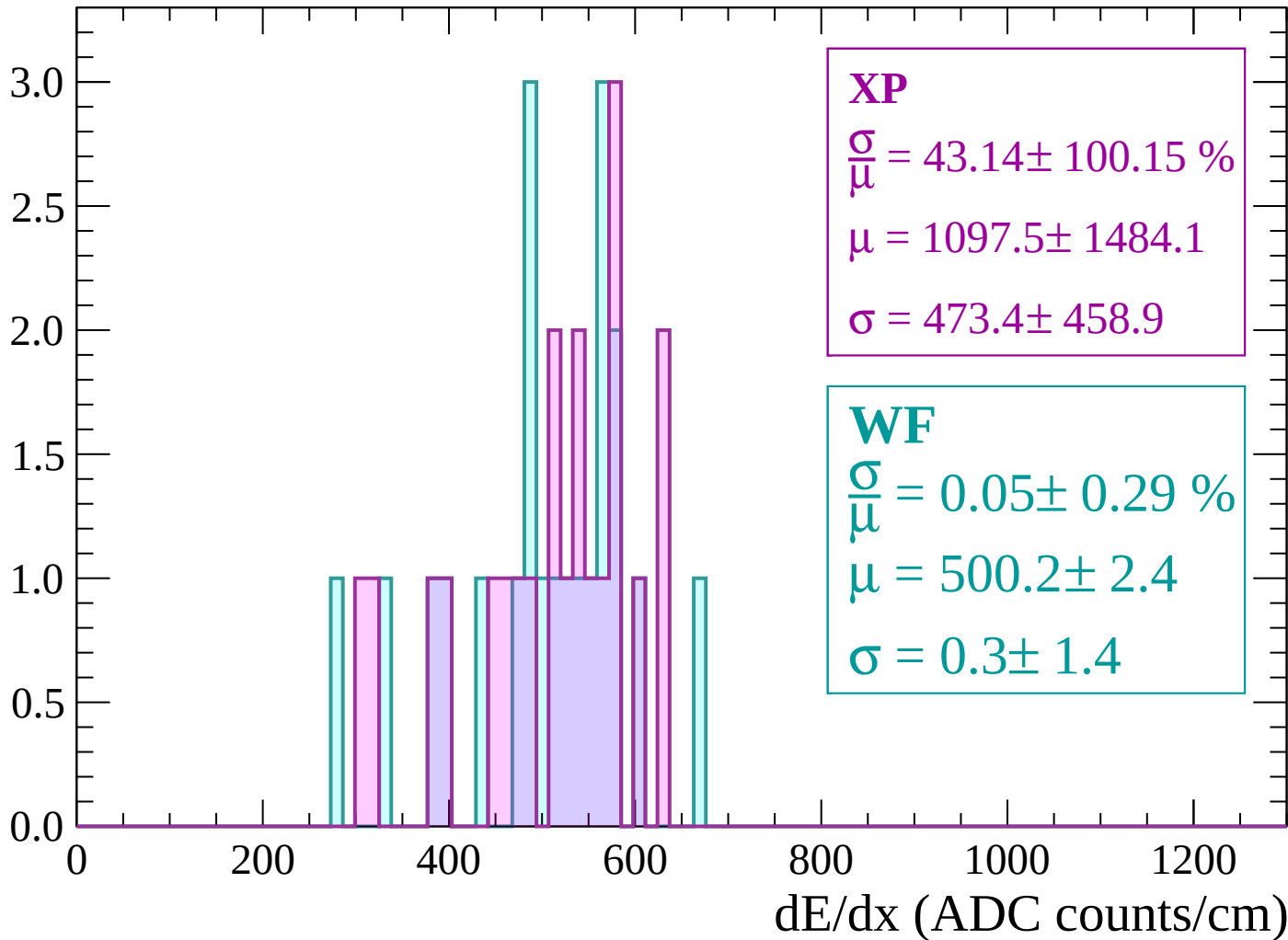
Energy loss | $900 < p < 960$

Count



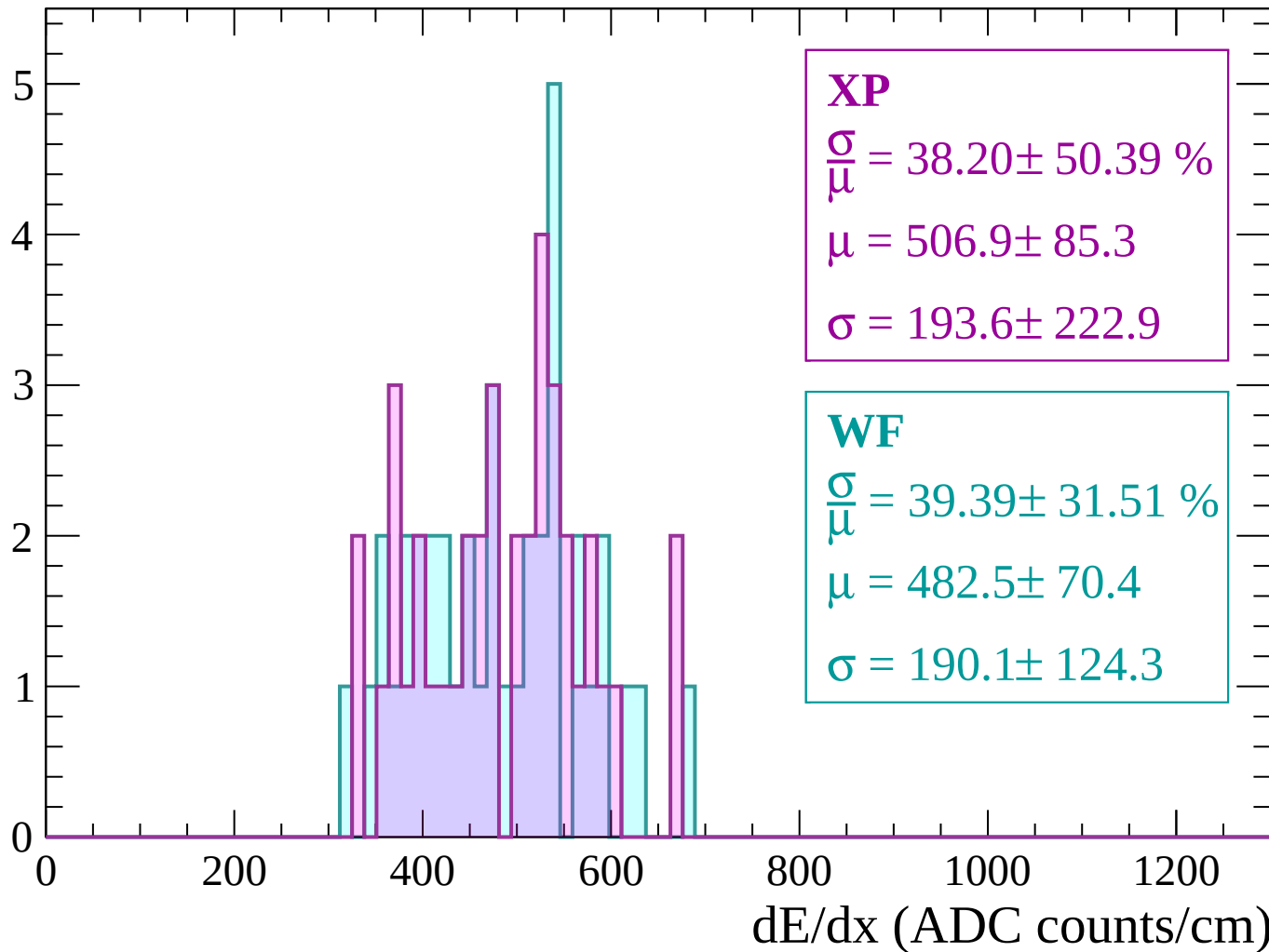
Energy loss | $960 < p < 1020$

Count



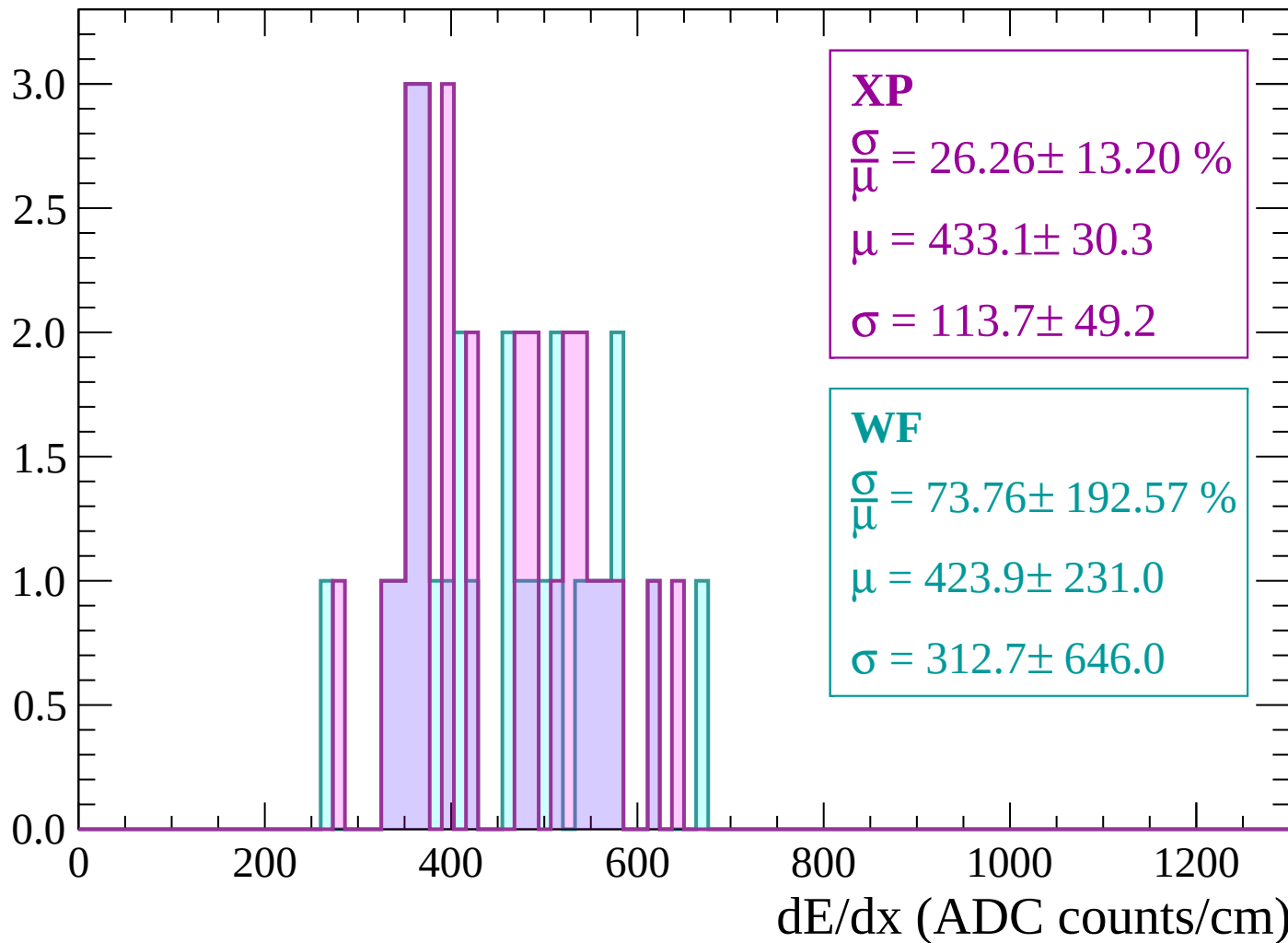
Energy loss | $1020 < p < 1080$

Count



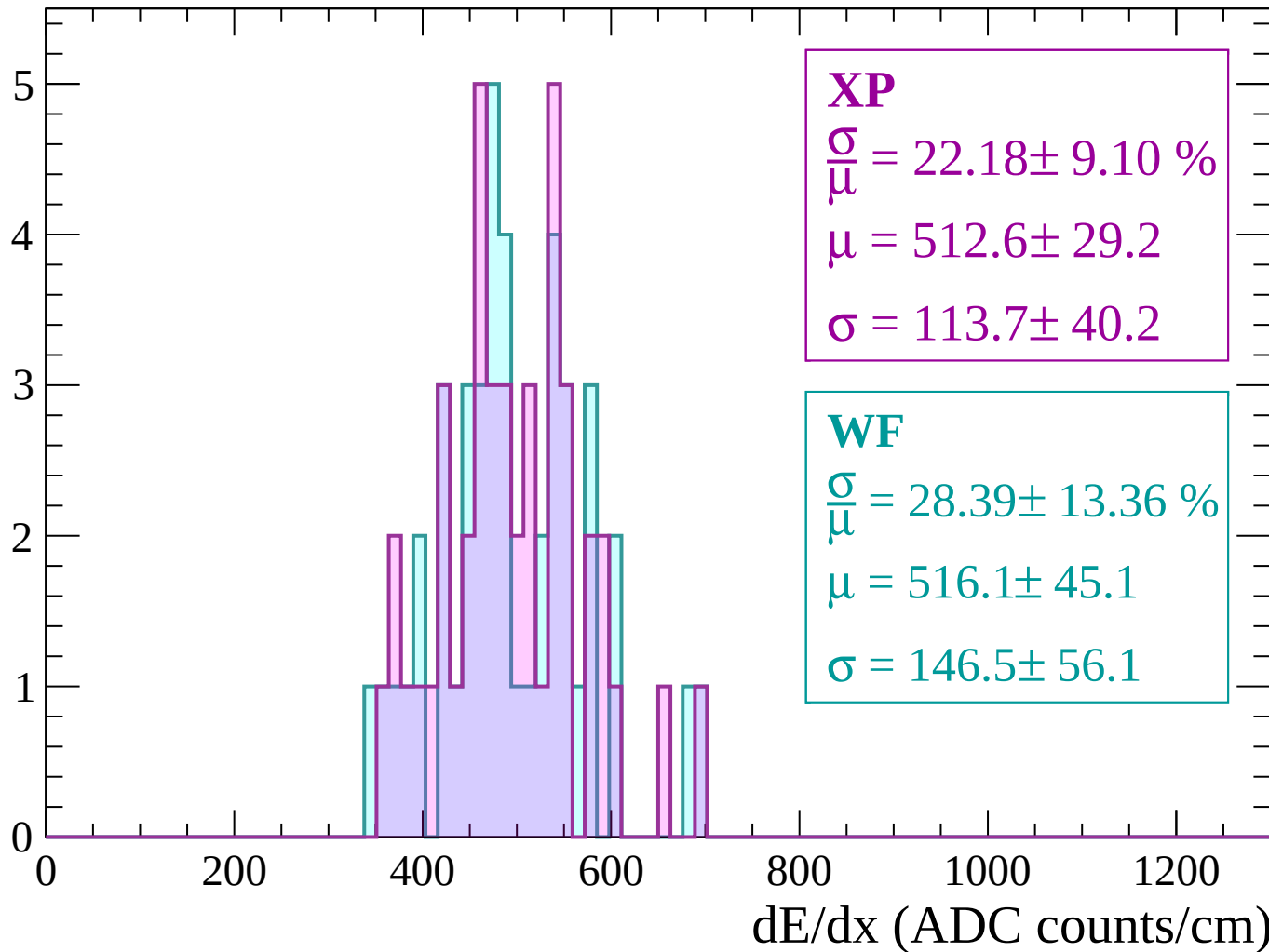
Energy loss | $1080 < p < 1140$

Count



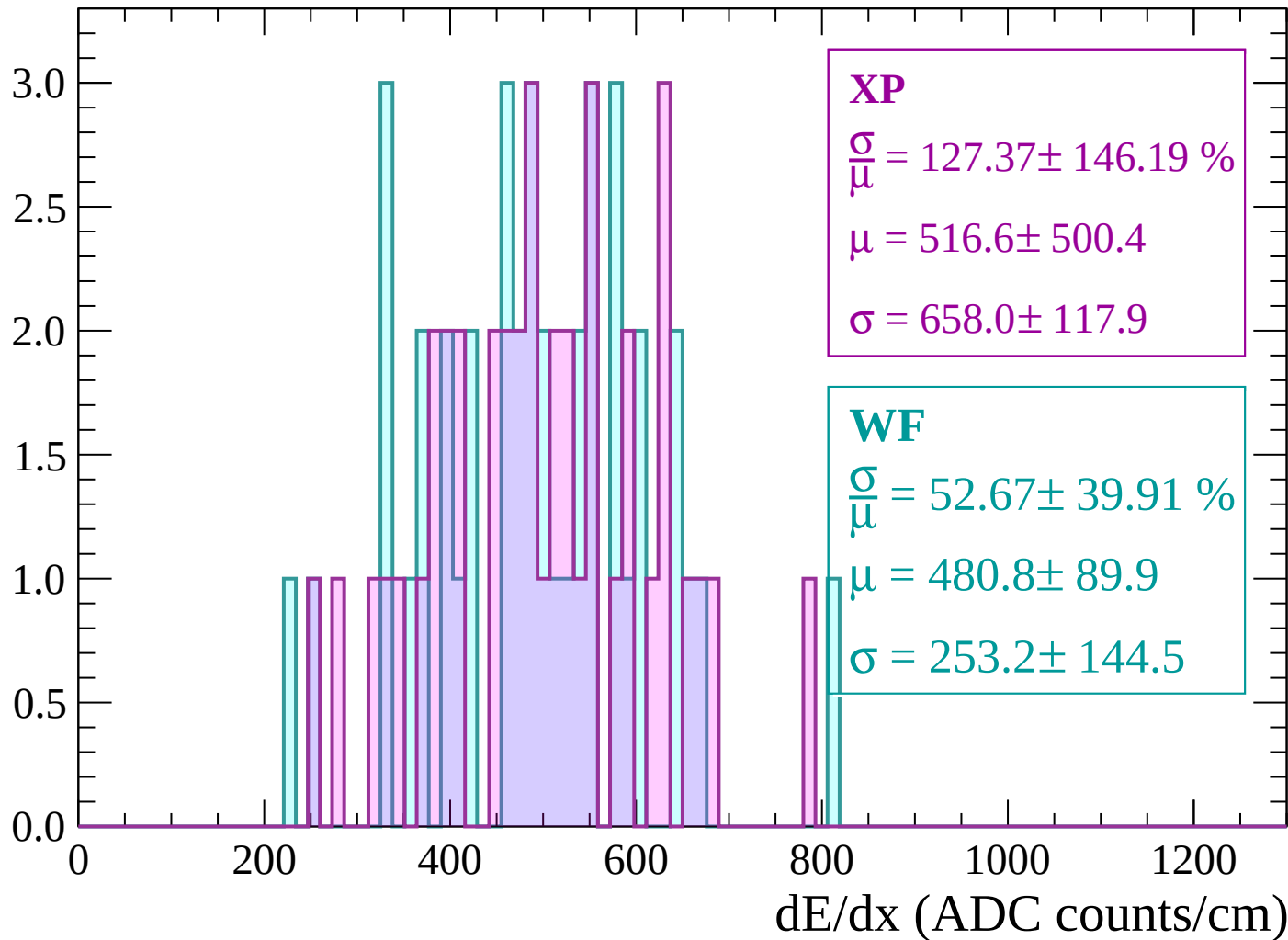
Energy loss | $1140 < p < 1200$

Count



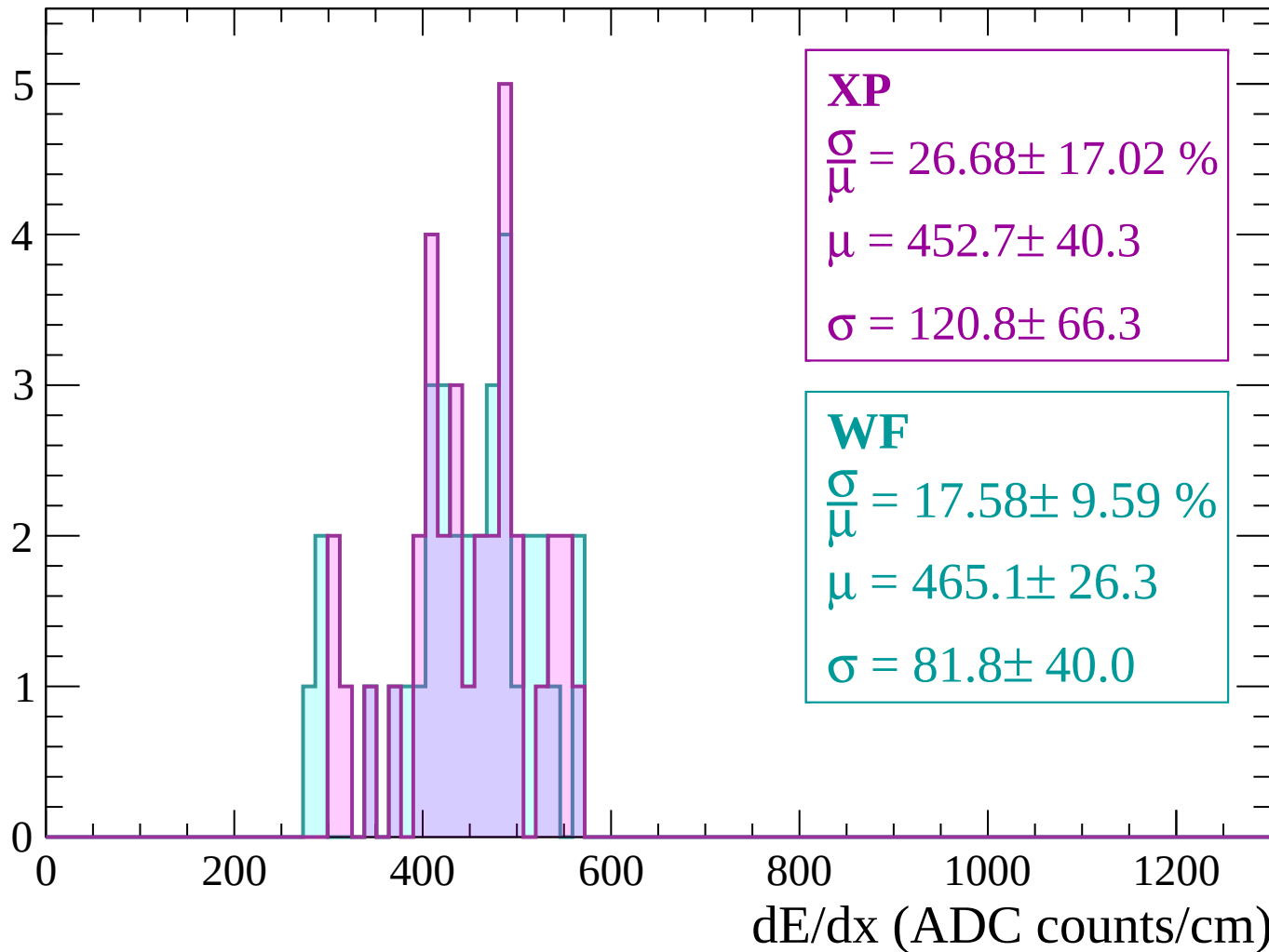
Energy loss | $1200 < p < 1260$

Count



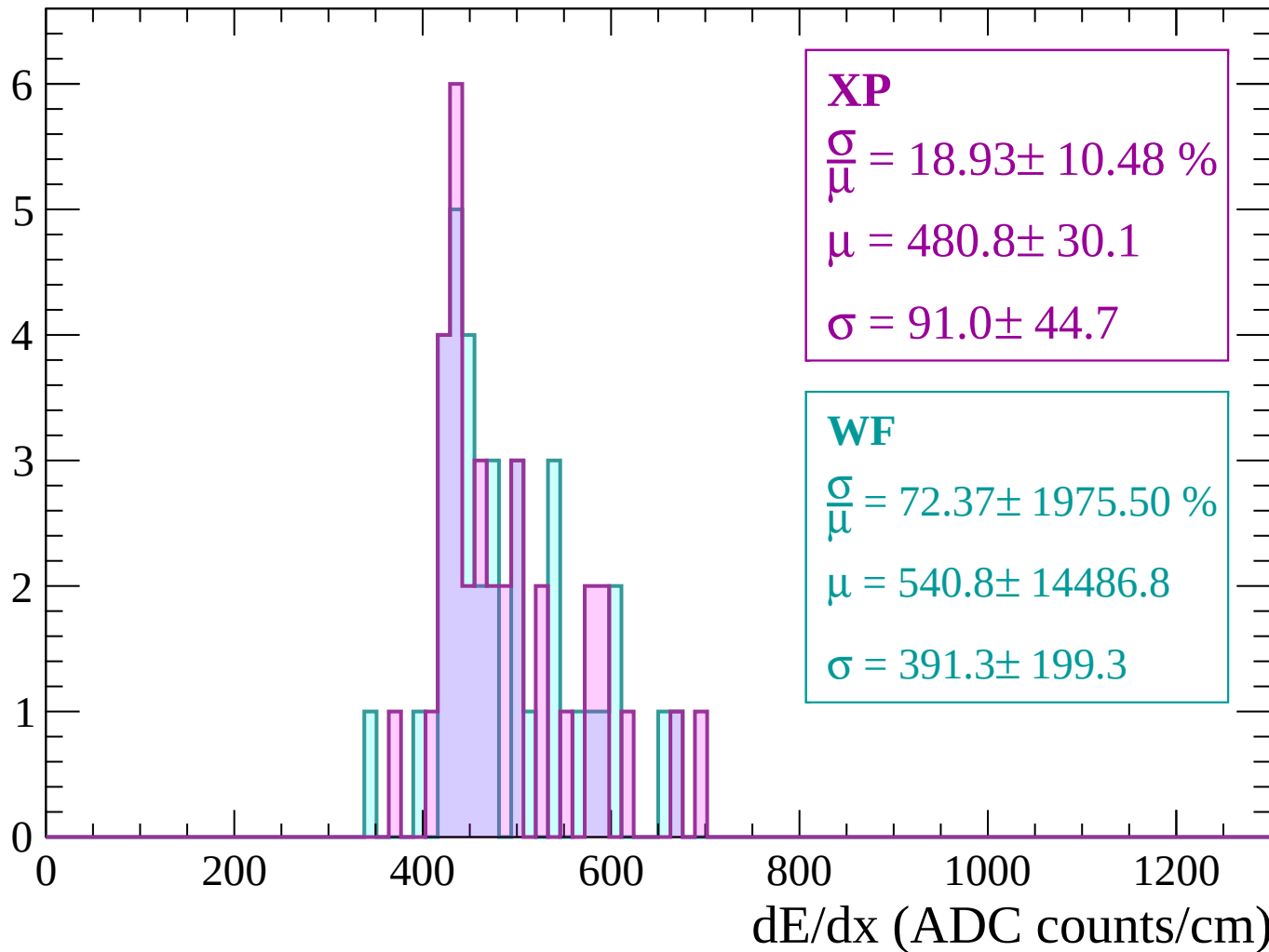
Energy loss | $1260 < p < 1320$

Count



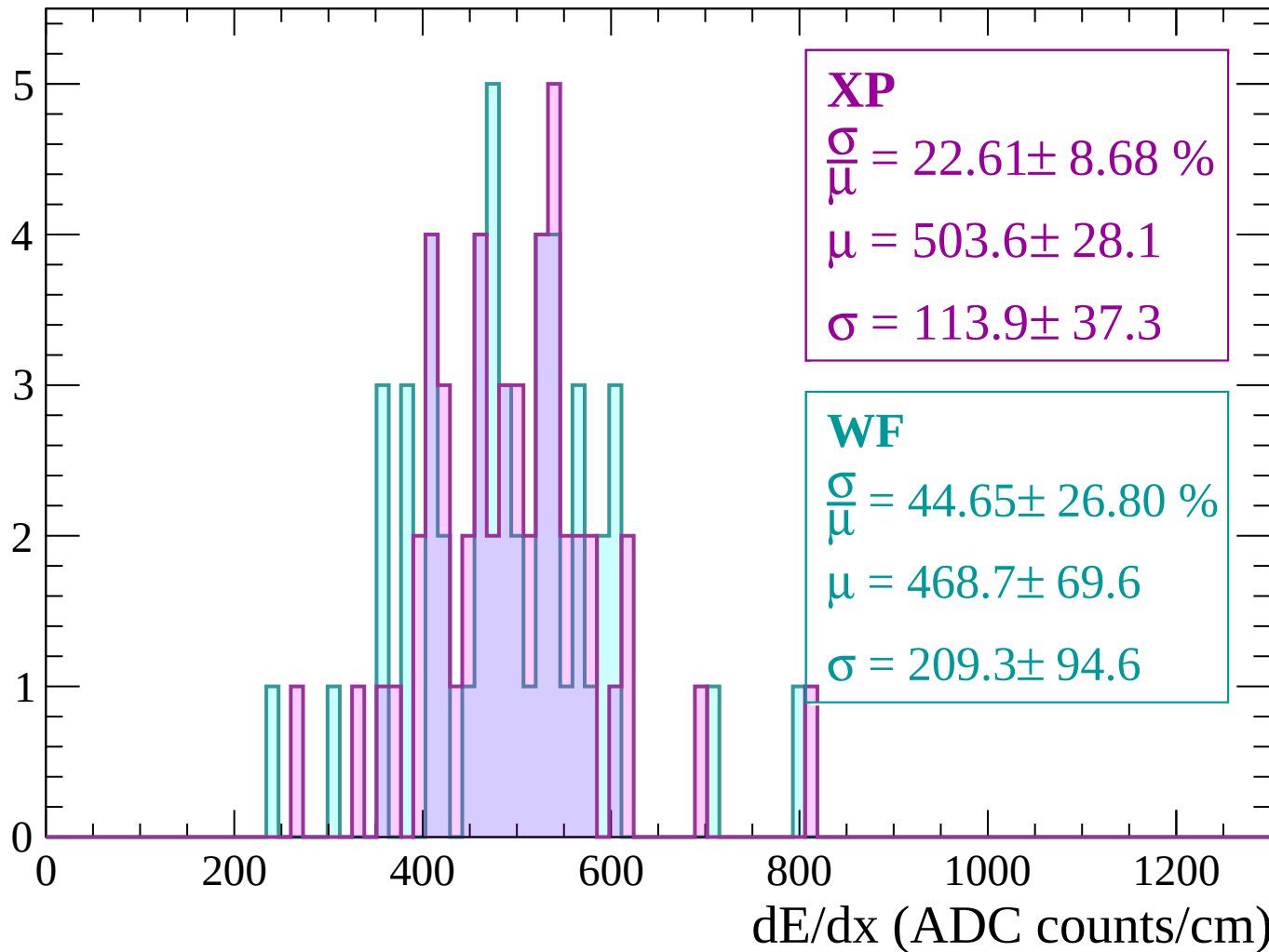
Energy loss | $1320 < p < 1380$

Count



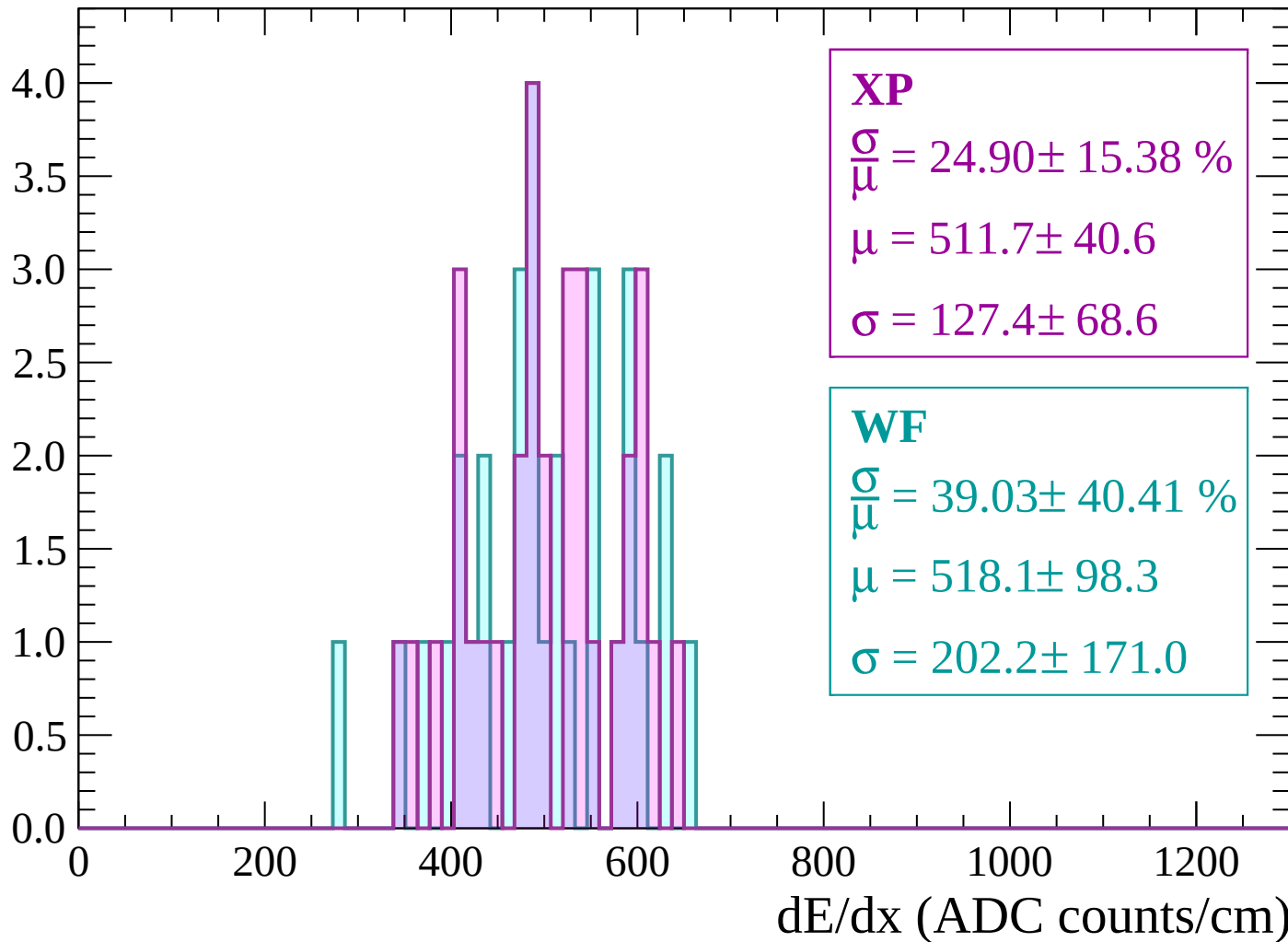
Energy loss | $1380 < p < 1440$

Count



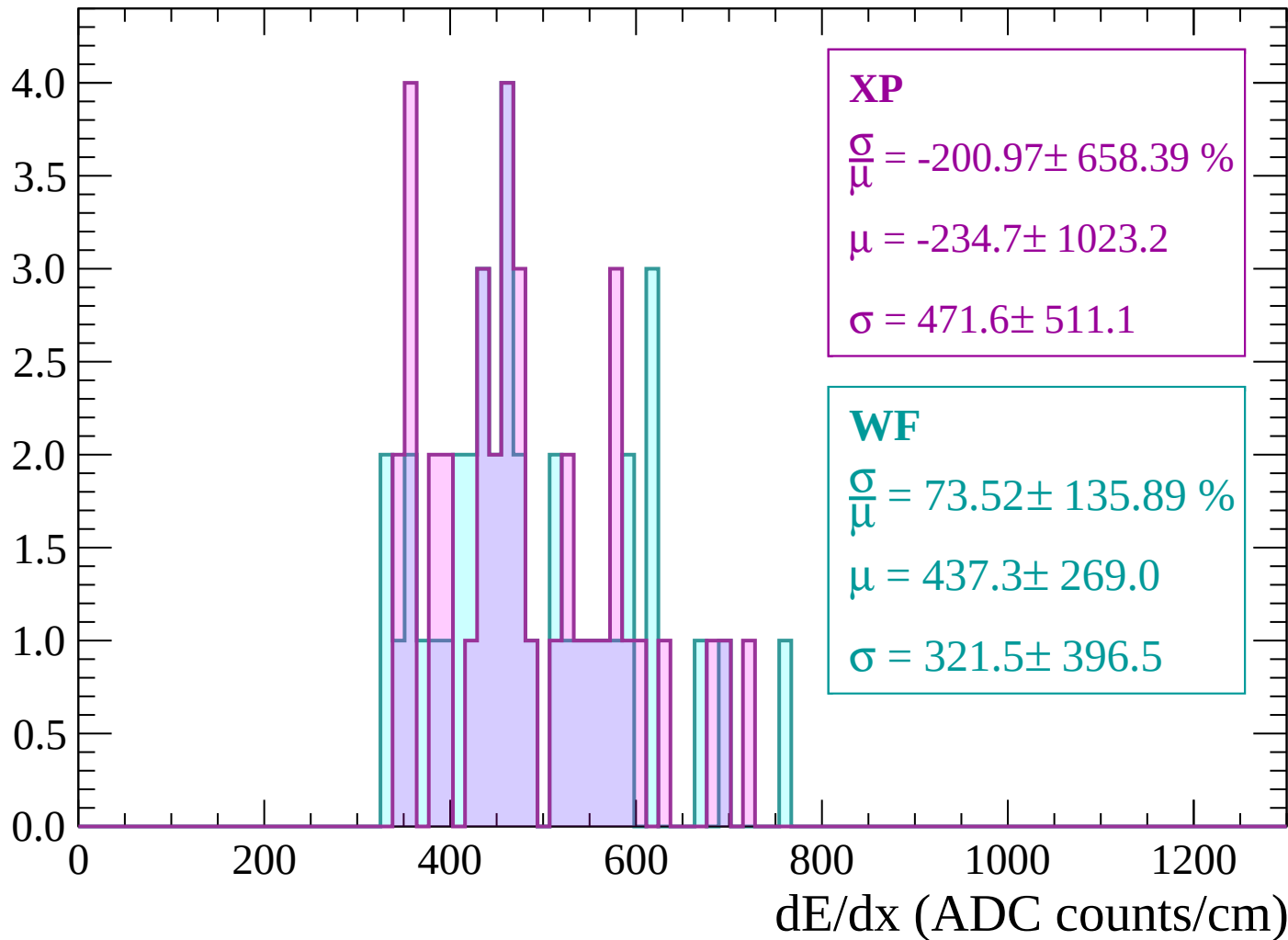
Energy loss | $1440 < p < 1500$

Count



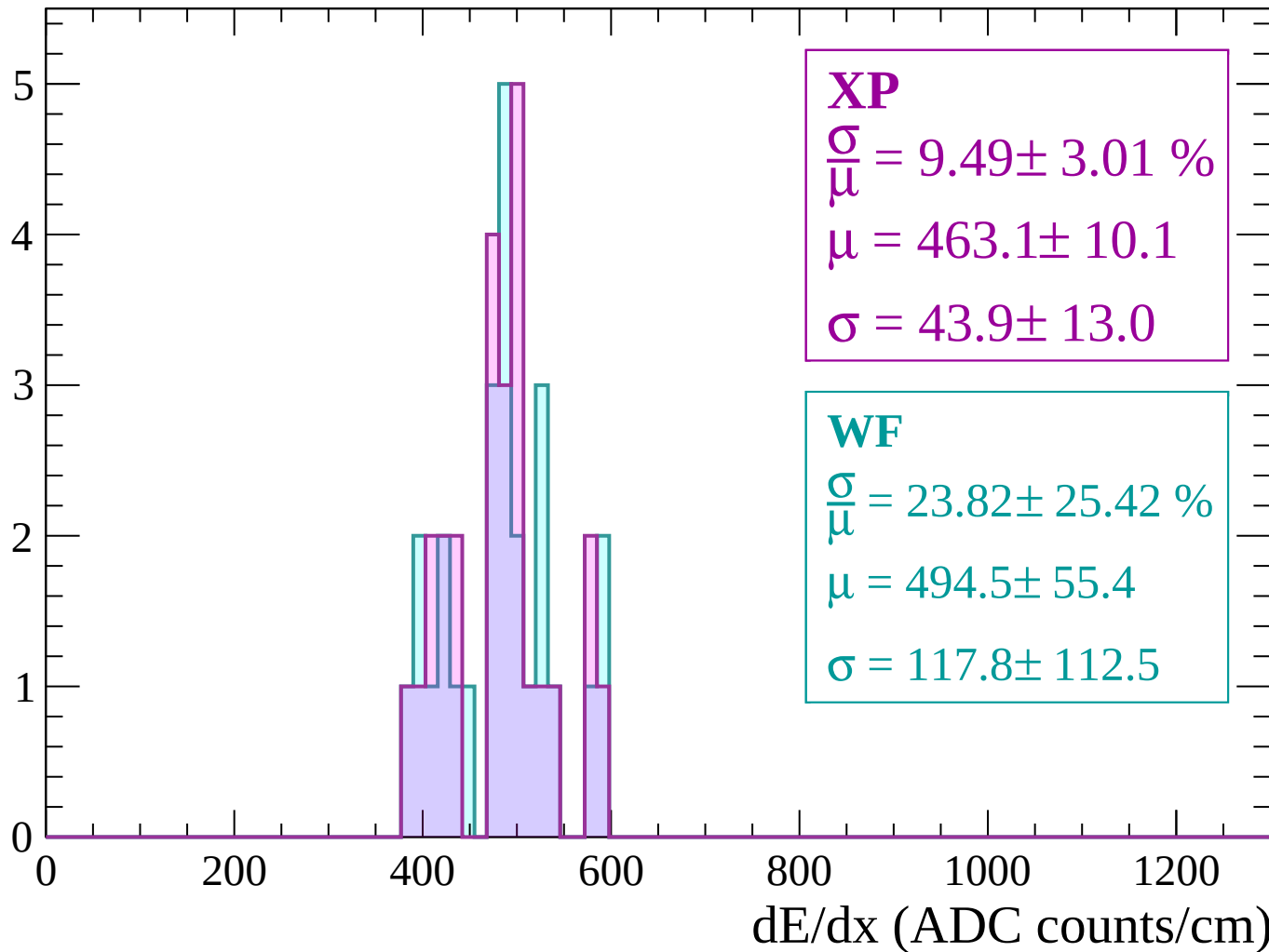
Energy loss | $1500 < p < 1560$

Count



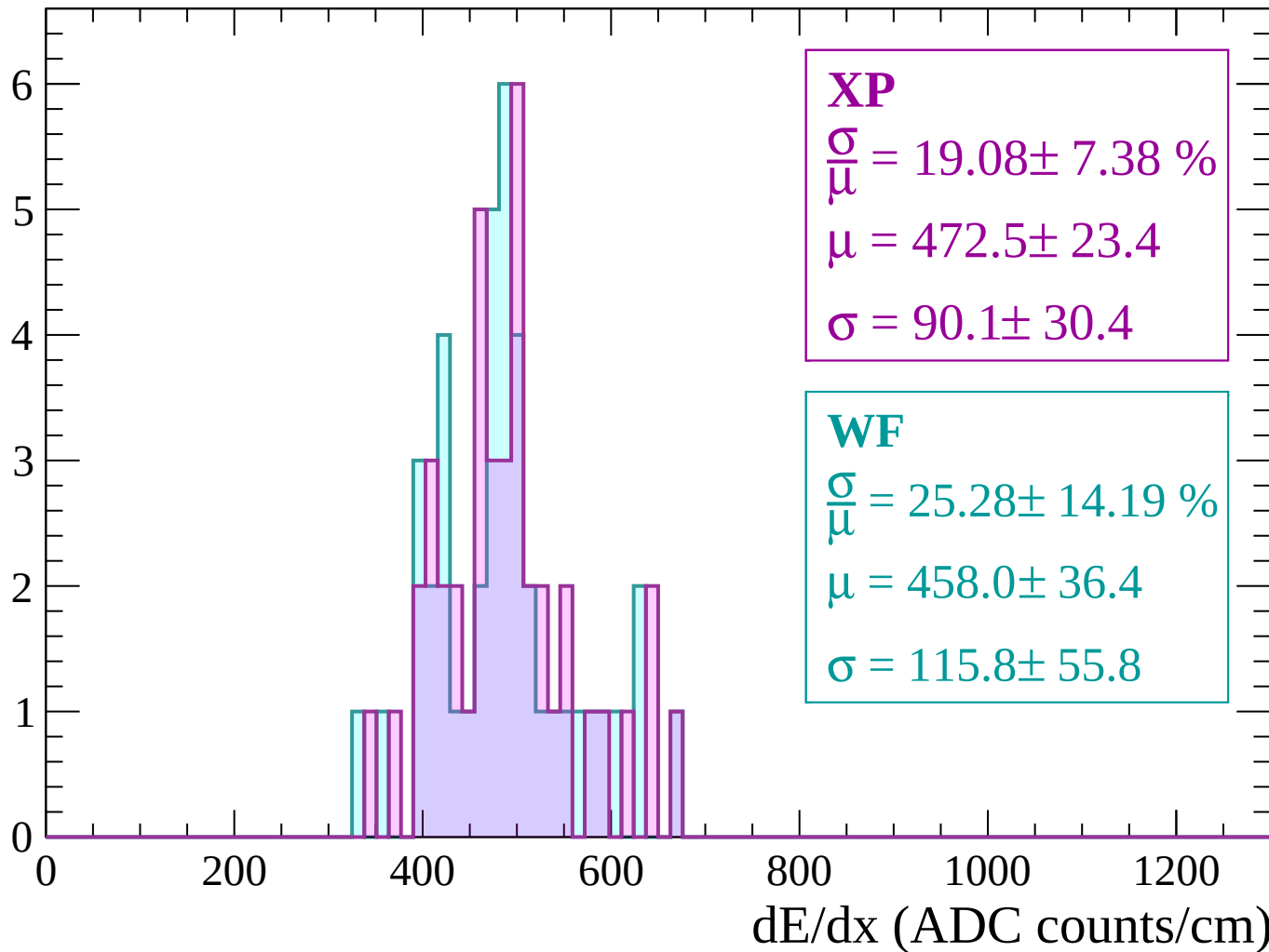
Energy loss | $1560 < p < 1620$

Count



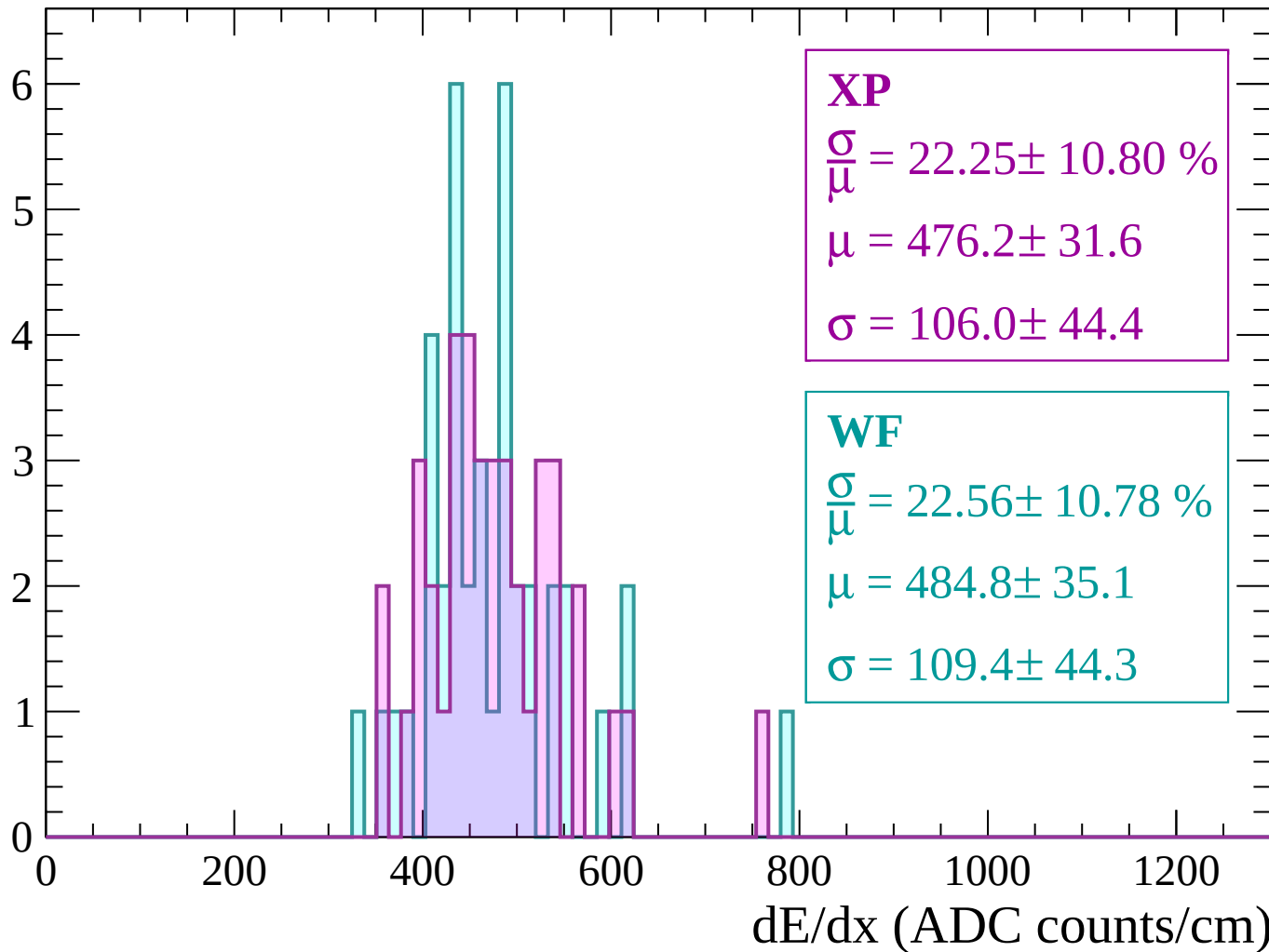
Energy loss | $1620 < p < 1680$

Count



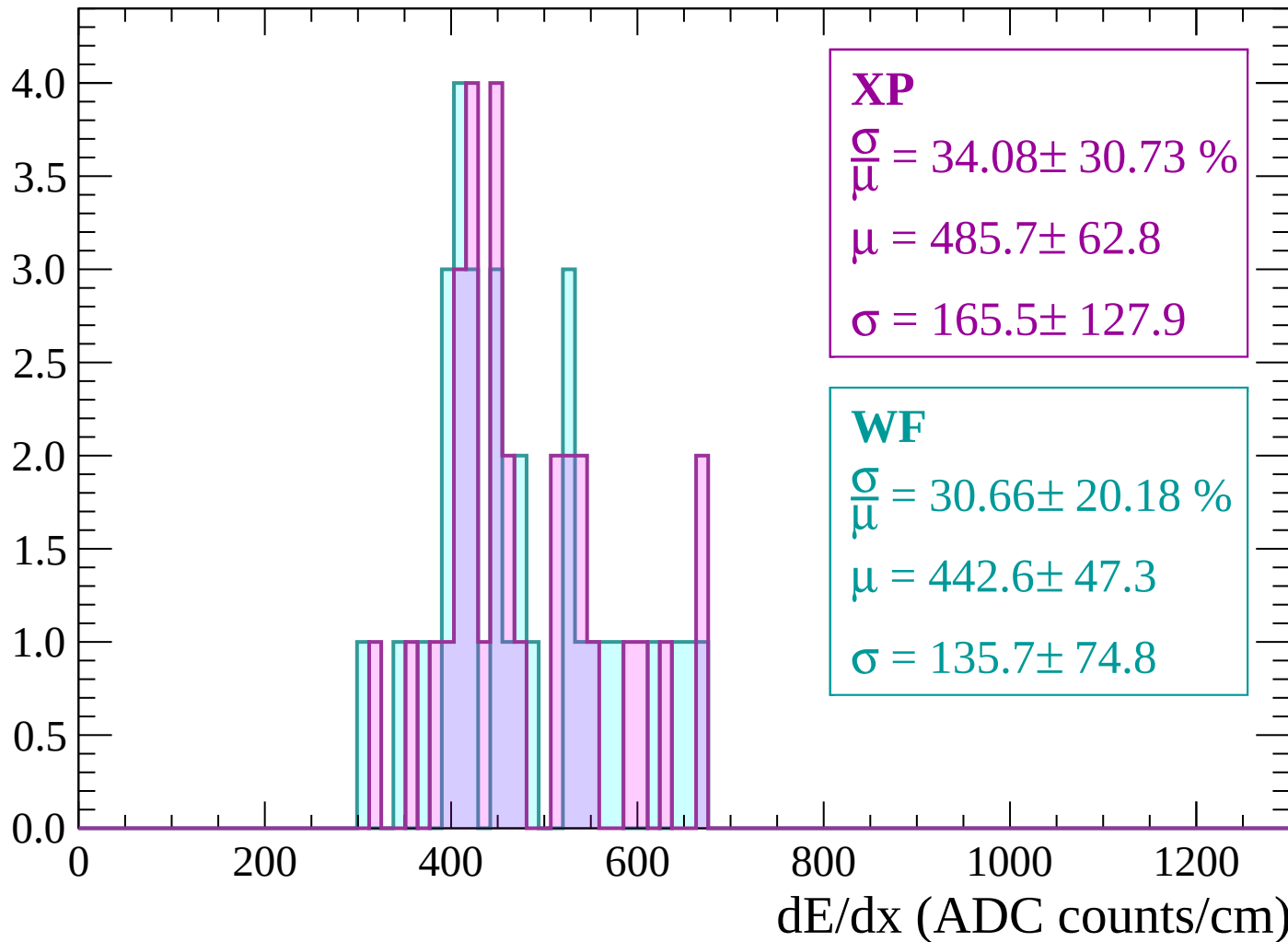
Energy loss | $1680 < p < 1740$

Count



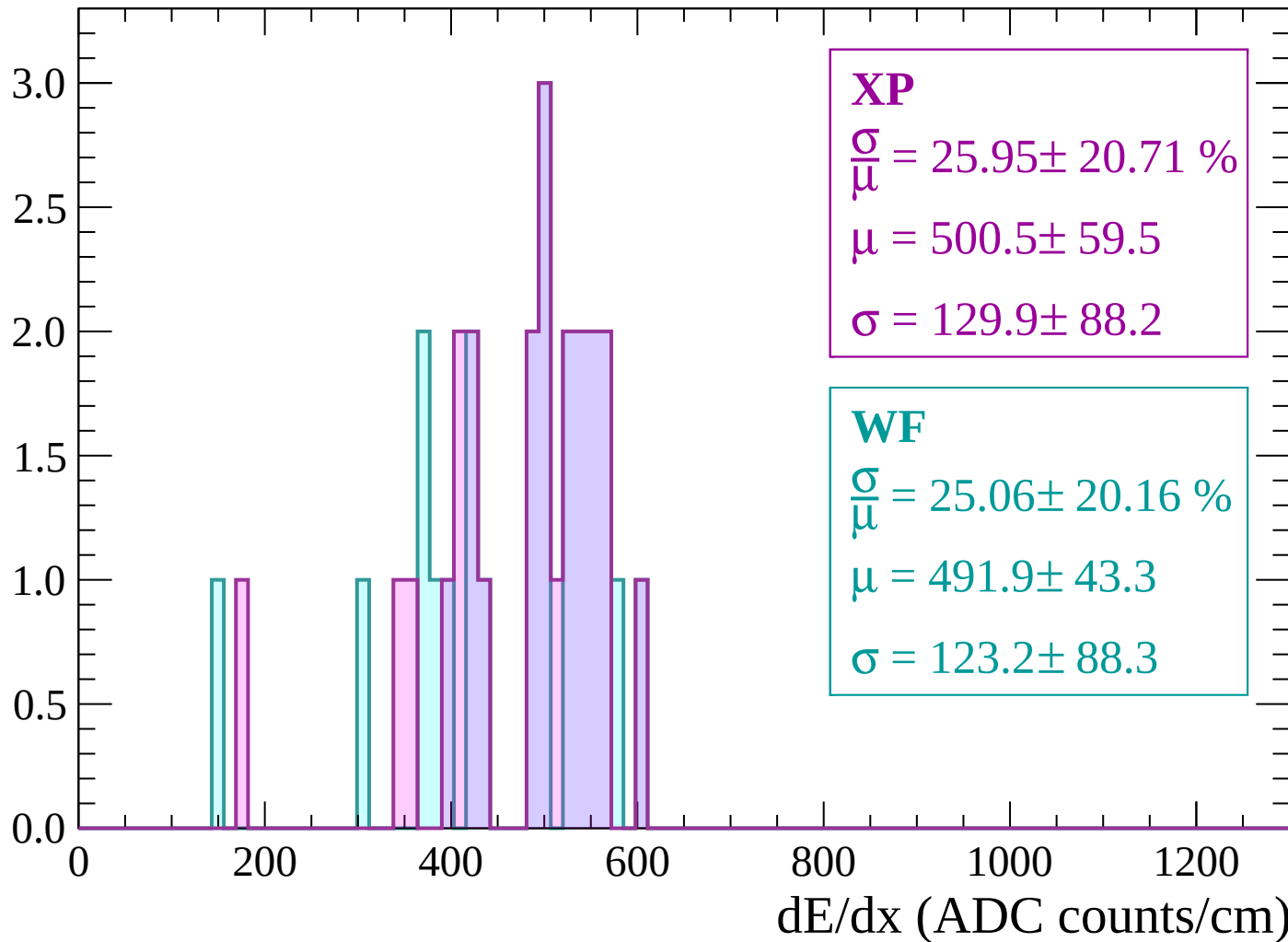
Energy loss | $1740 < p < 1800$

Count



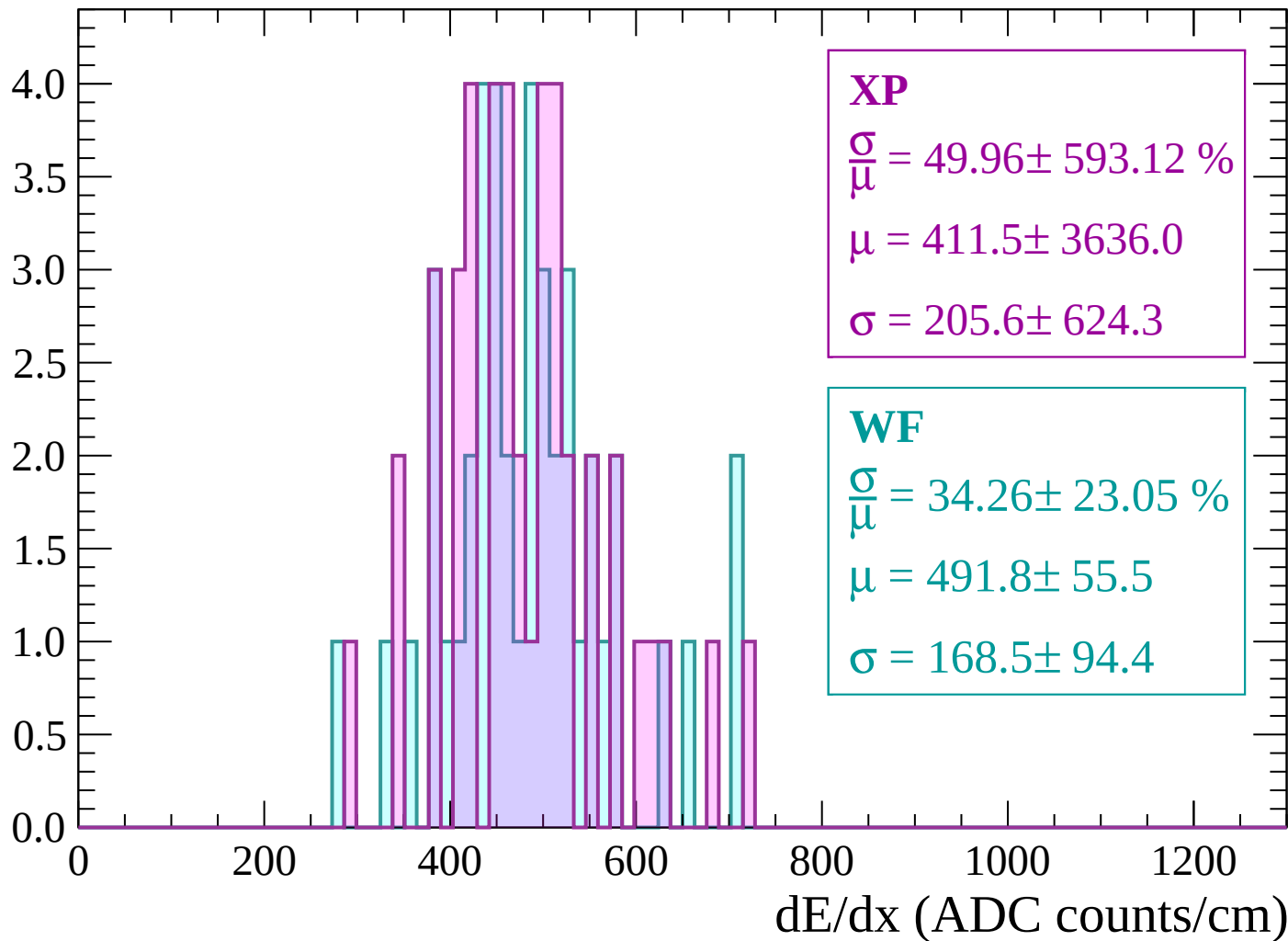
Energy loss | $1800 < p < 1860$

Count



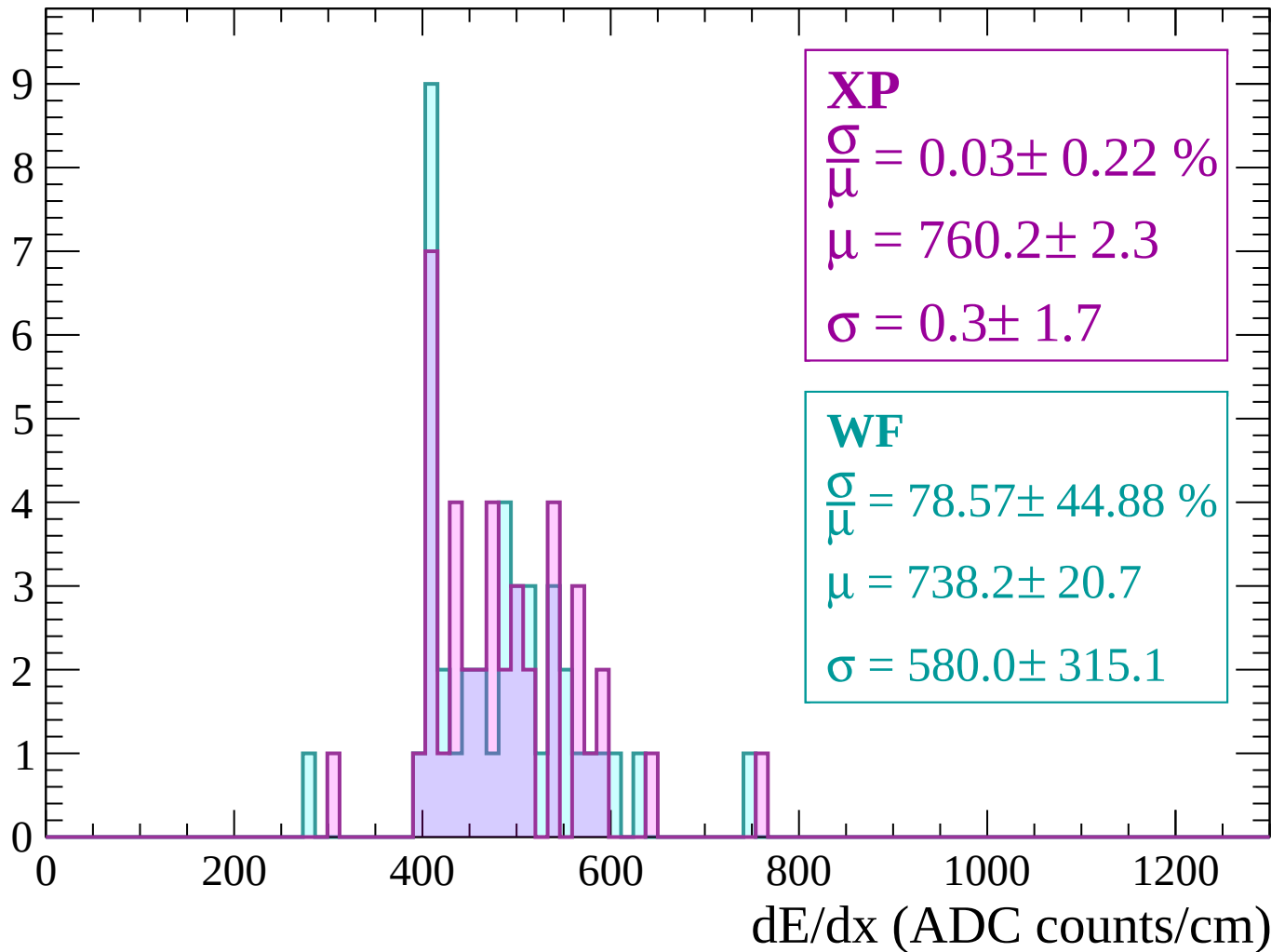
Energy loss | $1860 < p < 1920$

Count



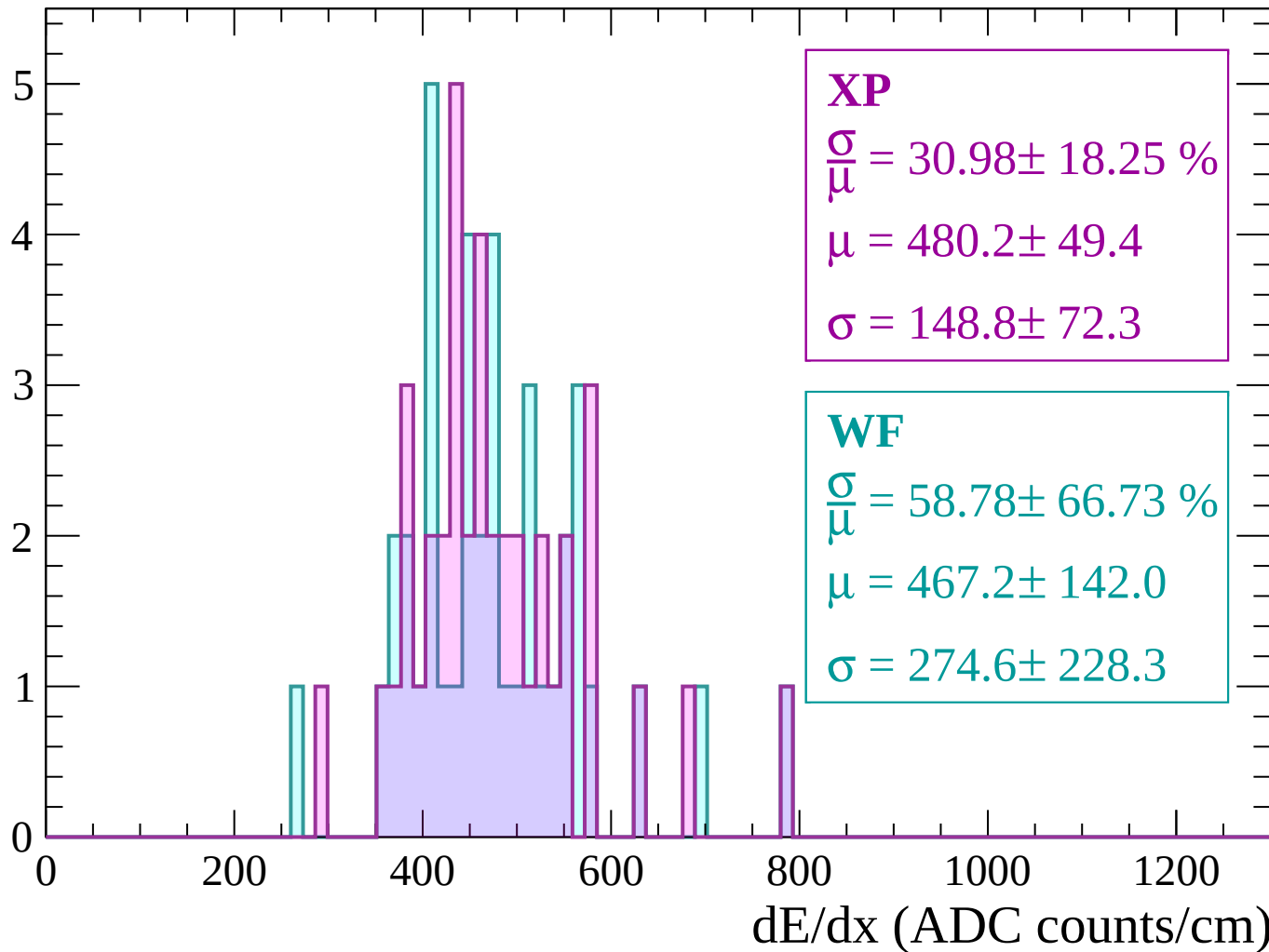
Energy loss | $1920 < p < 1980$

Count



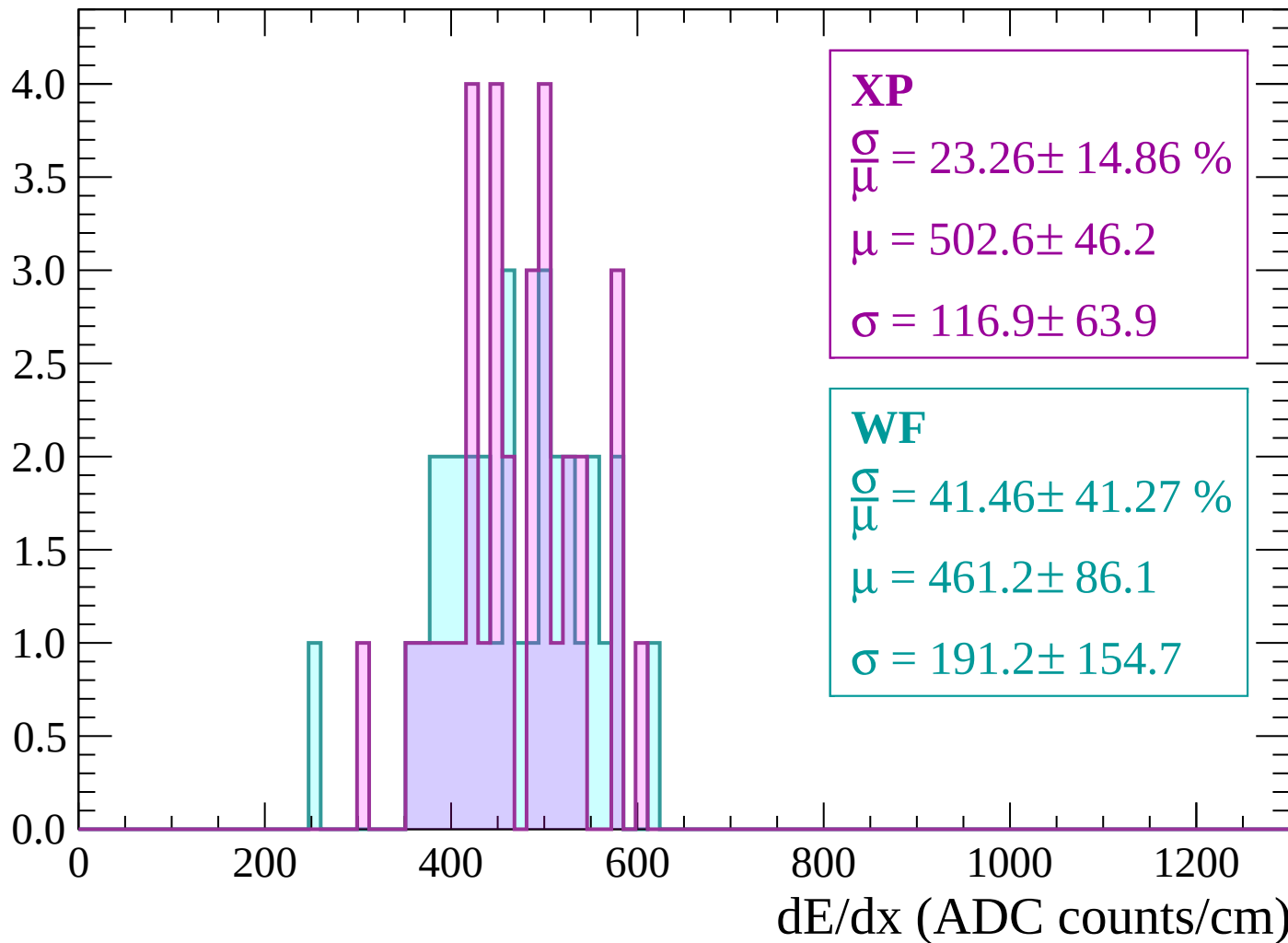
Energy loss | $1980 < p < 2040$

Count



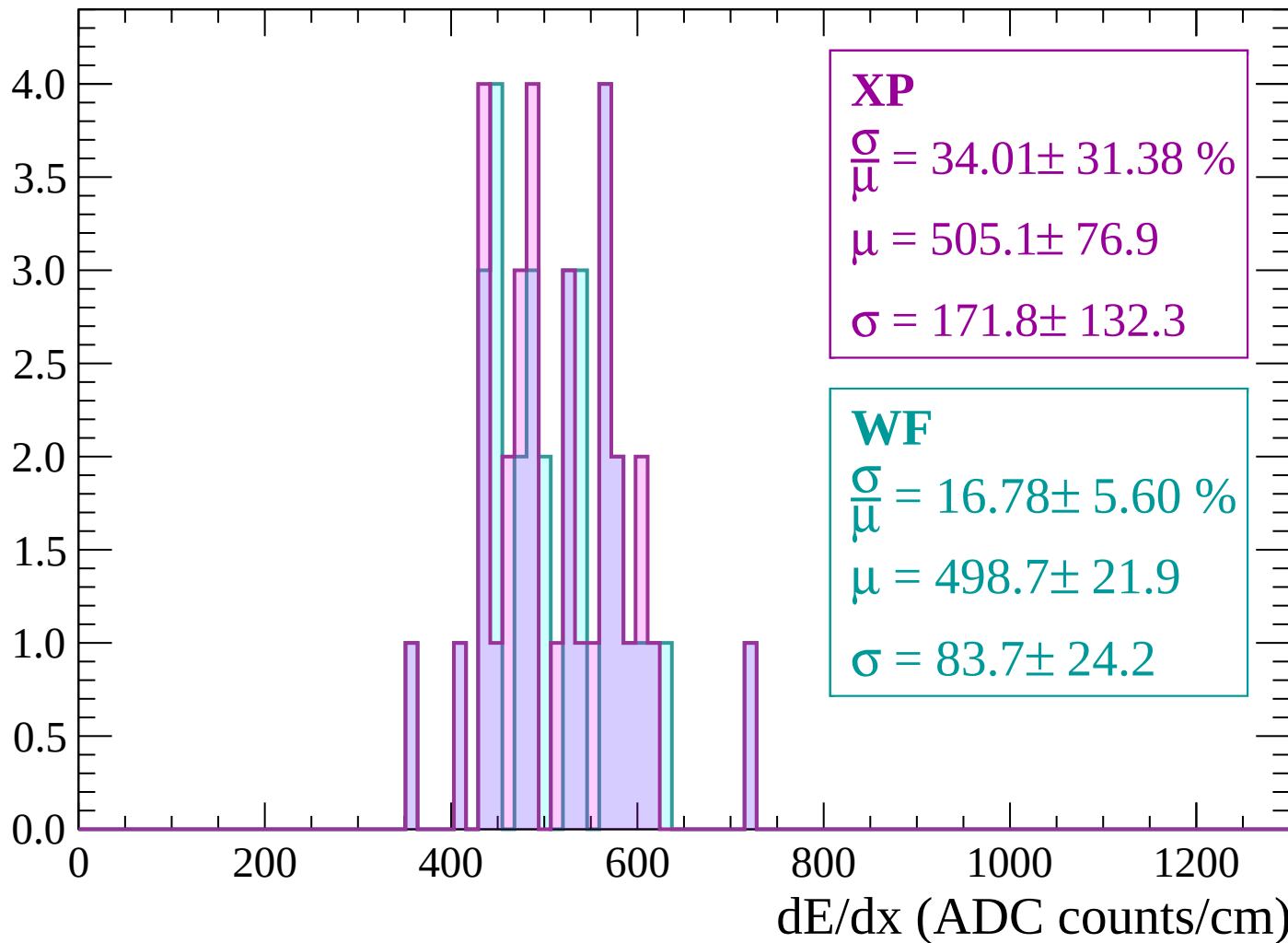
Energy loss | $2040 < p < 2100$

Count



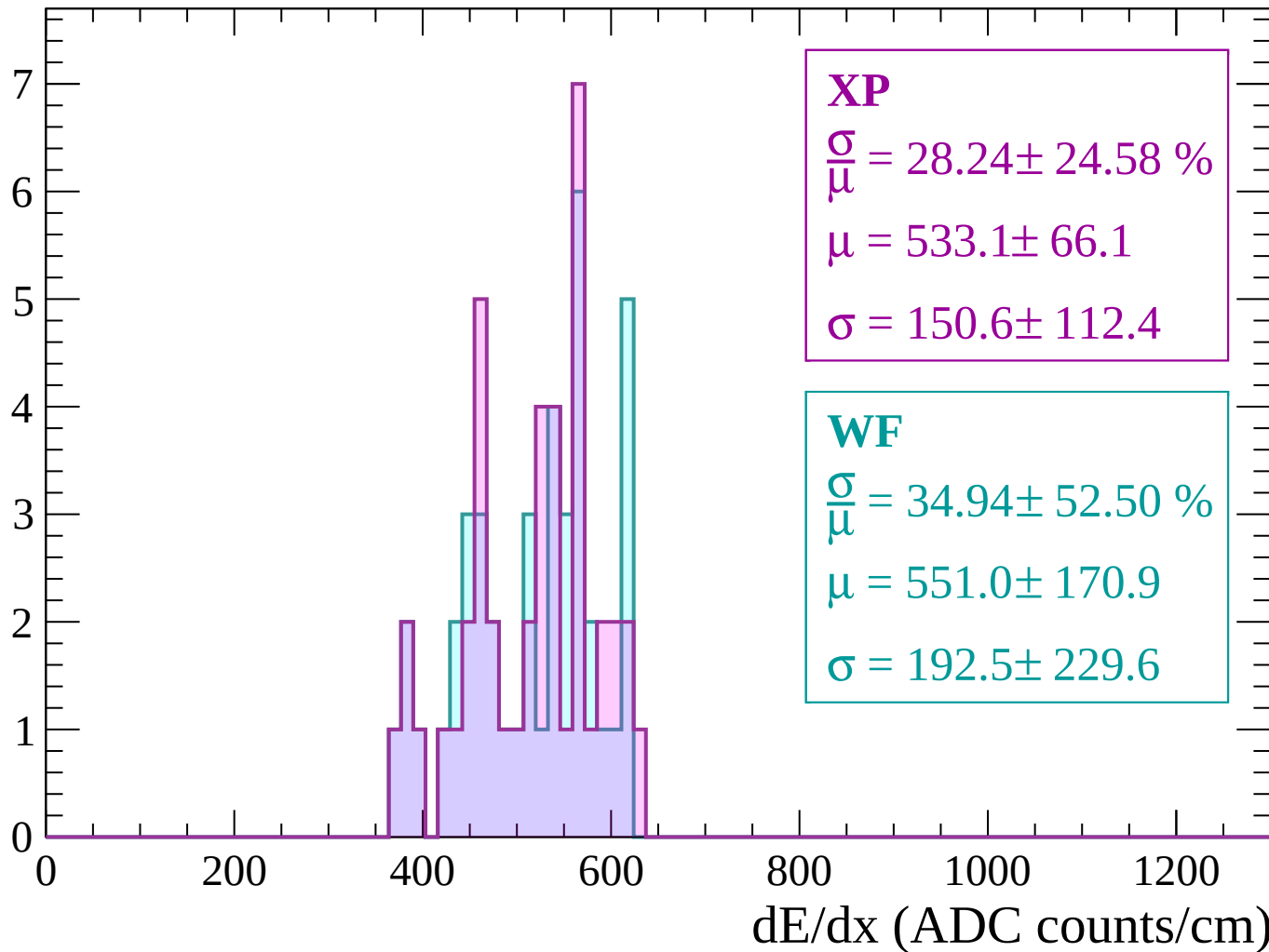
Energy loss | $2100 < p < 2160$

Count



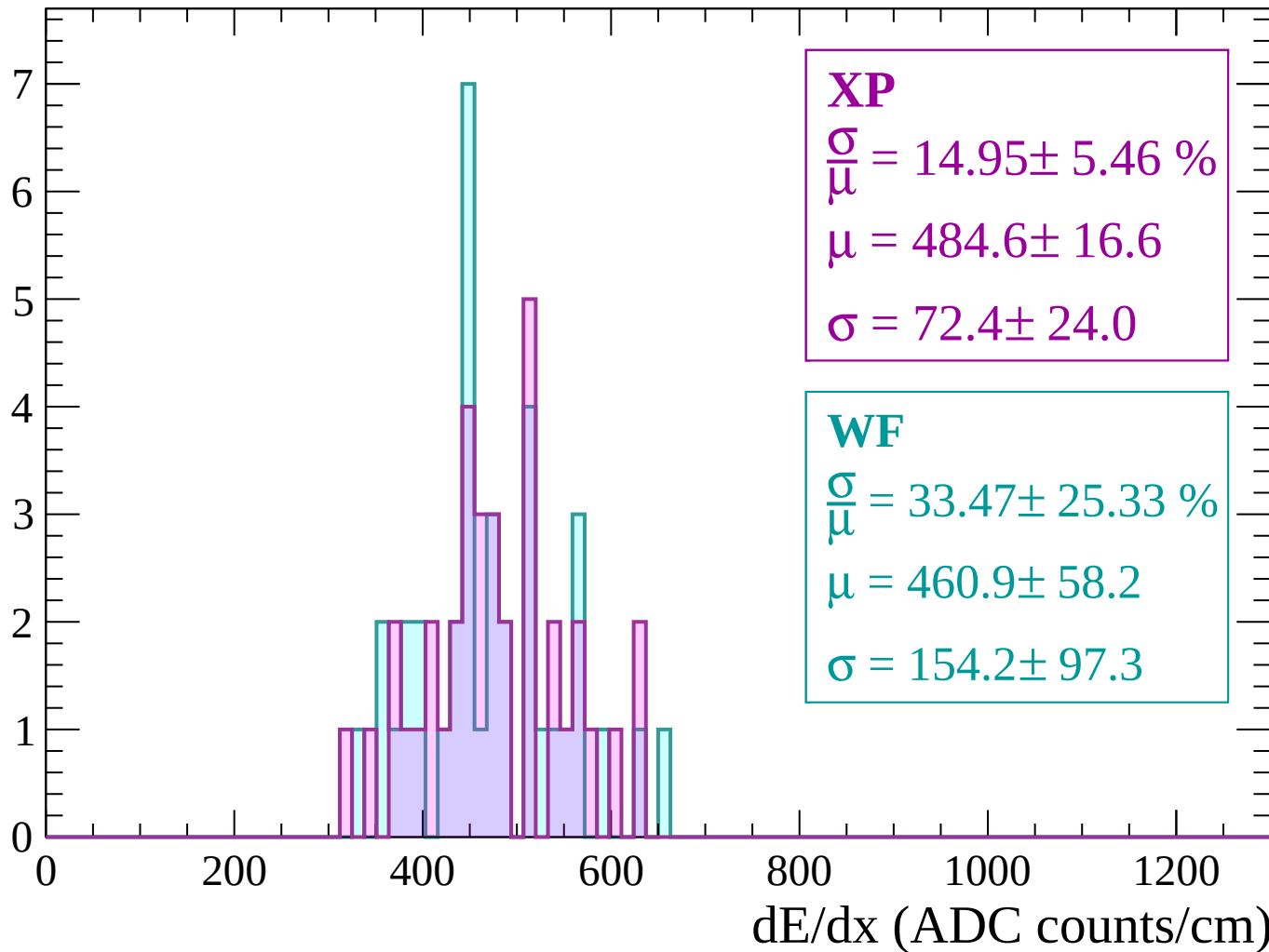
Energy loss | $2160 < p < 2220$

Count



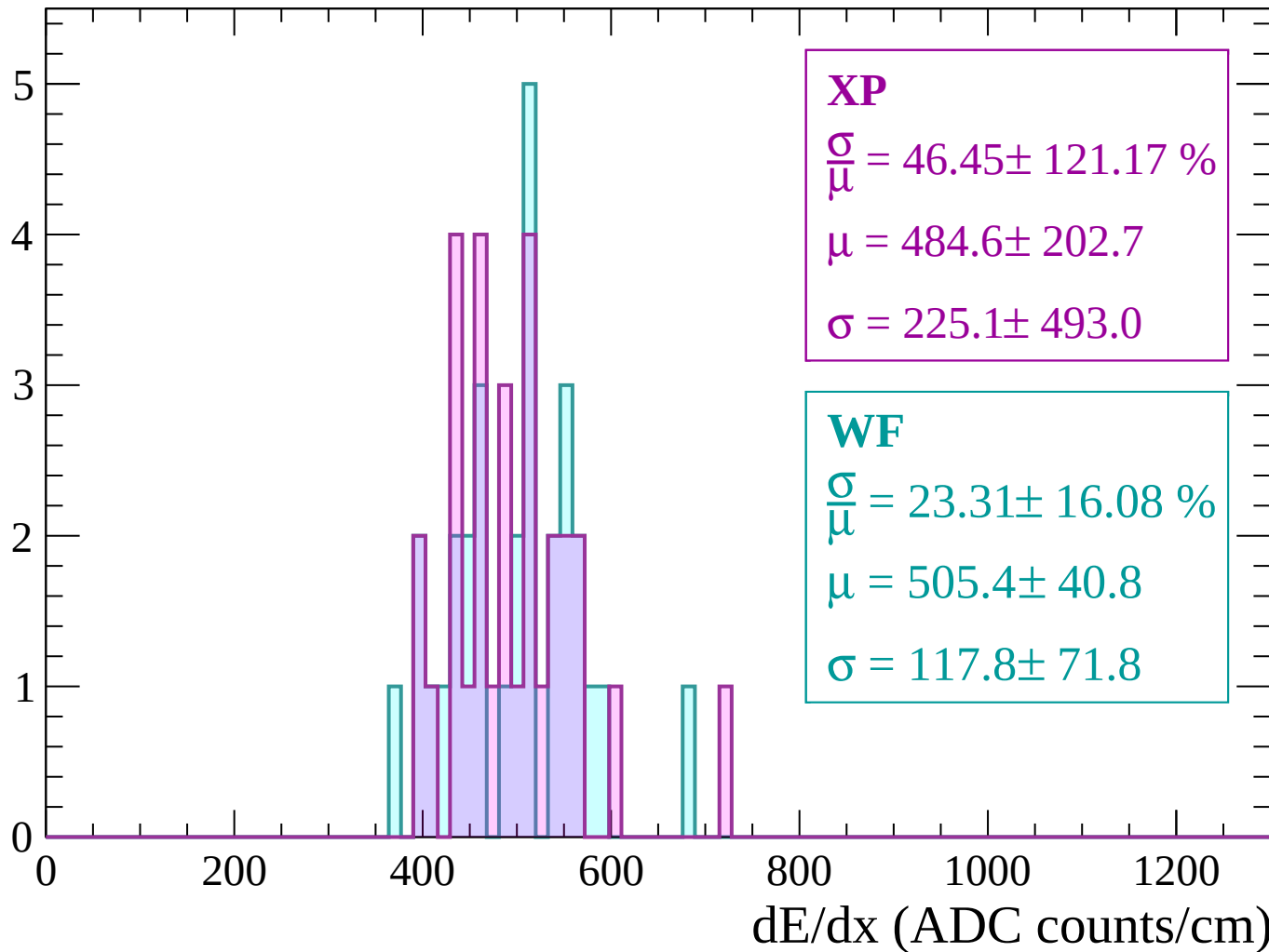
Energy loss | $2220 < p < 2280$

Count



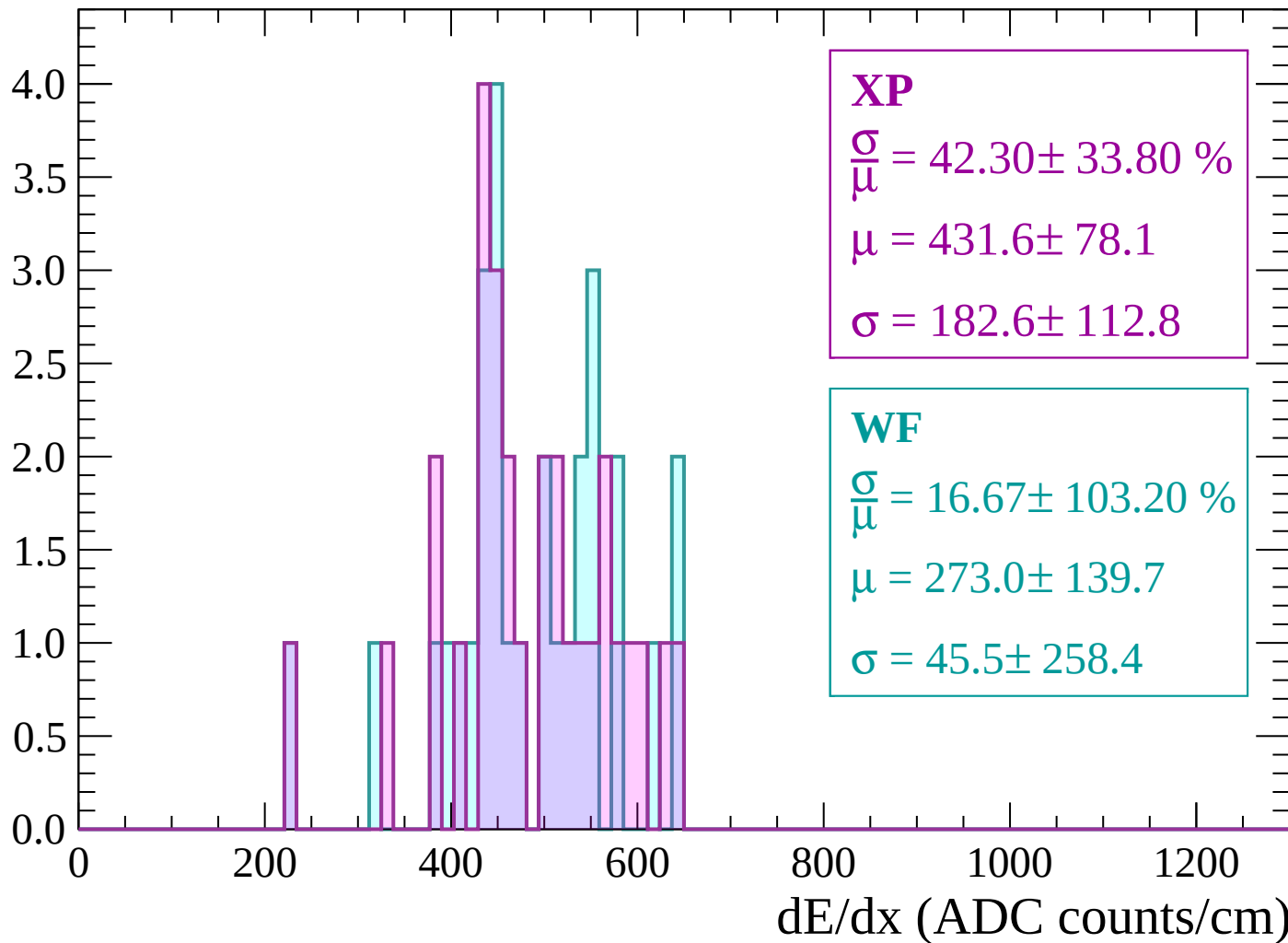
Energy loss | $2280 < p < 2340$

Count



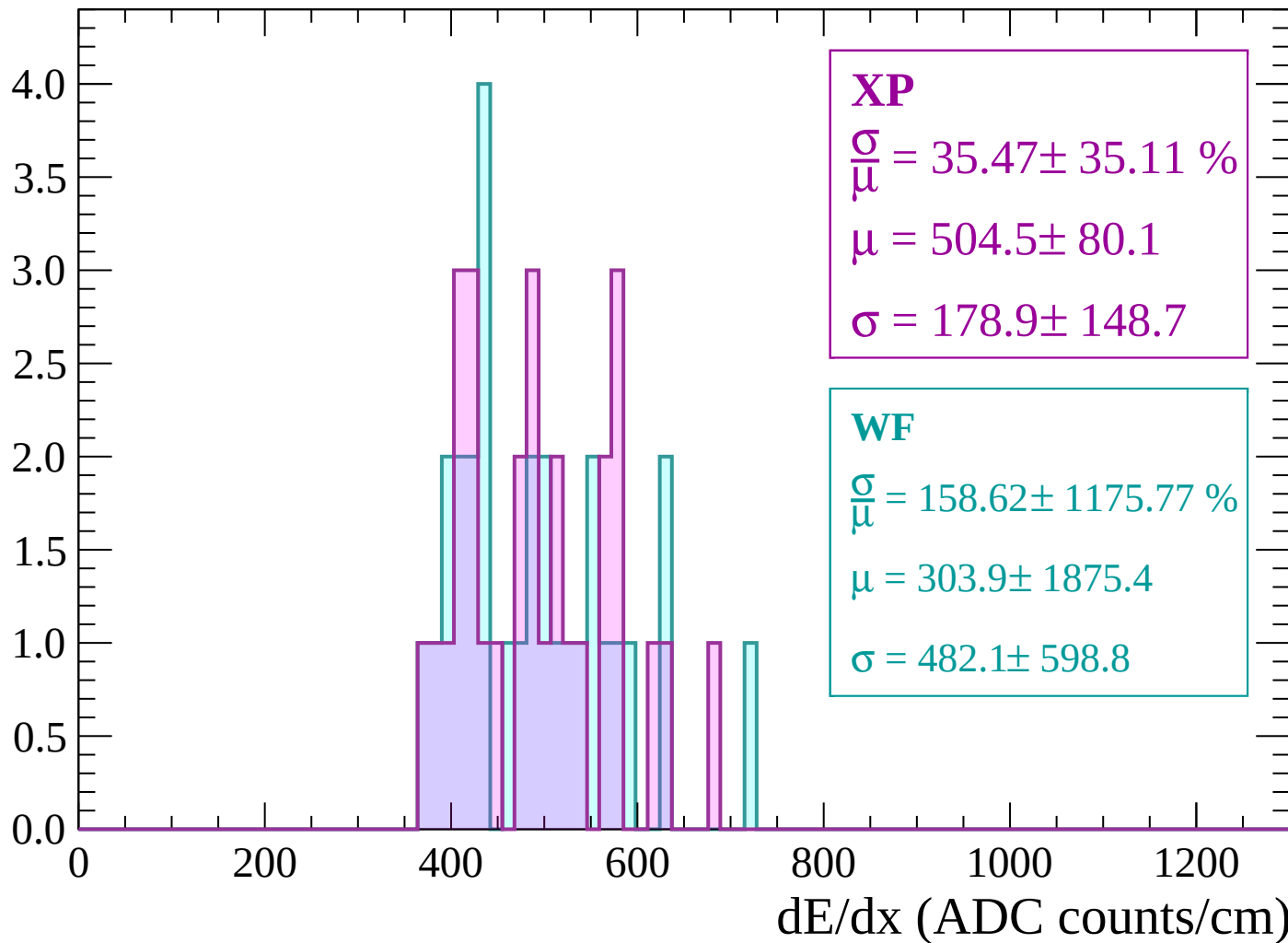
Energy loss | $2340 < p < 2400$

Count



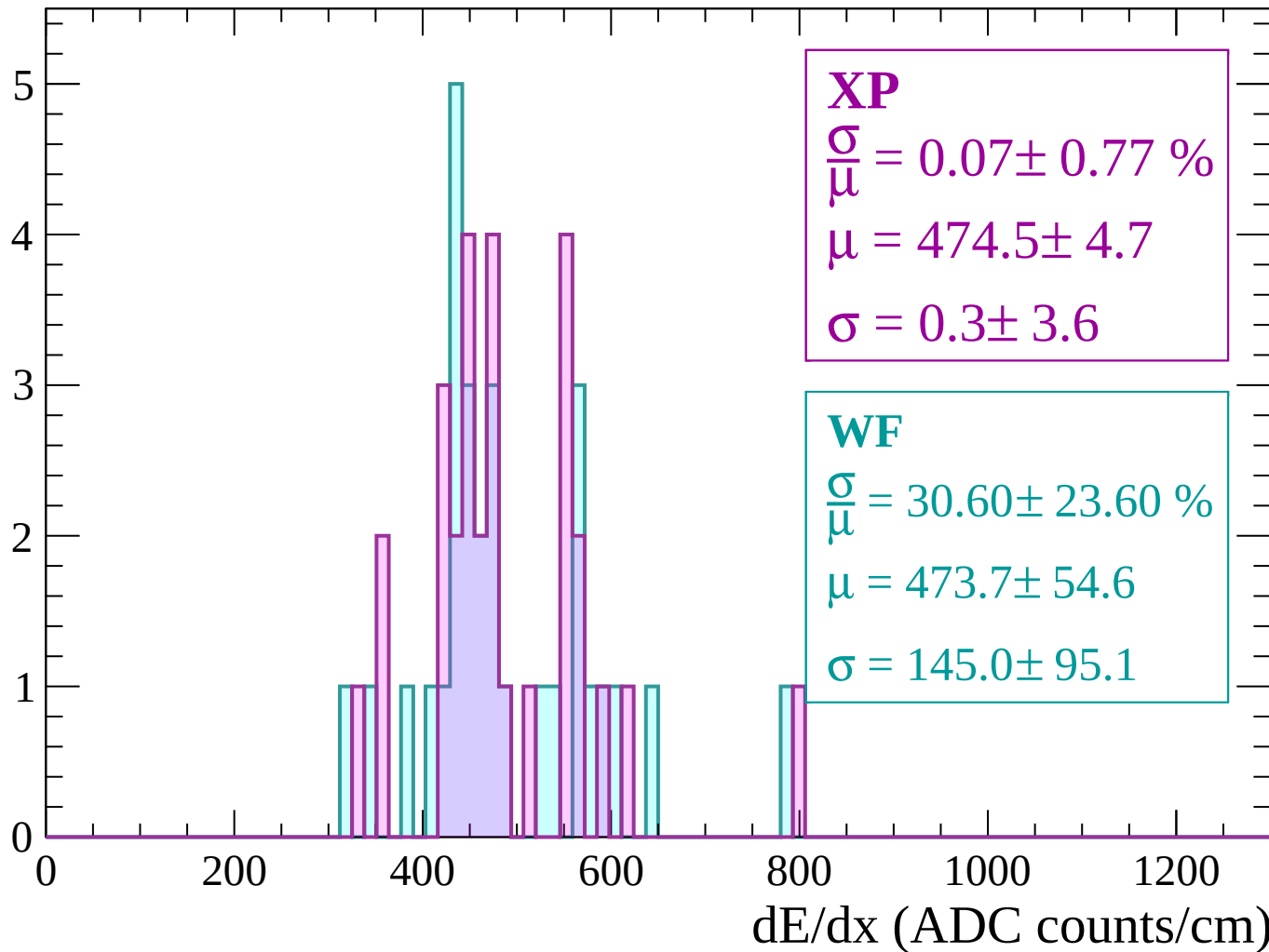
Energy loss | $2400 < p < 2460$

Count



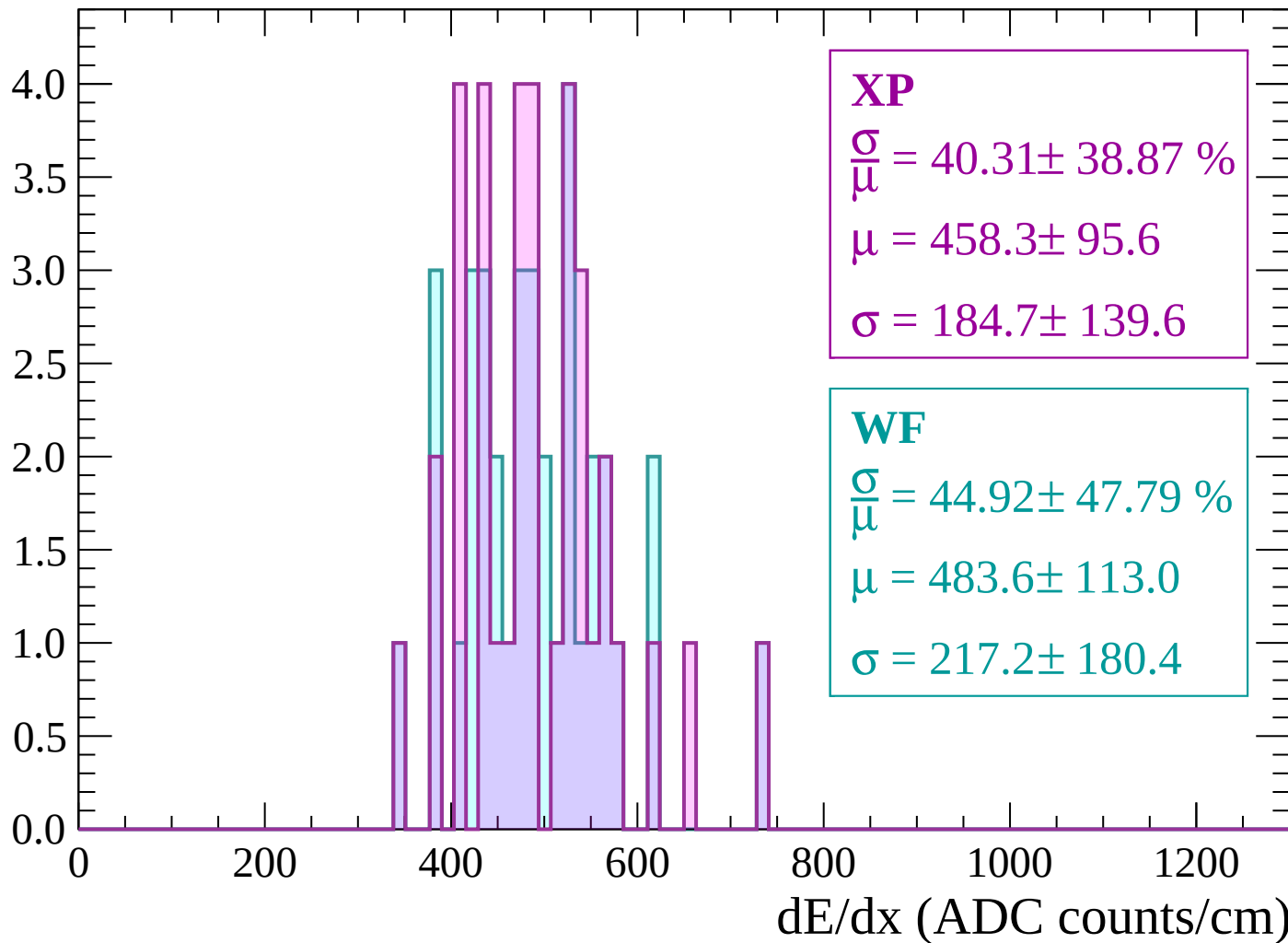
Energy loss | $2460 < p < 2520$

Count



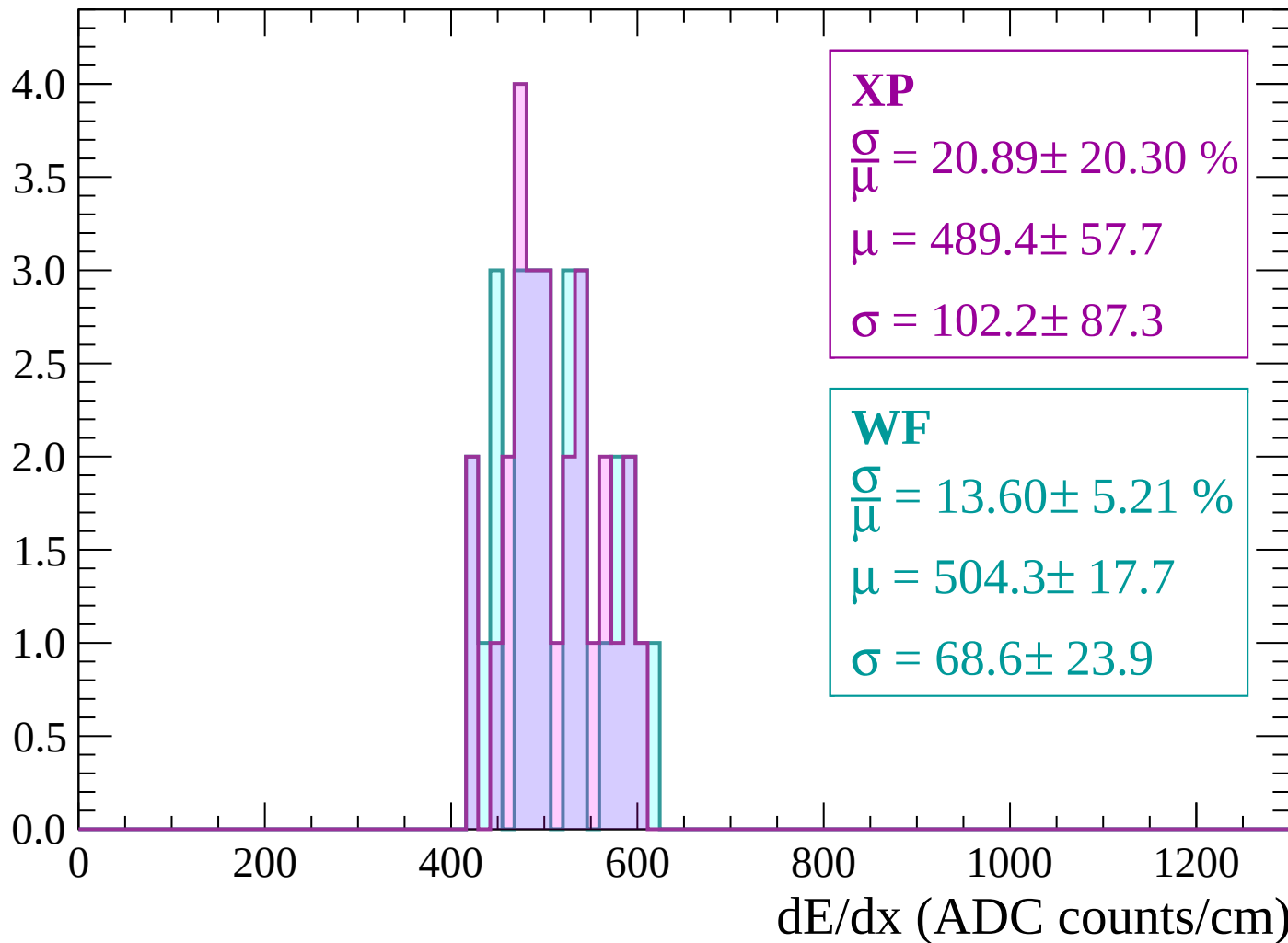
Energy loss | $2520 < p < 2580$

Count



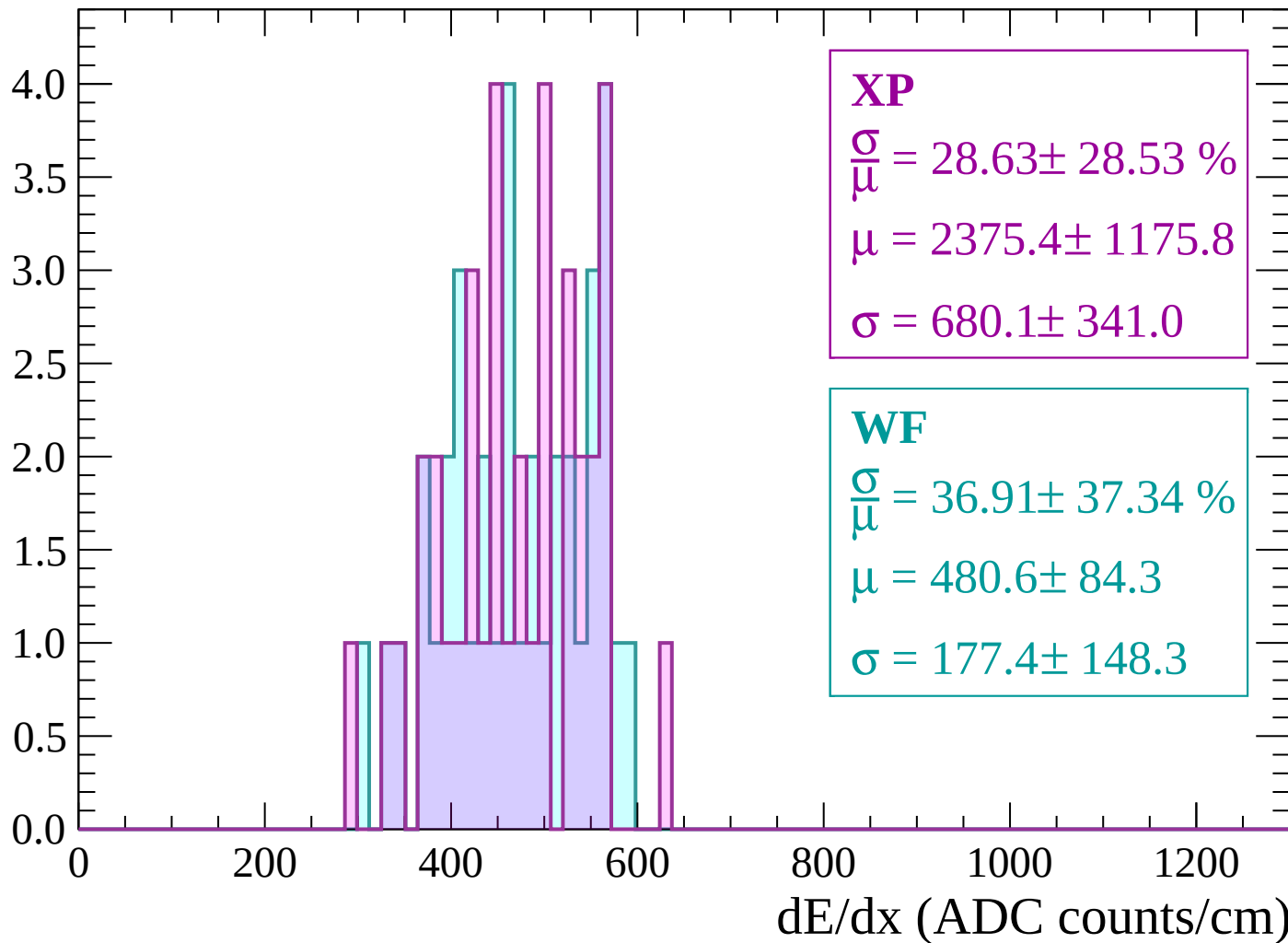
Energy loss | $2580 < p < 2640$

Count



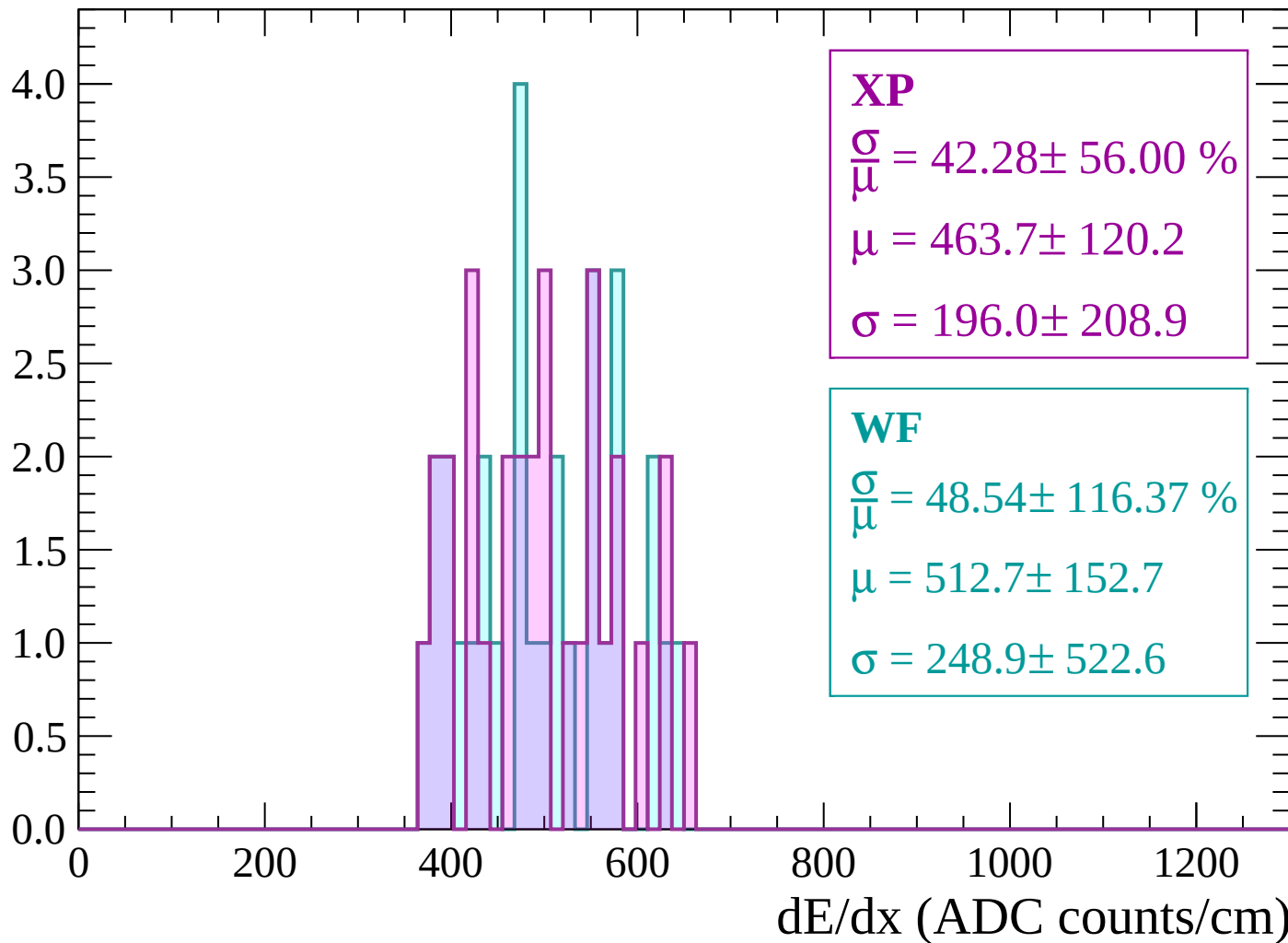
Energy loss | $2640 < p < 2700$

Count



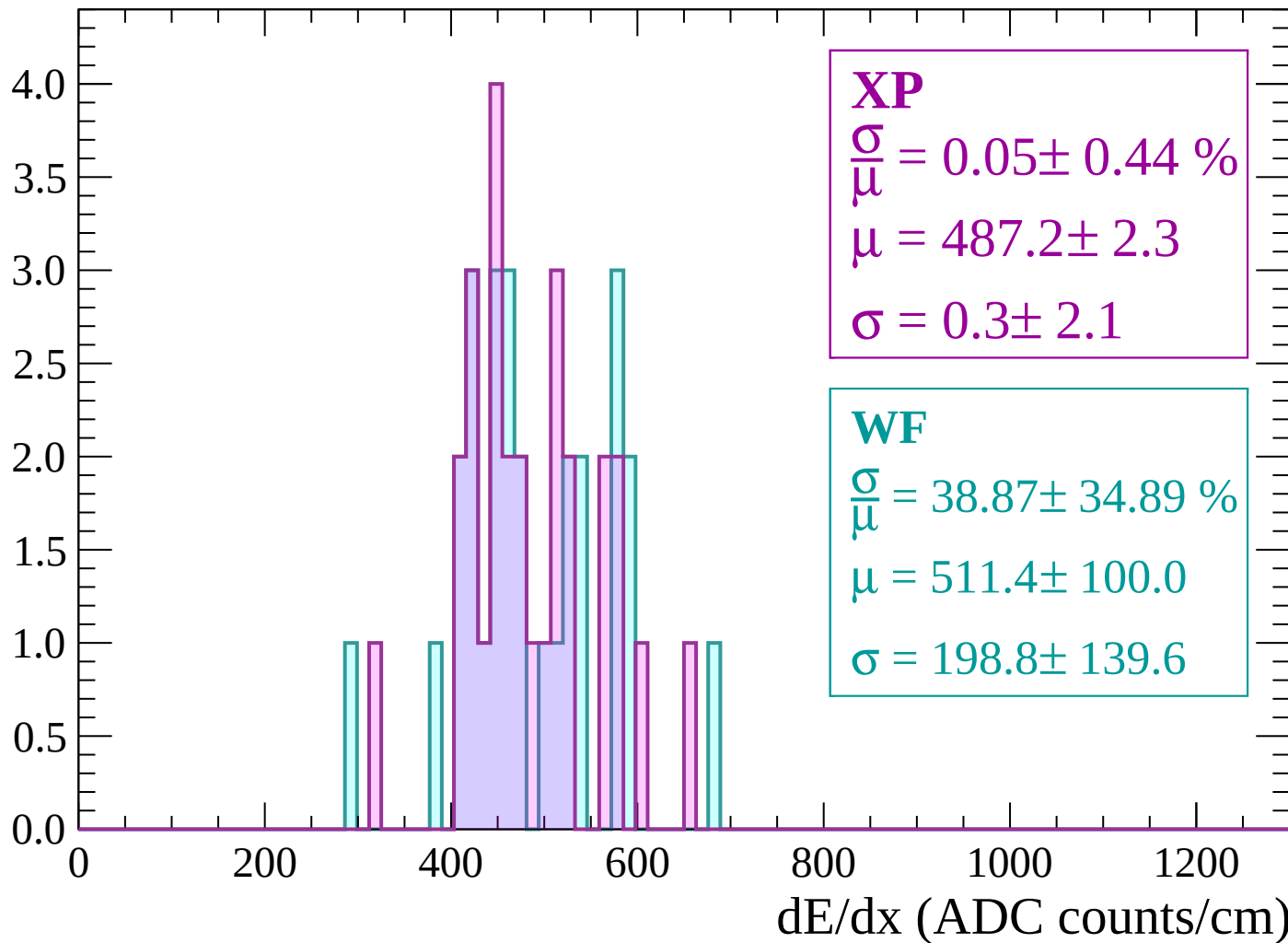
Energy loss | $2700 < p < 2760$

Count



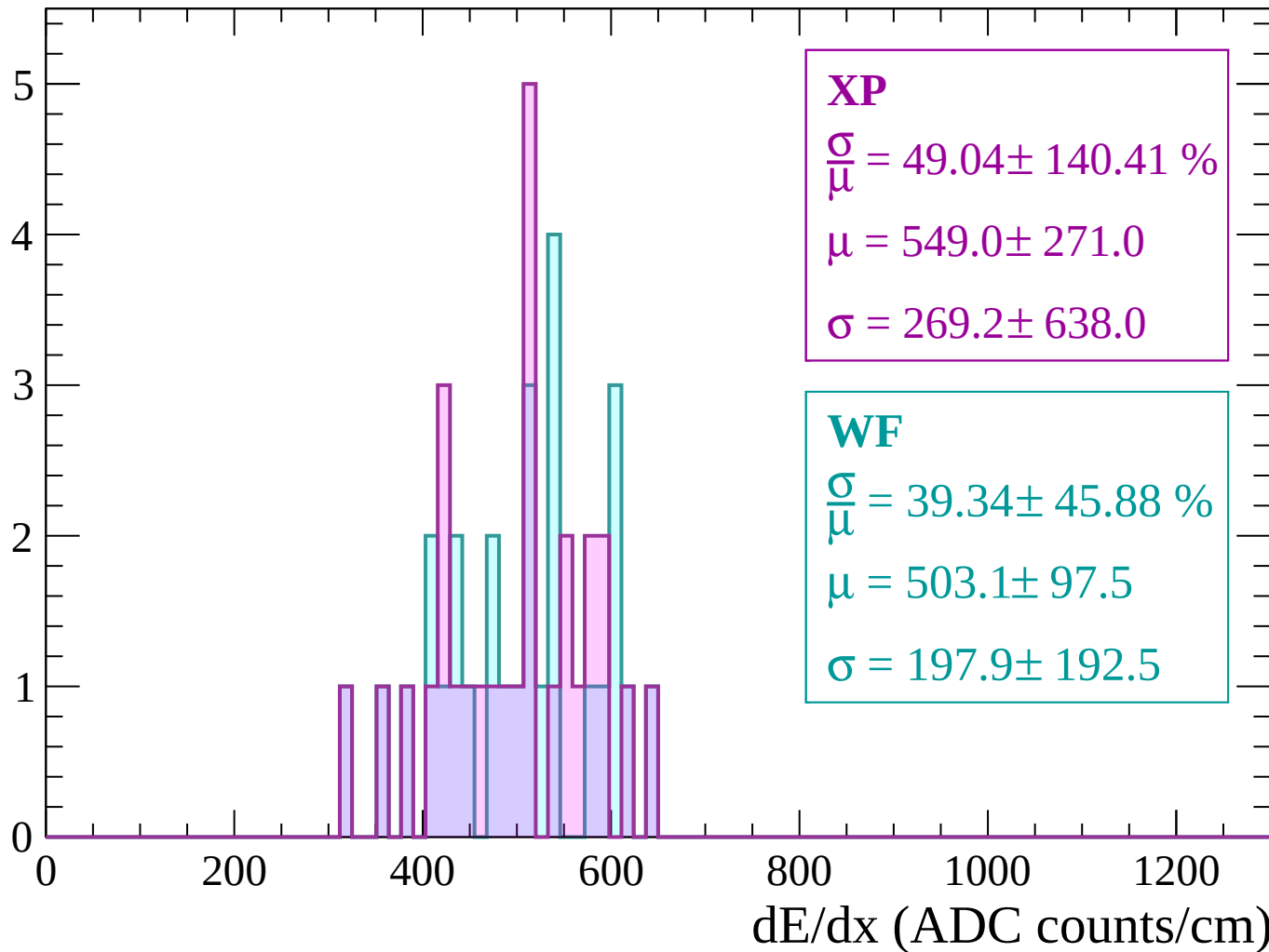
Energy loss | $2760 < p < 2820$

Count



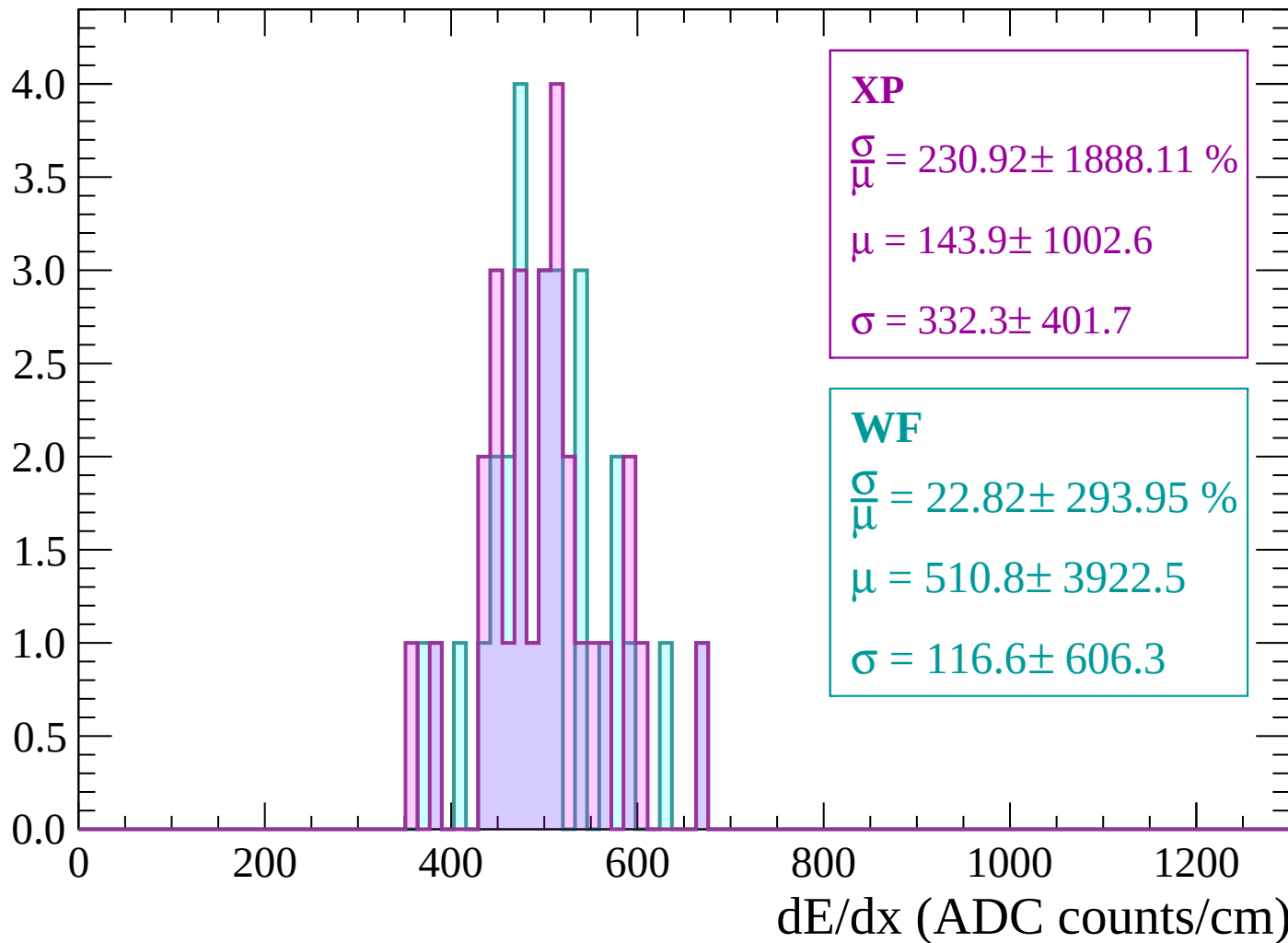
Energy loss | $2820 < p < 2880$

Count



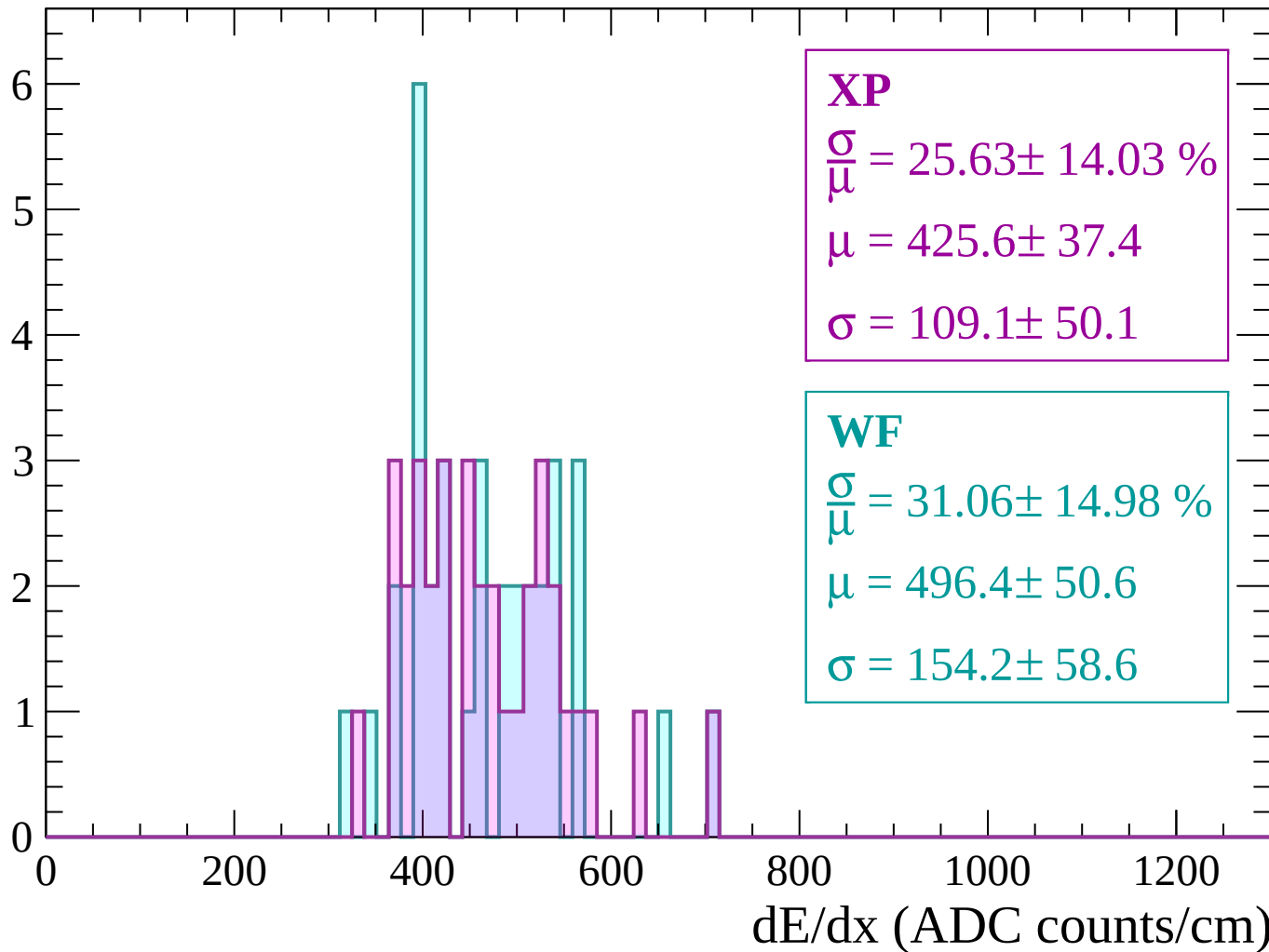
Energy loss | $2880 < p < 2940$

Count



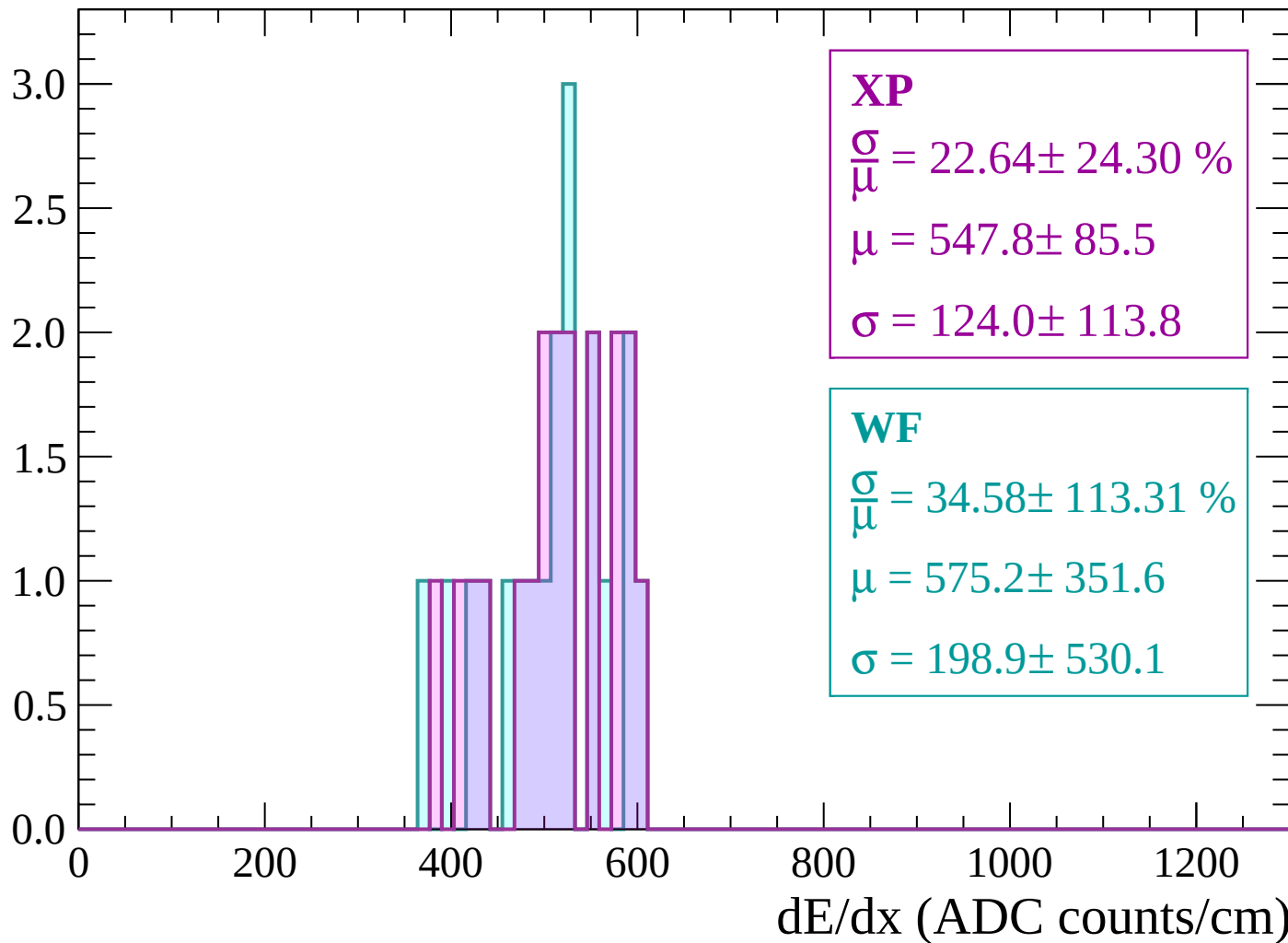
Energy loss | $2940 < p < 3000$

Count



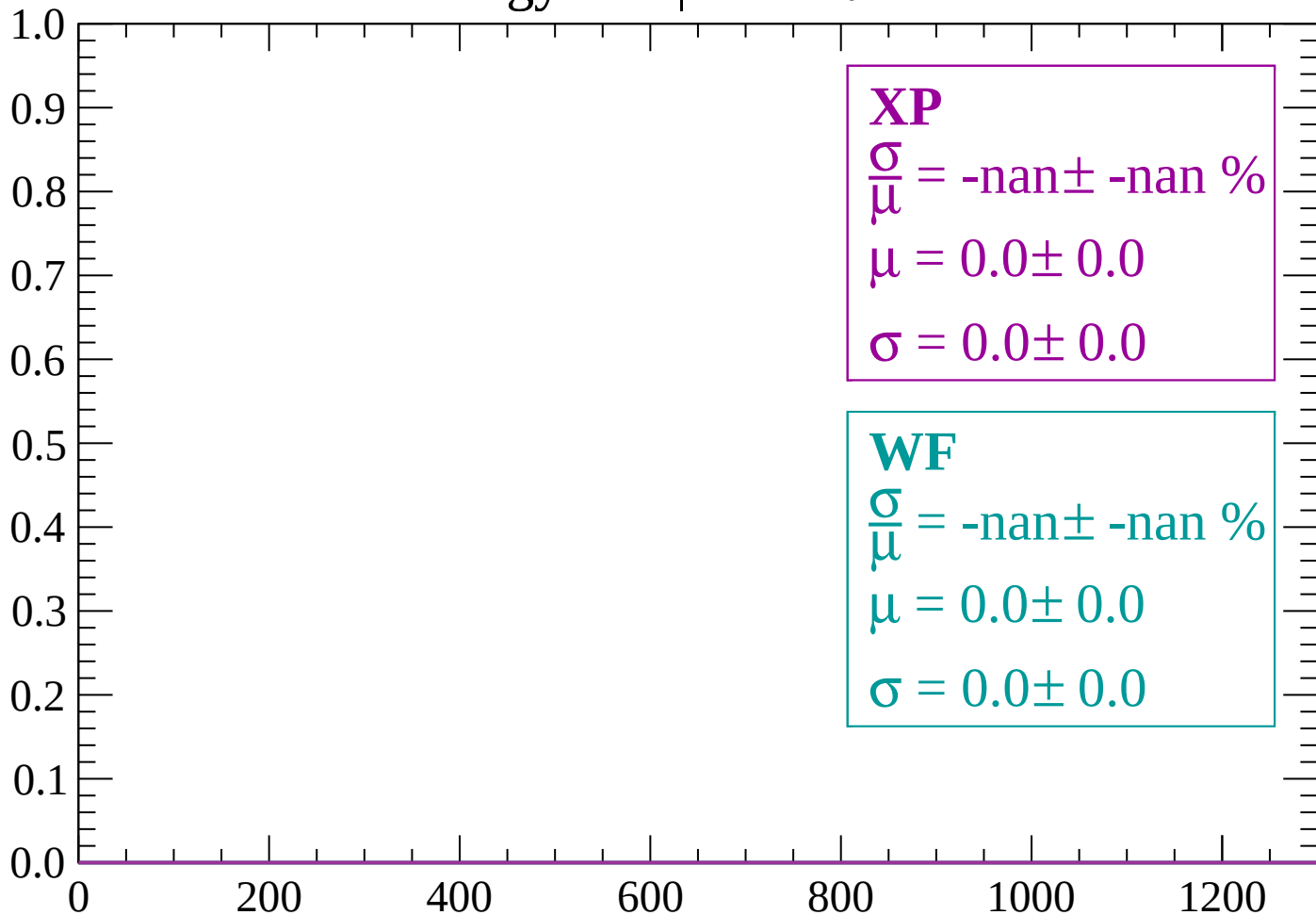
Energy loss | $3000 < p < 3060$

Count



Energy loss | $-90 < \theta < -88$

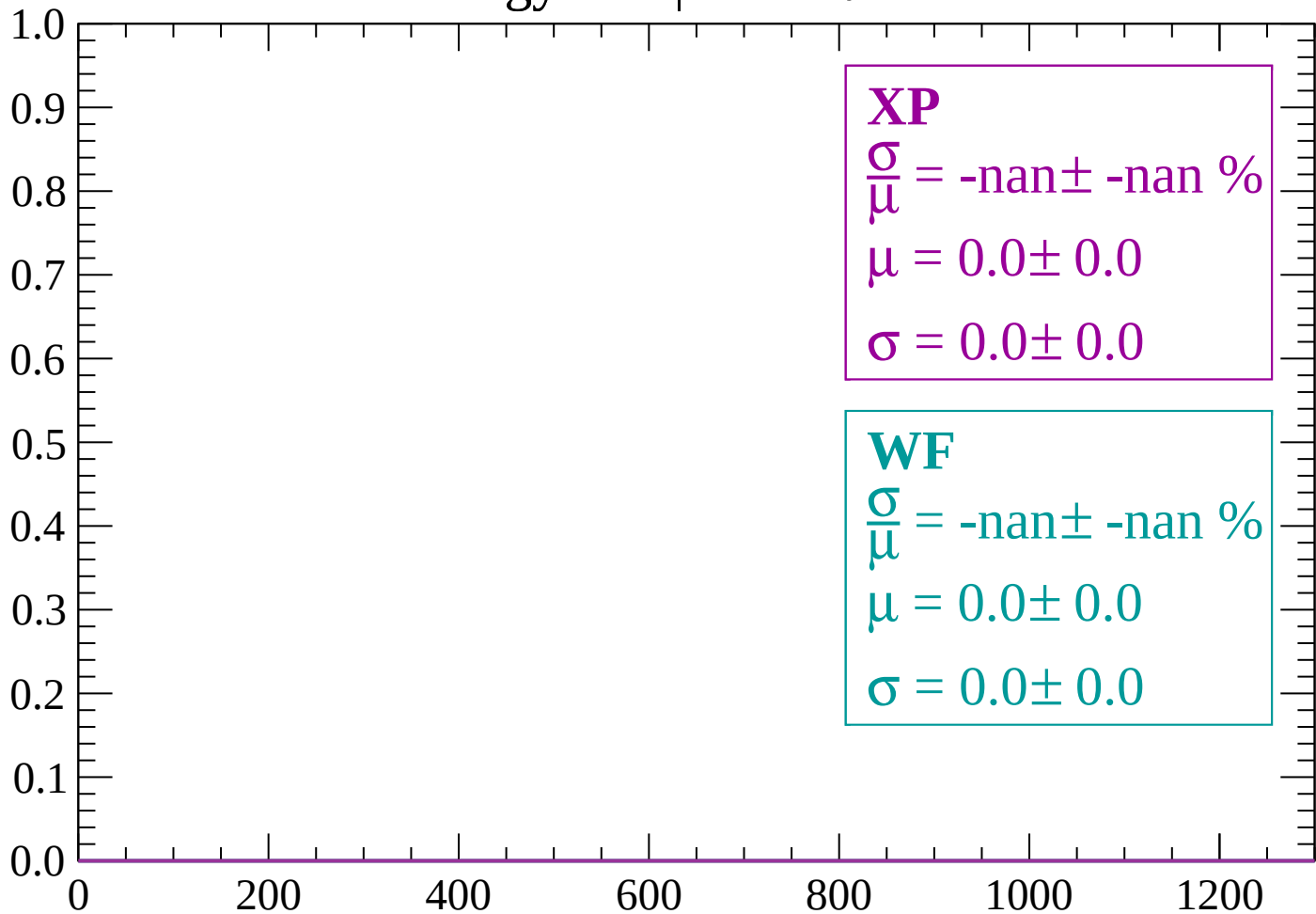
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

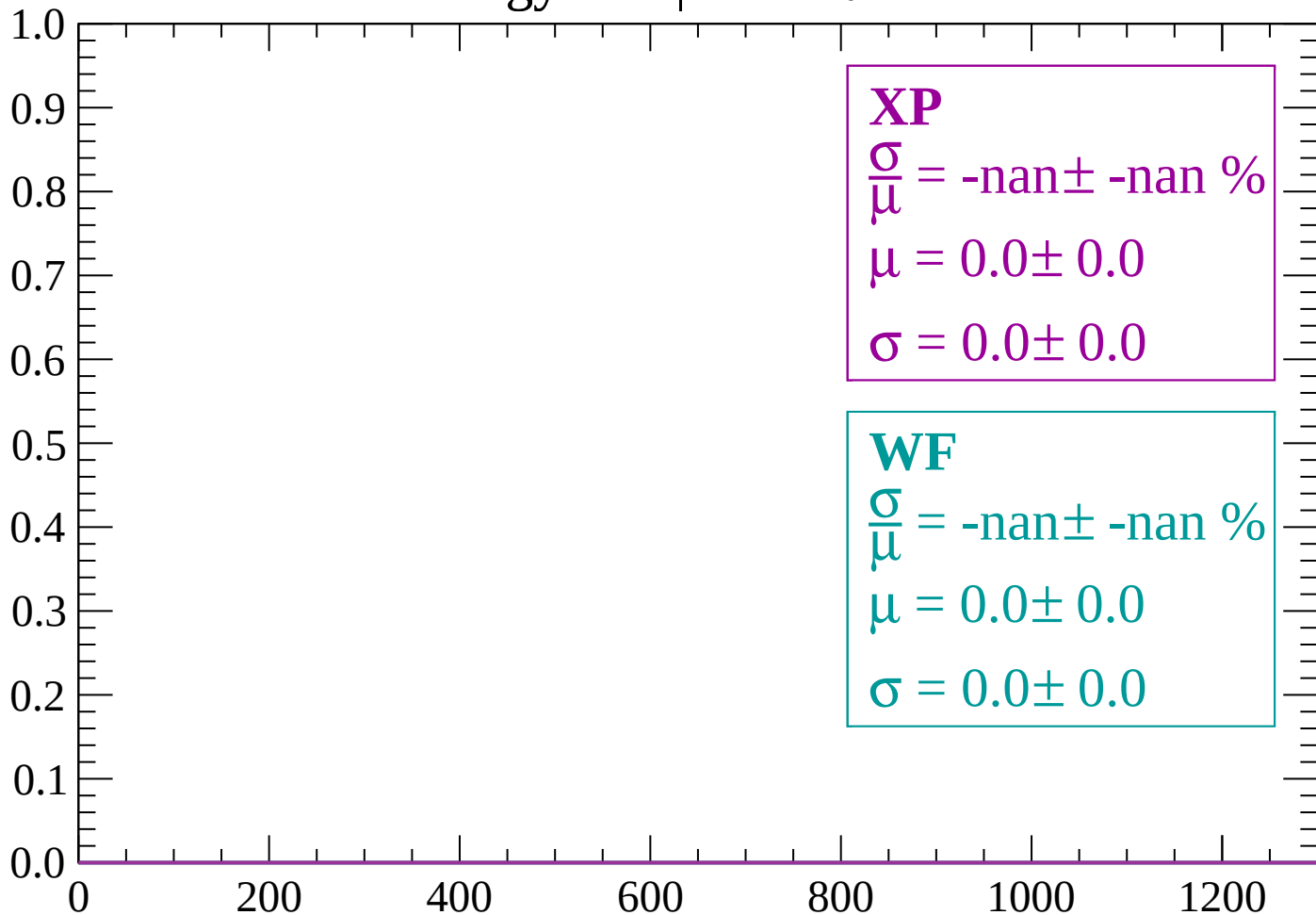
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

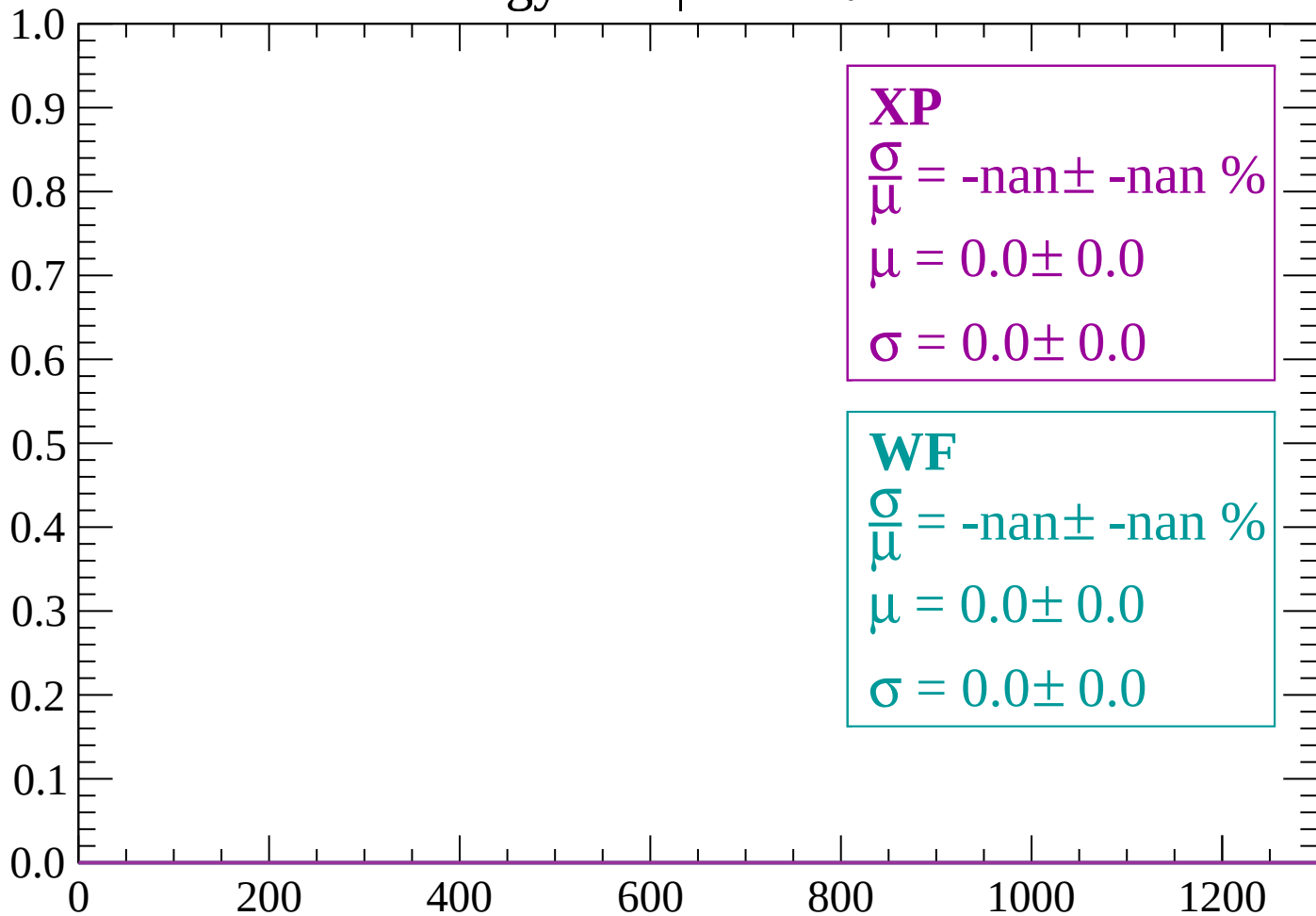
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

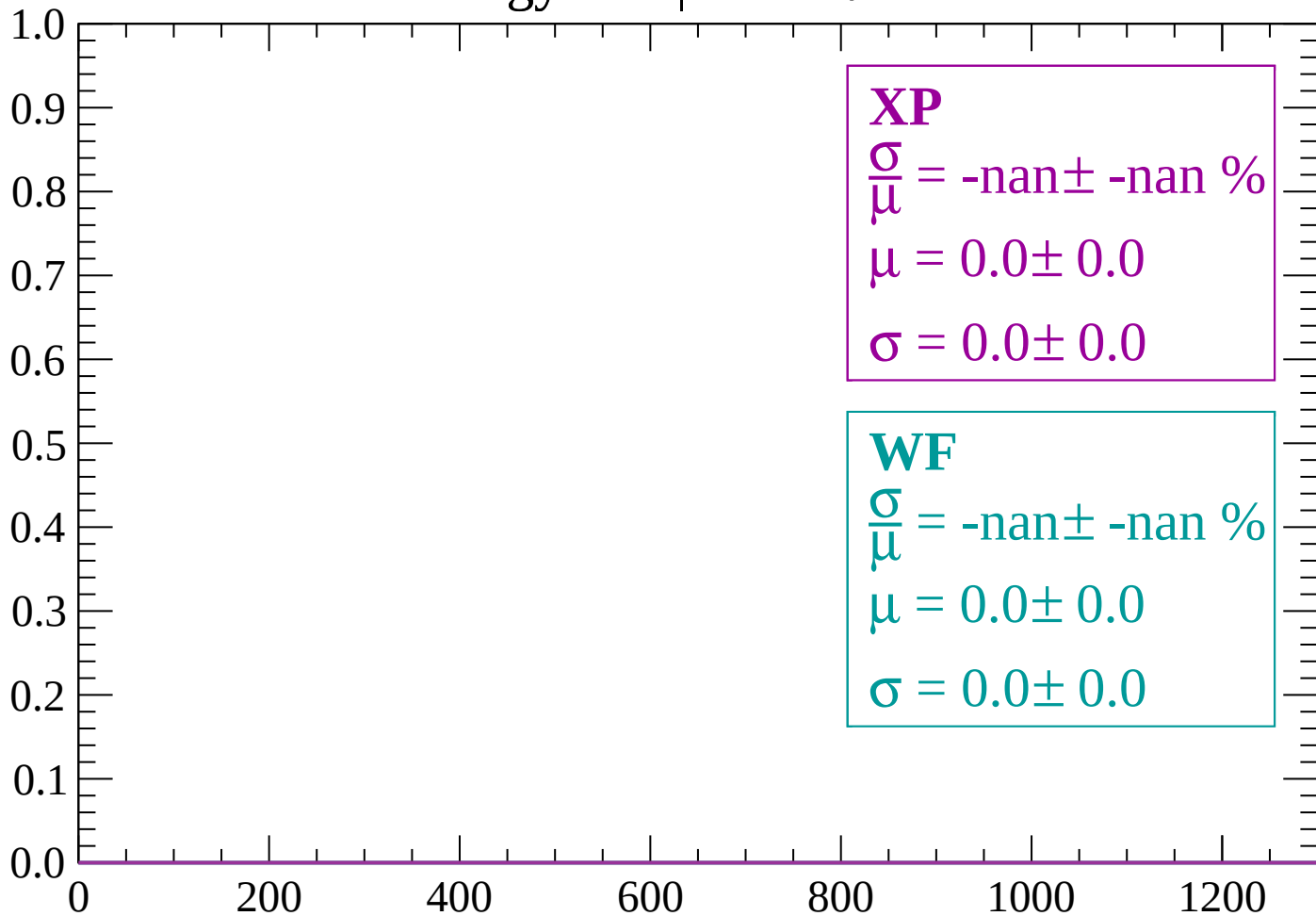
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $-80 < \theta < -78$

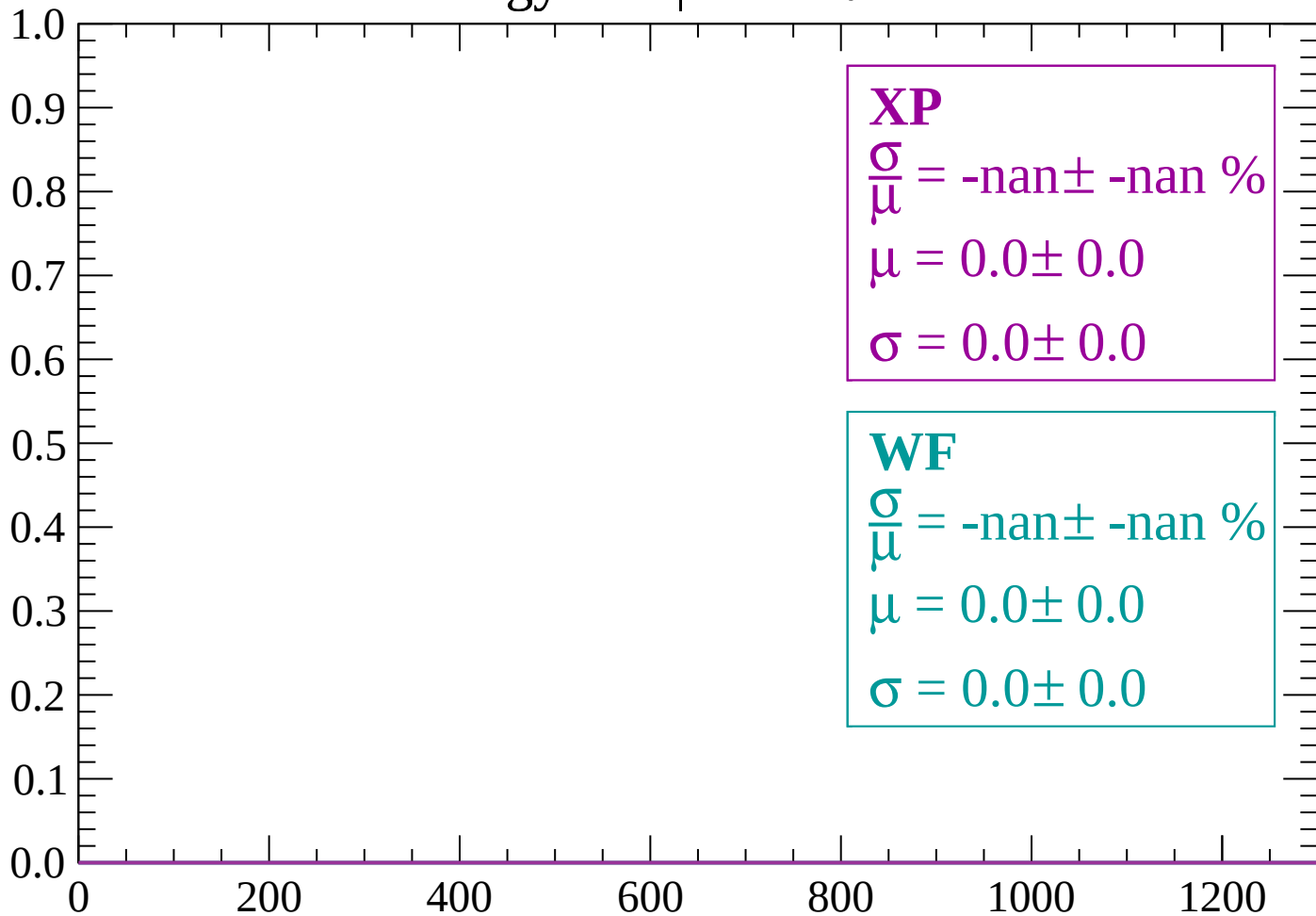
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

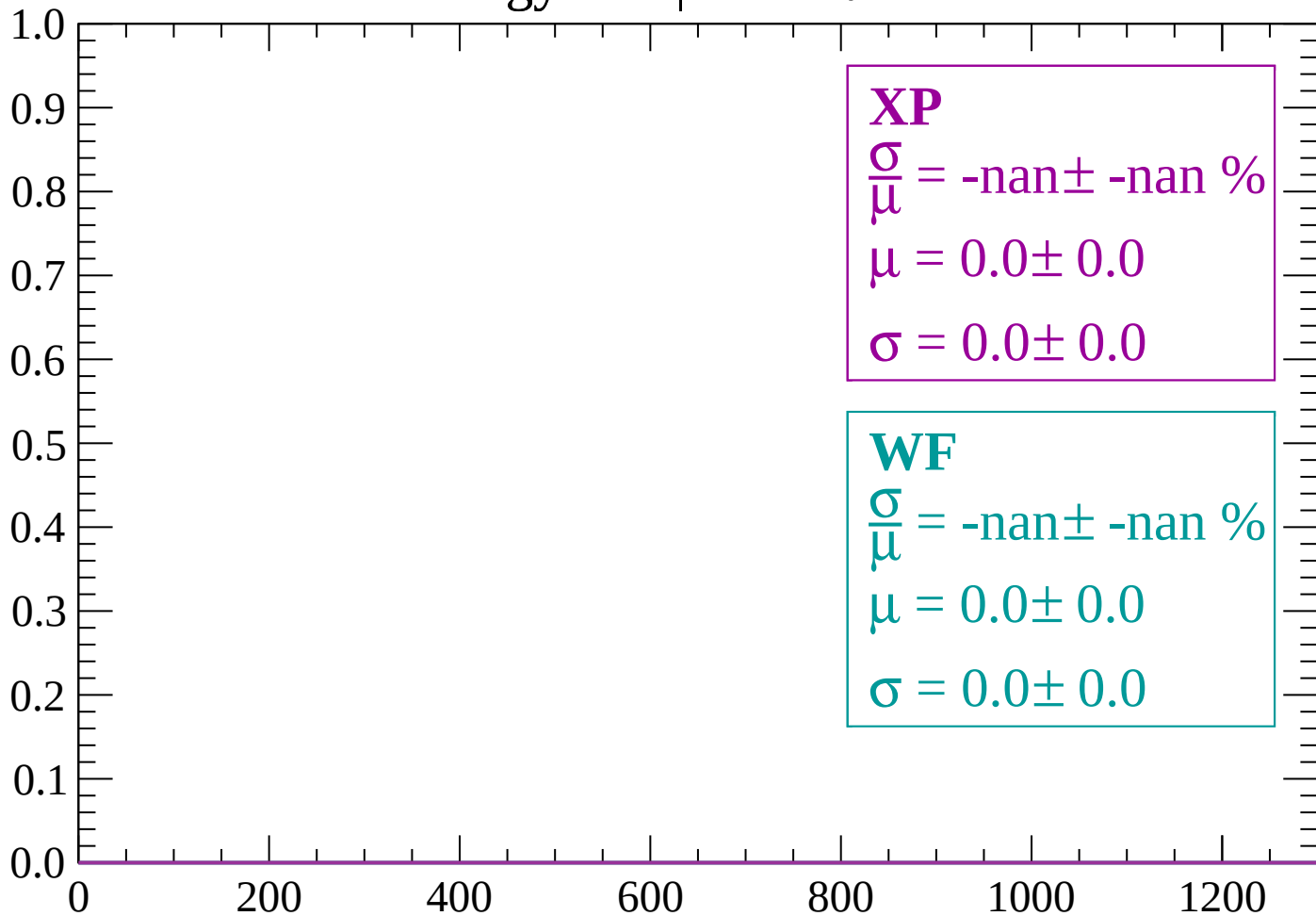
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

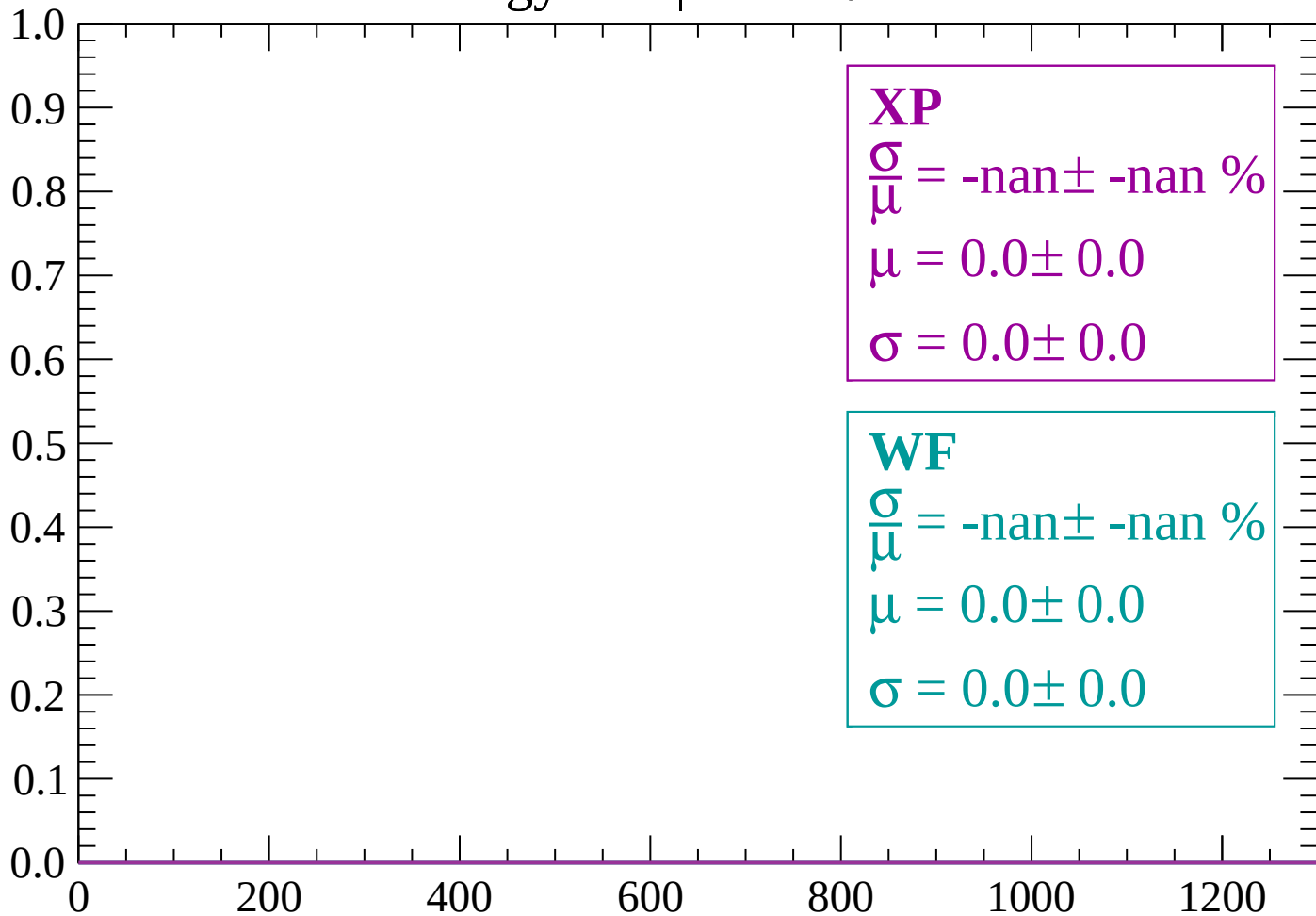
Count



dE/dx (ADC counts/cm)

Energy loss | $-74 < \theta < -72$

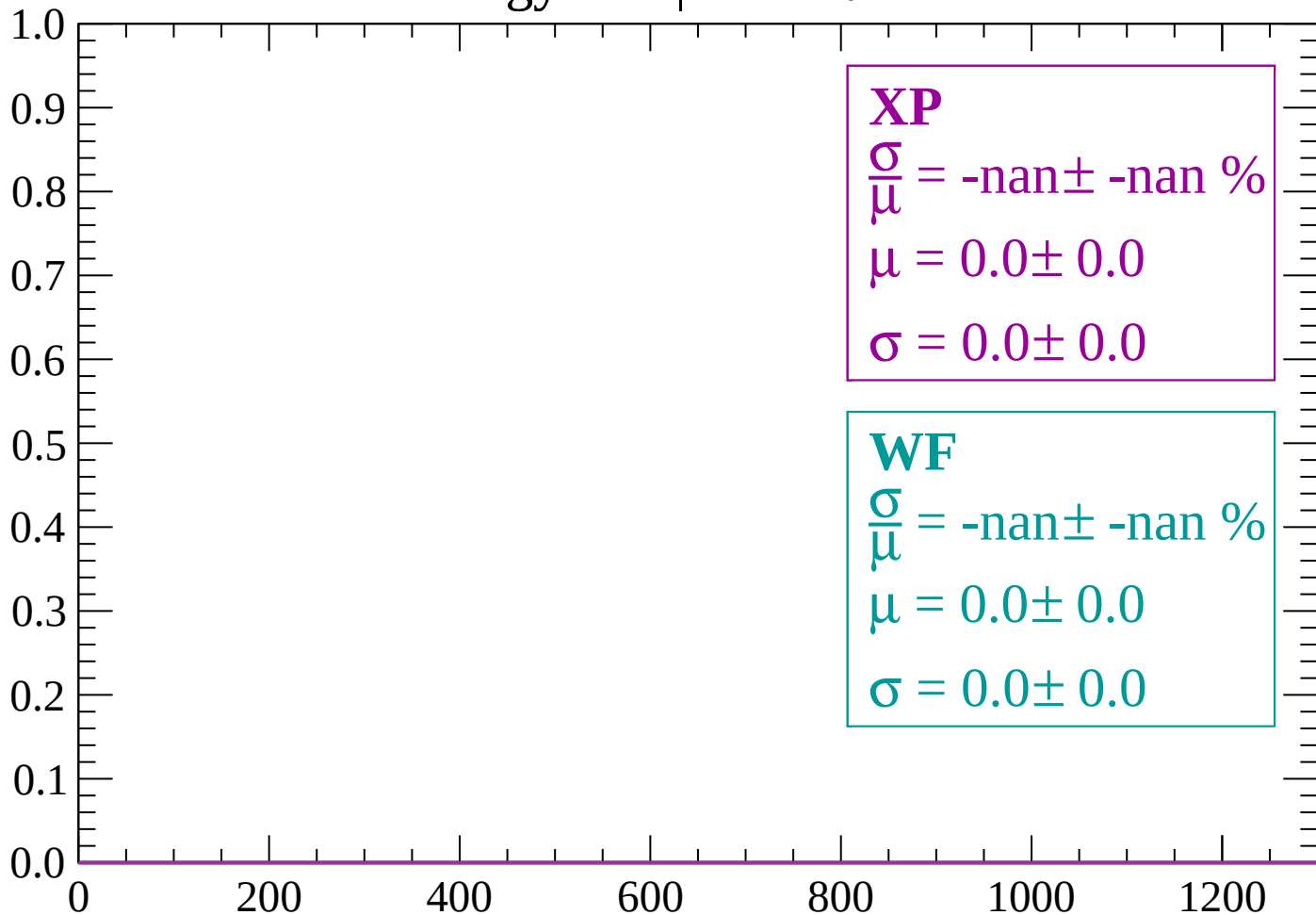
Count



dE/dx (ADC counts/cm)

Energy loss | $-72 < \theta < -70$

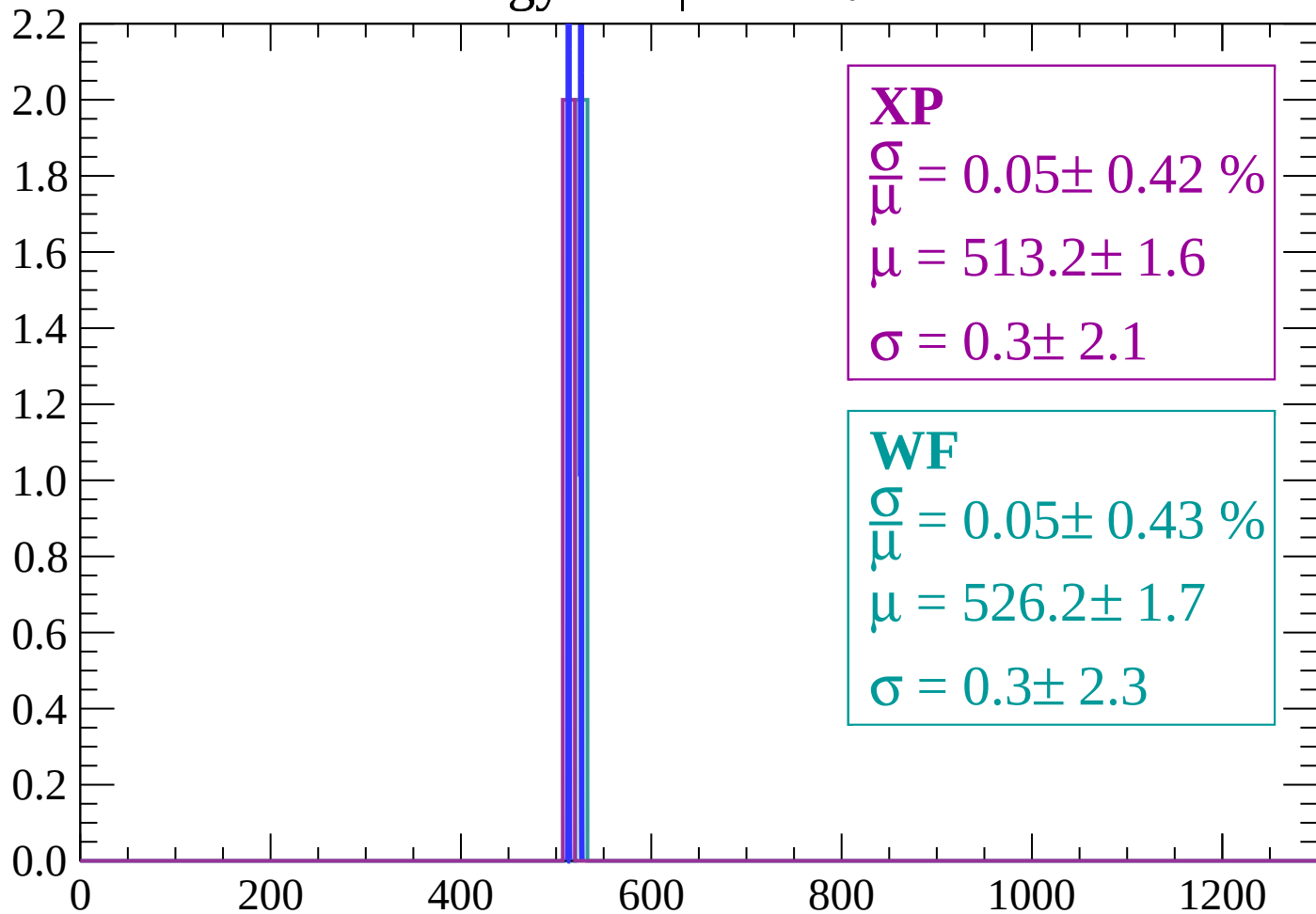
Count



dE/dx (ADC counts/cm)

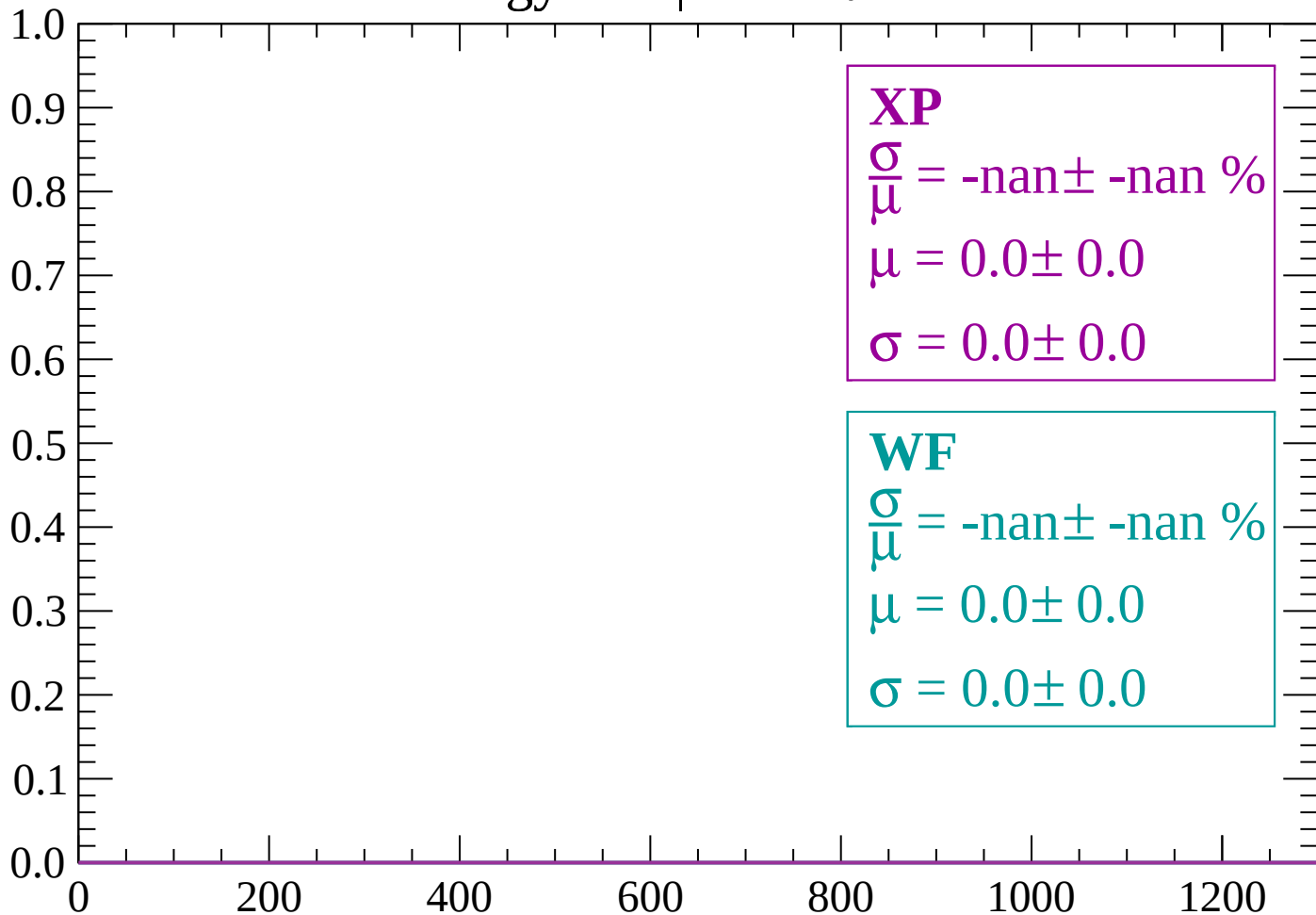
Energy loss | $-70 < \theta < -68$

Count



Energy loss | $-68 < \theta < -66$

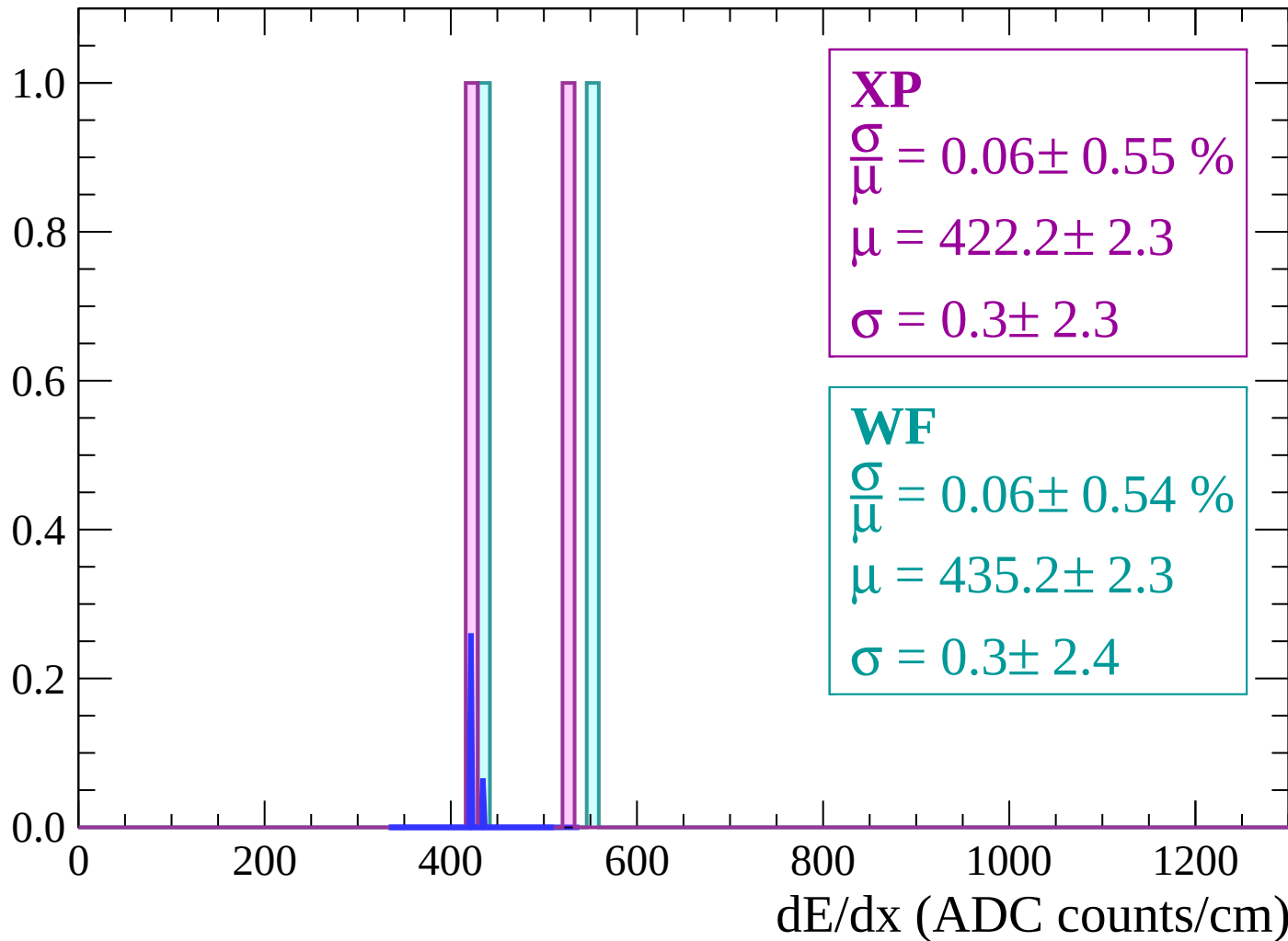
Count



dE/dx (ADC counts/cm)

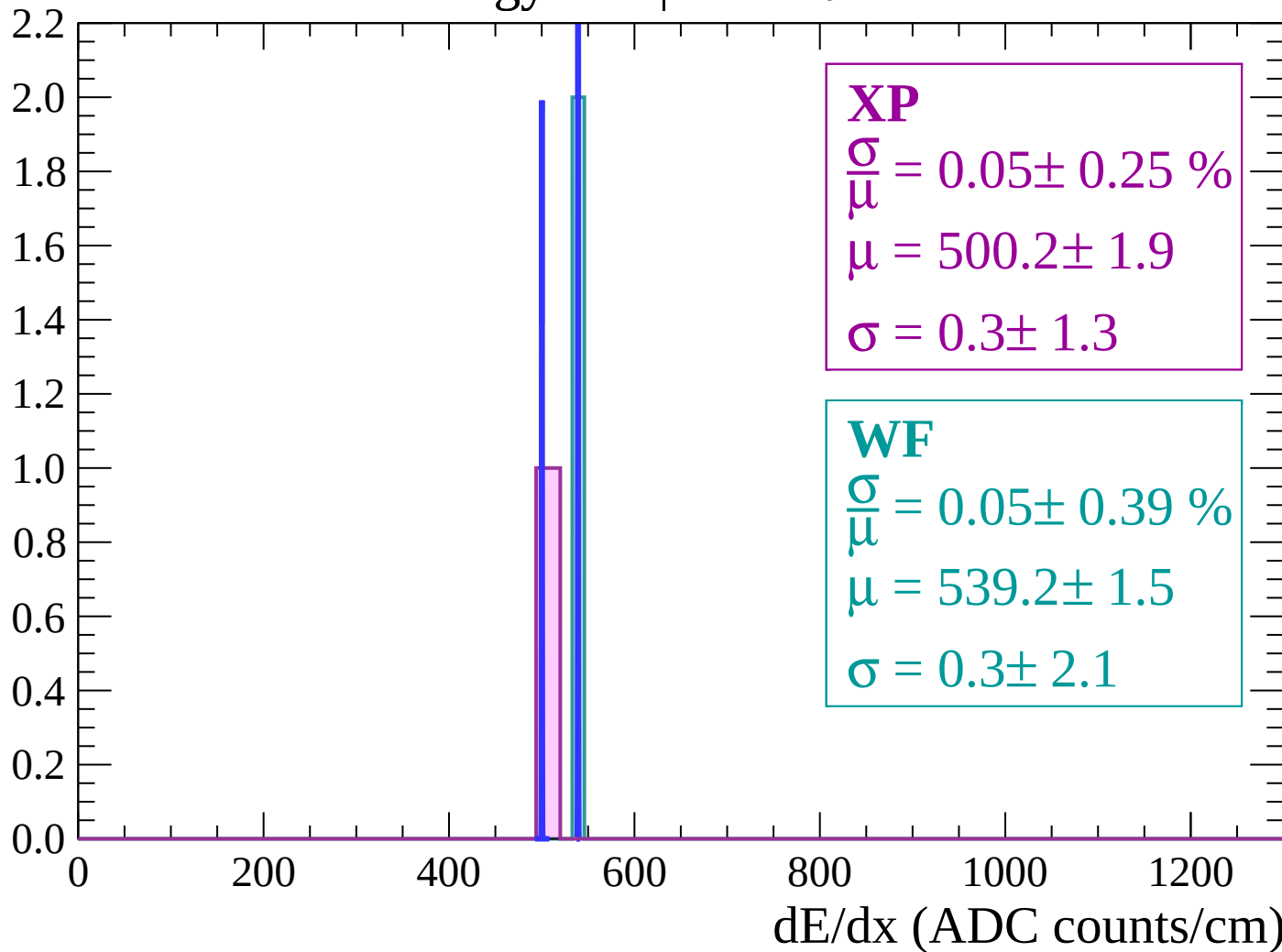
Energy loss | $-66 < \theta < -64$

Count



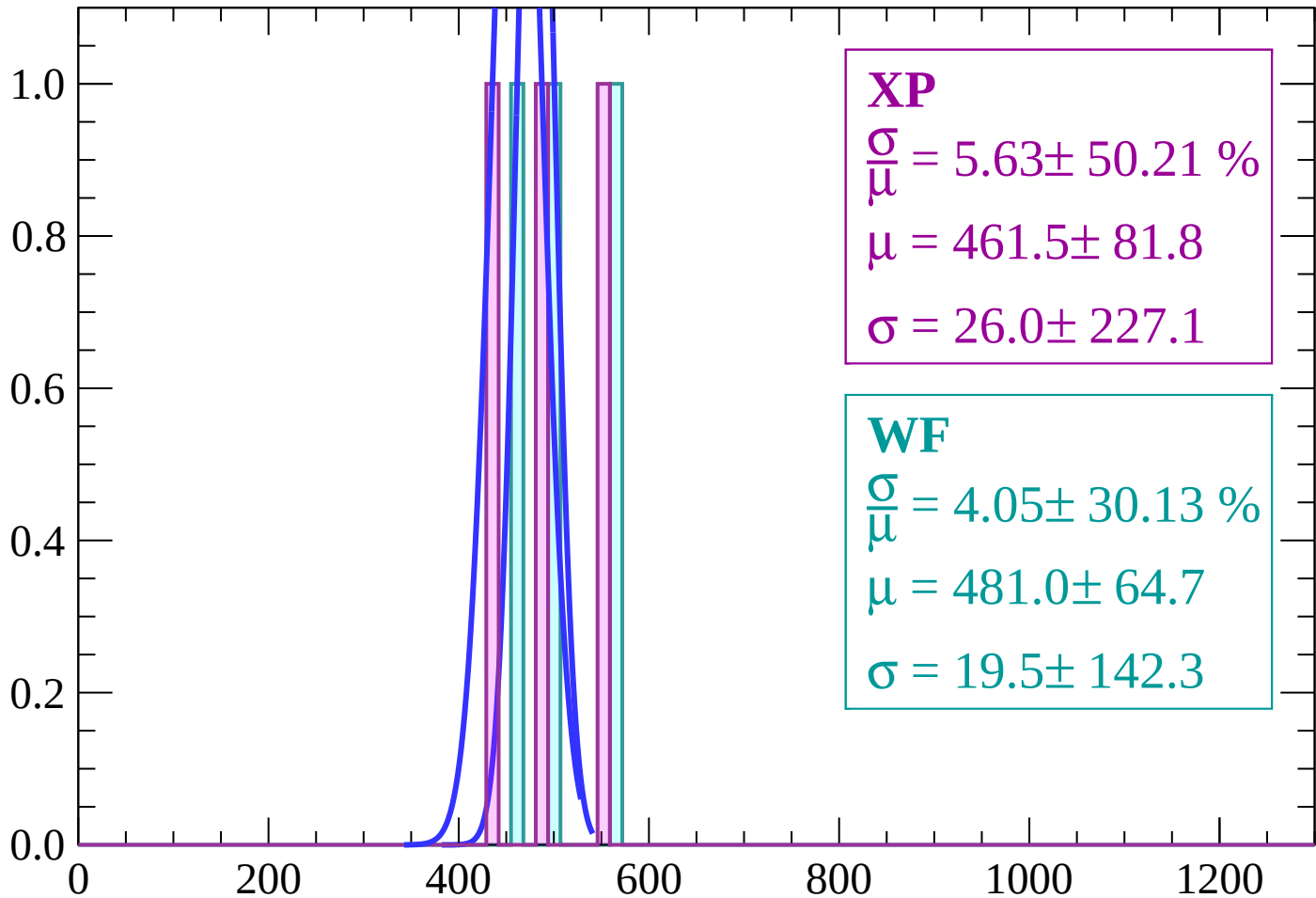
Energy loss | $-64 < \theta < -62$

Count



Energy loss | $-62 < \theta < -60$

Count



XP

$$\frac{\sigma}{\mu} = 5.63 \pm 50.21 \%$$

$$\mu = 461.5 \pm 81.8$$

$$\sigma = 26.0 \pm 227.1$$

WF

$$\frac{\sigma}{\mu} = 4.05 \pm 30.13 \%$$

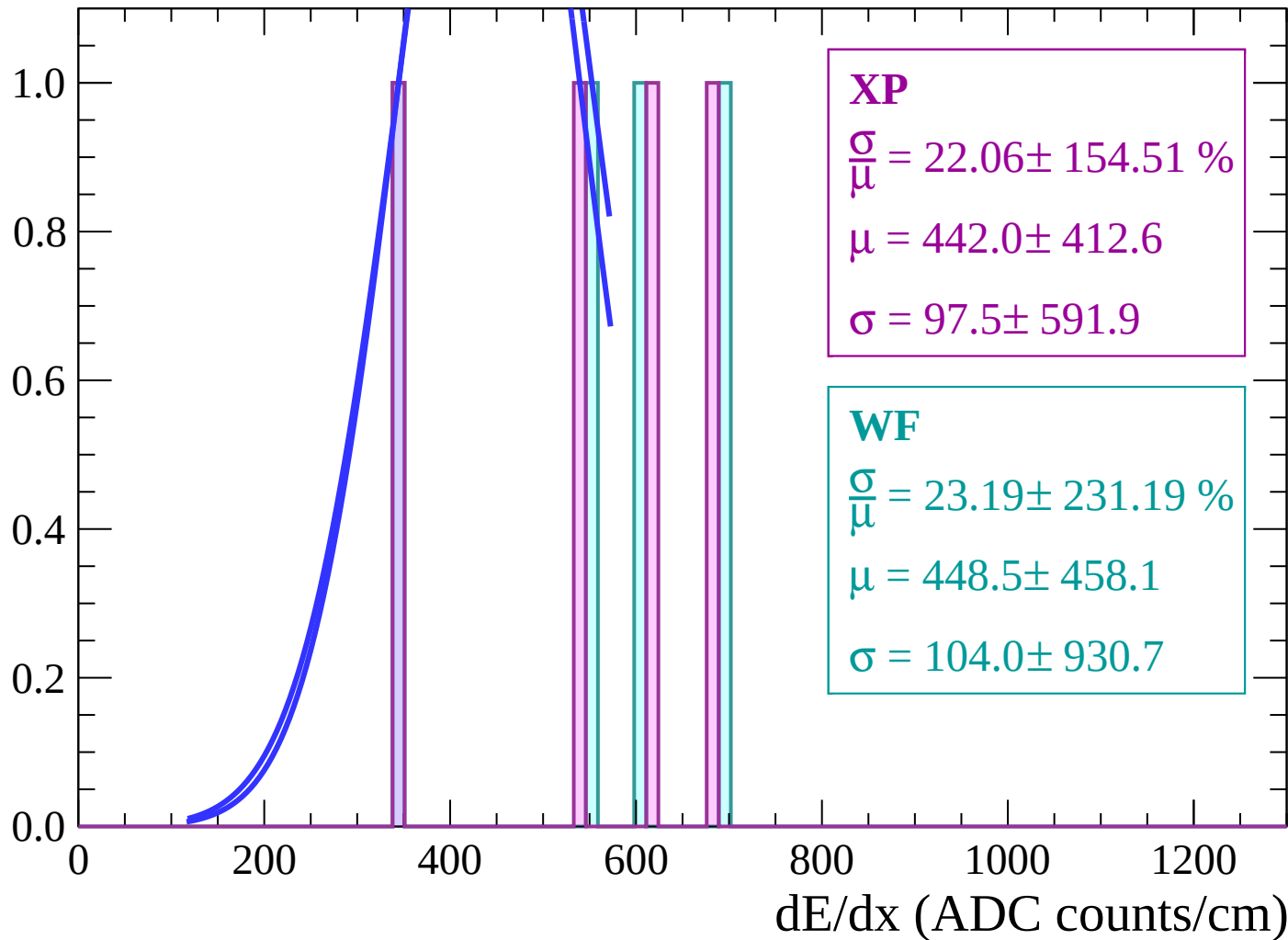
$$\mu = 481.0 \pm 64.7$$

$$\sigma = 19.5 \pm 142.3$$

dE/dx (ADC counts/cm)

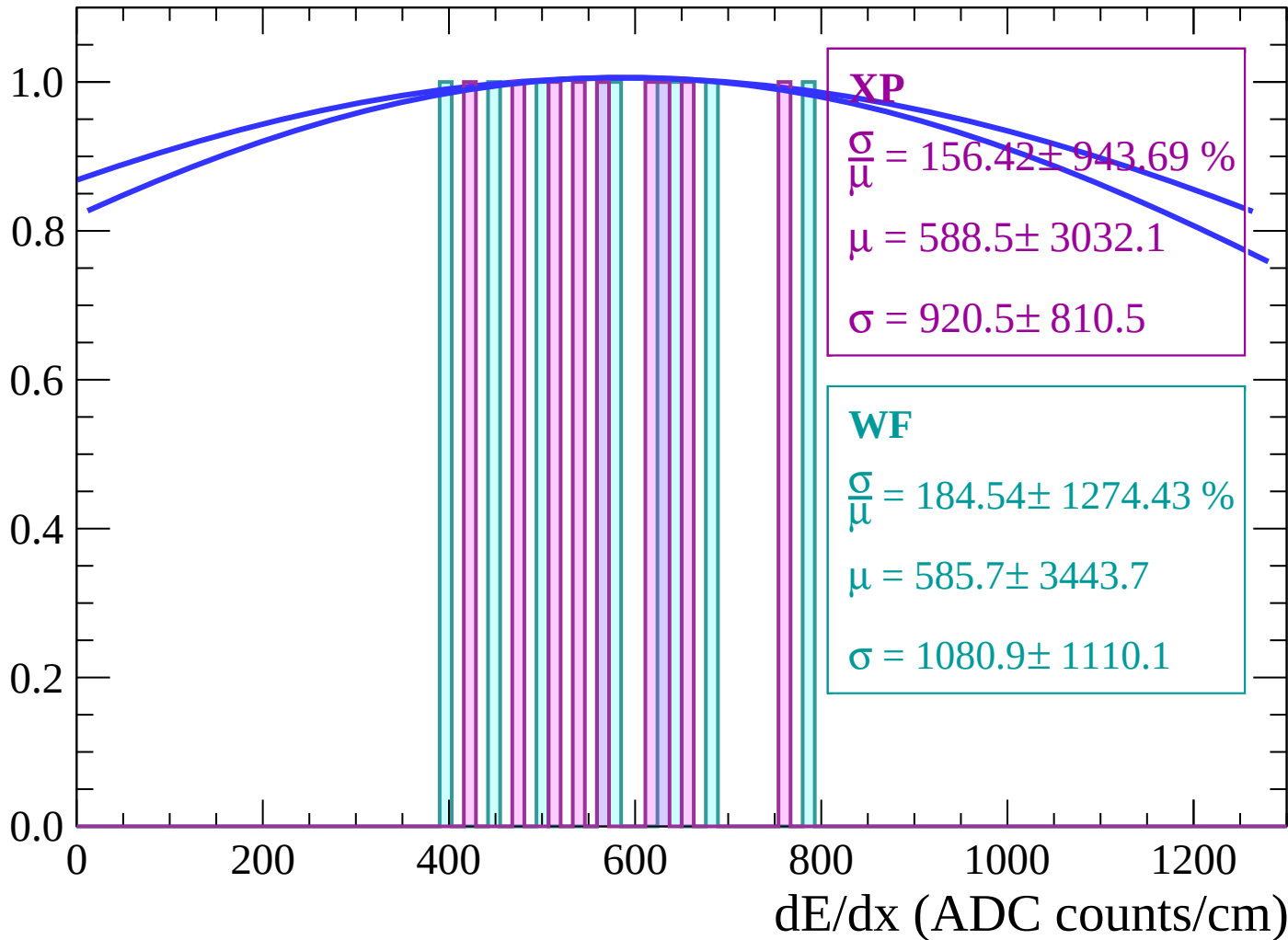
Energy loss | $-60 < \theta < -58$

Count



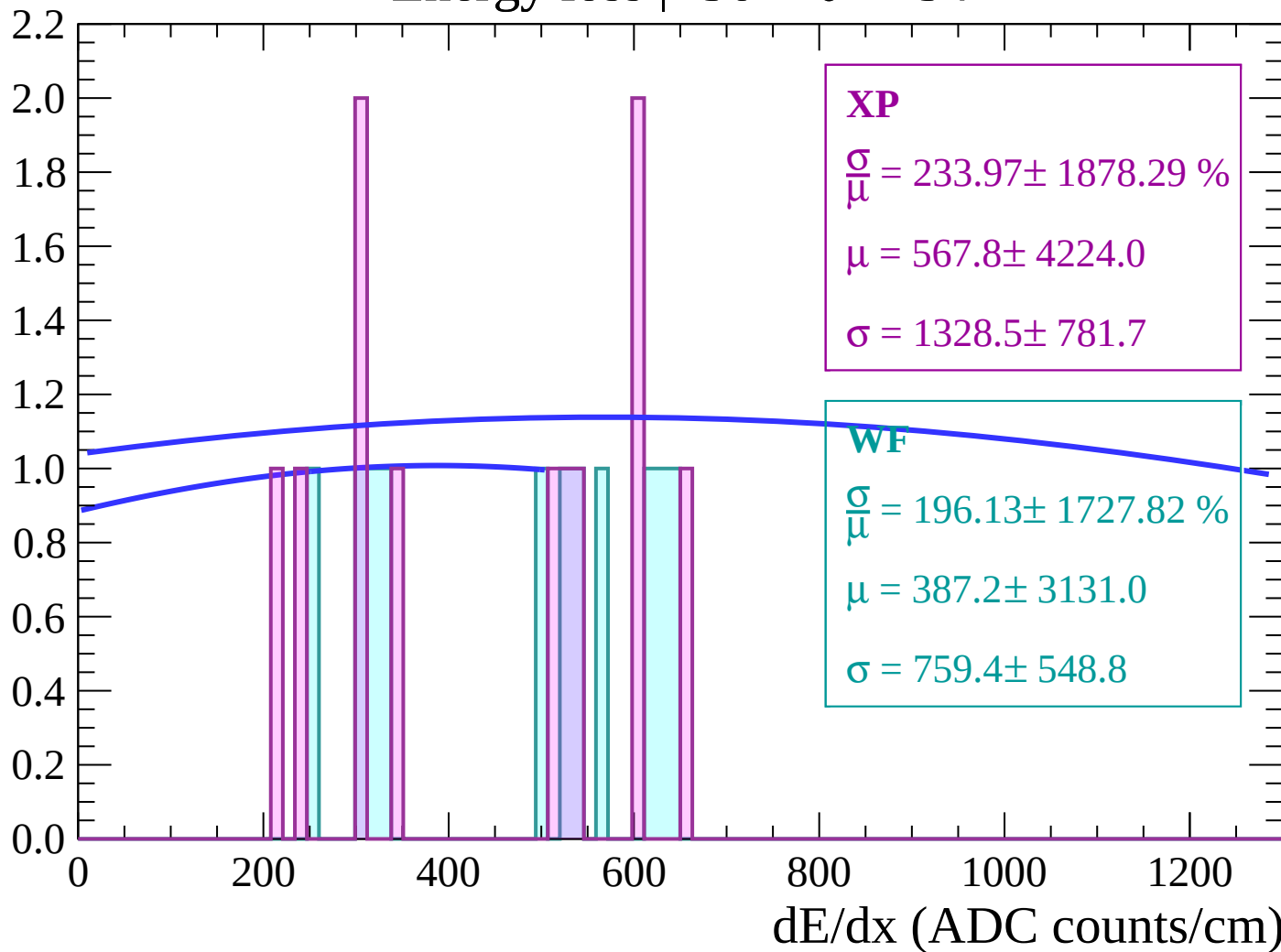
Energy loss | $-58 < \theta < -56$

Count



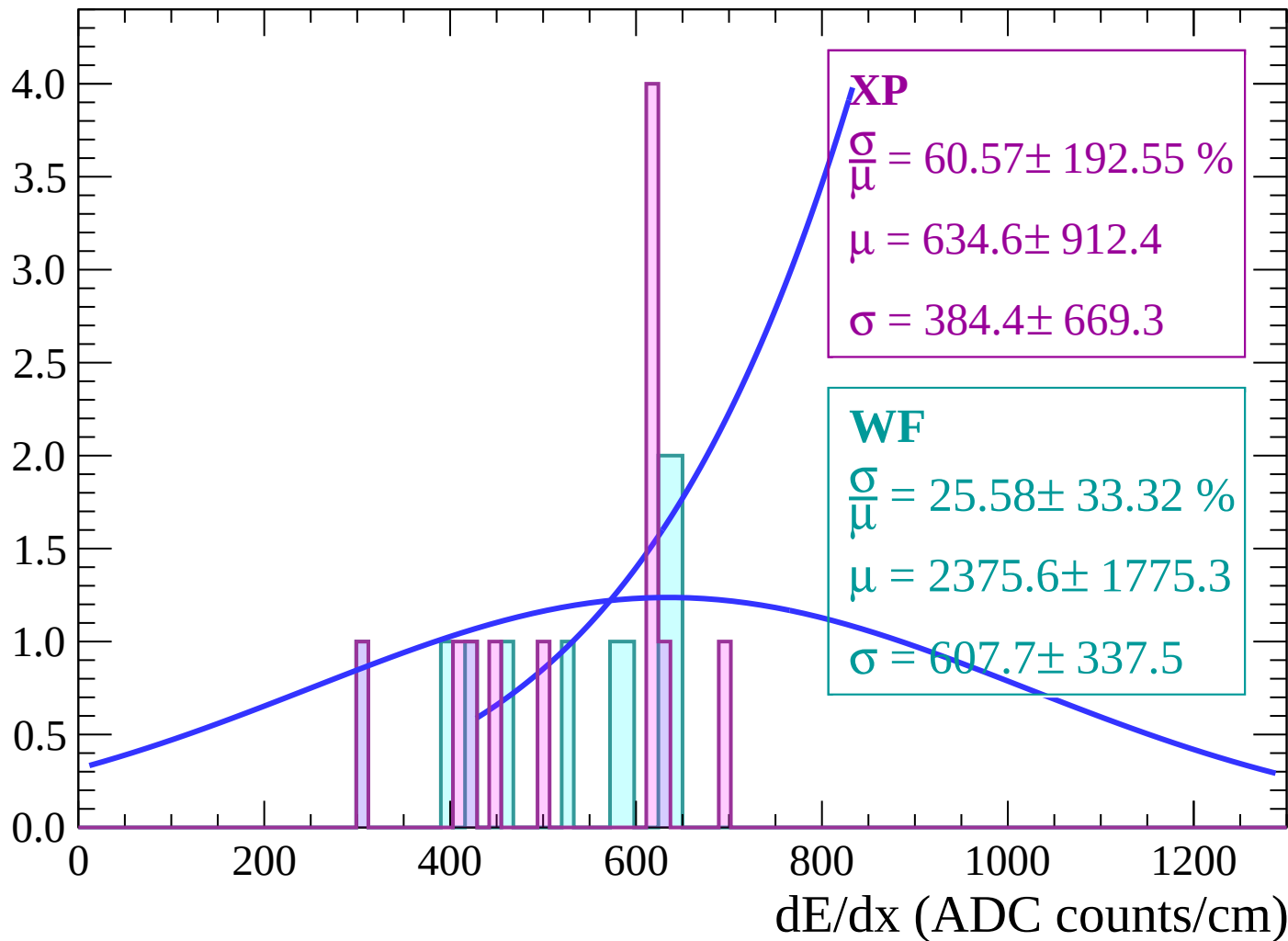
Energy loss | $-56 < \theta < -54$

Count



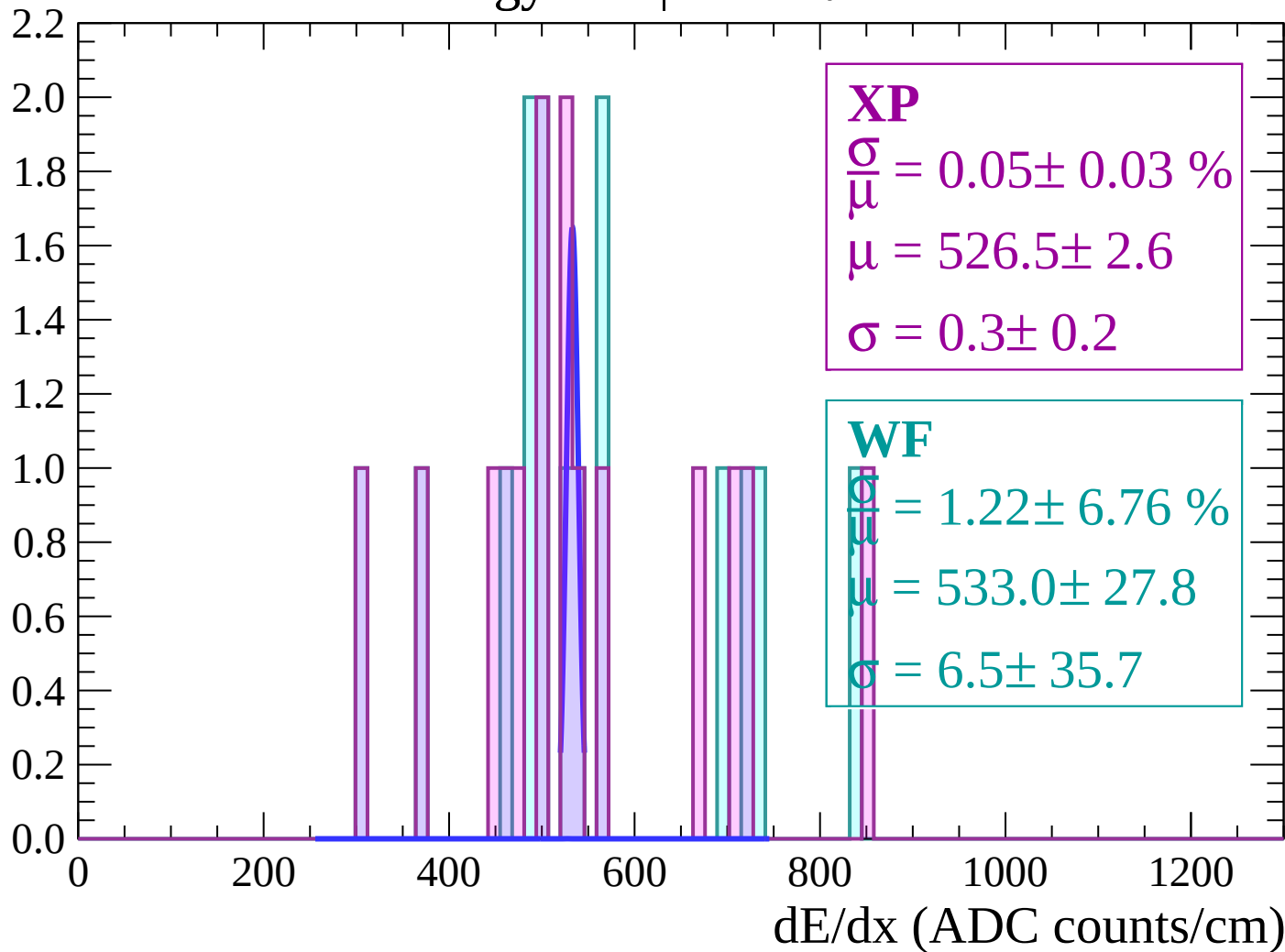
Energy loss | $-54 < \theta < -52$

Count



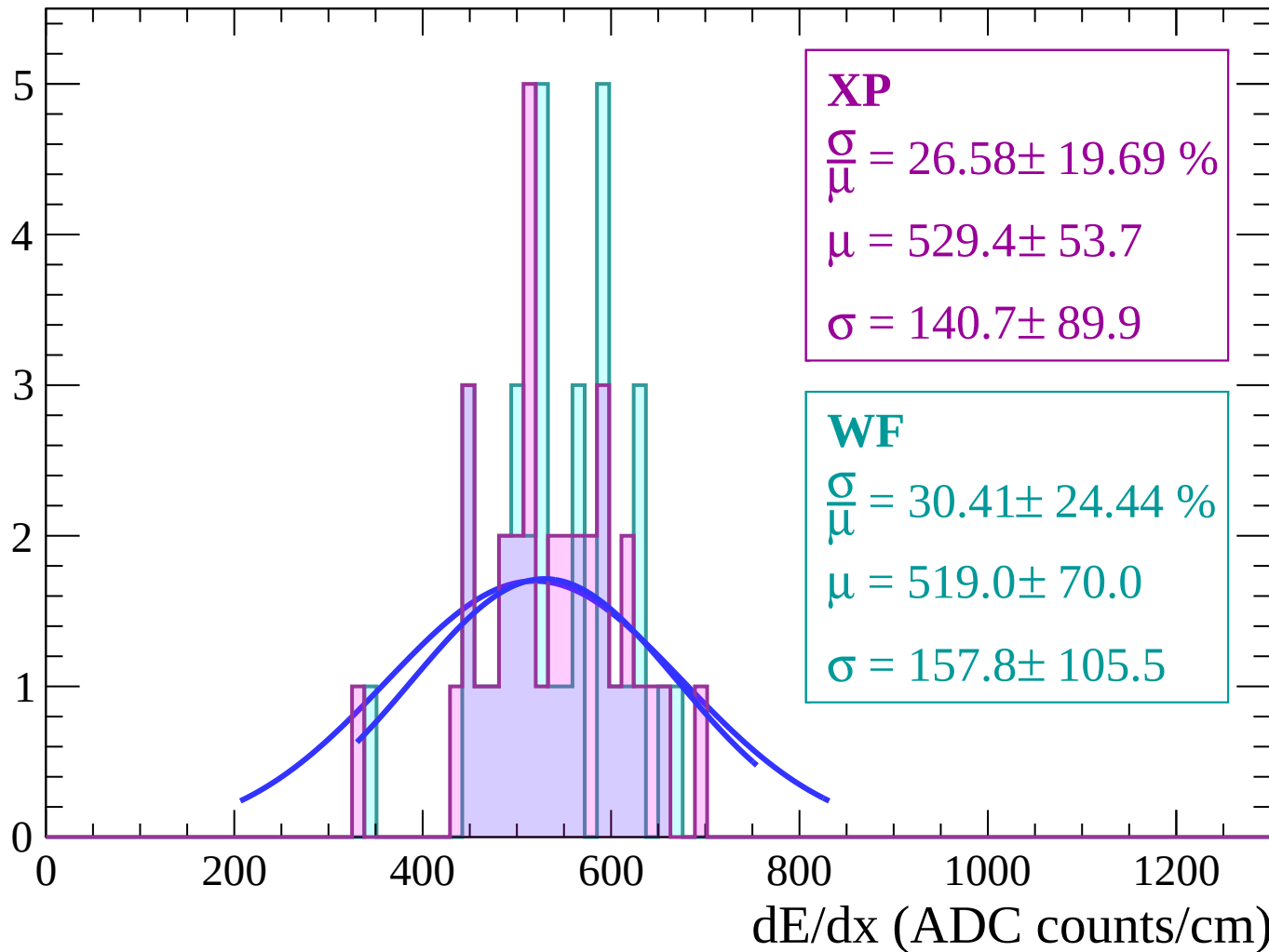
Energy loss | $-52 < \theta < -50$

Count



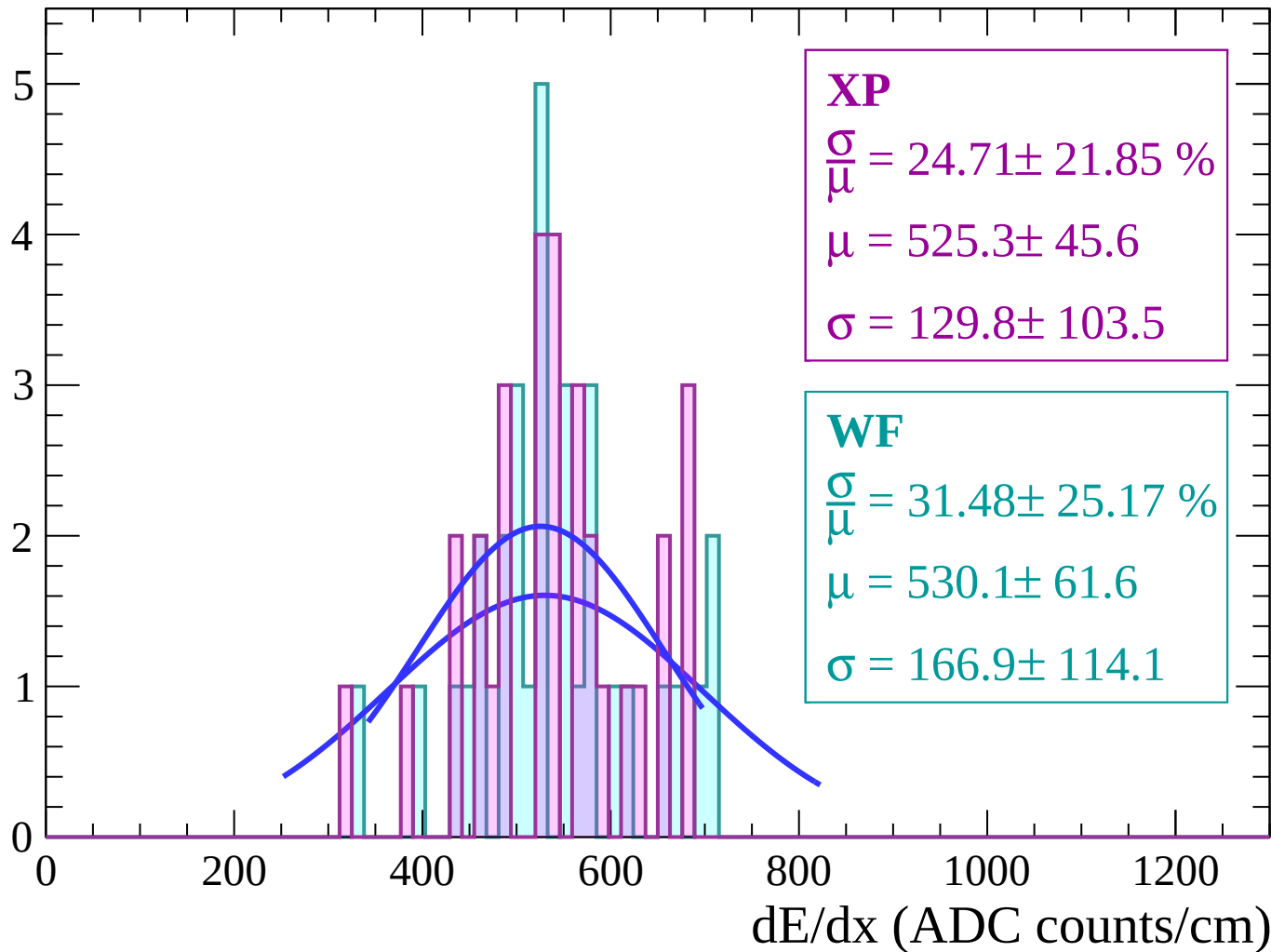
Energy loss | $-50 < \theta < -48$

Count



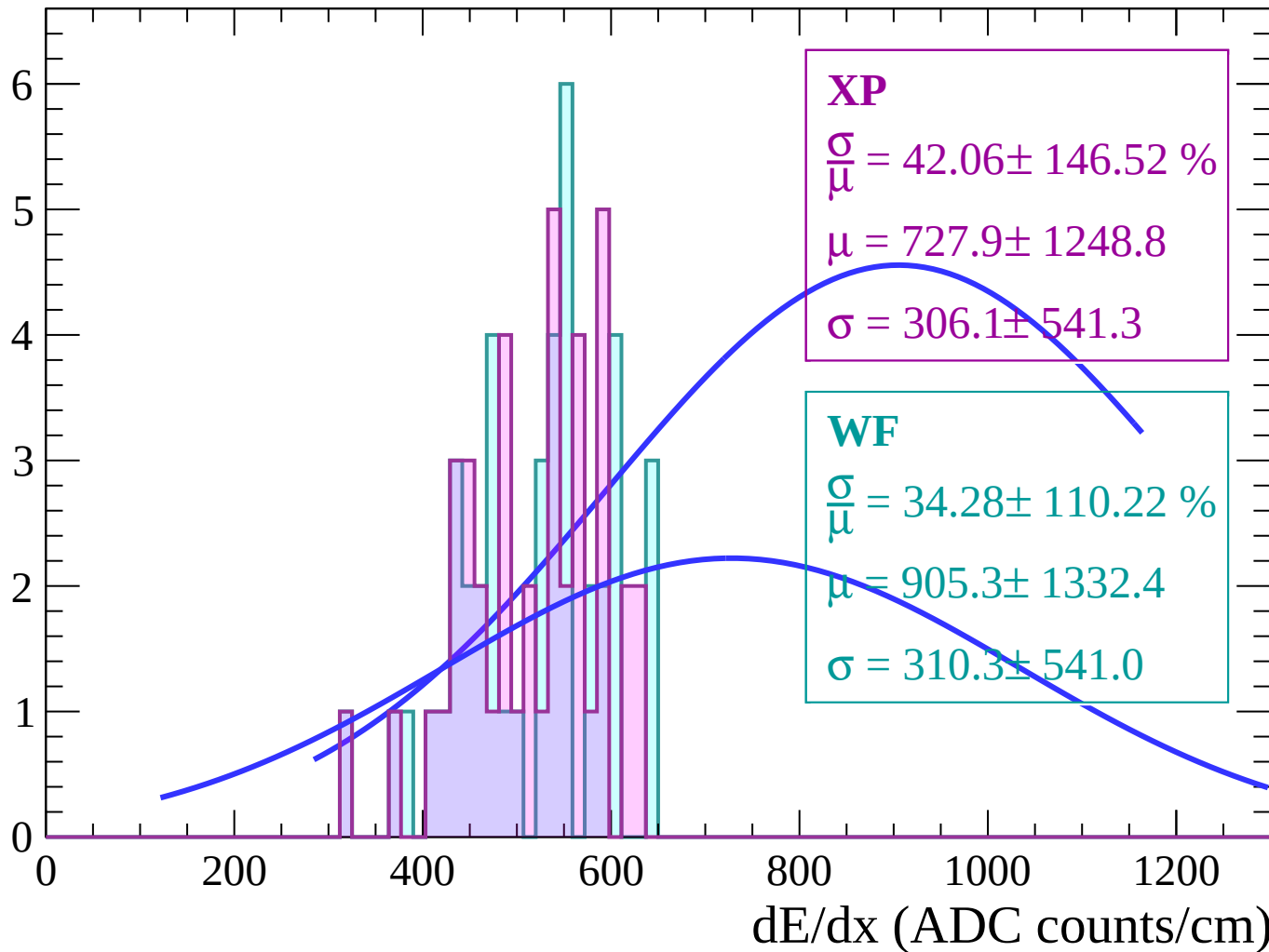
Energy loss | $-48 < \theta < -46$

Count



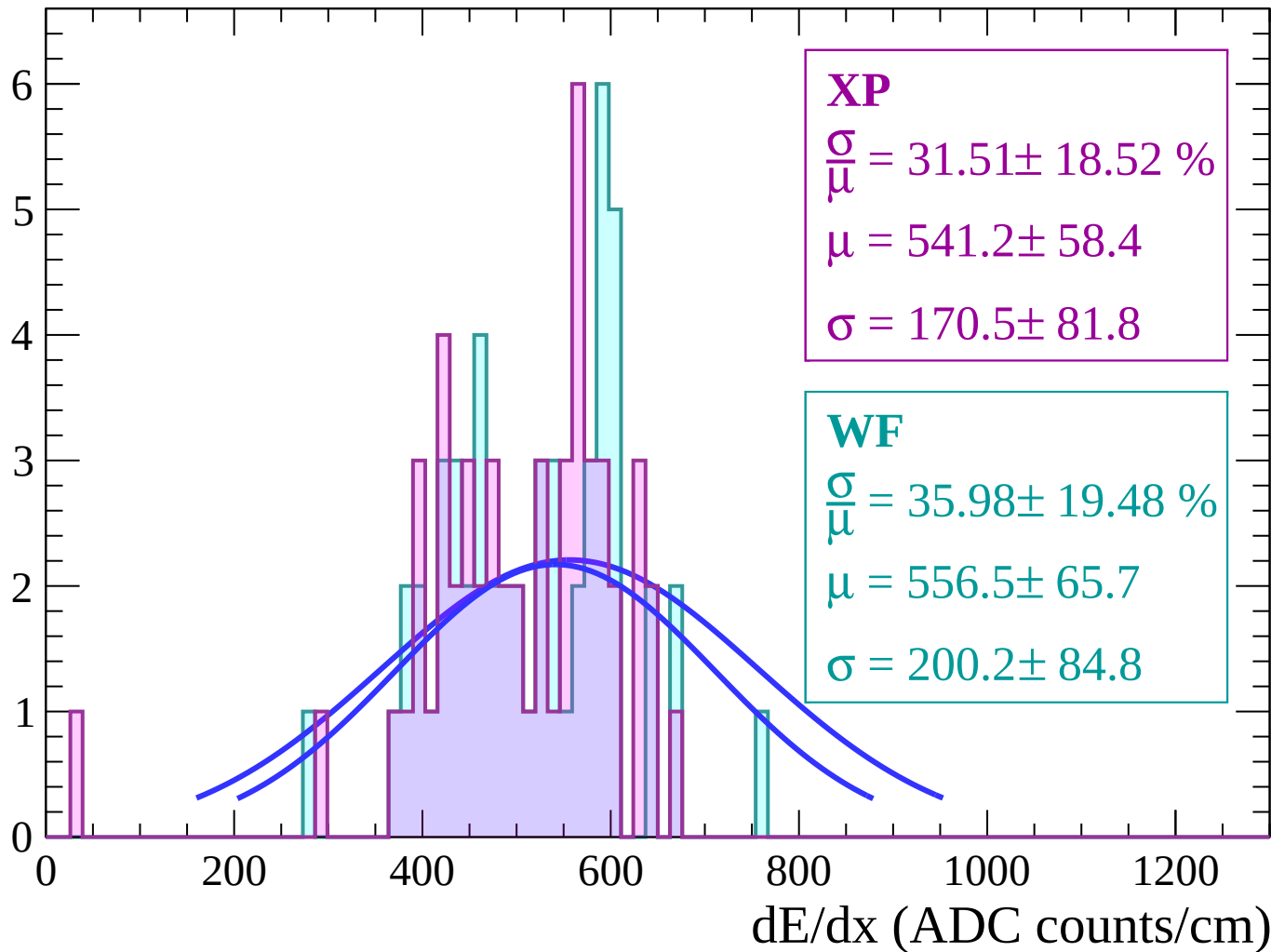
Energy loss | $-46 < \theta < -44$

Count



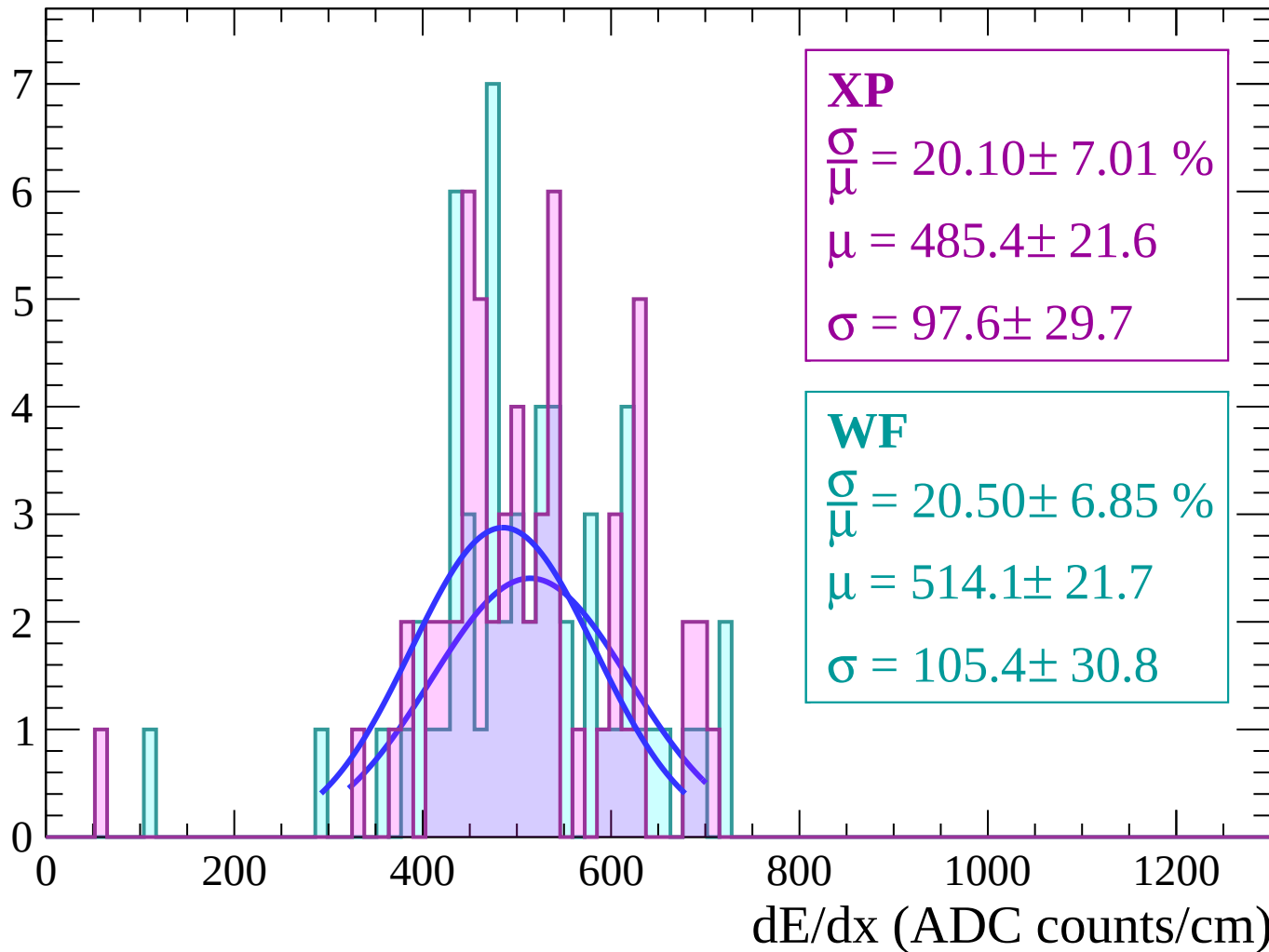
Energy loss | $-44 < \theta < -42$

Count



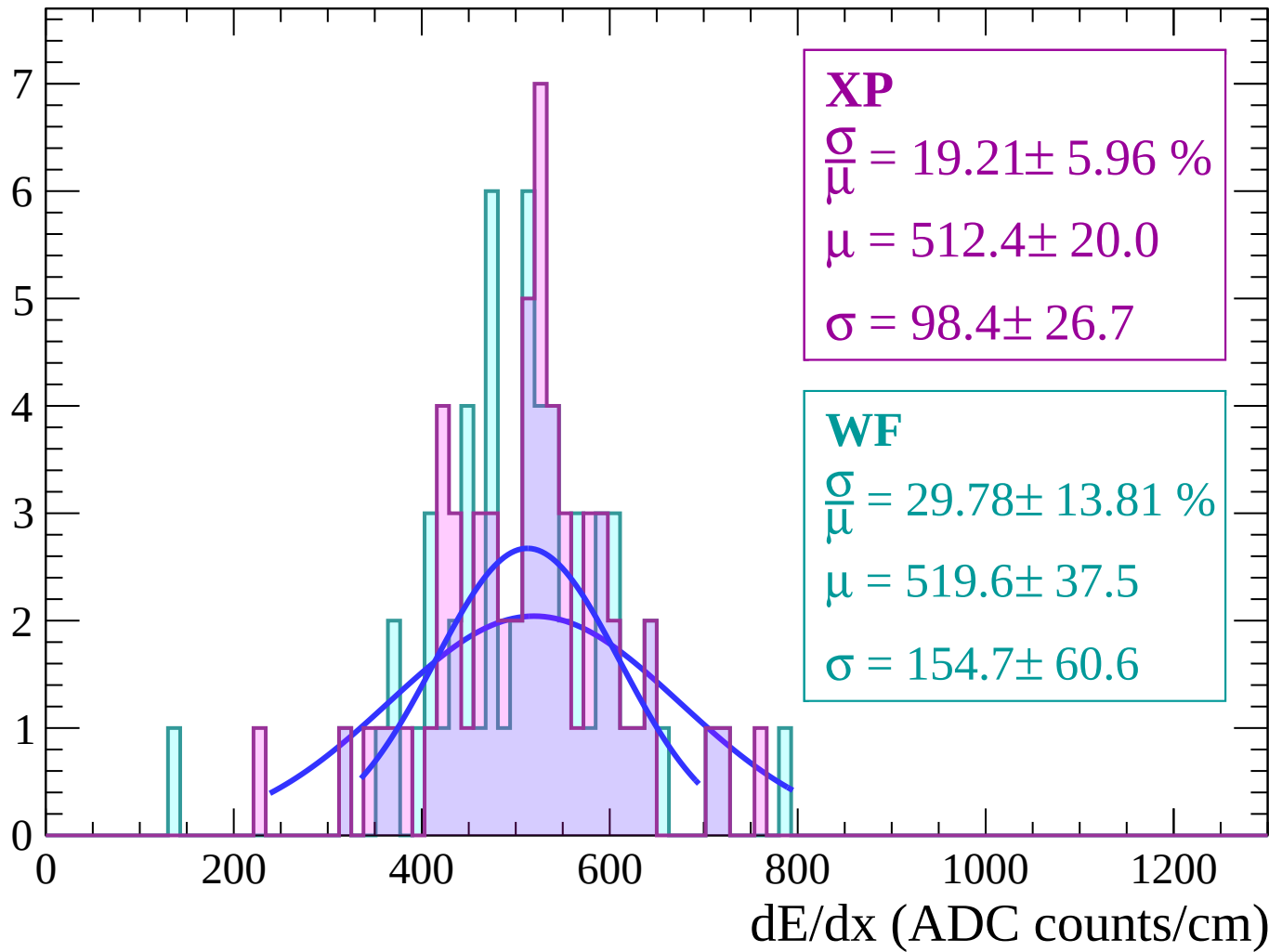
Energy loss | $-42 < \theta < -40$

Count



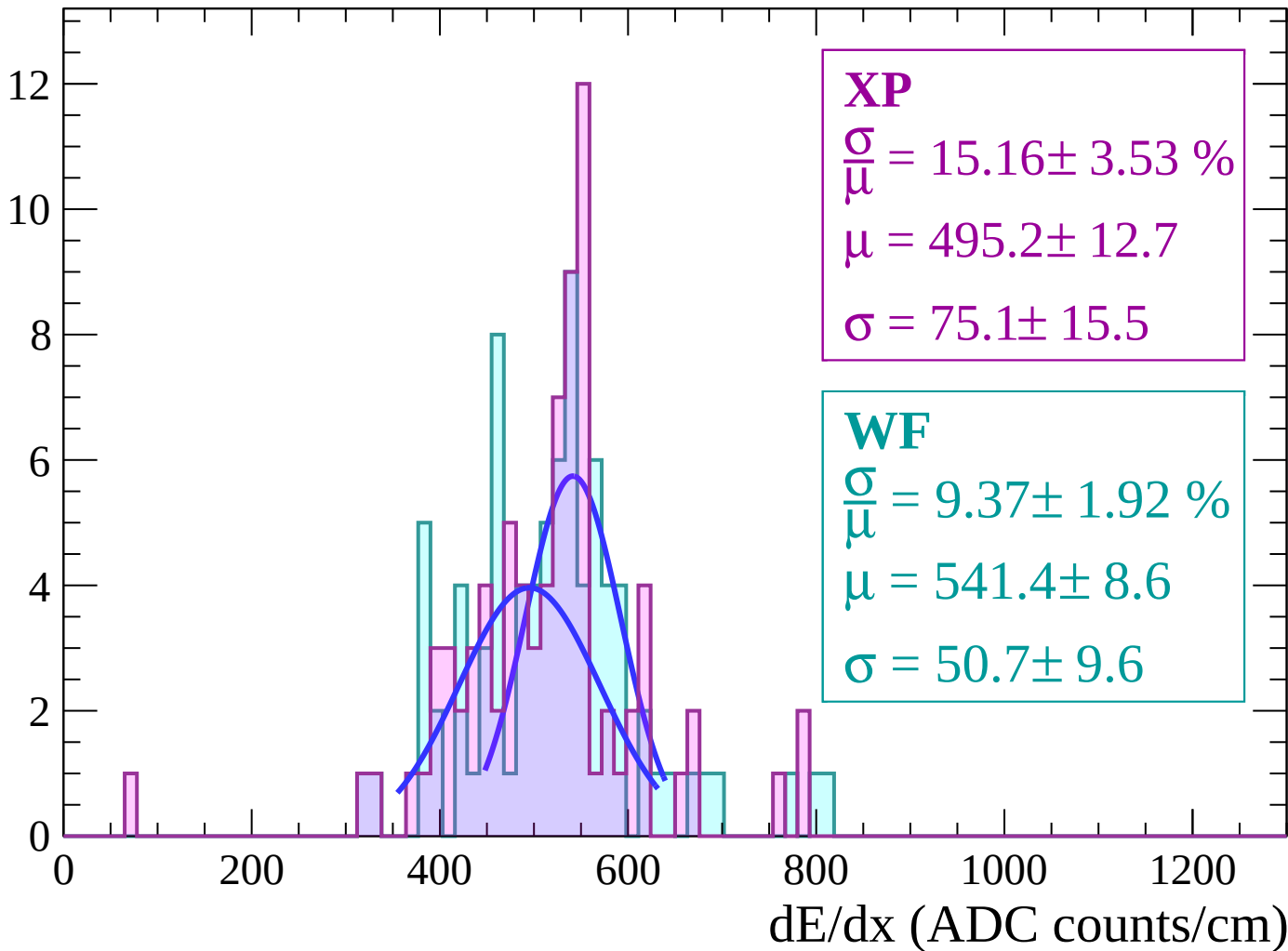
Energy loss | $-40 < \theta < -38$

Count



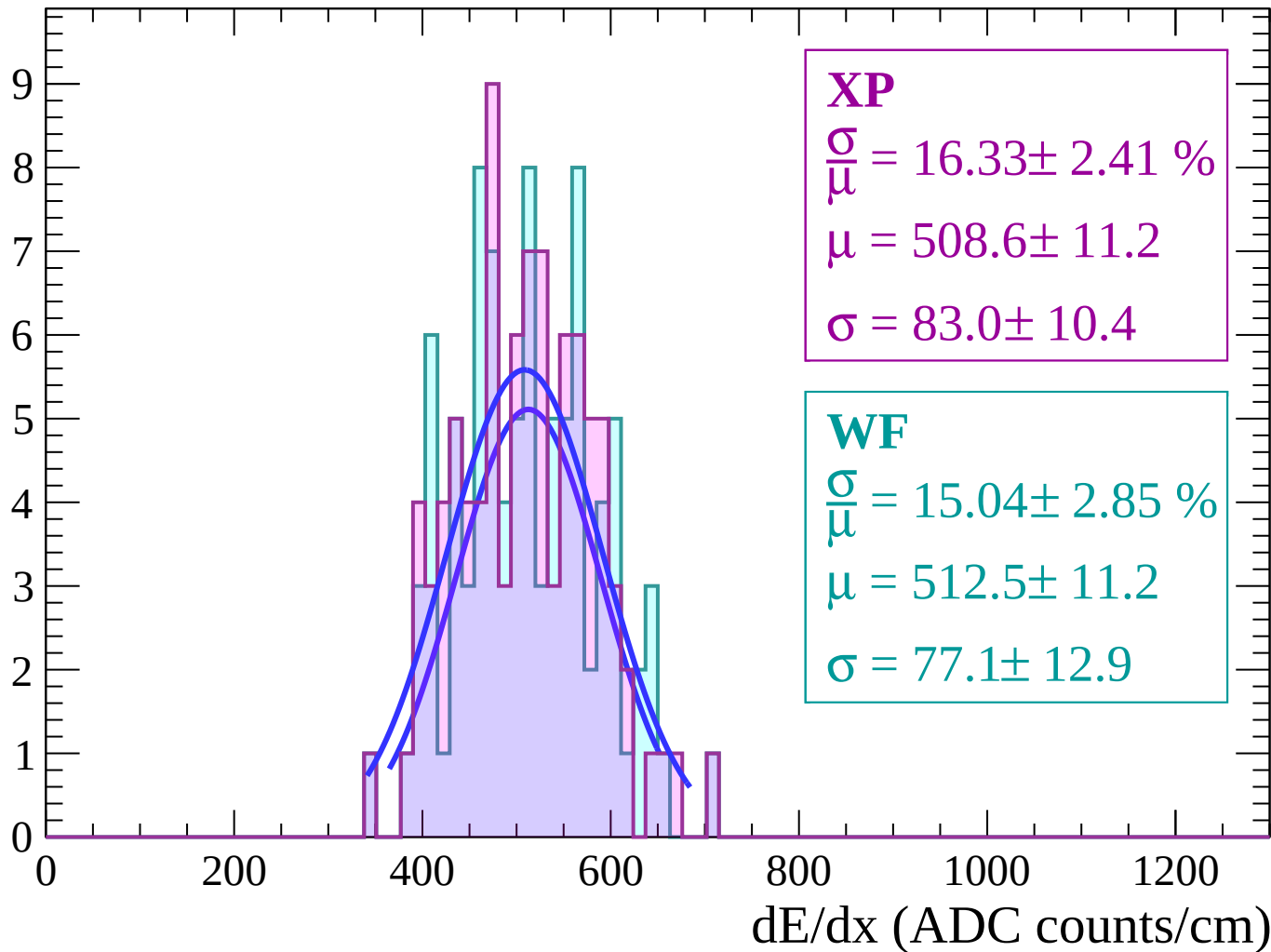
Energy loss | $-38 < \theta < -36$

Count



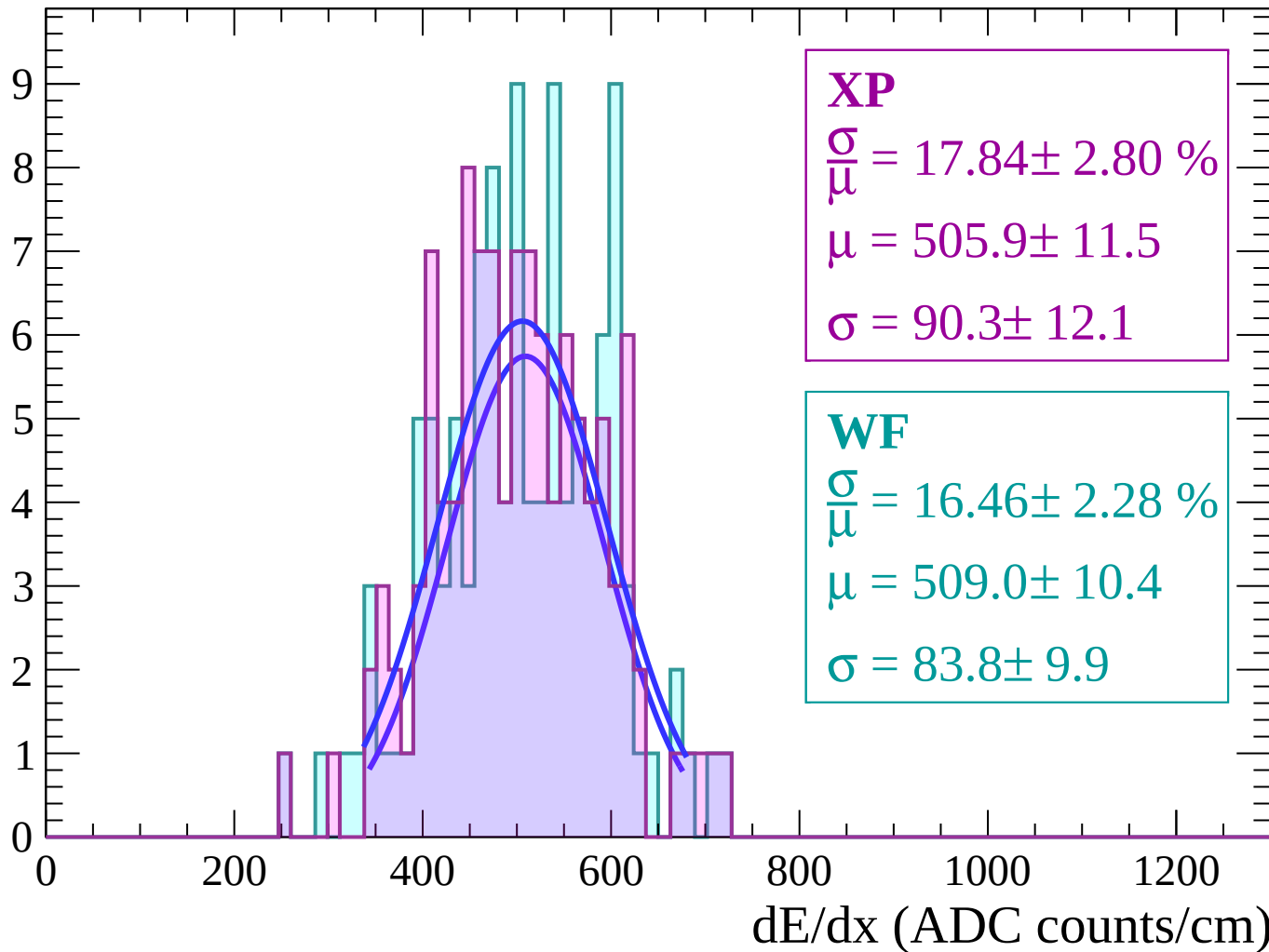
Energy loss | $-36 < \theta < -34$

Count



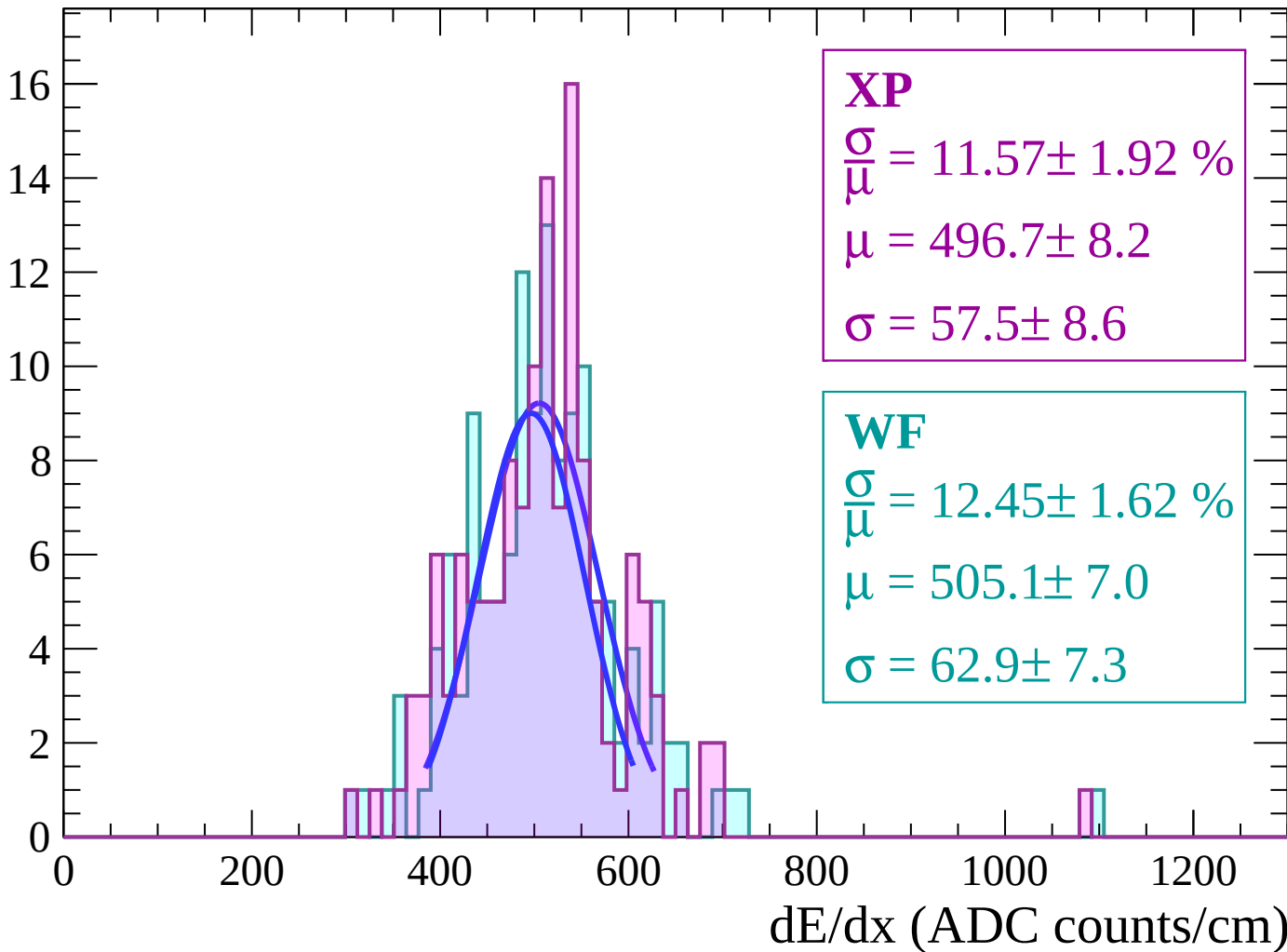
Energy loss | $-34 < \theta < -32$

Count



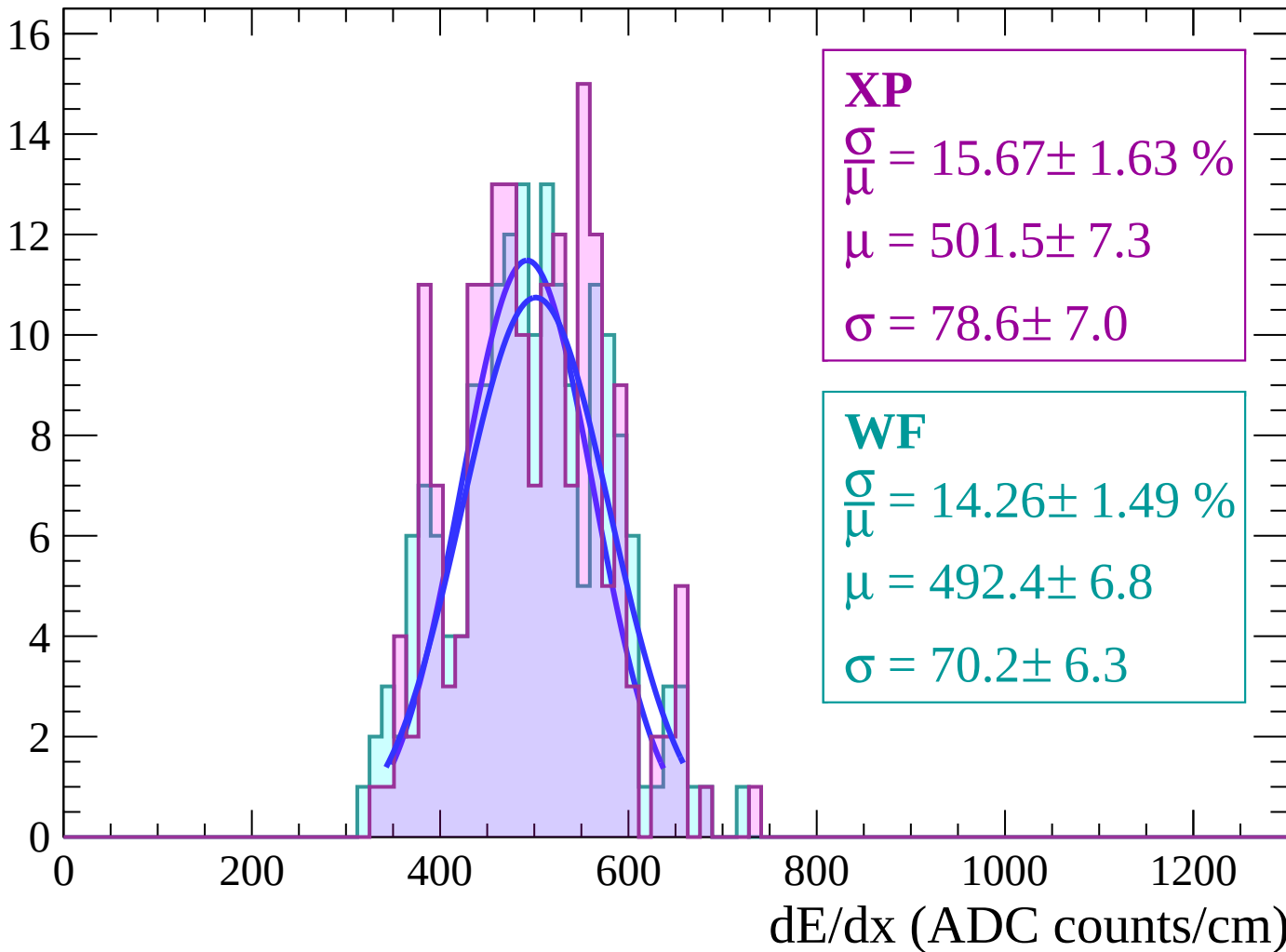
Energy loss | $-32 < \theta < -30$

Count



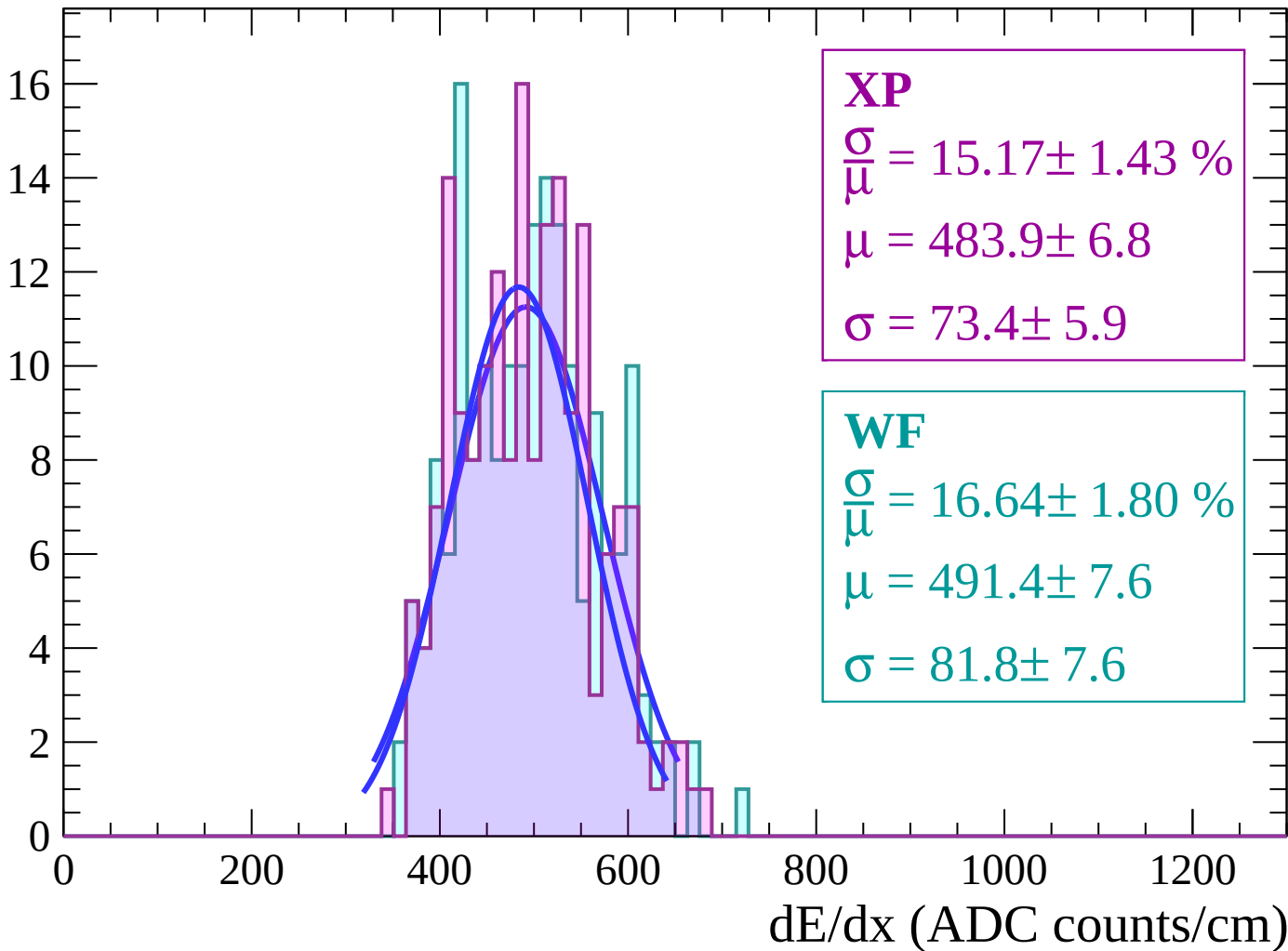
Energy loss | $-30 < \theta < -28$

Count



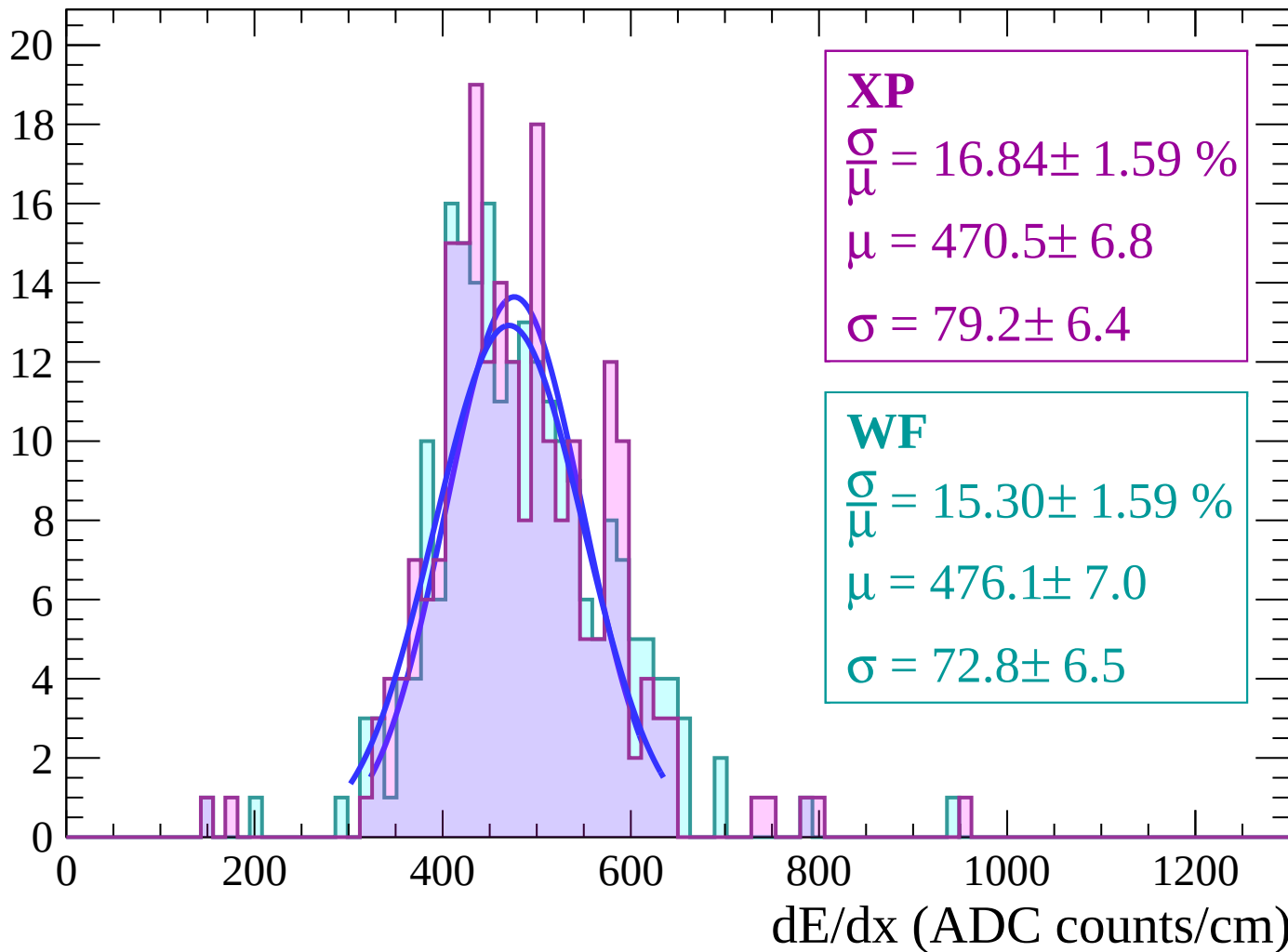
Energy loss | $-28 < \theta < -26$

Count



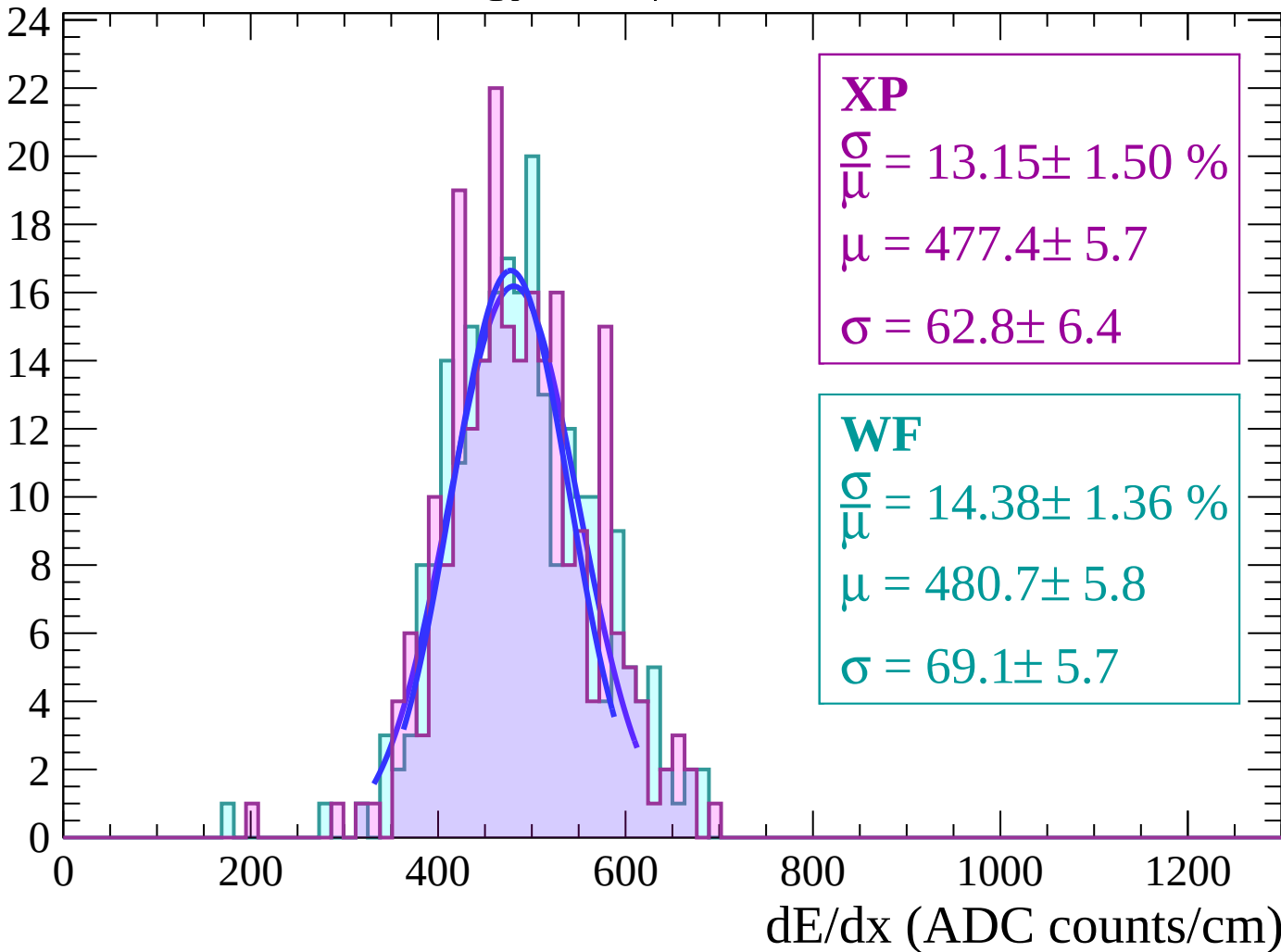
Energy loss | $-26 < \theta < -24$

Count



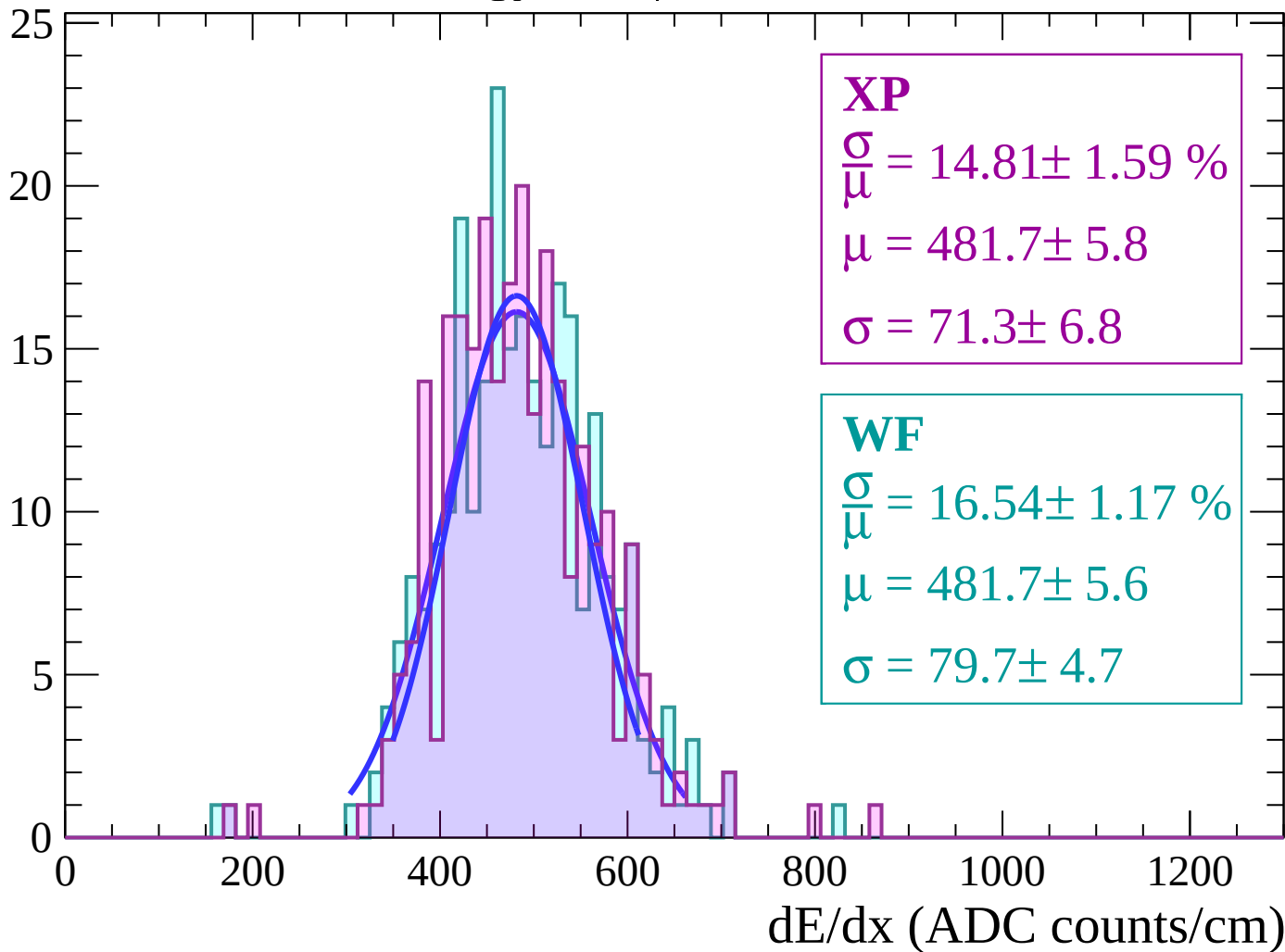
Energy loss | $-24 < \theta < -22$

Count



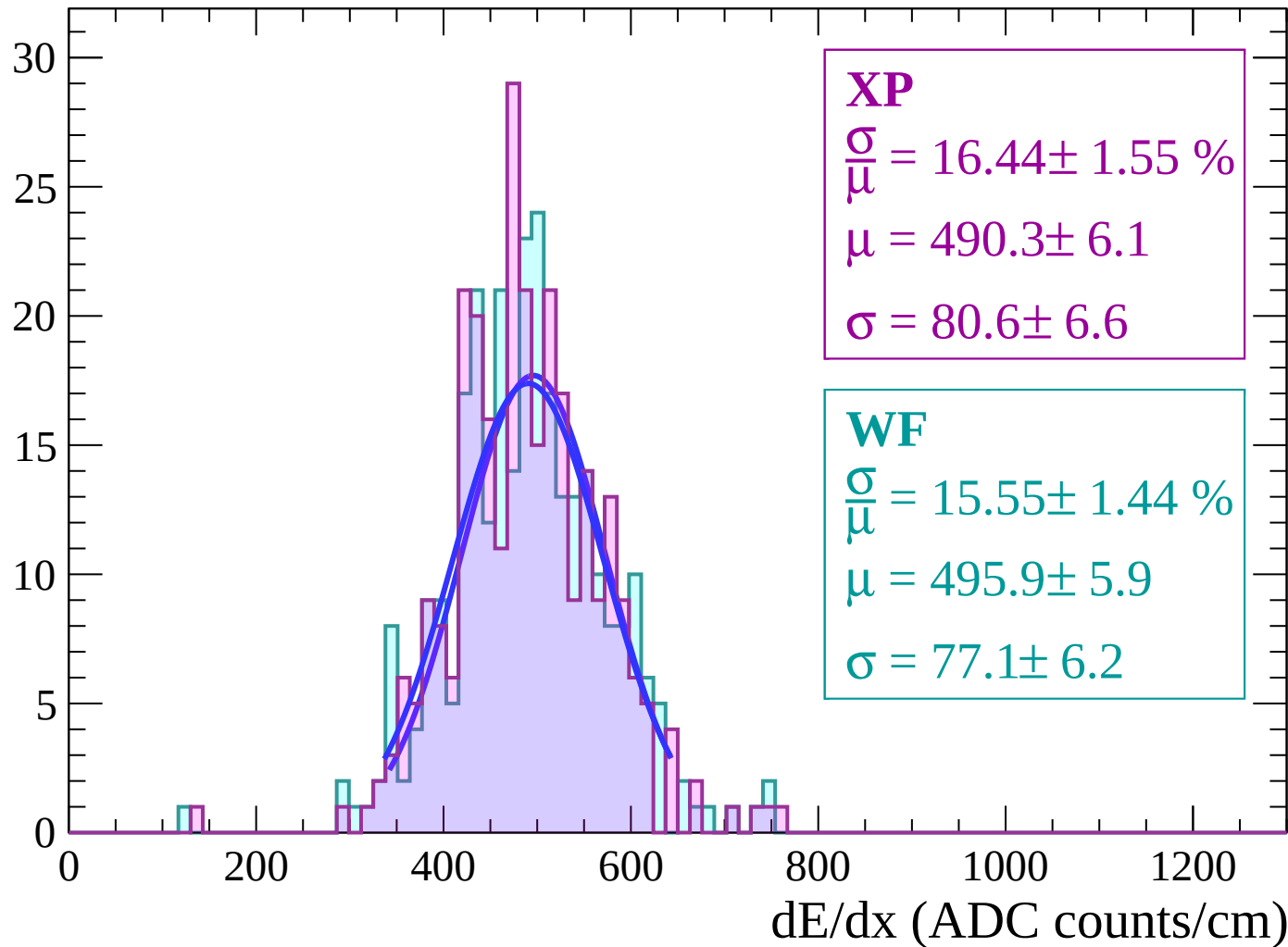
Energy loss | $-22 < \theta < -20$

Count



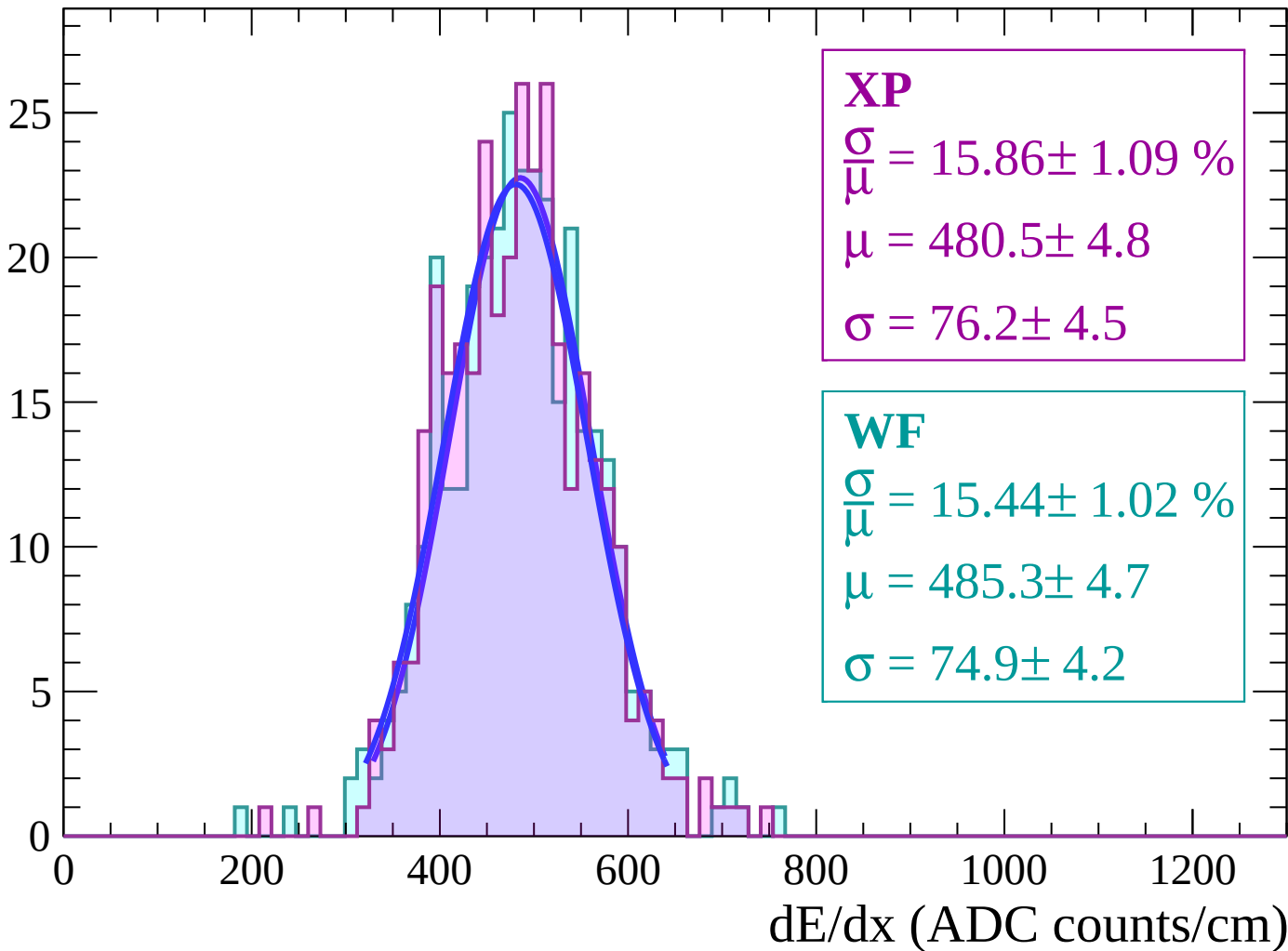
Energy loss | $-20 < \theta < -18$

Count



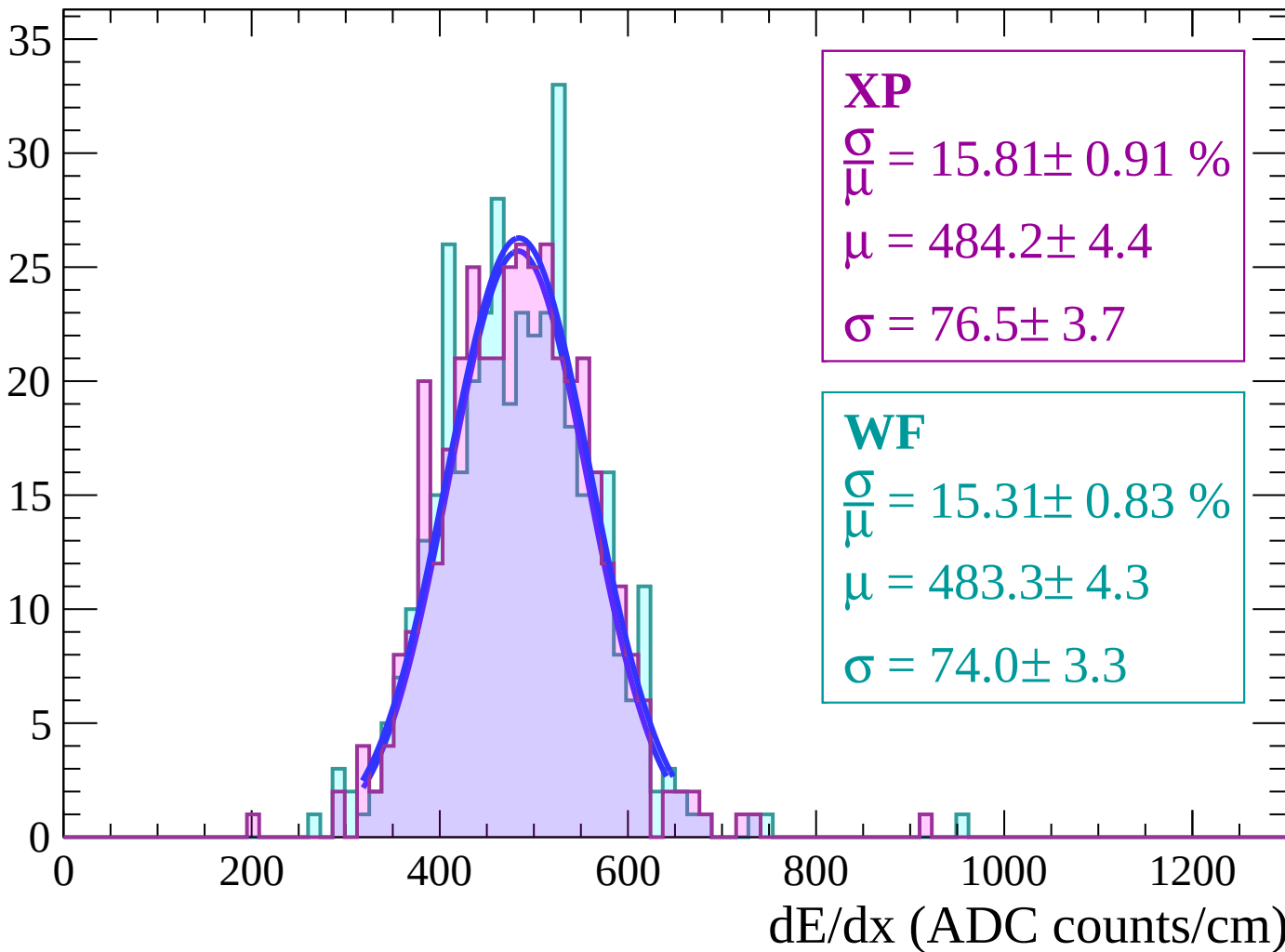
Energy loss | $-18 < \theta < -16$

Count



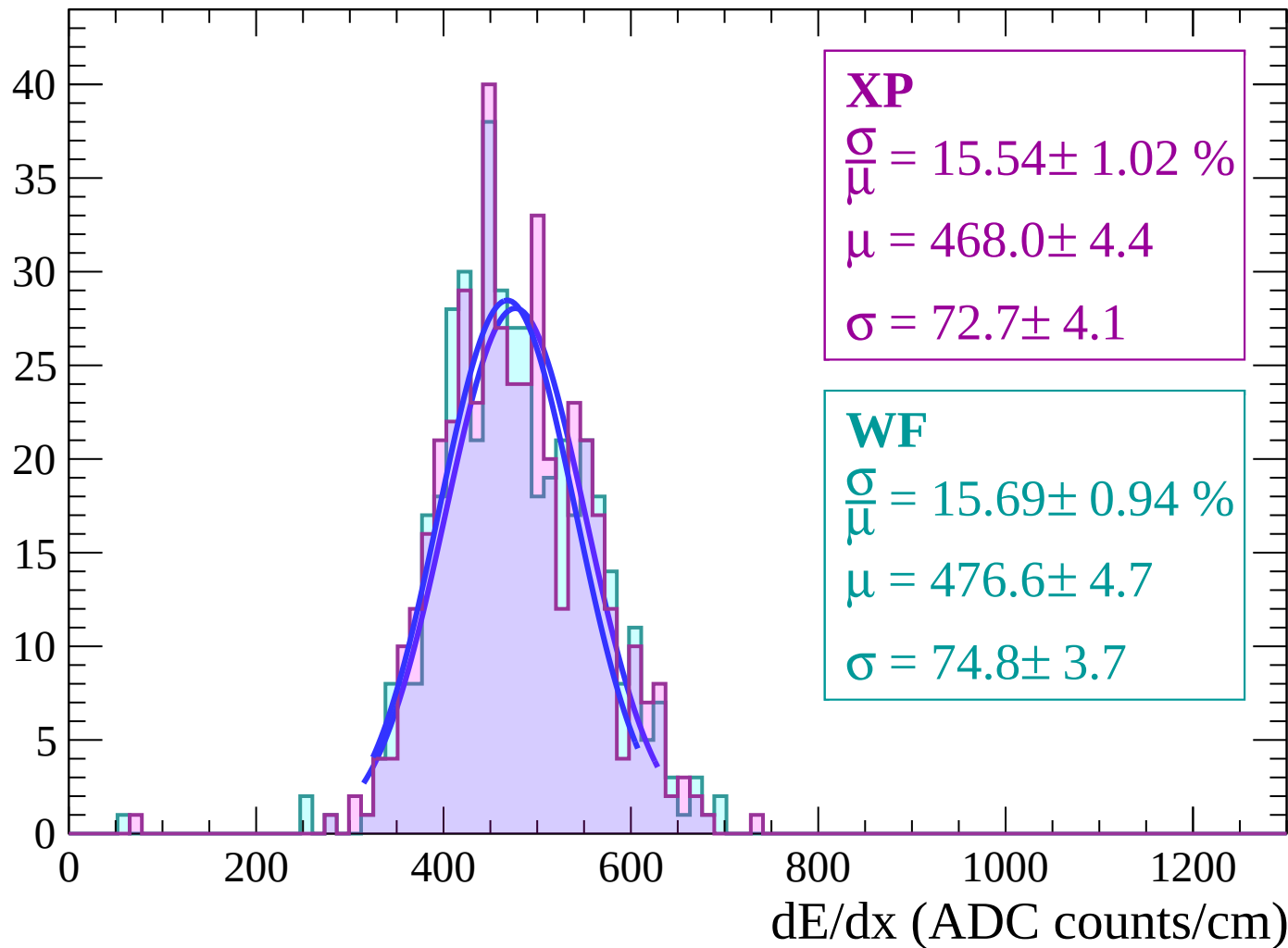
Energy loss | $-16 < \theta < -14$

Count



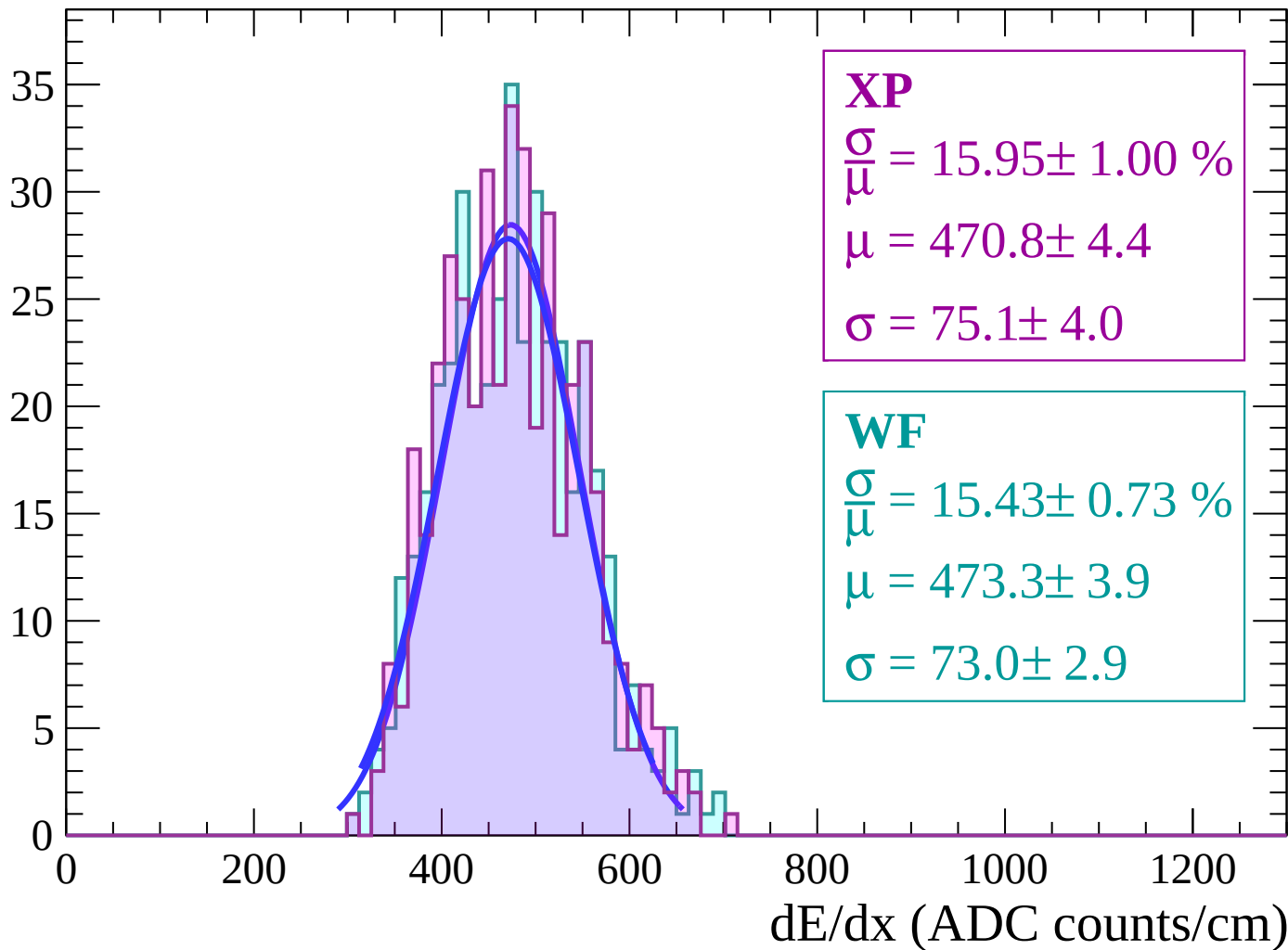
Energy loss | $-14 < \theta < -12$

Count



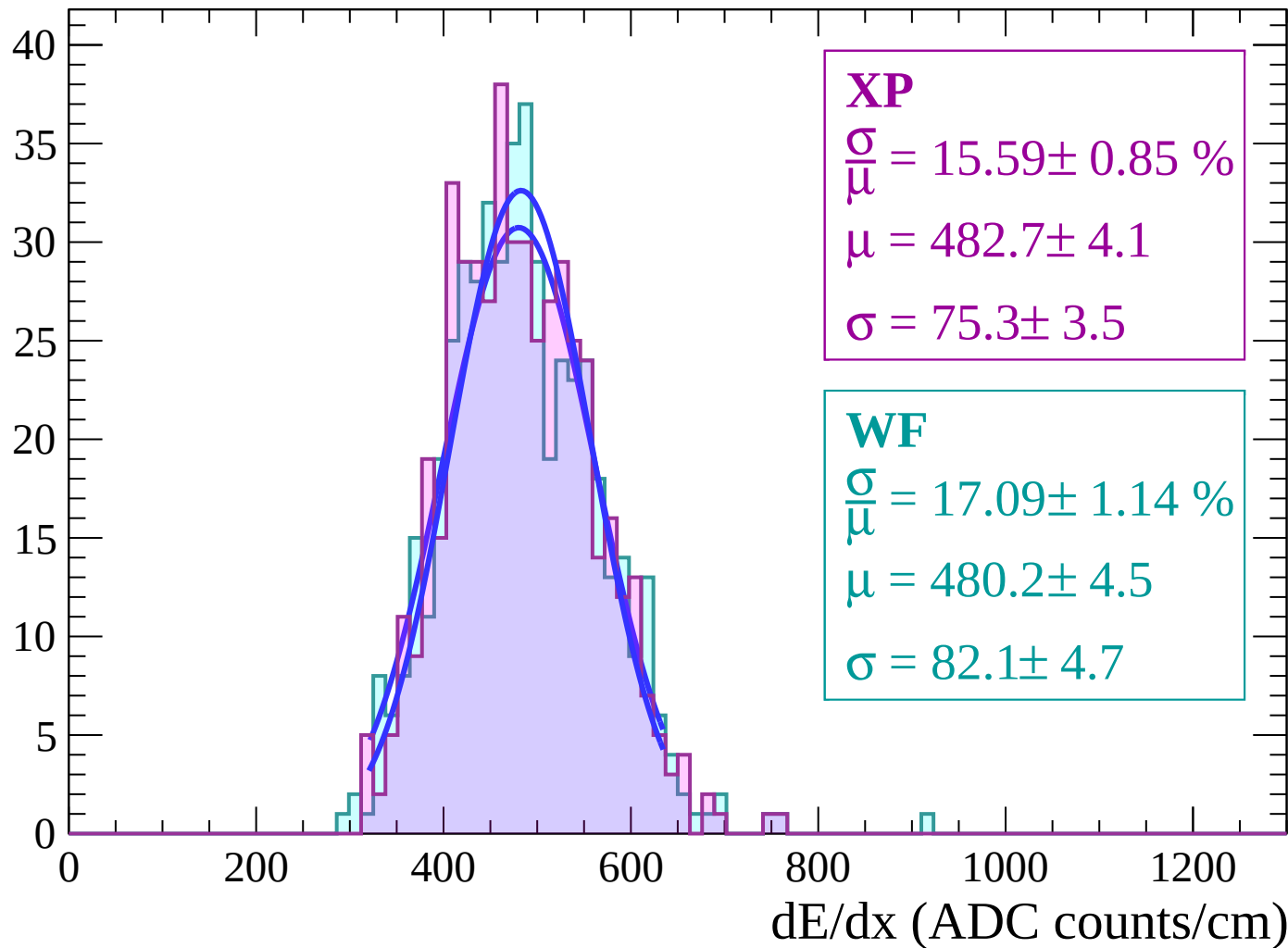
Energy loss | $-12 < \theta < -10$

Count



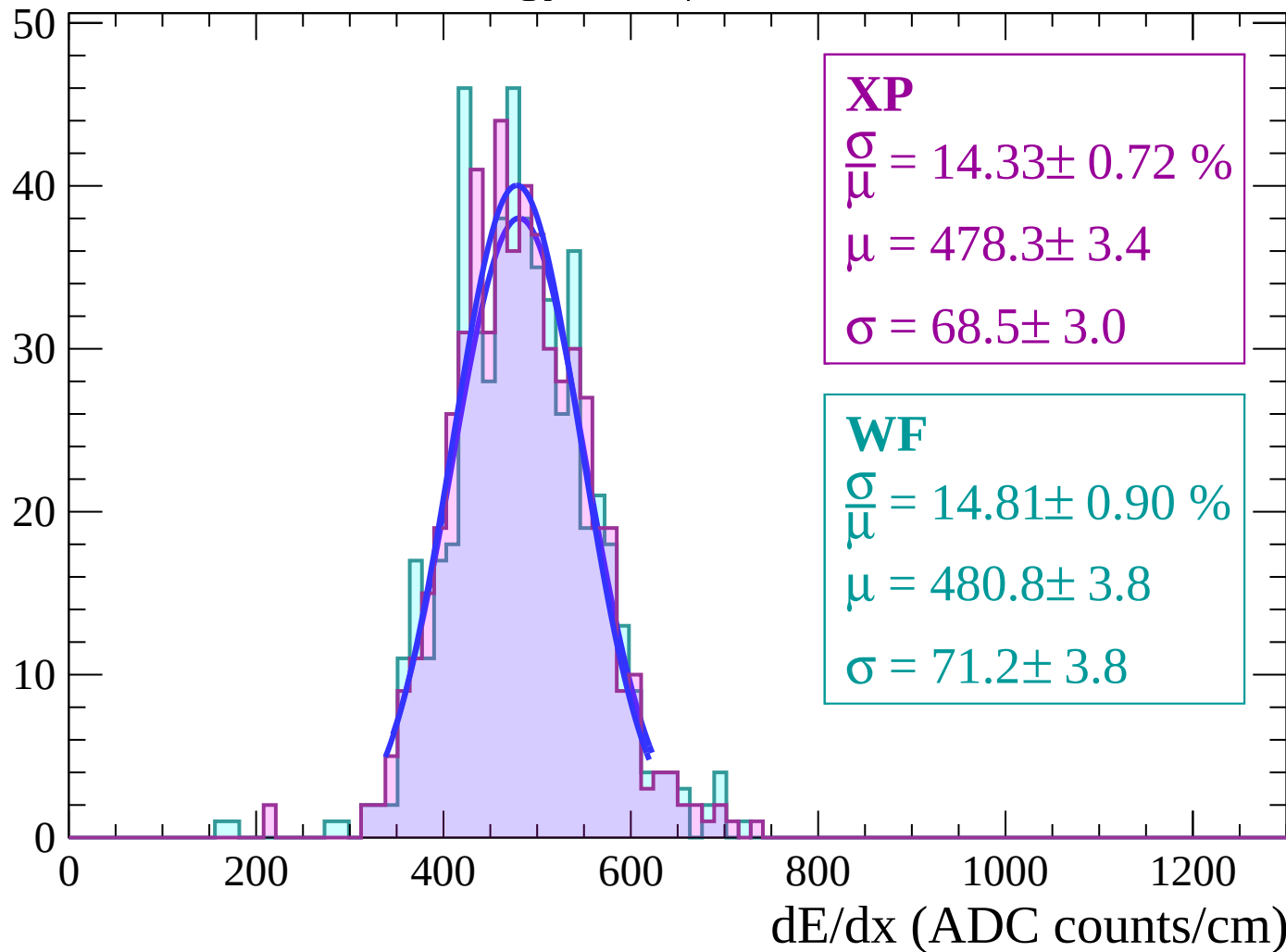
Energy loss | $-10 < \theta < -8$

Count



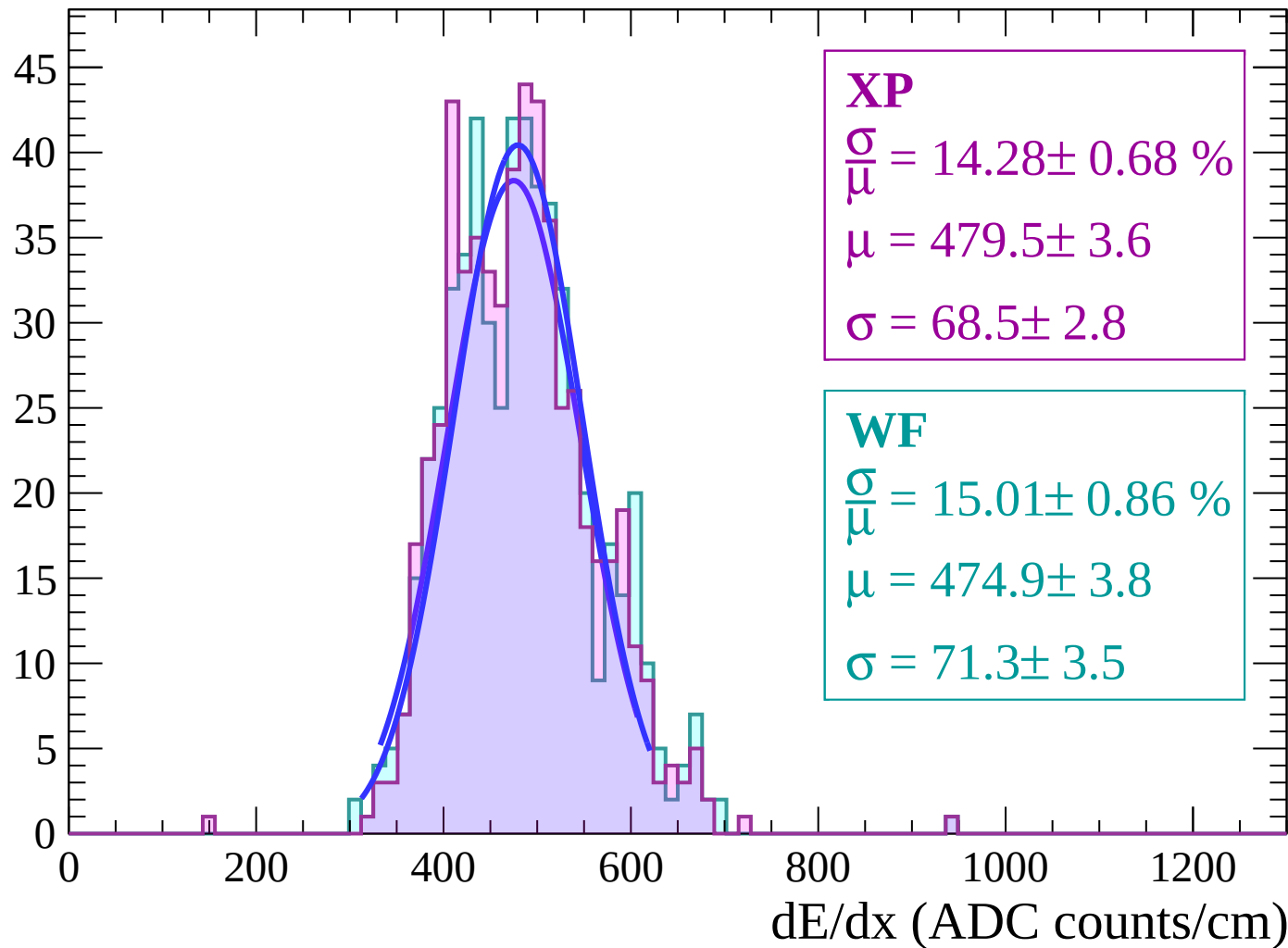
Energy loss | $-8 < \theta < -6$

Count



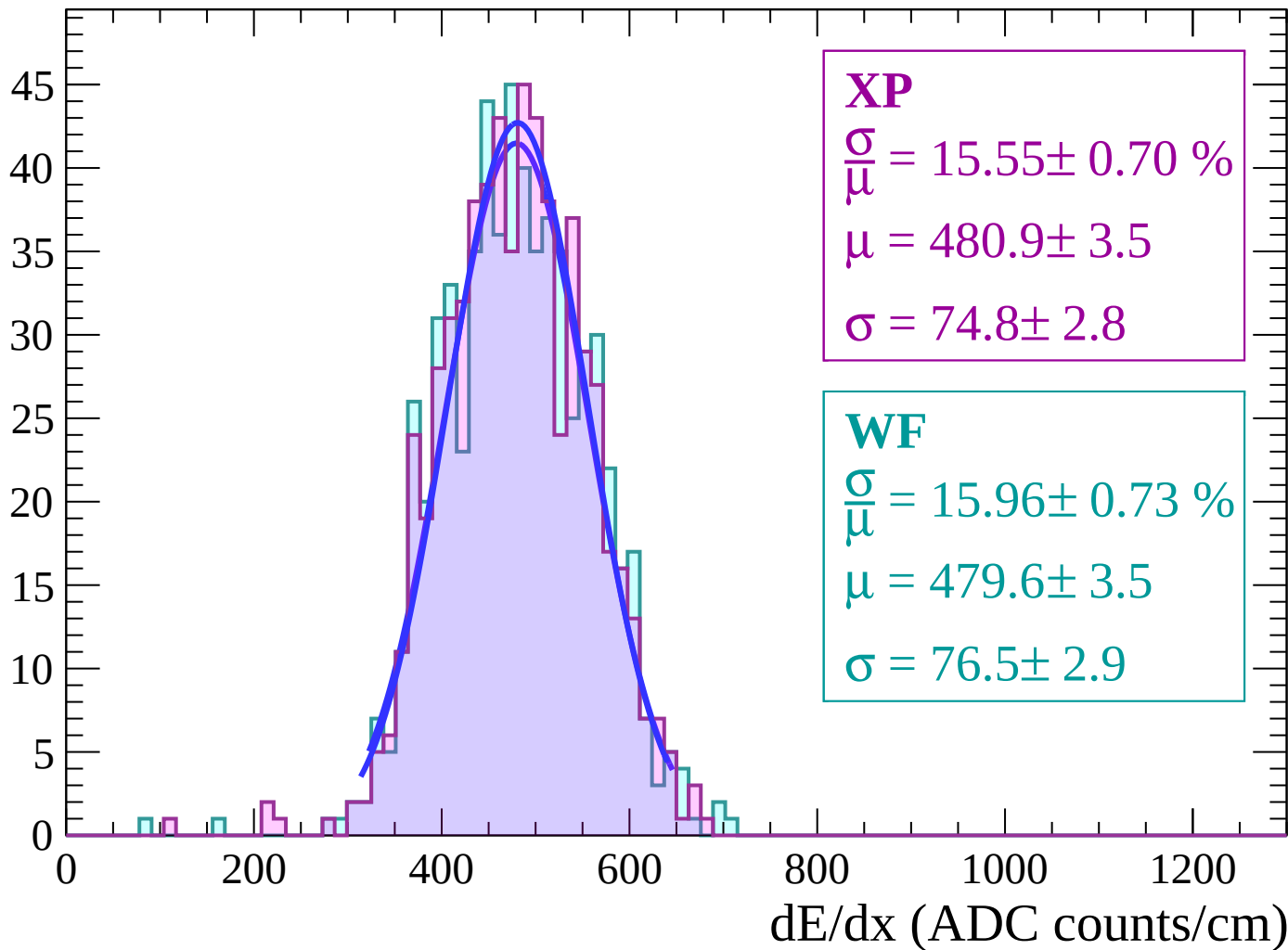
Energy loss | $-6 < \theta < -4$

Count



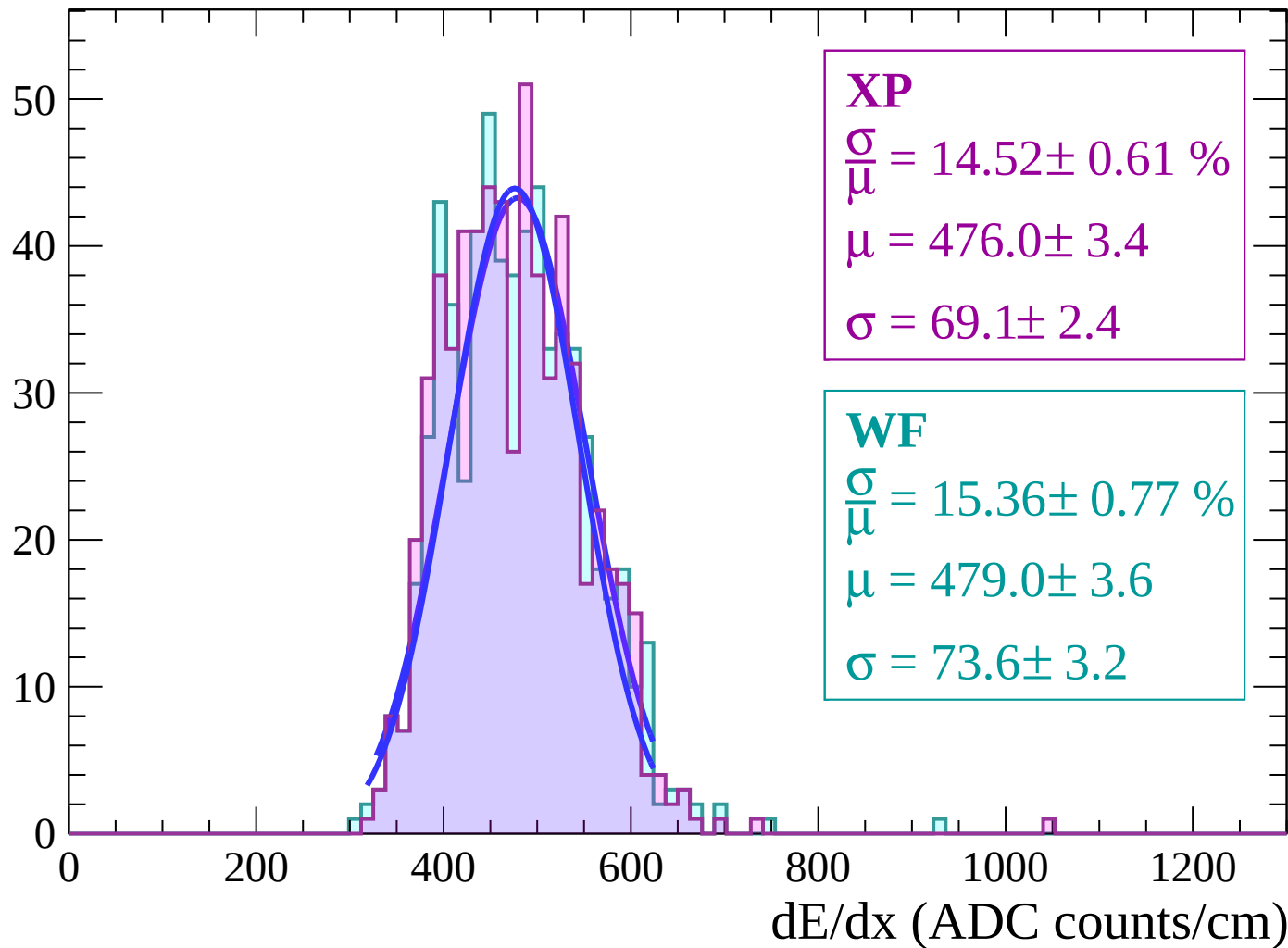
Energy loss | $-4 < \theta < -2$

Count



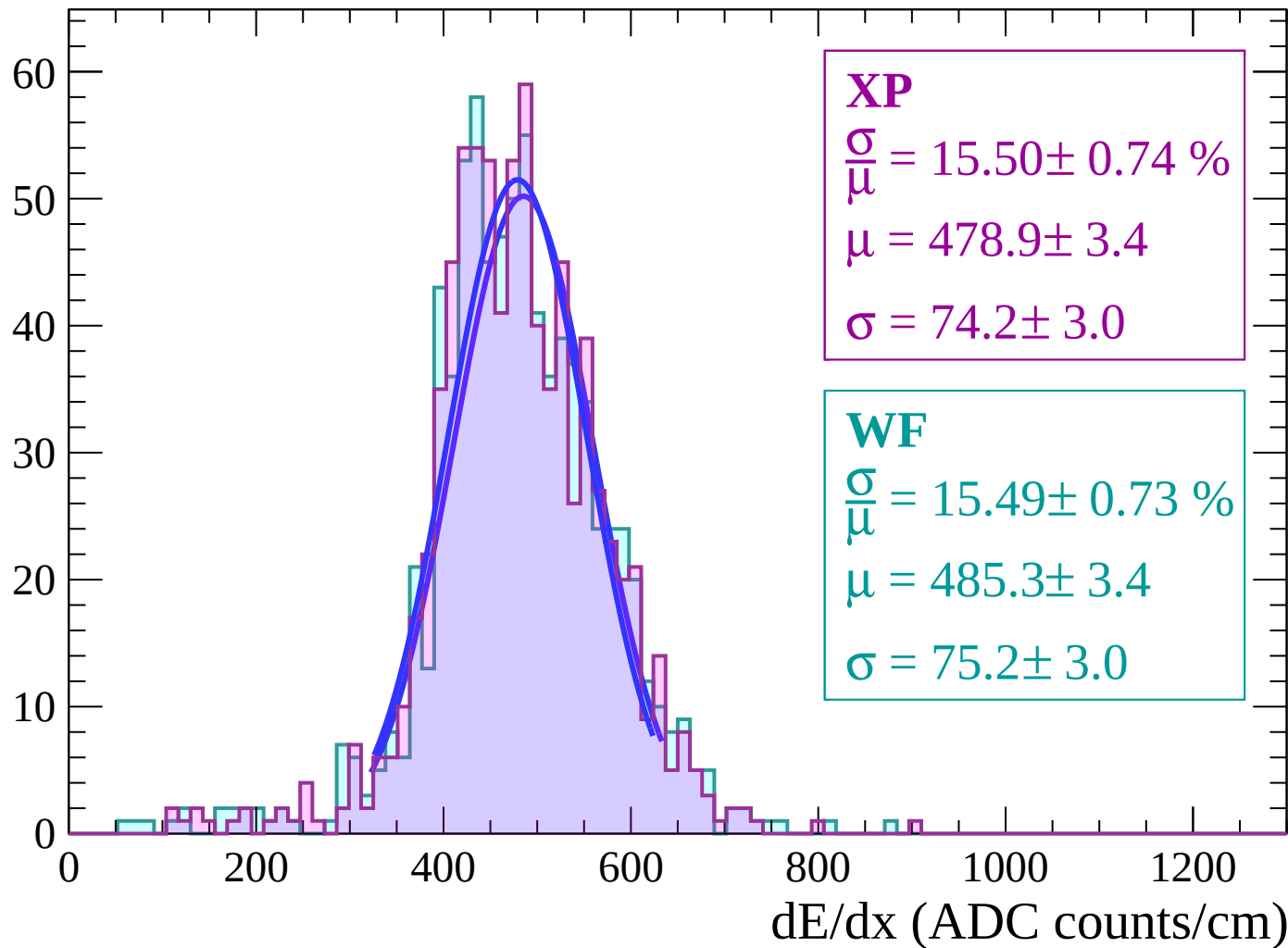
Energy loss | $-2 < \theta < 0$

Count



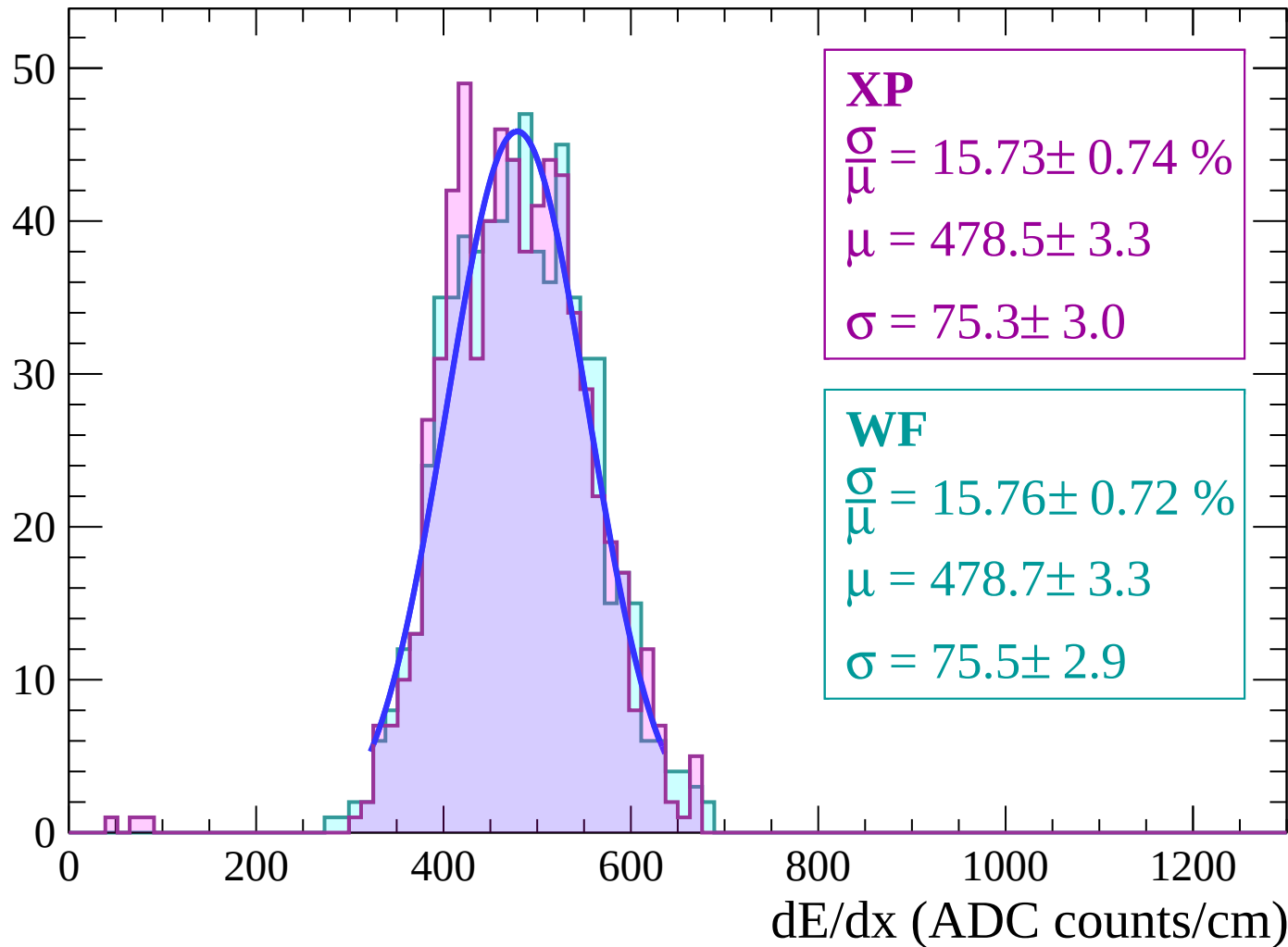
Energy loss | $0 < \theta < 2$

Count



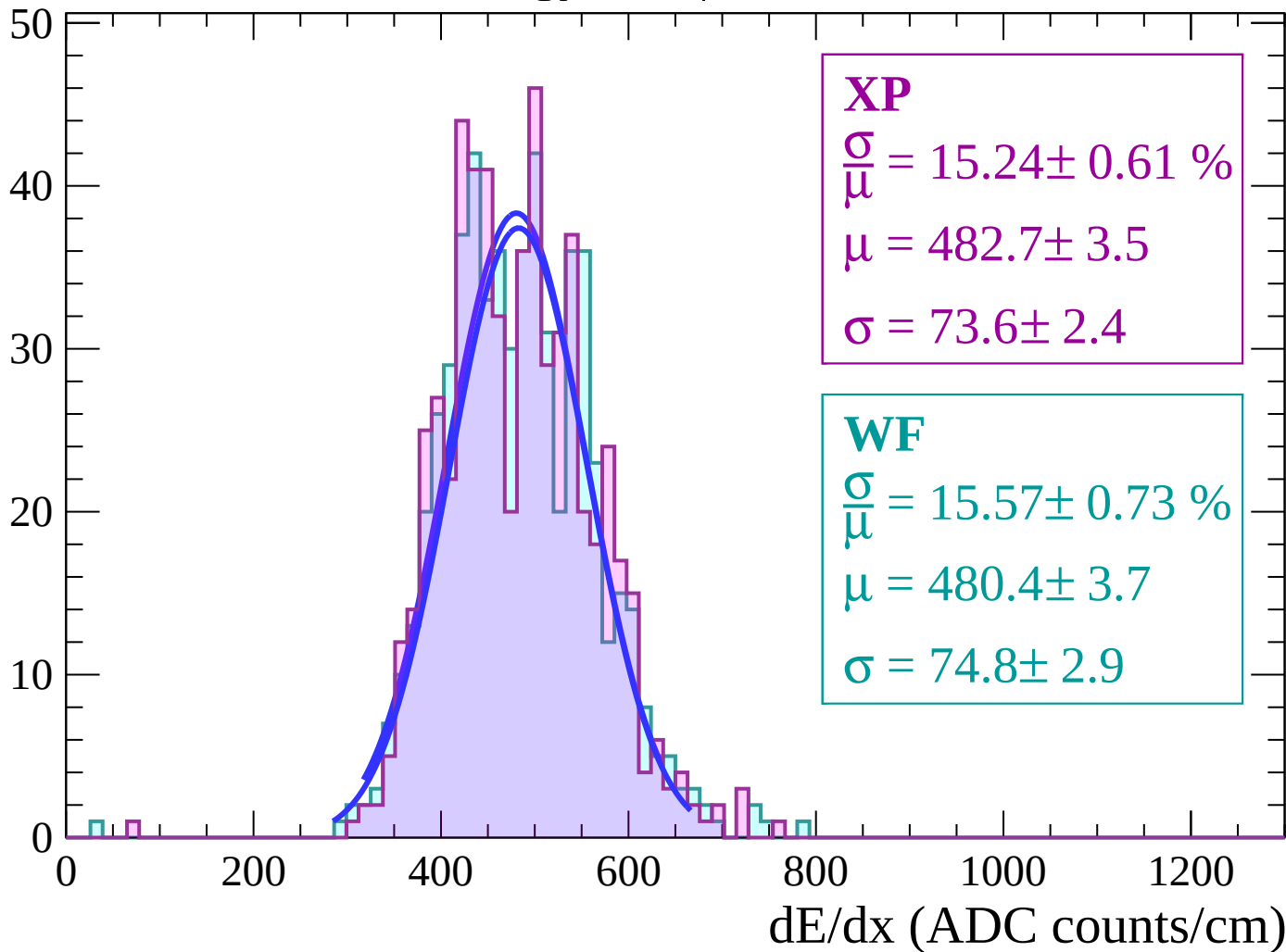
Energy loss | $2 < \theta < 4$

Count



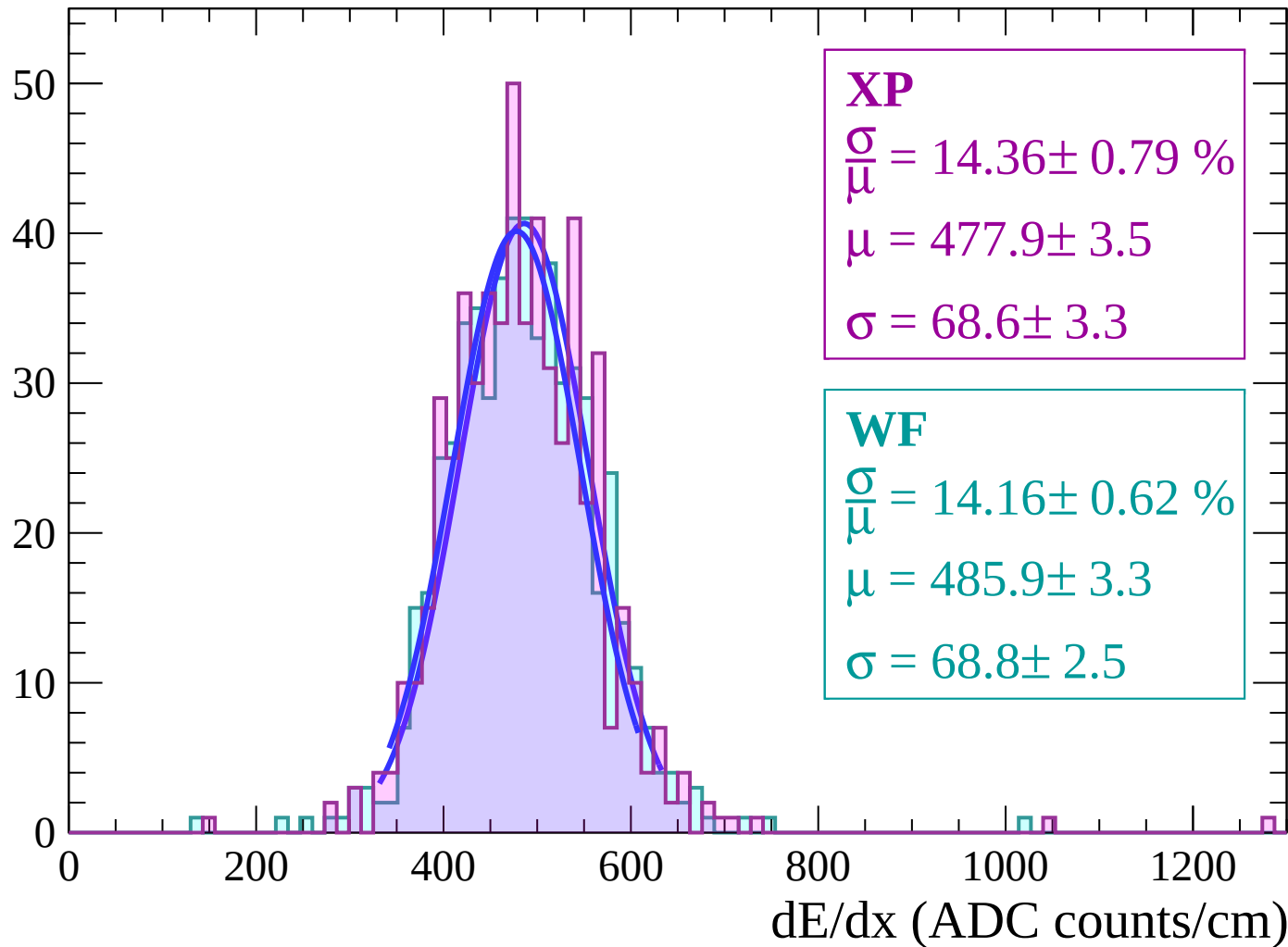
Energy loss | $4 < \theta < 6$

Count



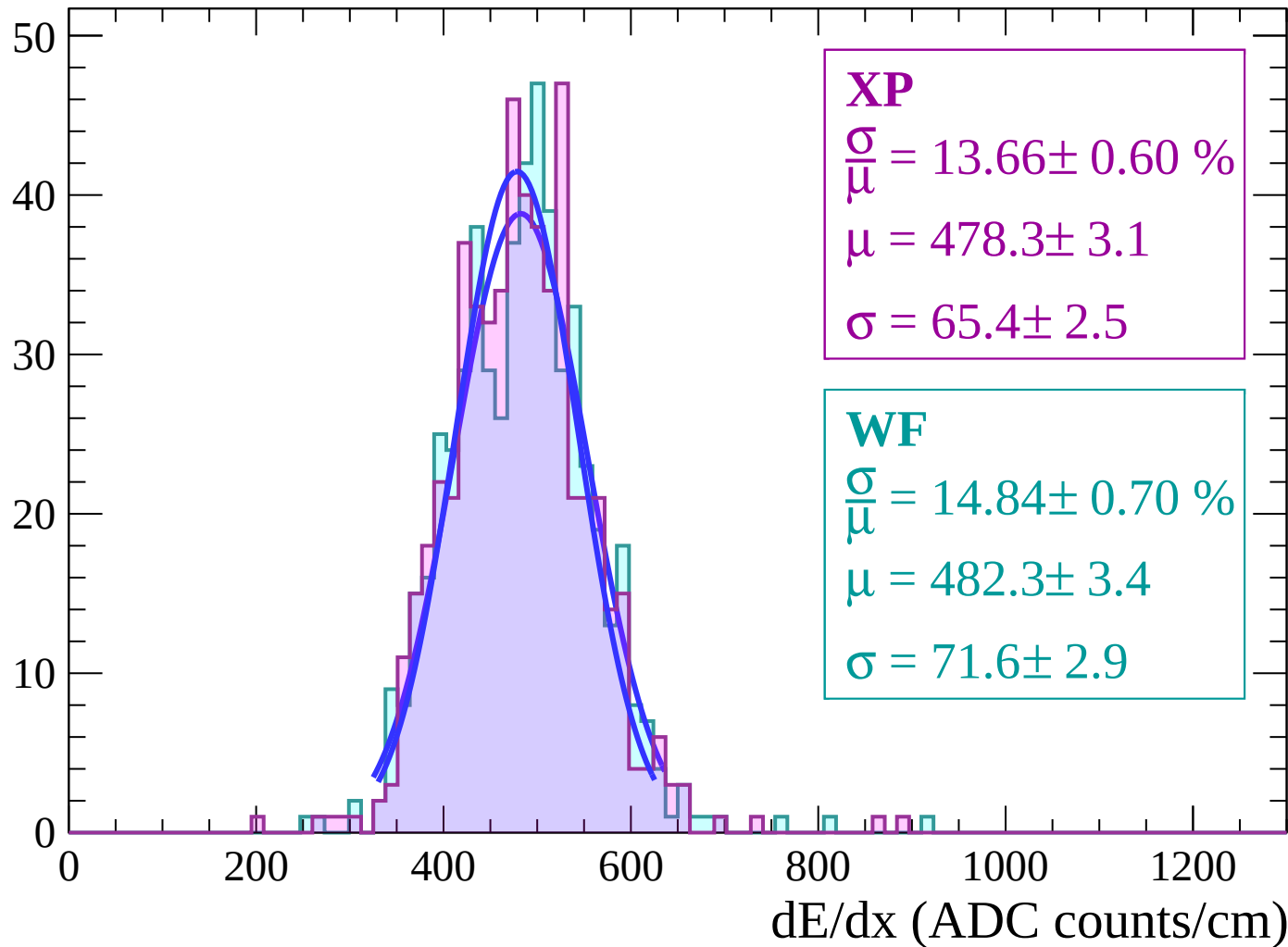
Energy loss | $6 < \theta < 8$

Count



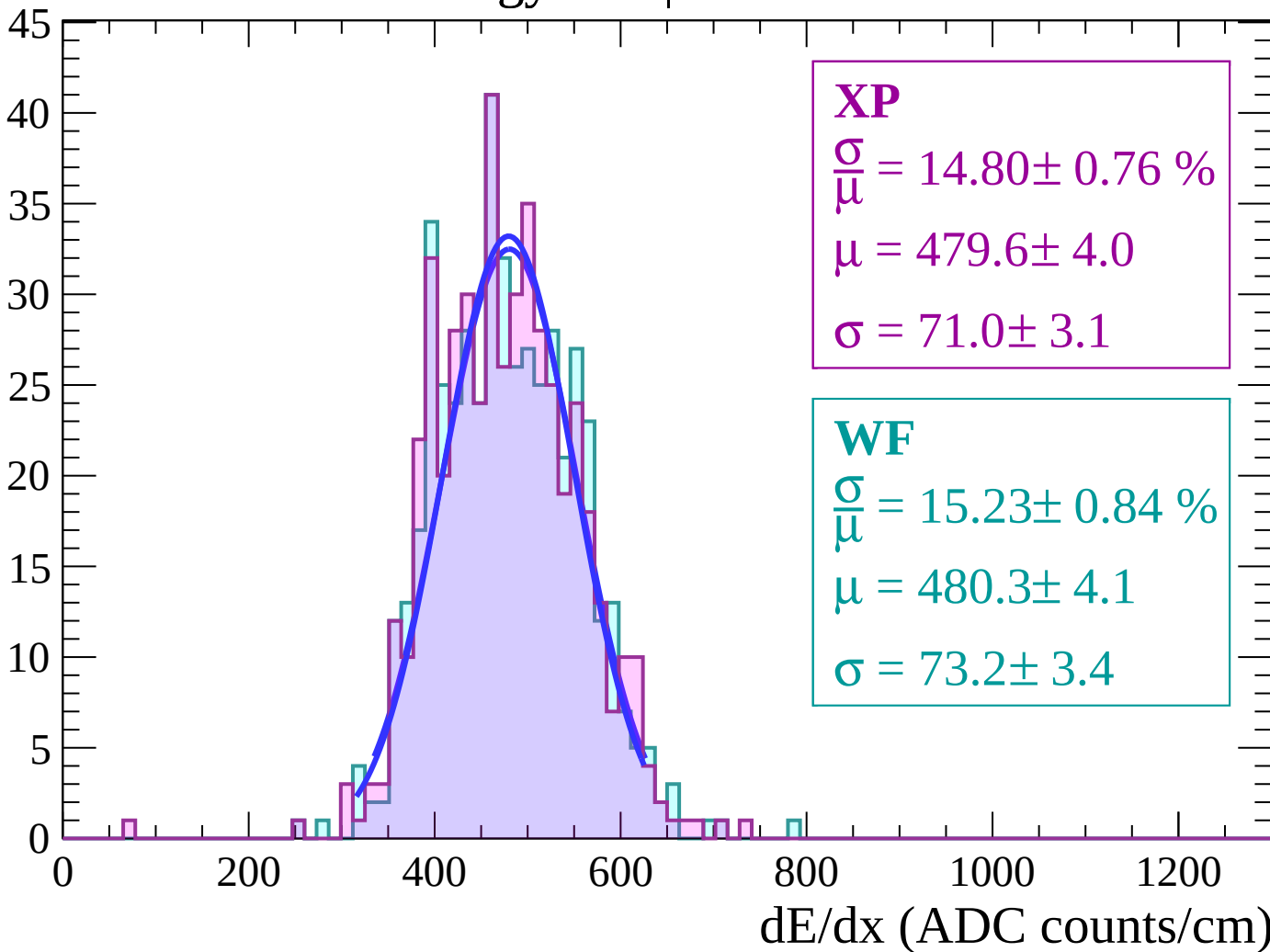
Energy loss | $8 < \theta < 10$

Count



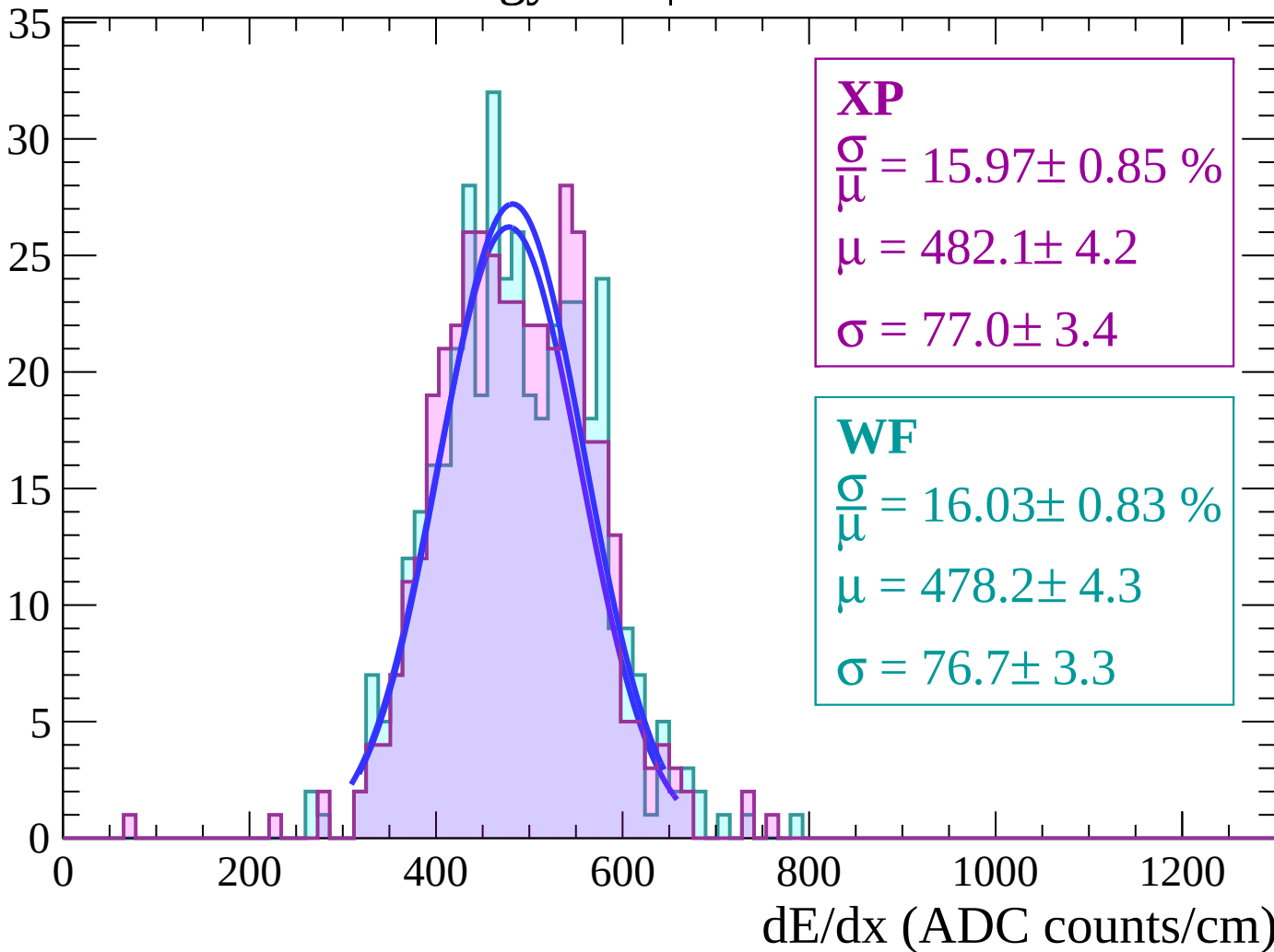
Energy loss | $10 < \theta < 12$

Count



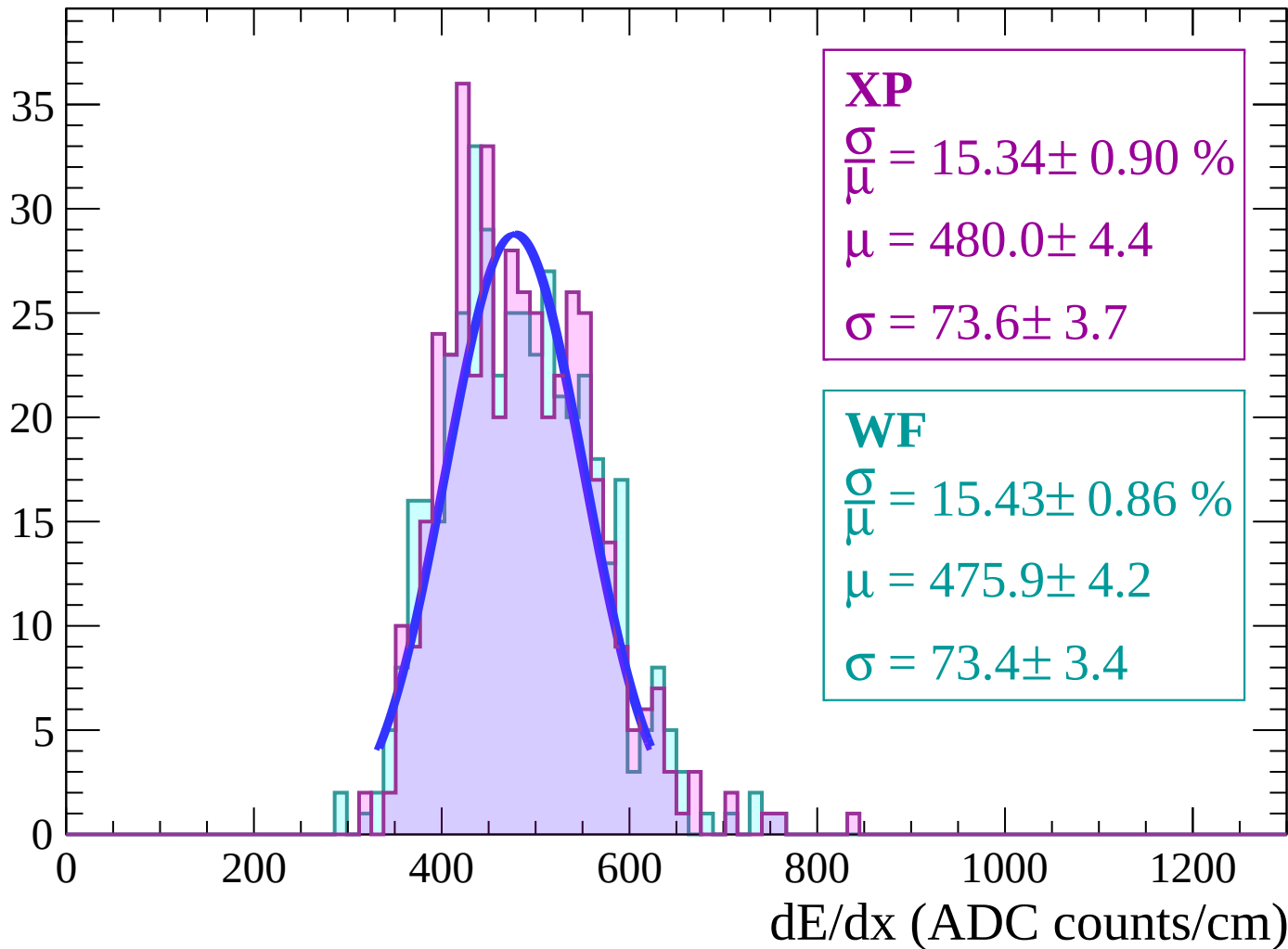
Energy loss | $12 < \theta < 14$

Count



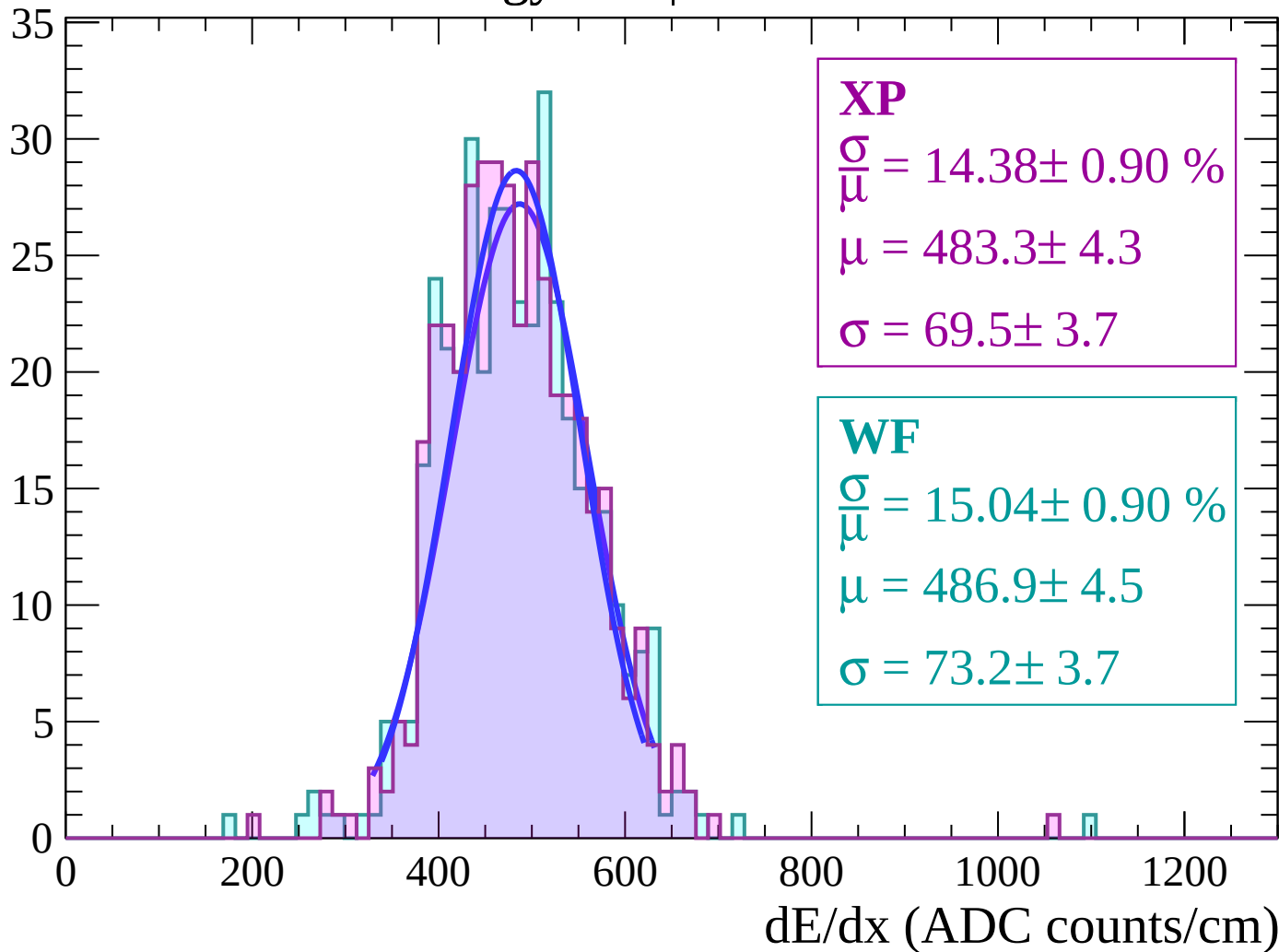
Energy loss | $14 < \theta < 16$

Count



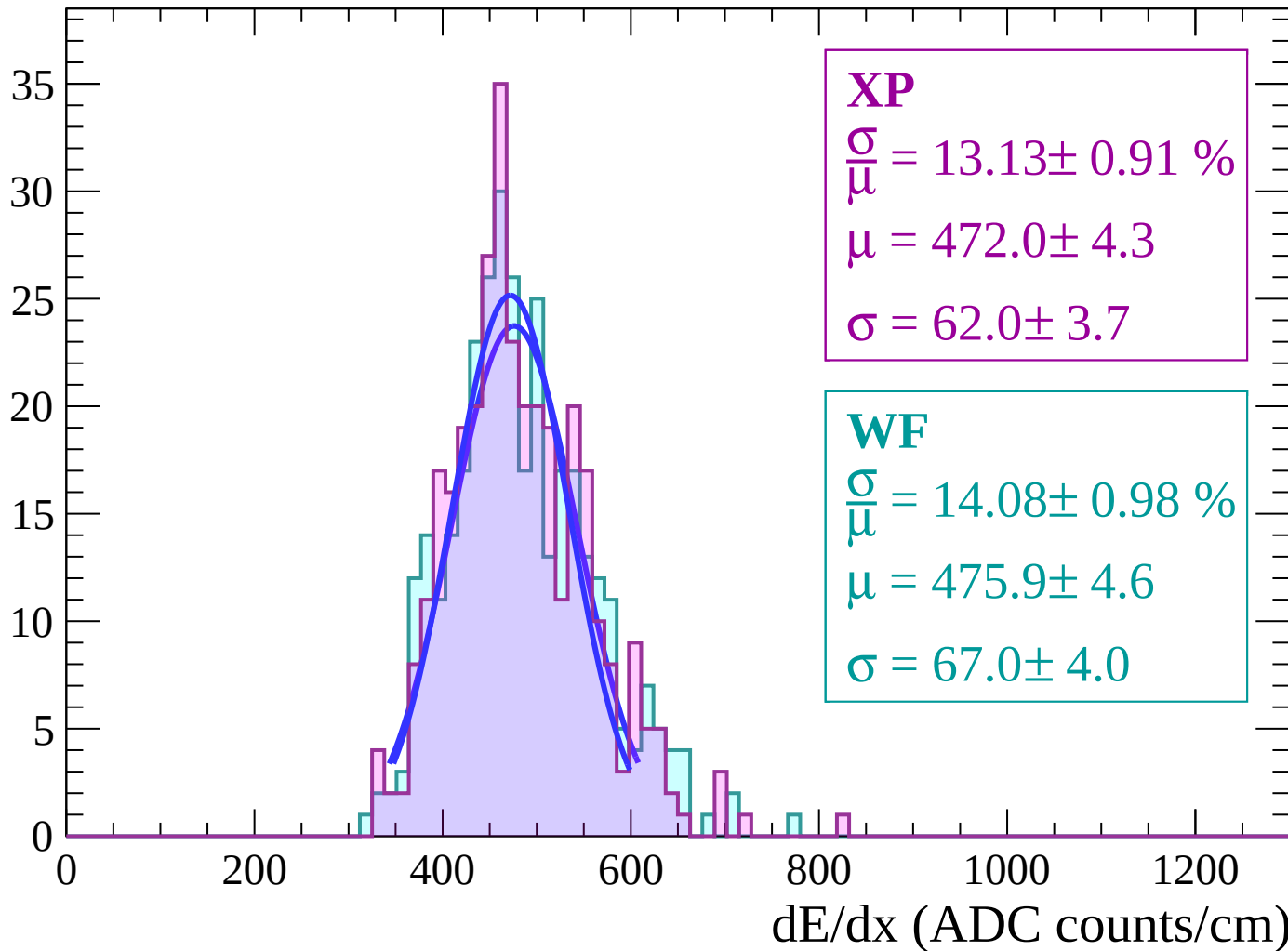
Energy loss | $16 < \theta < 18$

Count



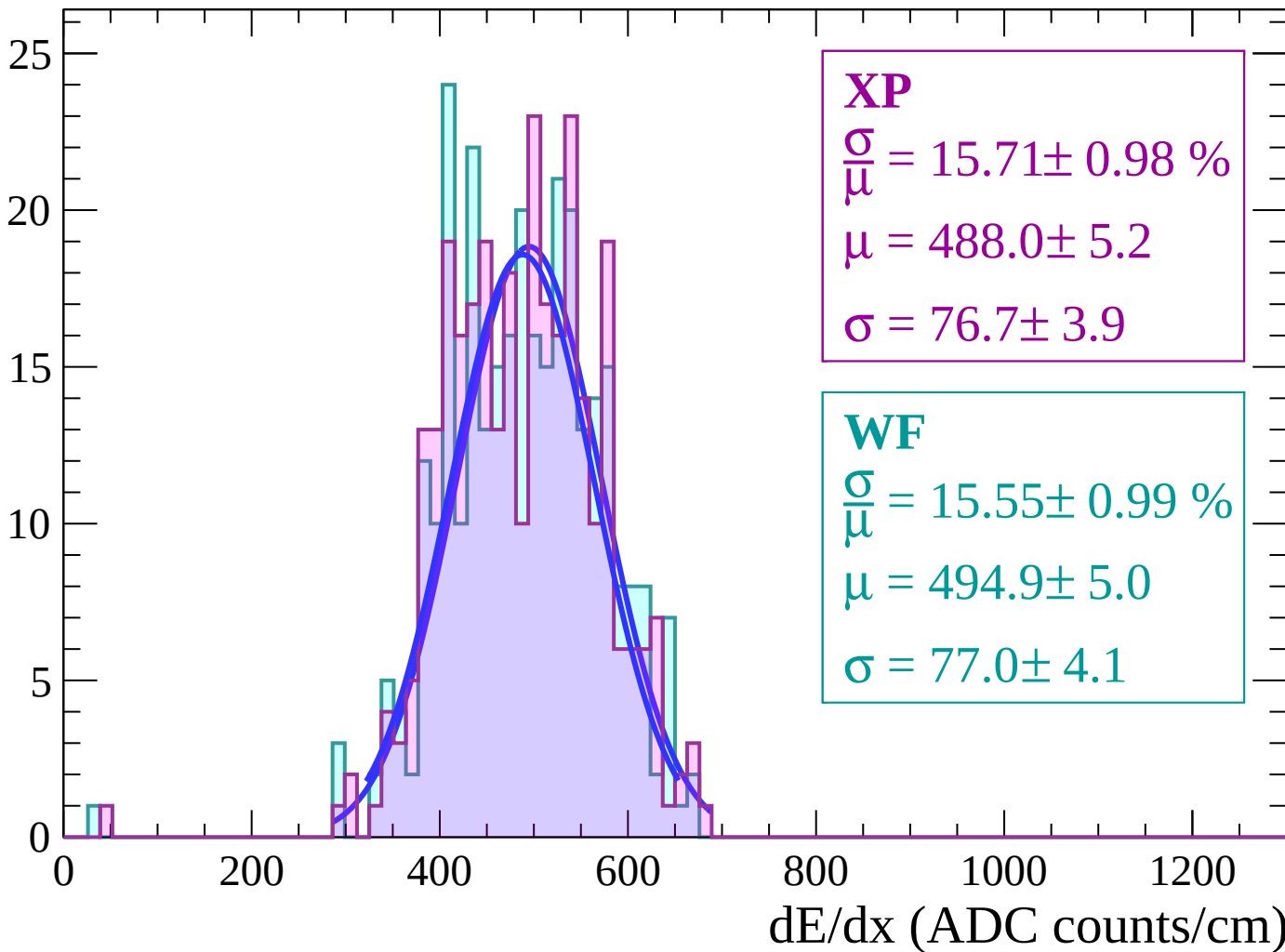
Energy loss | $18 < \theta < 20$

Count



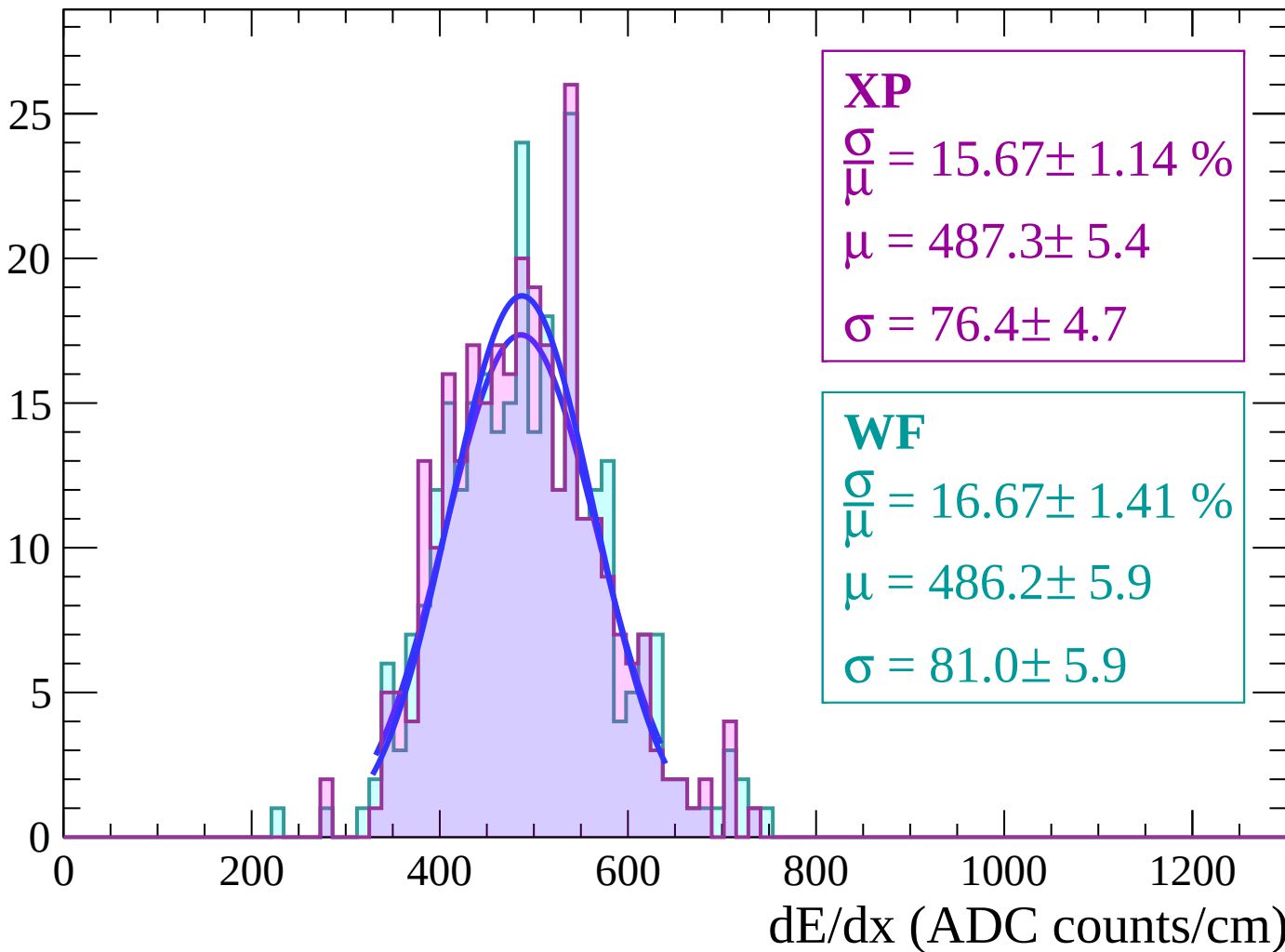
Energy loss | $20 < \theta < 22$

Count



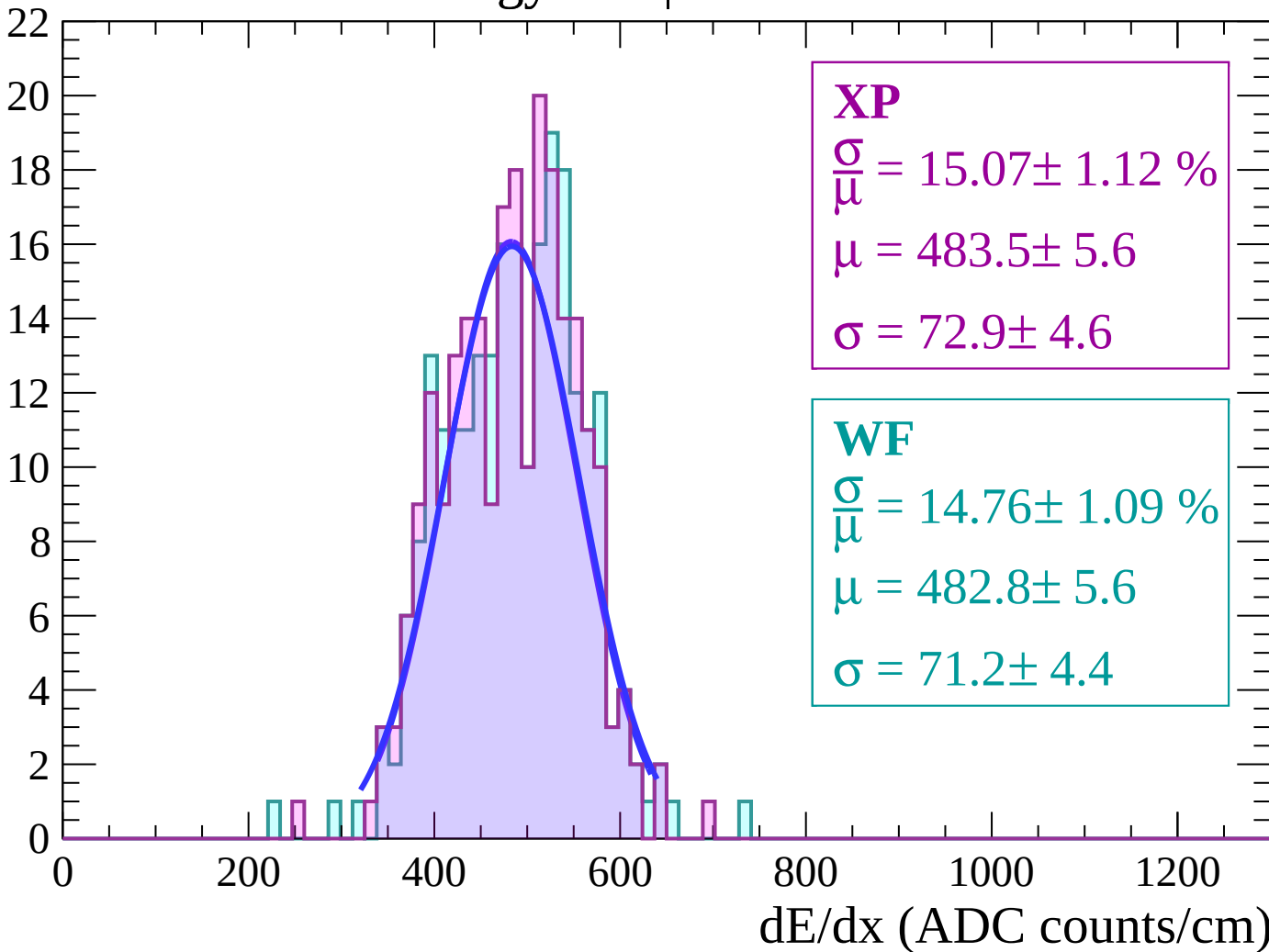
Energy loss | $22 < \theta < 24$

Count



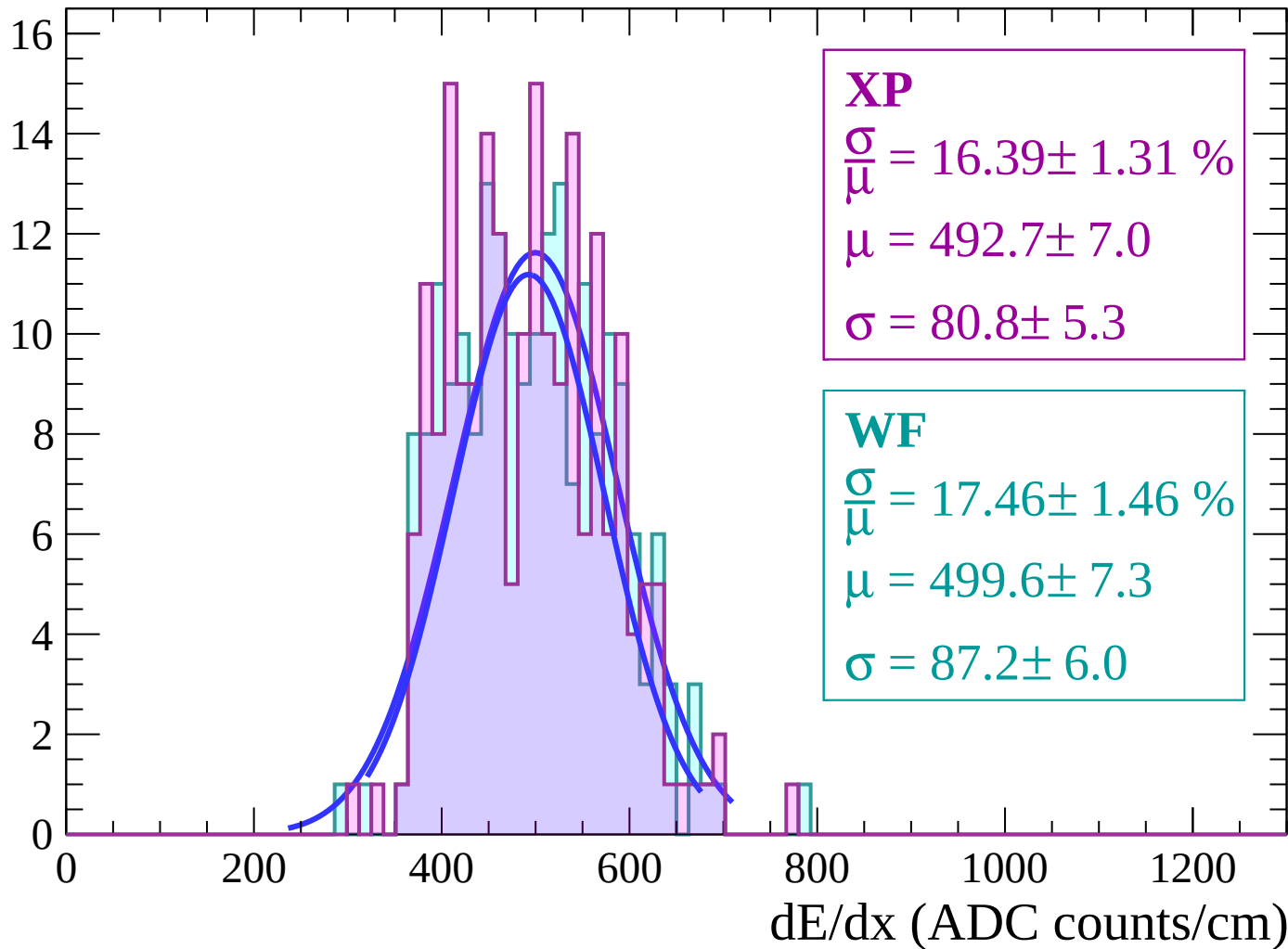
Energy loss | $24 < \theta < 26$

Count



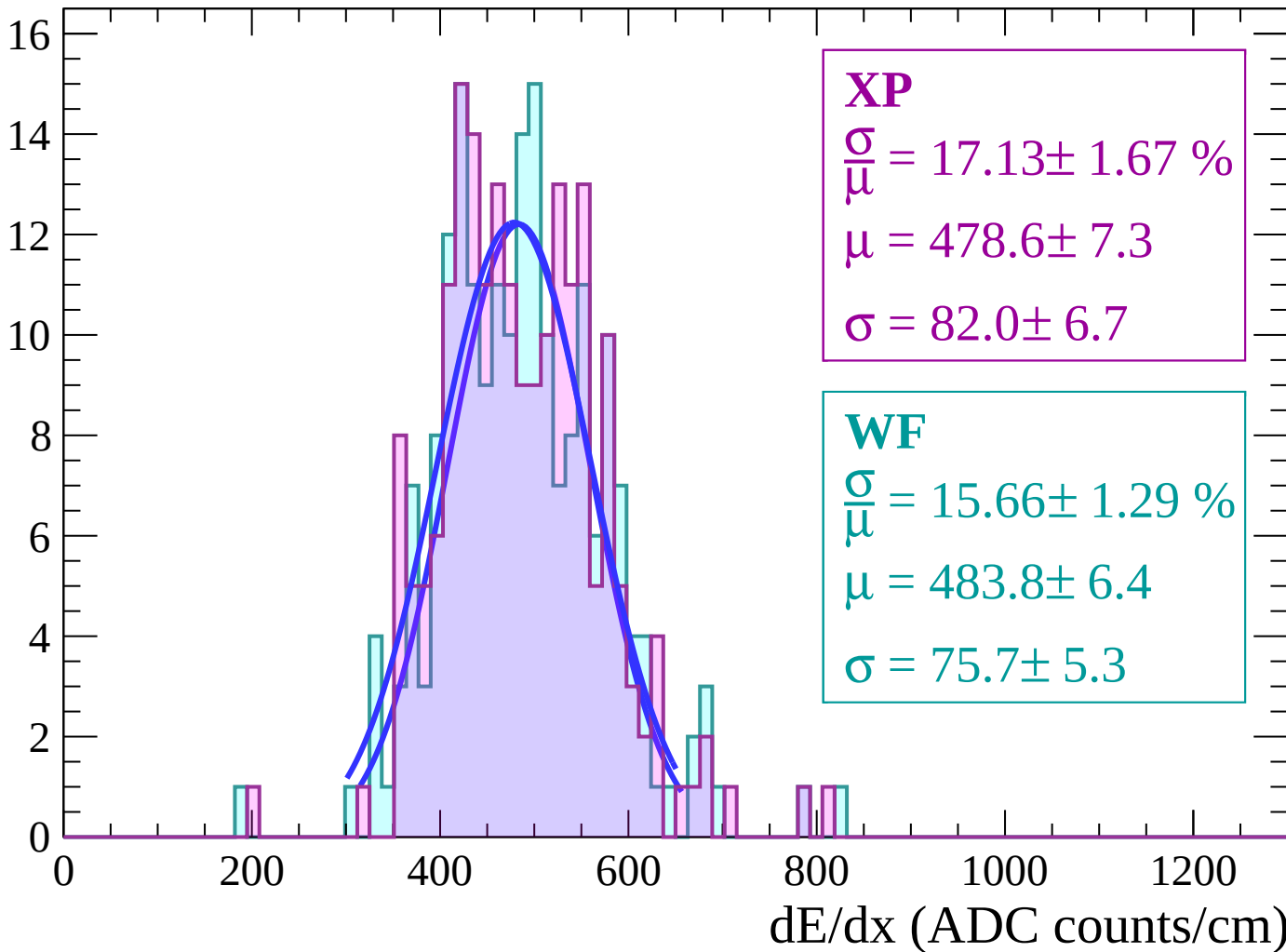
Energy loss | $26 < \theta < 28$

Count



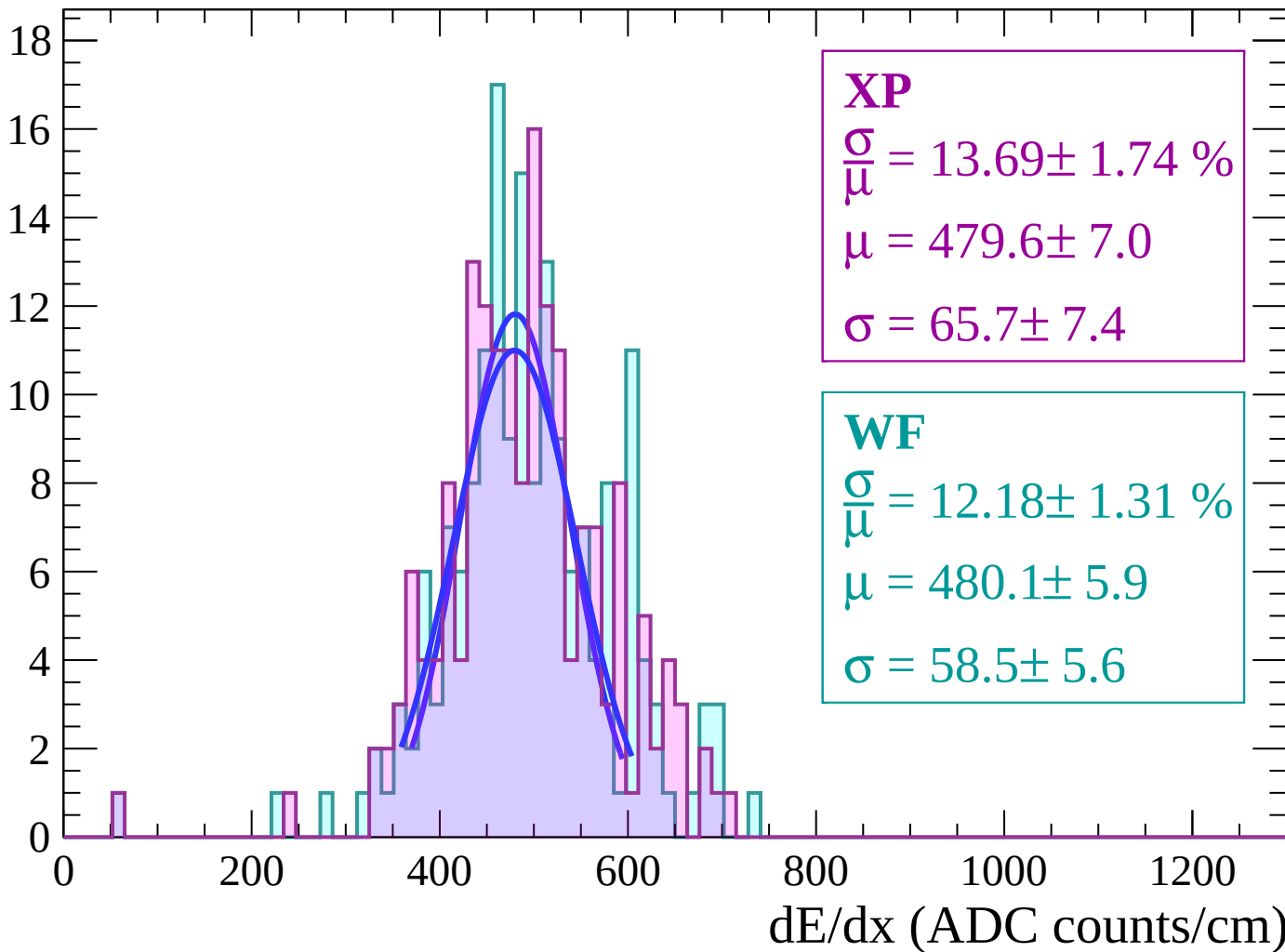
Energy loss | $28 < \theta < 30$

Count



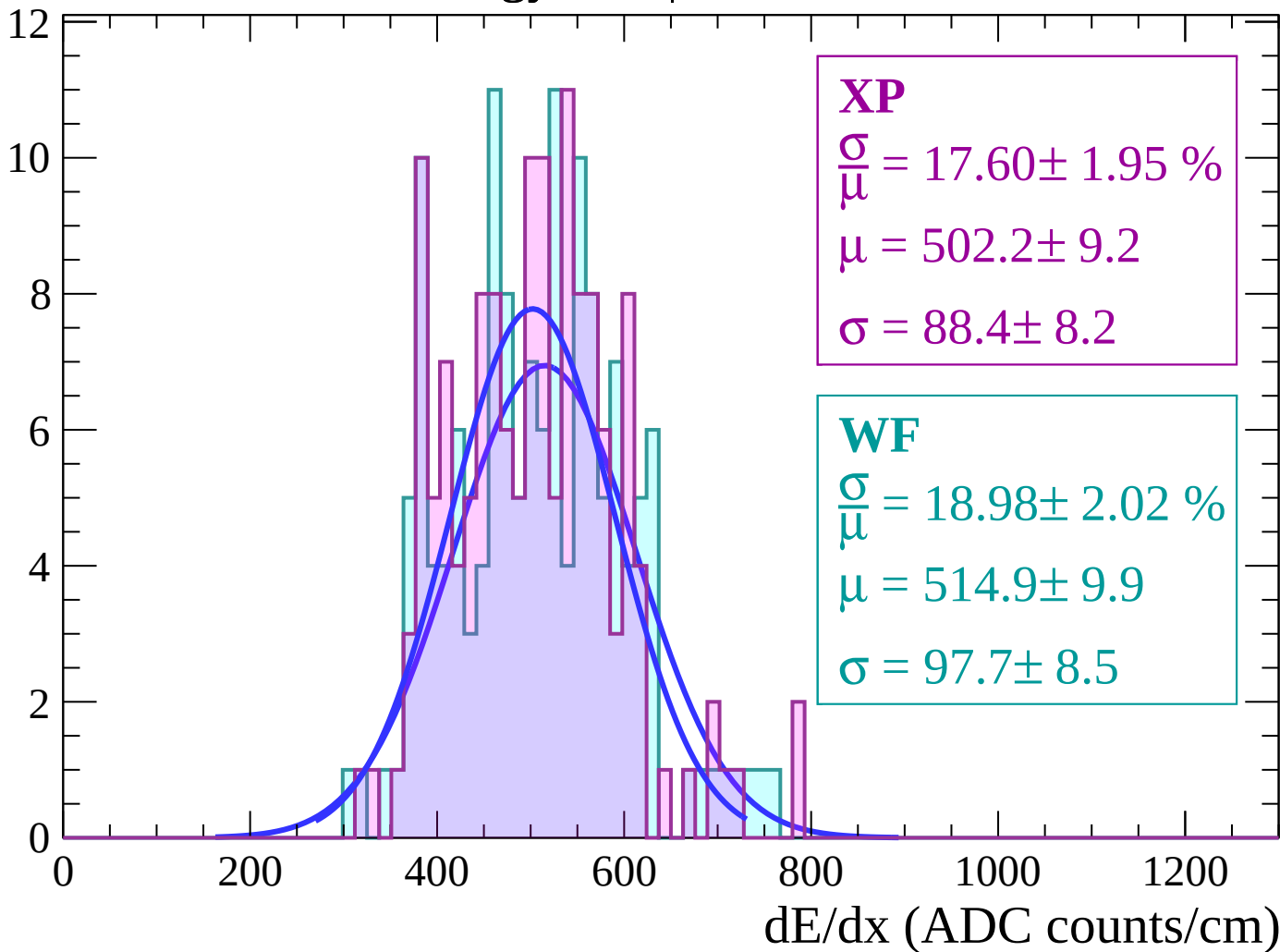
Energy loss | $30 < \theta < 32$

Count



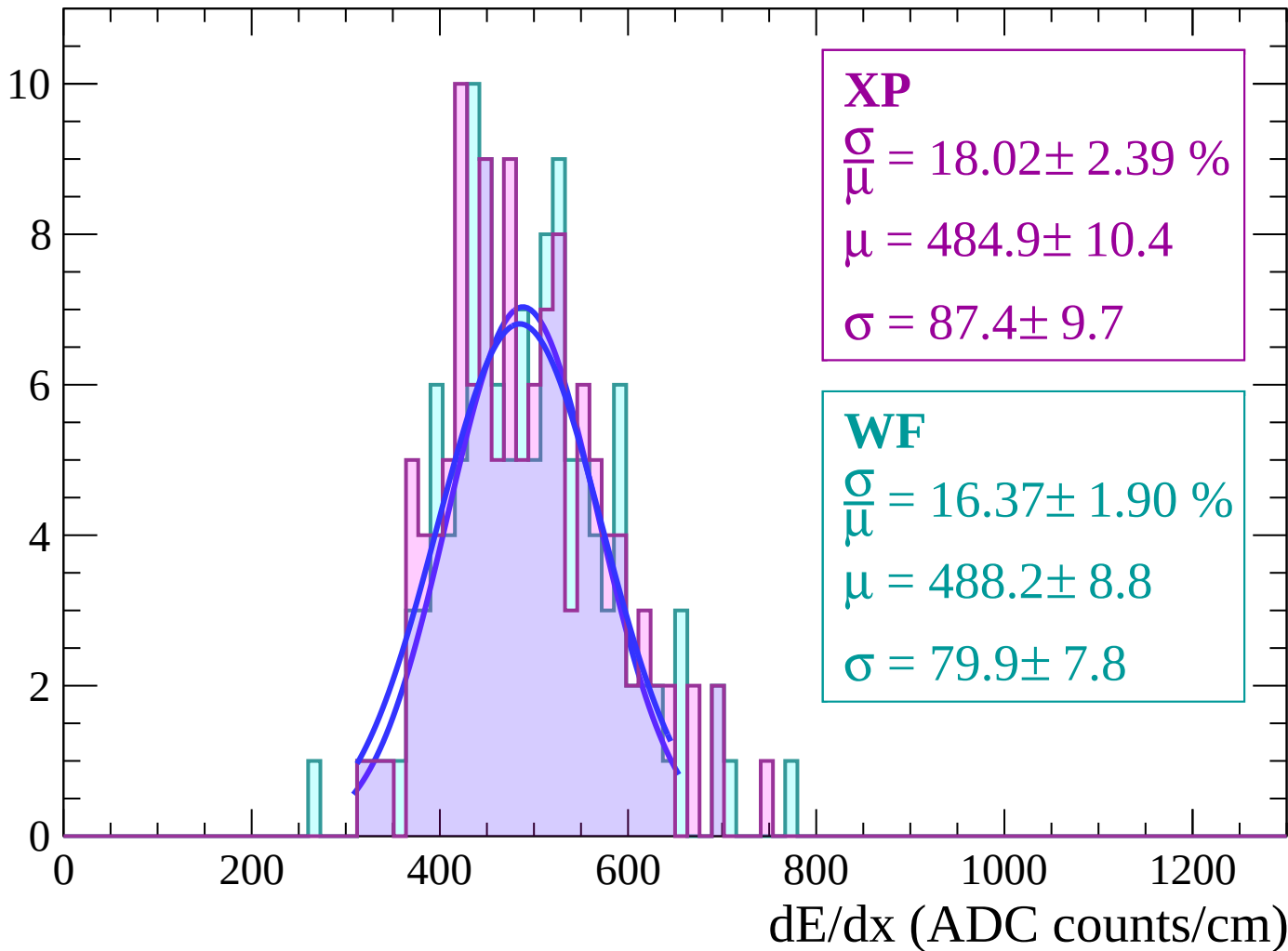
Energy loss | $32 < \theta < 34$

Count



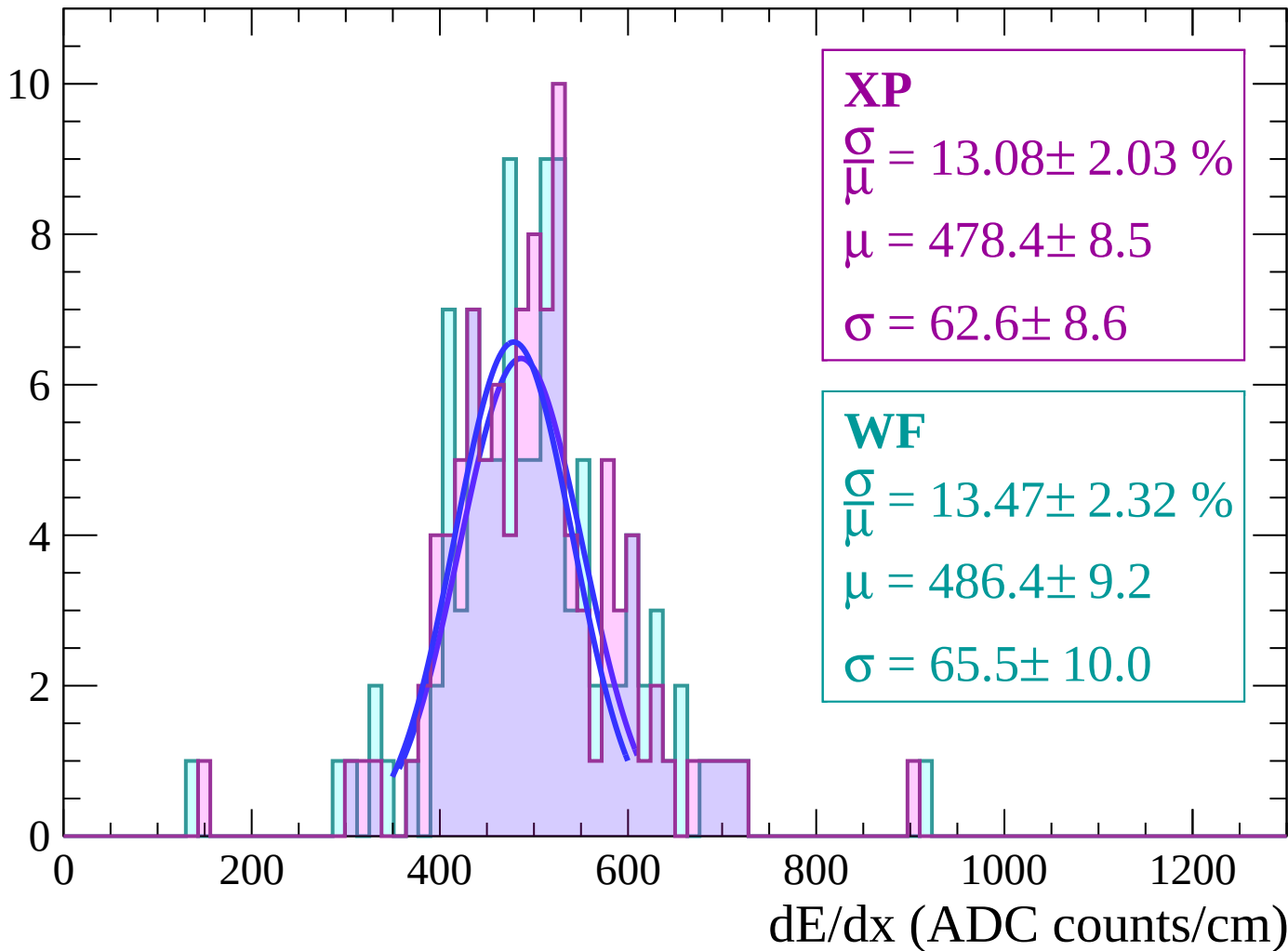
Energy loss | $34 < \theta < 36$

Count



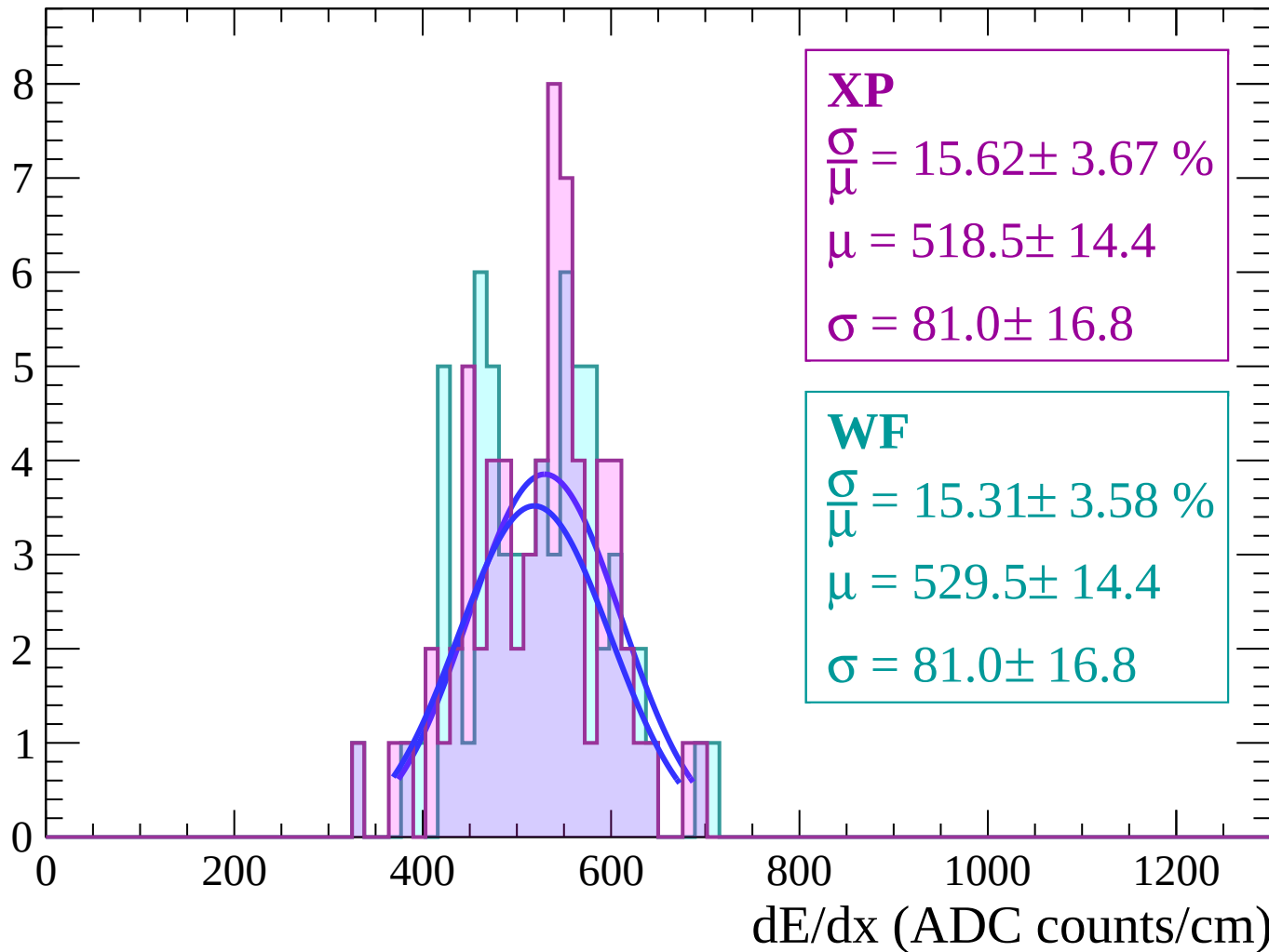
Energy loss | $36 < \theta < 38$

Count



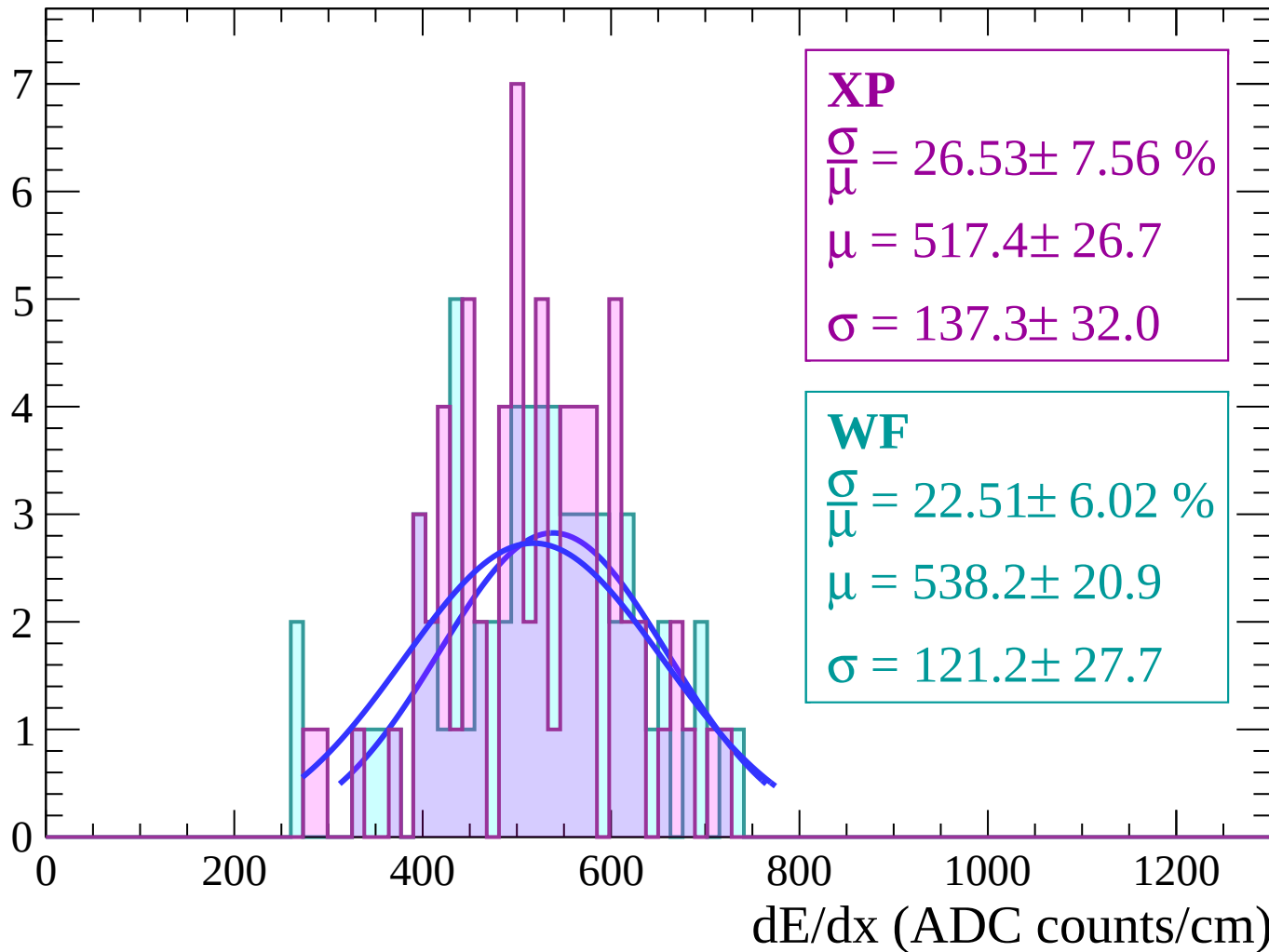
Energy loss | $38 < \theta < 40$

Count



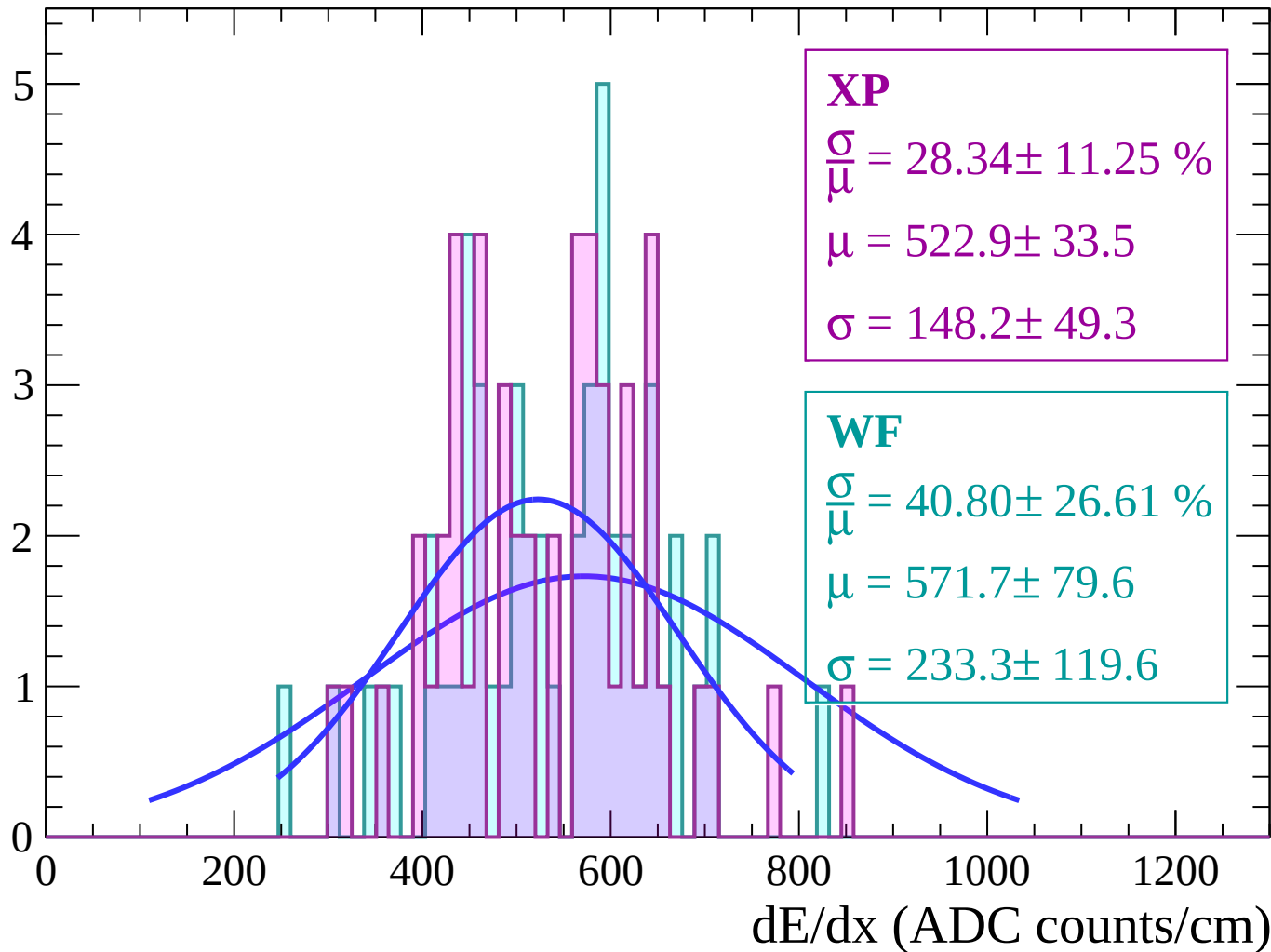
Energy loss | $40 < \theta < 42$

Count



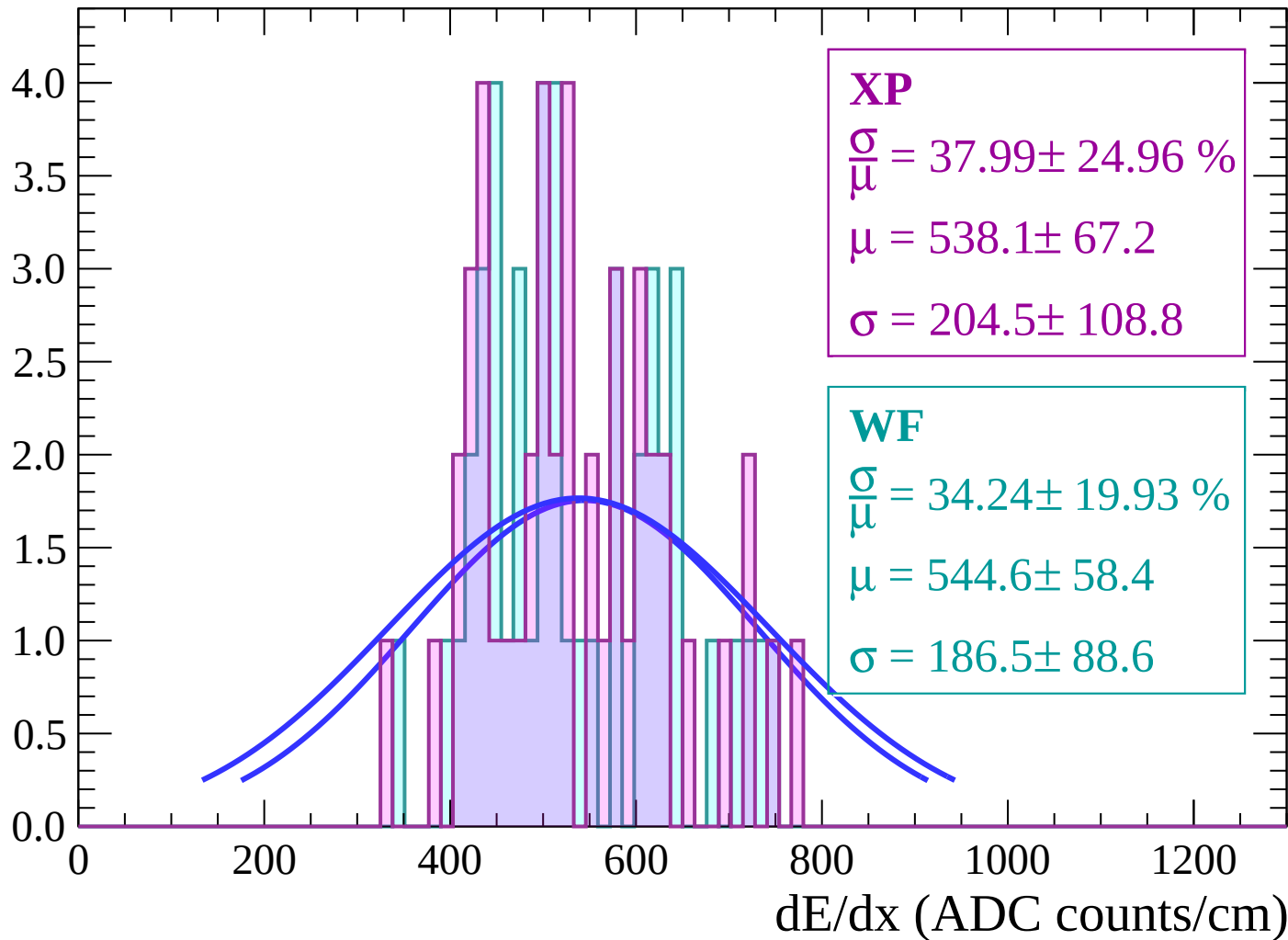
Energy loss | $42 < \theta < 44$

Count



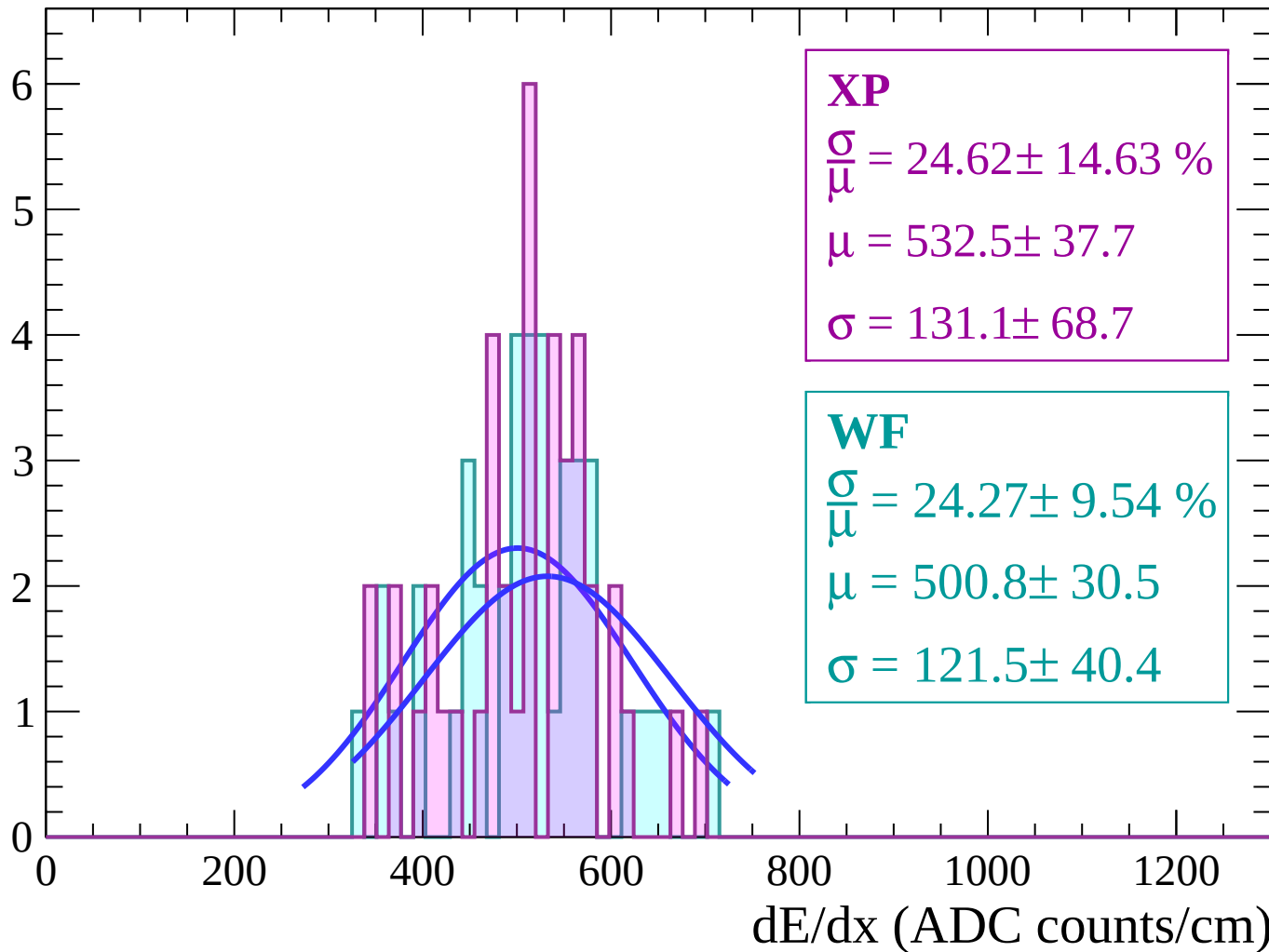
Energy loss | $44 < \theta < 46$

Count



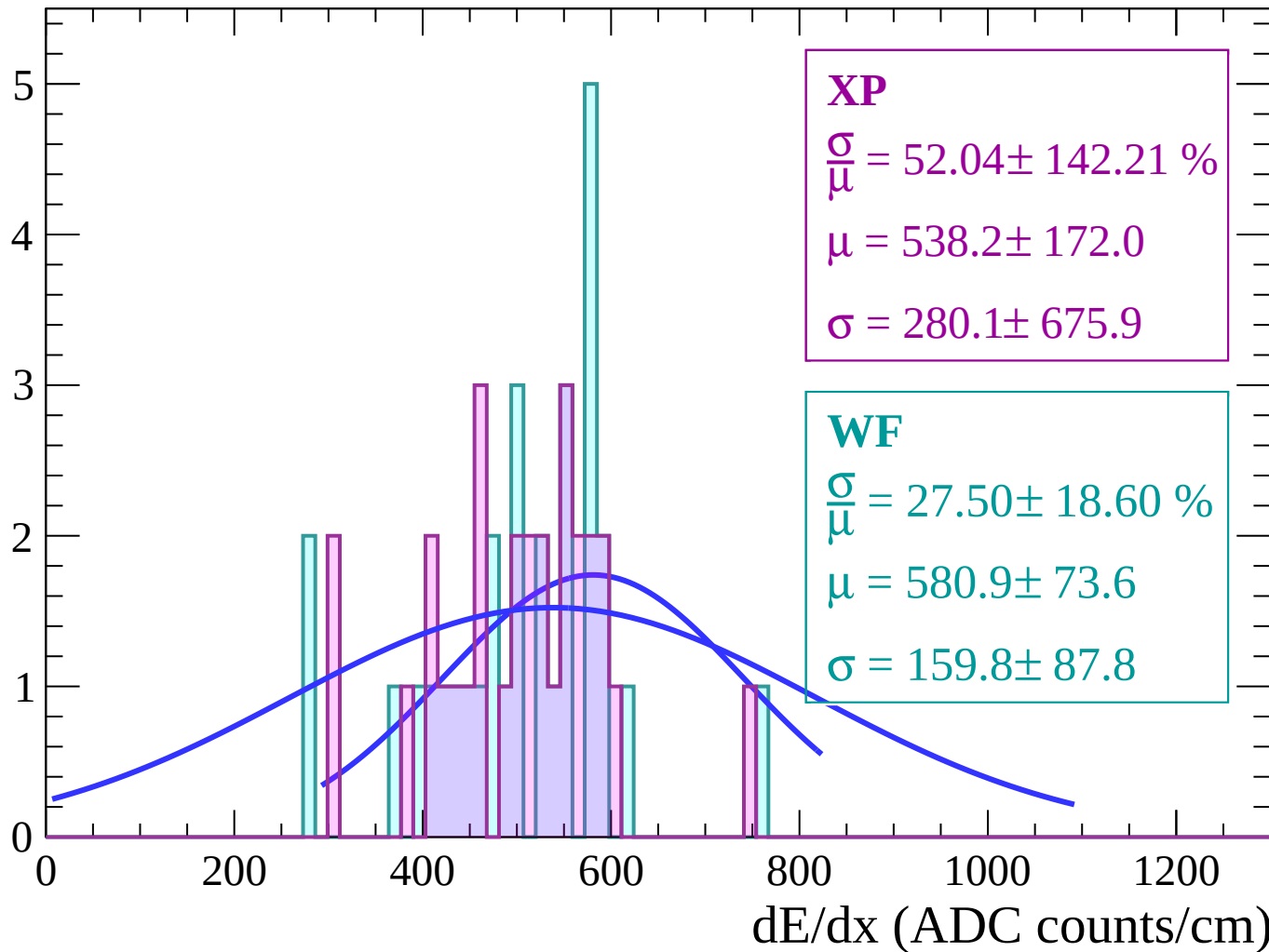
Energy loss | $46 < \theta < 48$

Count



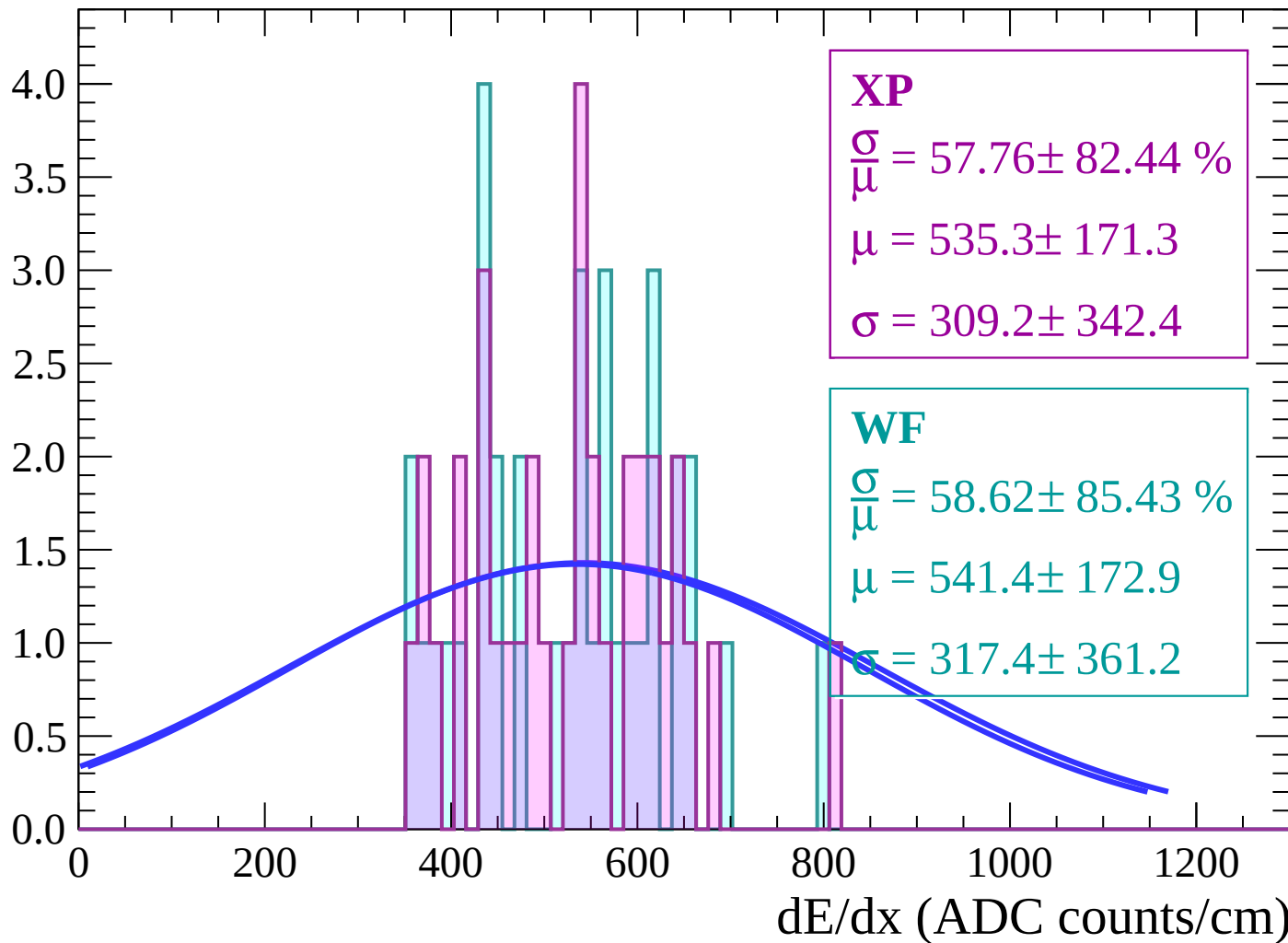
Energy loss | $48 < \theta < 50$

Count



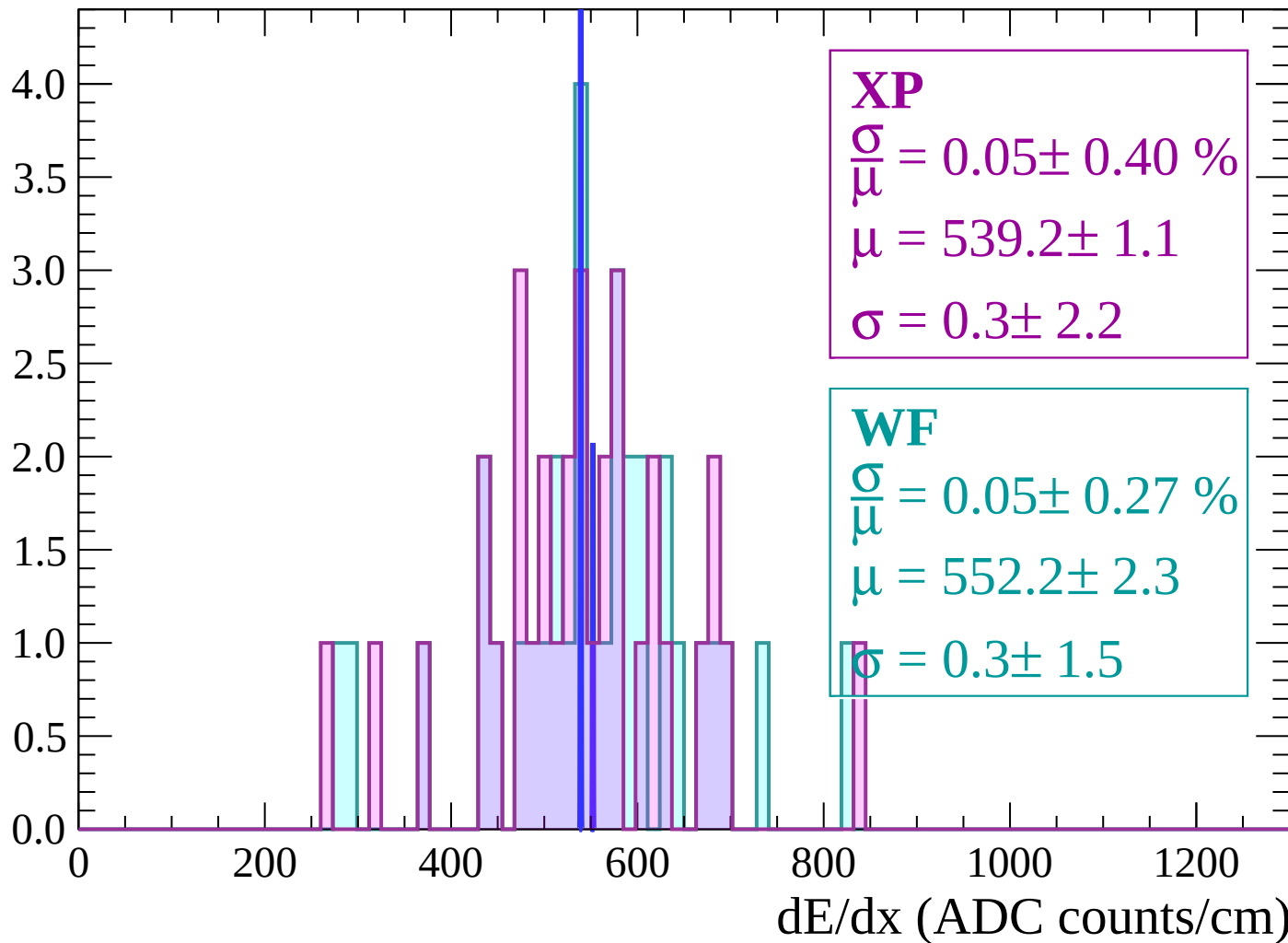
Energy loss | $50 < \theta < 52$

Count



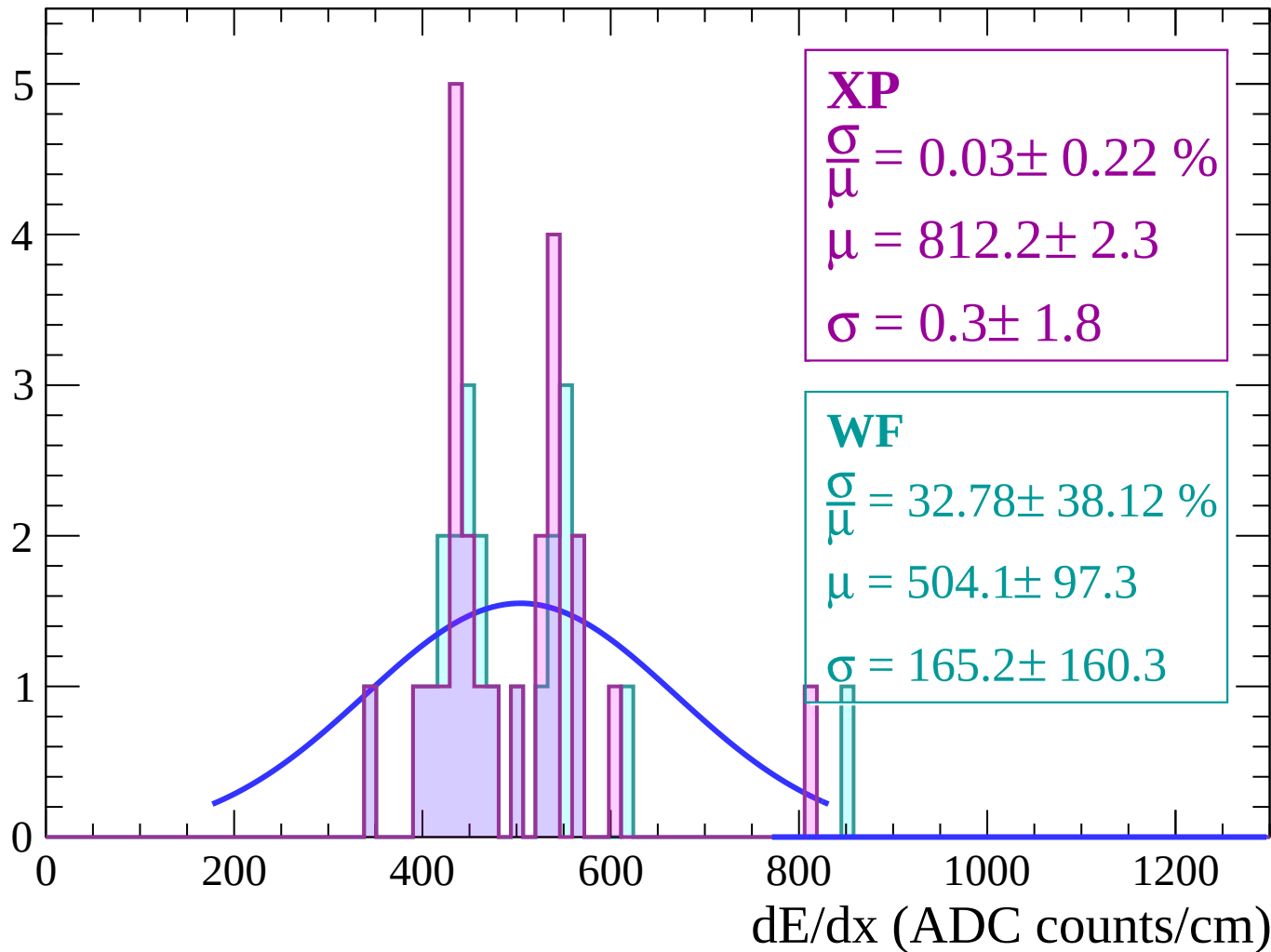
Energy loss | $52 < \theta < 54$

Count



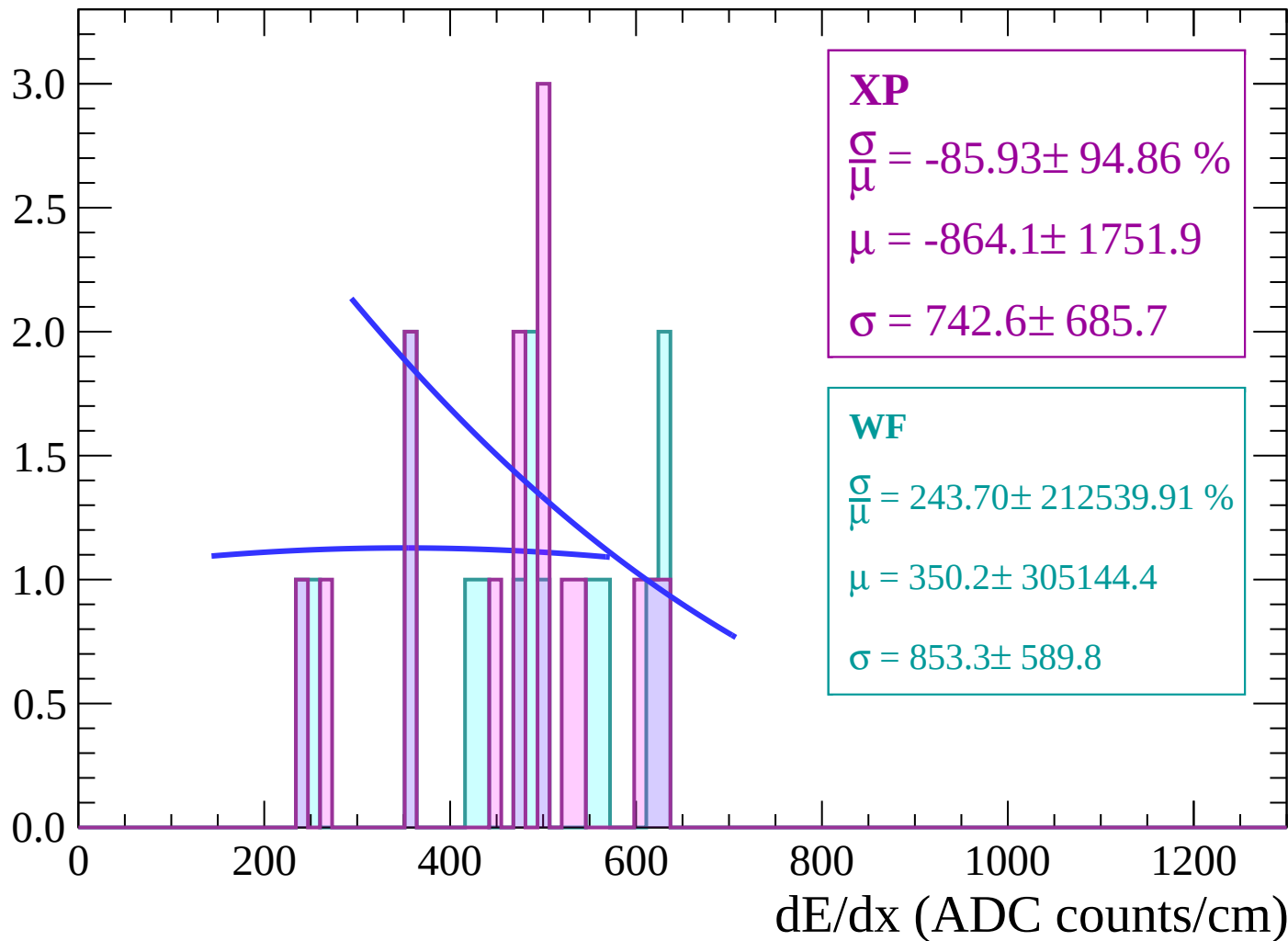
Energy loss | $54 < \theta < 56$

Count



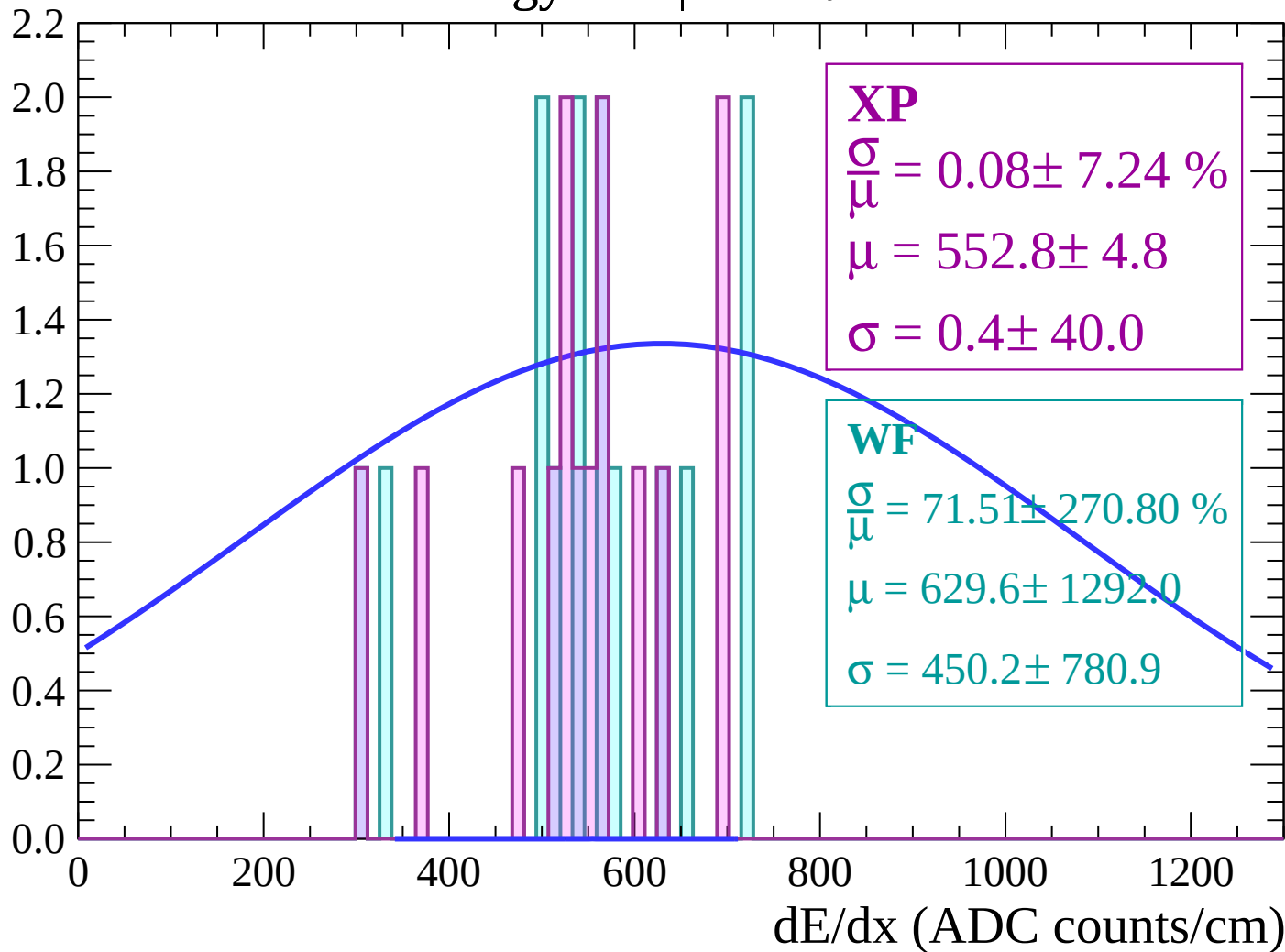
Energy loss | $56 < \theta < 58$

Count



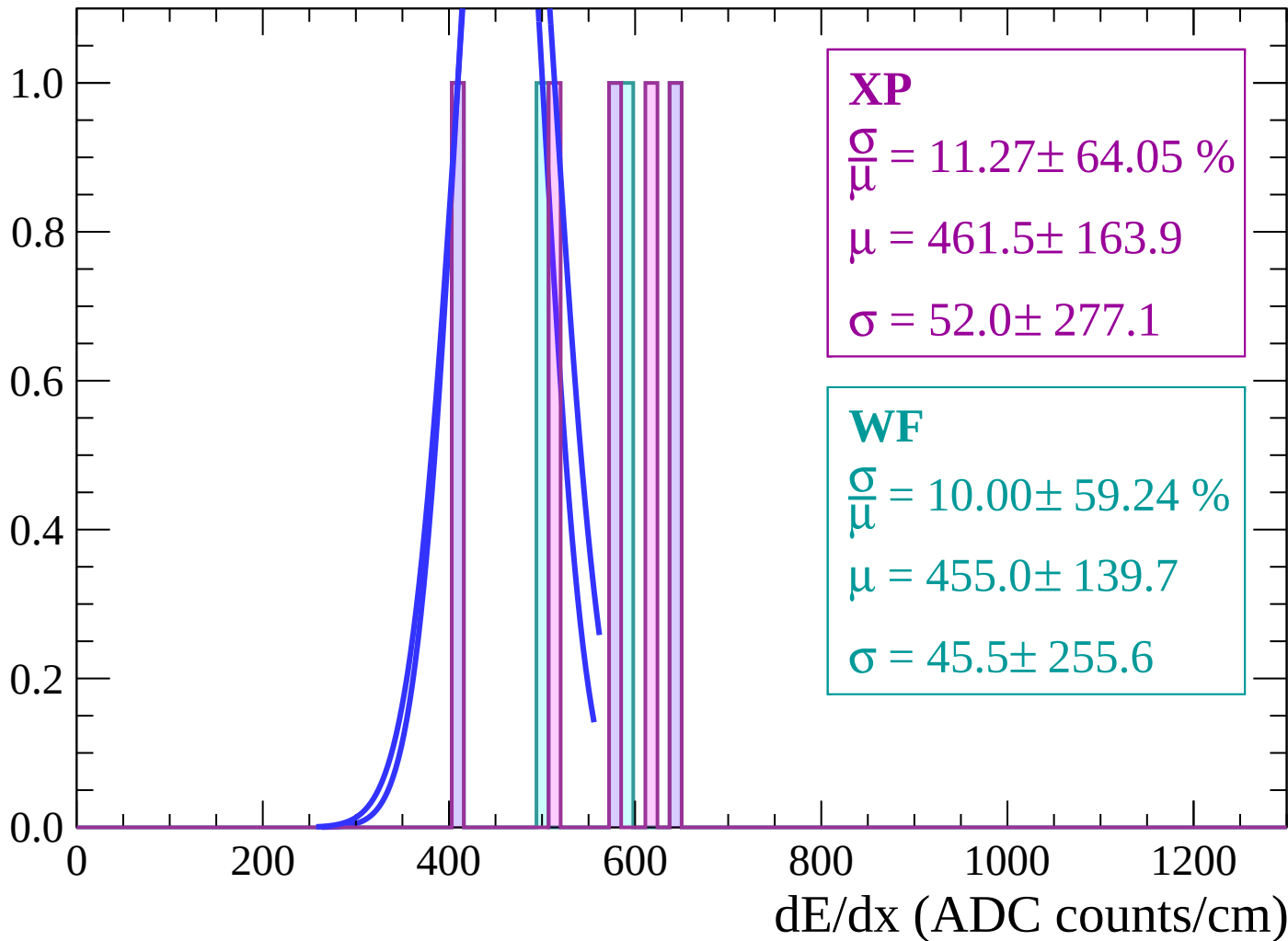
Energy loss | $58 < \theta < 60$

Count



Energy loss | $60 < \theta < 62$

Count



Energy loss | $62 < \theta < 64$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 0.08 \pm 0.55 \%$$

$$\mu = 331.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 0.08 \pm 0.57 \%$$

$$\mu = 318.2 \pm 2.4$$

$$\sigma = 0.3 \pm 1.8$$

Energy loss | $64 < \theta < 66$

Count

1.0

0.8

0.6

0.4

0.2

0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

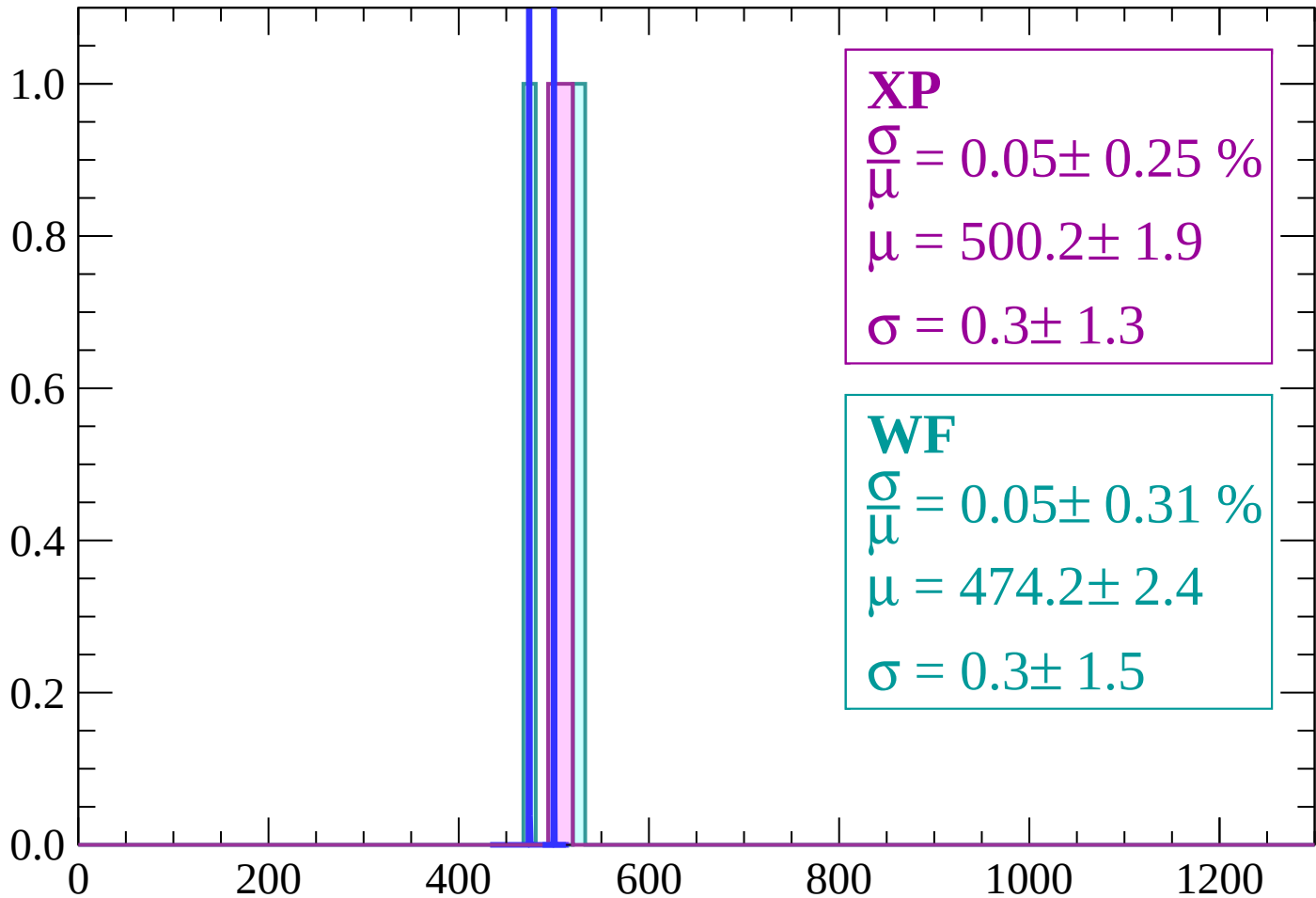
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $66 < \theta < 68$

Count



Energy loss | $68 < \theta < 70$

Count

1.0

0.8

0.6

0.4

0.2

0.0

XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

0

200

400

600

800

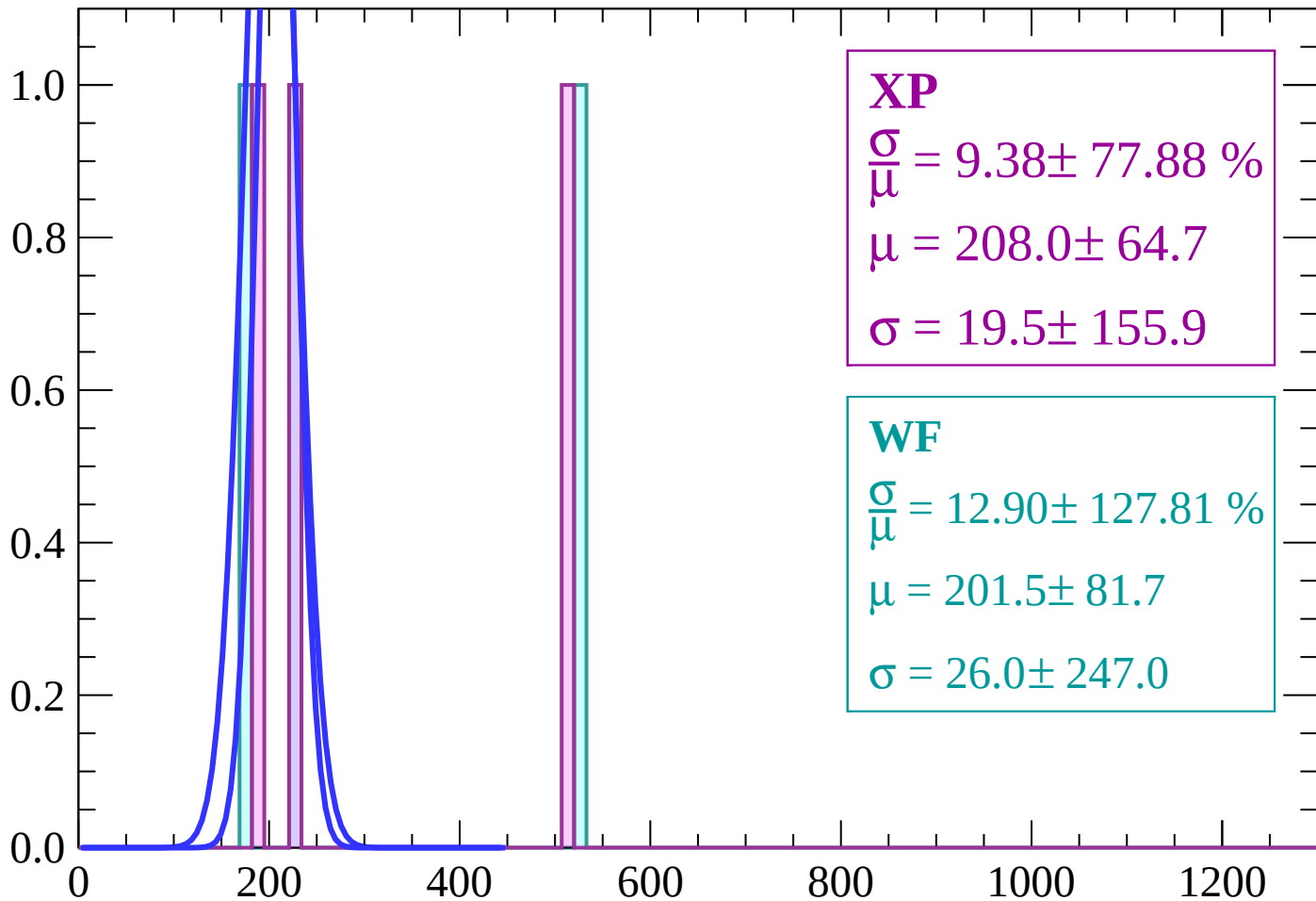
1000

1200

dE/dx (ADC counts/cm)

Energy loss | $70 < \theta < 72$

Count



Energy loss | $72 < \theta < 74$

Count

1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

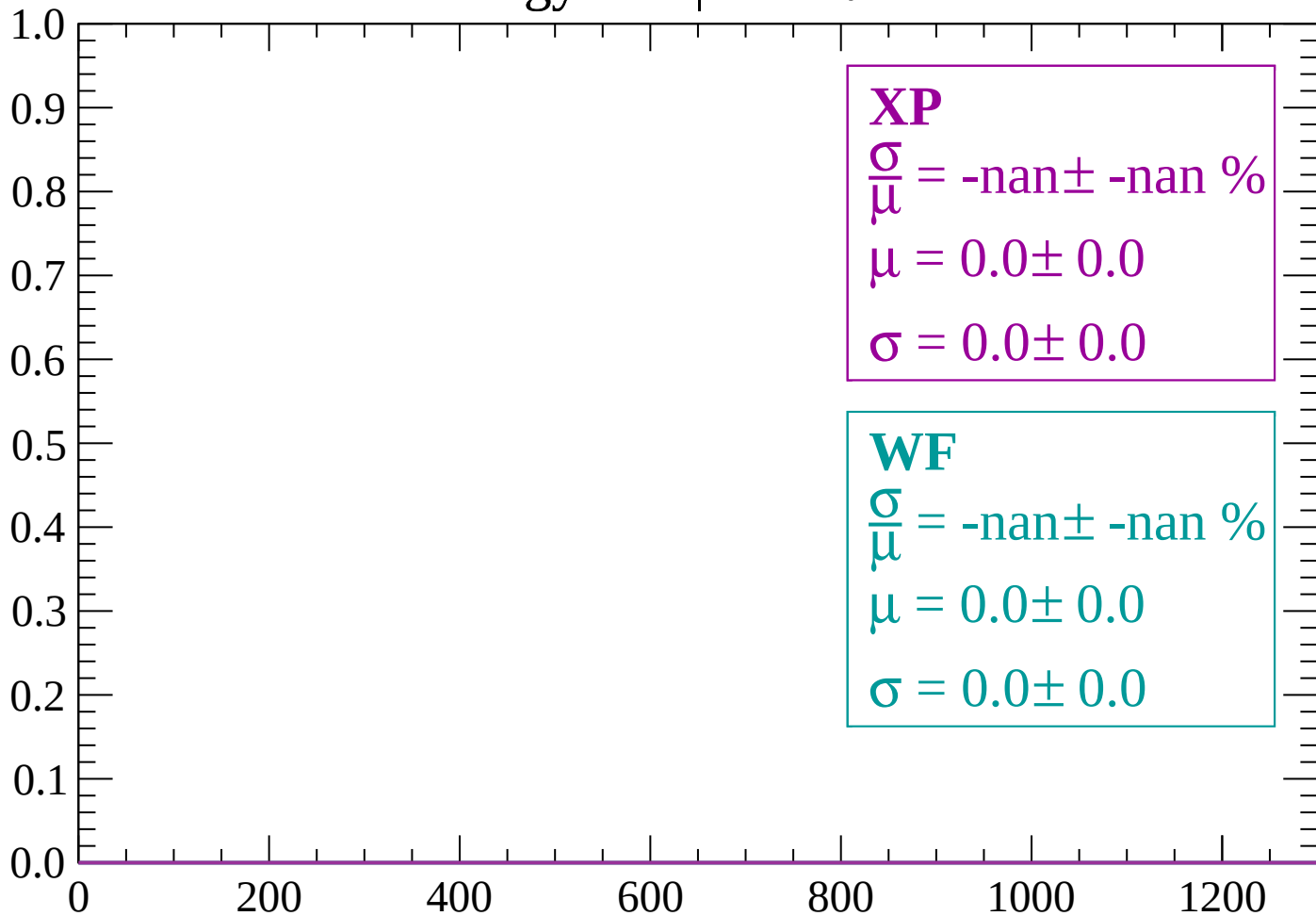
$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

Energy loss | $74 < \theta < 76$

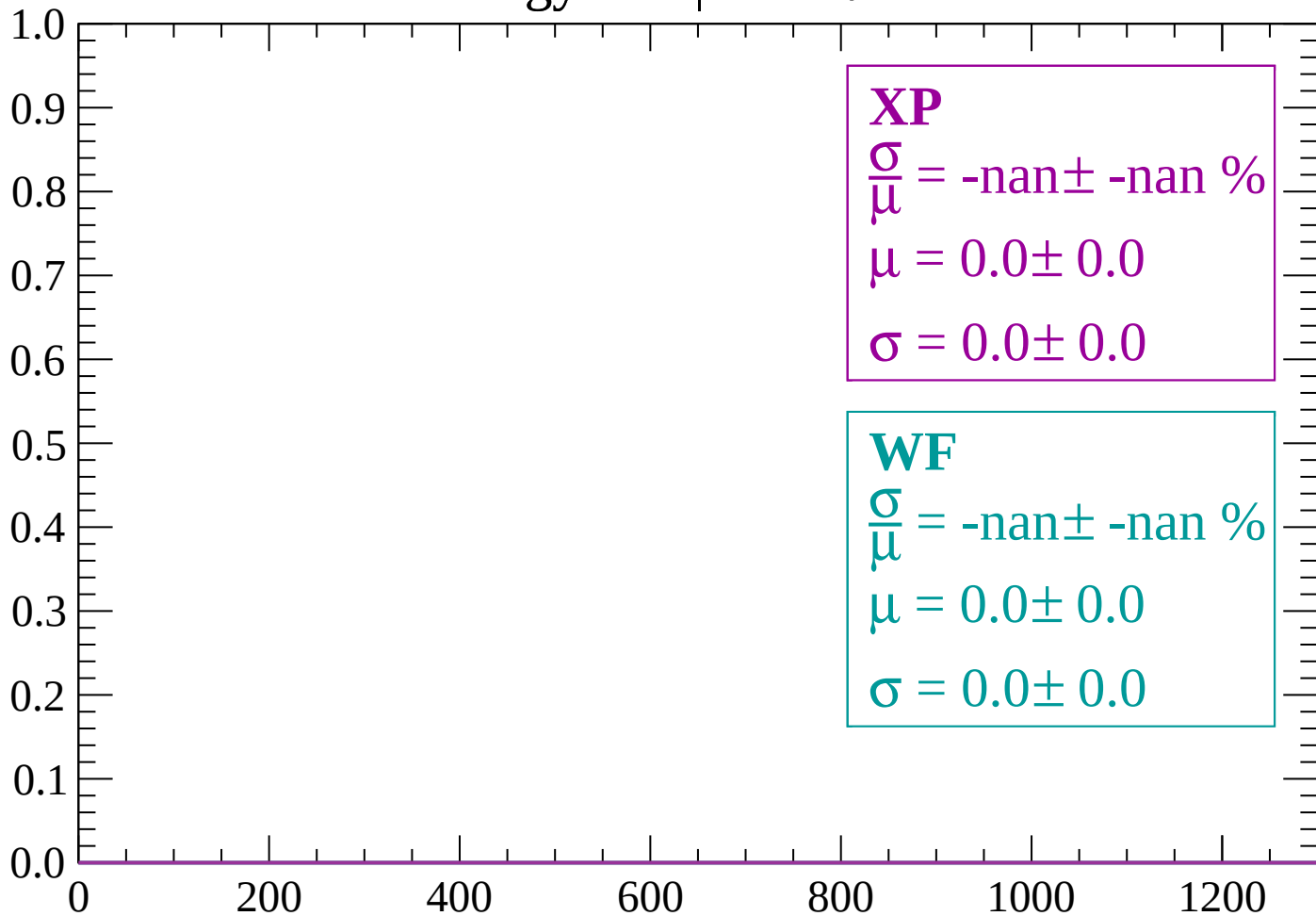
Count



dE/dx (ADC counts/cm)

Energy loss | $76 < \theta < 78$

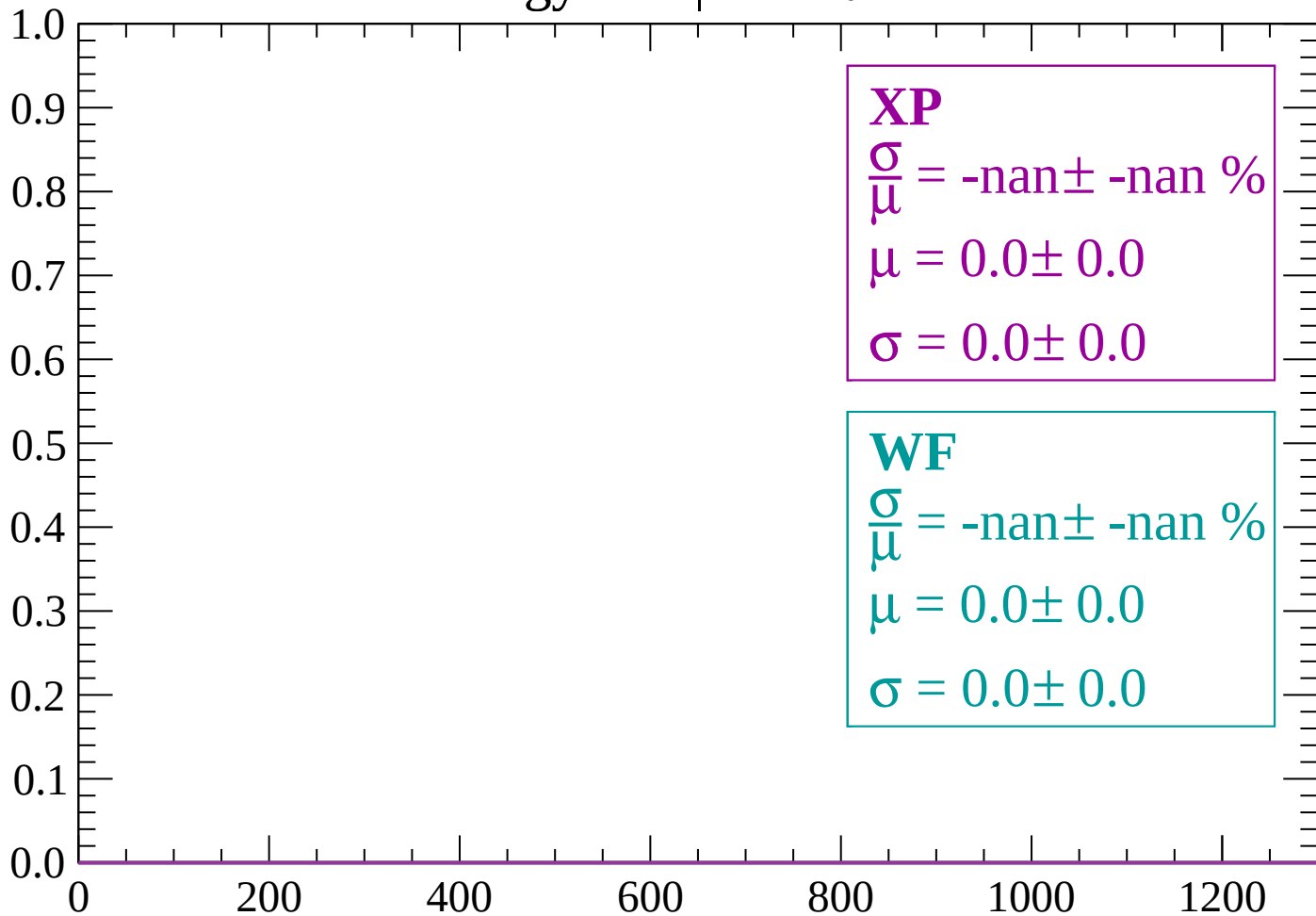
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 80$

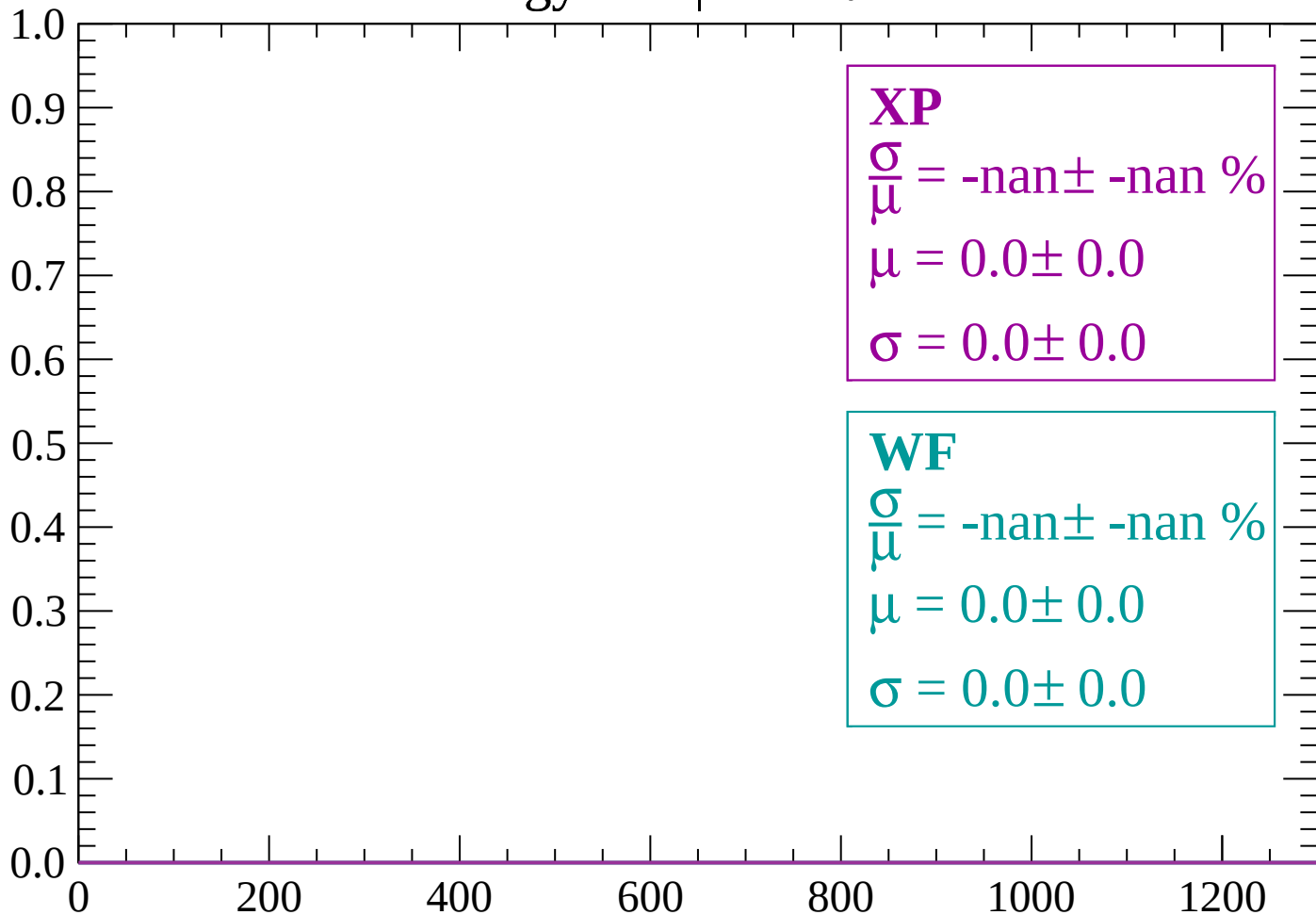
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

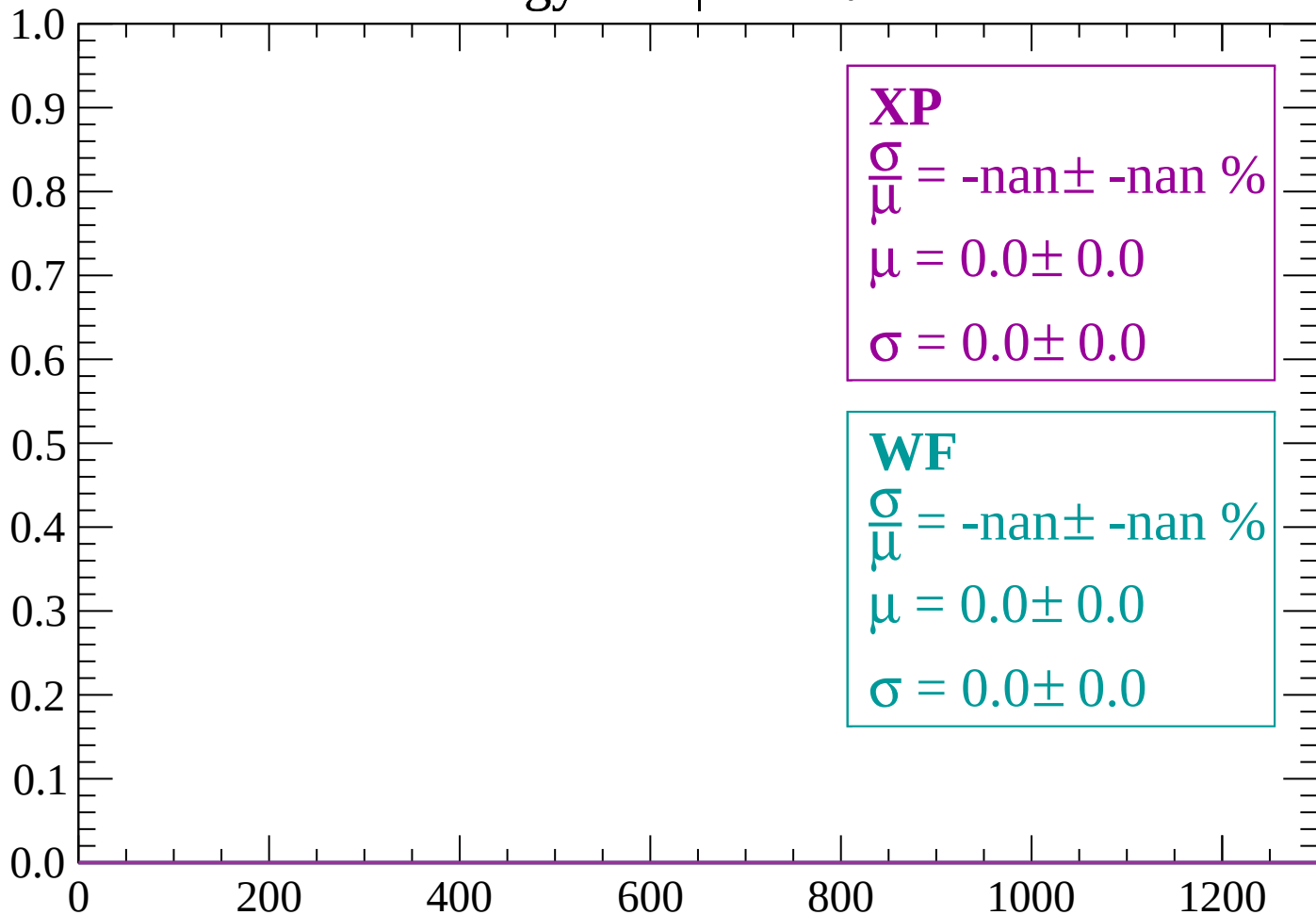
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

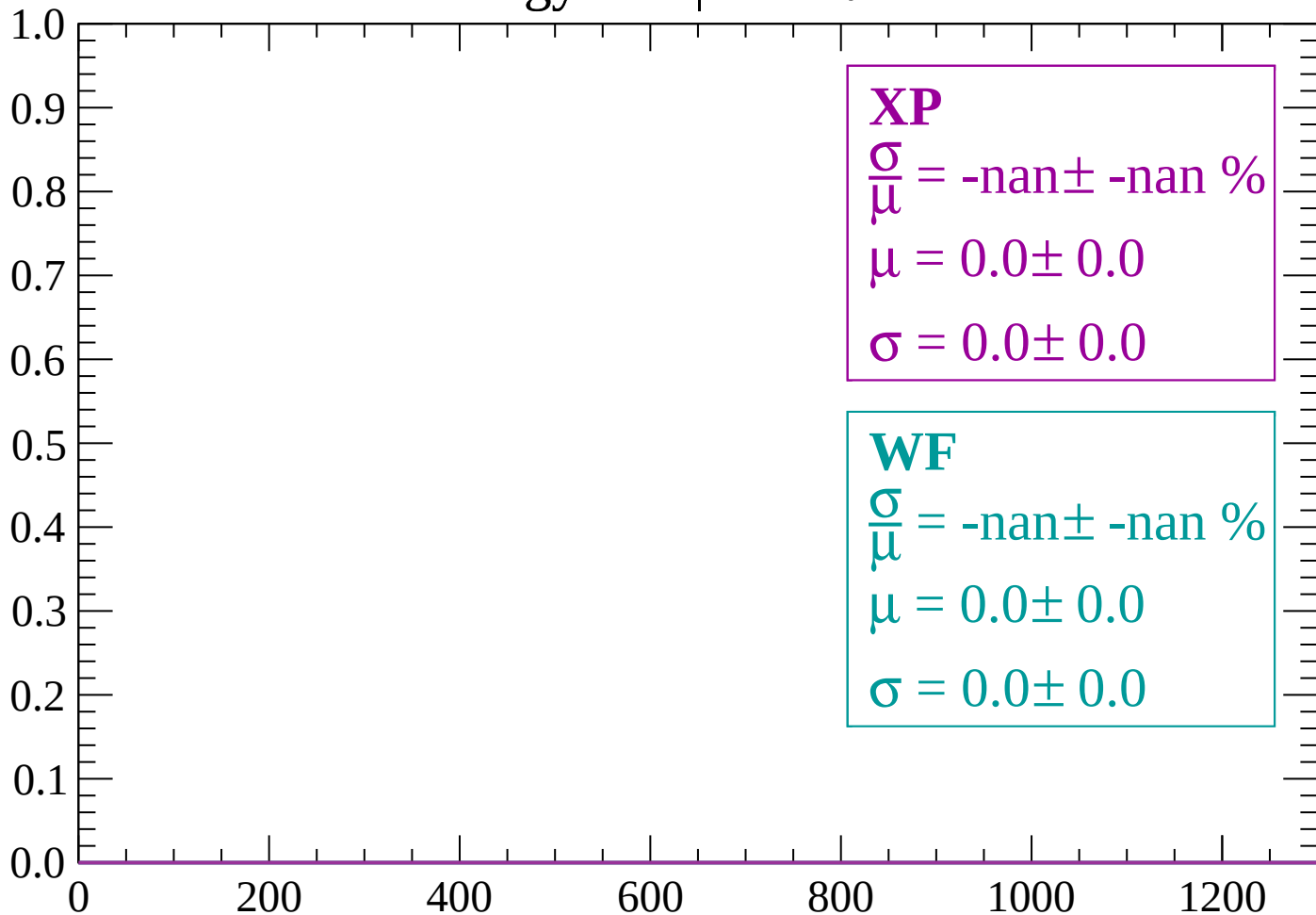
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

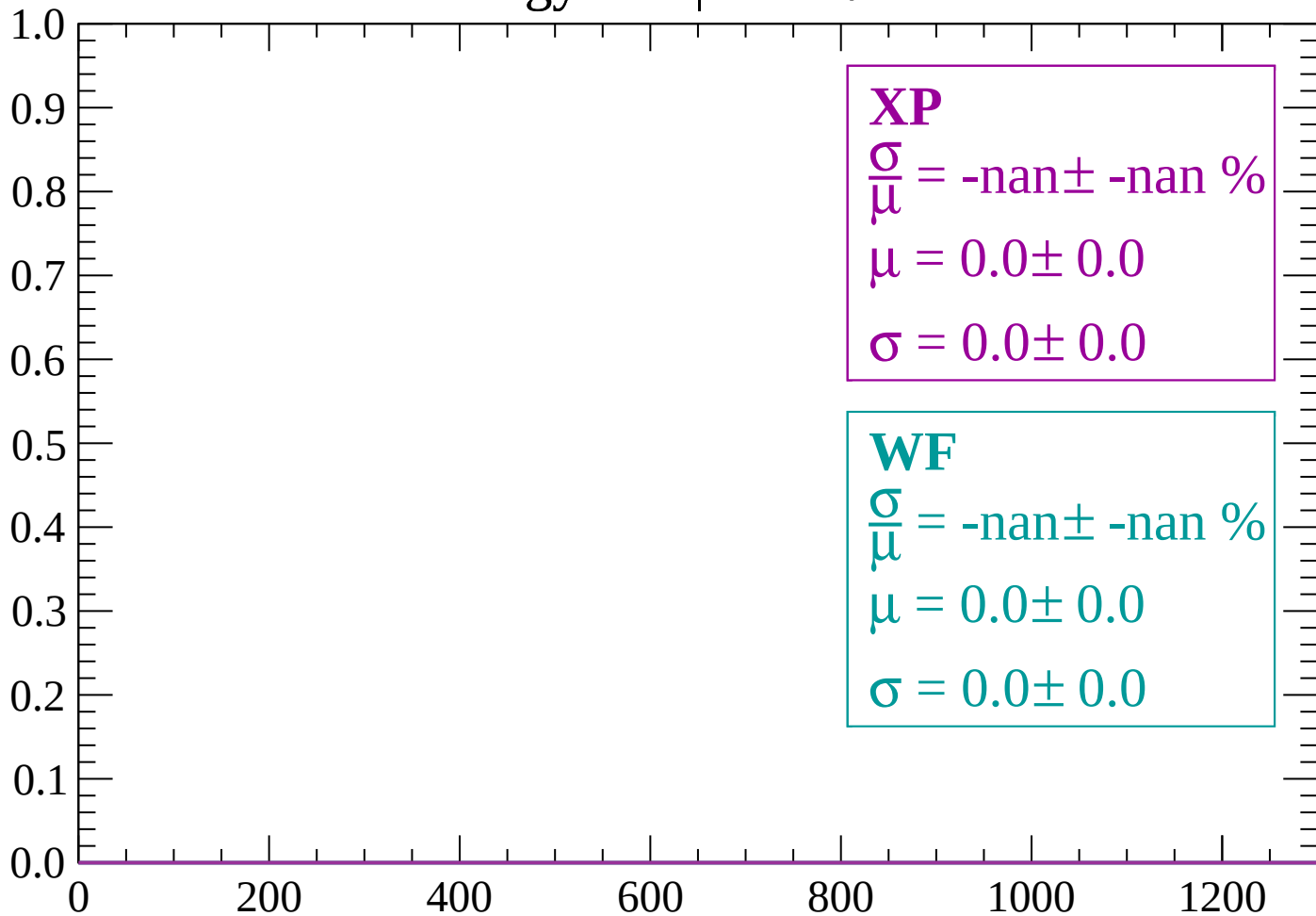
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

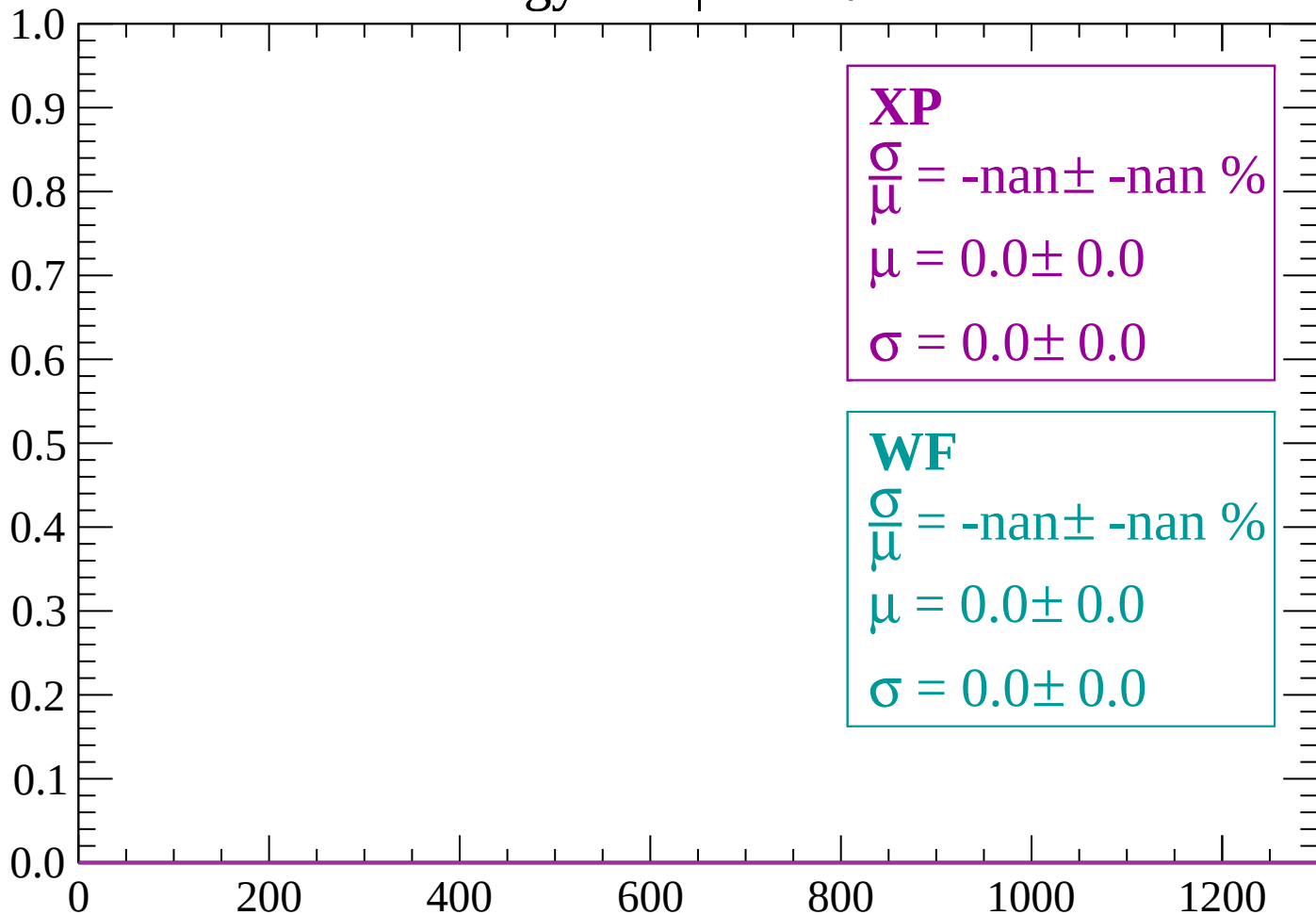
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

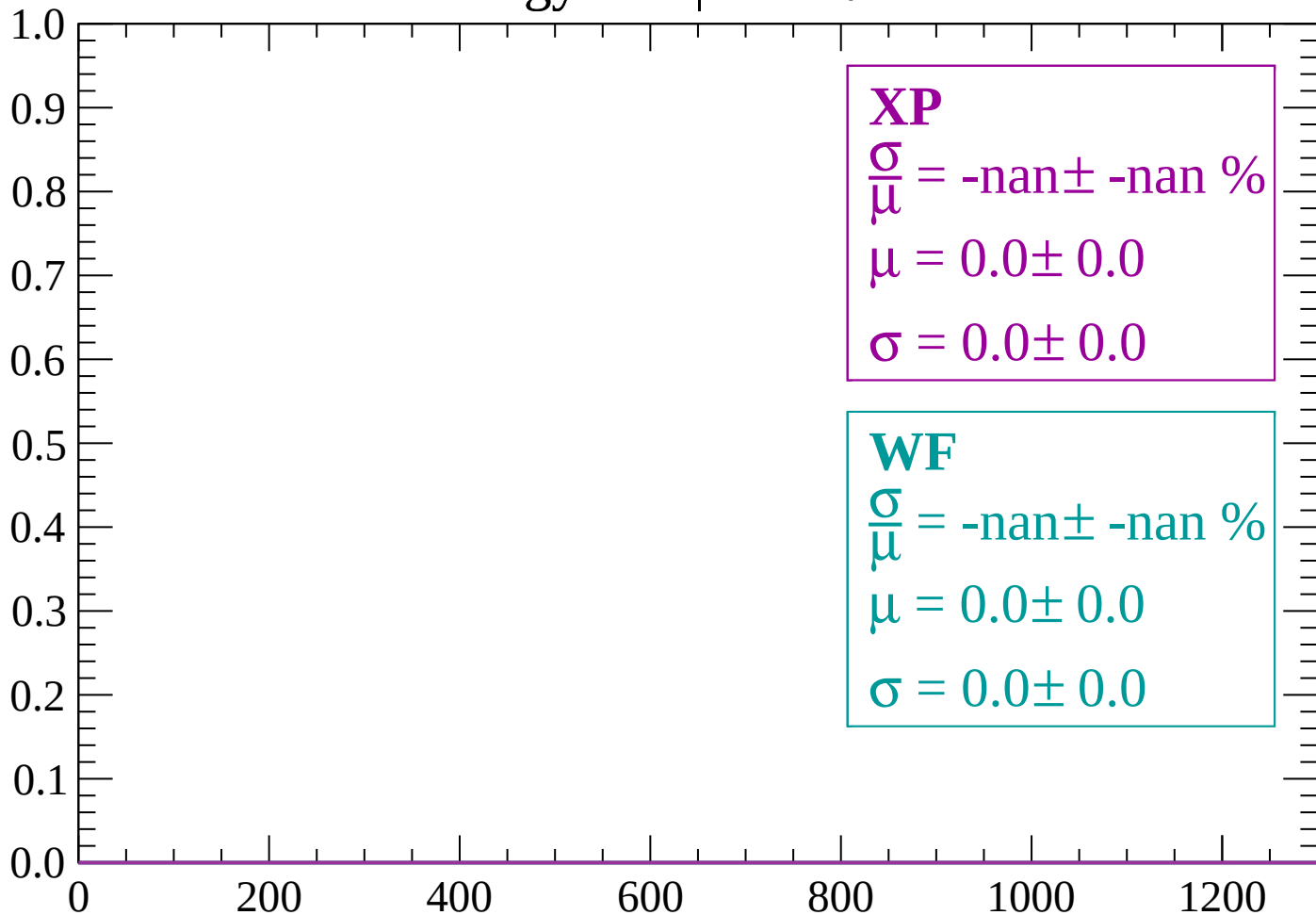
Count



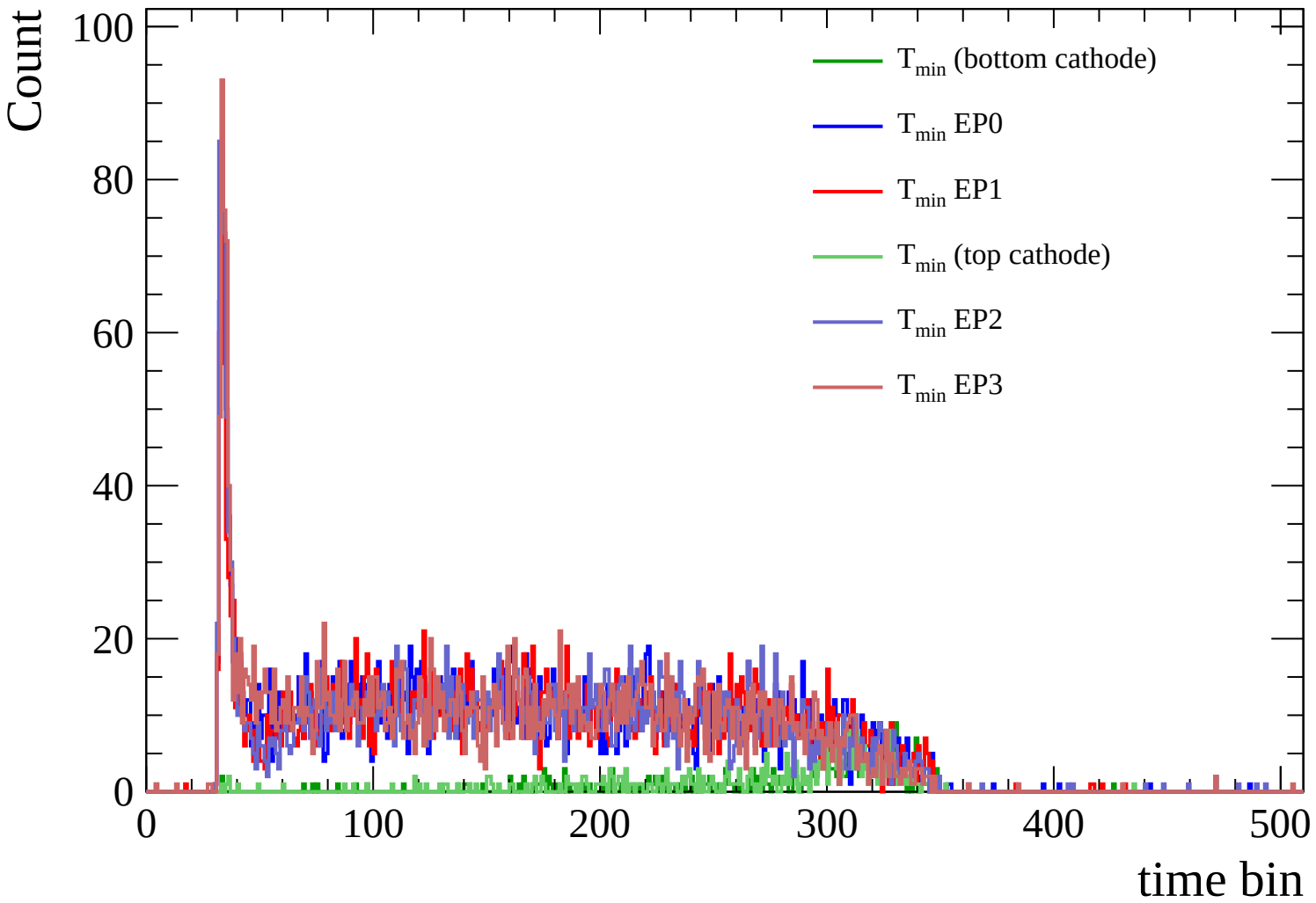
dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)



Count

